

DAT BIOLOGY

2026 EDITION

Everything you
need to know
to get a great
DAT biology score

High-Yield Biology Notes



“How Do I Get a Good Bio Score on the DAT?”

Bootcamp has multiple ways to study bio based on how you learn best. We have High-Yield Bio Notes, Biology Videos, Bio Bites, question banks, and more.

For content review: If you like reading, use these notes. They serve as a concise reference to everything you need to know for the DAT. If you prefer videos, watch the Bio Videos. They cover the same info and incorporate practice questions.

Most importantly, you have to **practice** what you’ve learned. You can practice with the Bio Question Bank and Bio Bites as you progress through biology. The bio practice tests will tie everything together and the performance page will identify areas where you need further studying.

If you’re short on time, **my #1 recommendation** is finishing and reviewing all the **DAT Bootcamp practice tests 1-10** and focusing on learning from the explanations. These are the most representative and high-yield questions you’re likely to see on the DAT. **Be sure to learn everything in the explanations in the practice tests before you take your DAT.**

Lastly, I want DAT Bootcamp to be perfect for you.

If you have any feedback or questions, please email us at team@bootcamp.com! Your feedback is invaluable to improving these notes for future generations of students.

Happy studying! 😊

-Dr. Ari and the DAT Bootcamp team

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Ch. 1: Molecules and Fundamentals of Biology

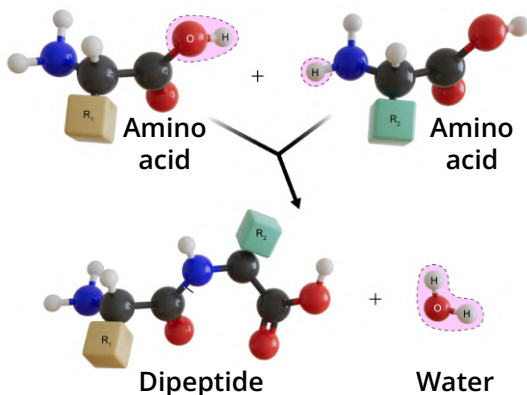
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Biological Chemistry

- **Matter:** Anything that takes up space and has mass.
- **Element:** A pure substance that has specific physical/chemical properties and can't be broken down into a simpler substance.
- **Atom:** The smallest unit of matter that still retains the chemical properties of the element.
- **Molecule:** Two or more atoms joined together.
- **Intramolecular forces:** Attractive forces that act on atoms *within* a molecule.
- **Intermolecular forces:** Forces that exist *between* molecules and affect physical properties of the substance.
- **Monomers:** Single molecules that can **polymerize**, or bond with one another.
- **Polymers:** Substances made up of many monomers joined together in chains.
- **Dehydration (condensation) reaction:** Creates a covalent bond between monomers and releases water.
- **Hydrolysis:** A reaction that breaks a covalent bond by using water.

Dehydration Reaction: Proteins



Carbohydrates

Carbohydrates are used as fuel and structural support. They contain carbon, hydrogen, and oxygen atoms (CHO). They can come in the form of monosaccharides, disaccharides, and polysaccharides.

Monosaccharides are carbohydrate monomers.

- **Ribose:** A five-carbon monosaccharide.
- **Fructose:** A six-carbon monosaccharide.
- **Glucose:** A six-carbon monosaccharide.

Glucose and fructose are **isomers** of each other (same chemical formula, different arrangement of atoms).

Disaccharides consist of two monosaccharides joined by a **glycosidic bond**. It is the result of a **dehydration (condensation)** reaction. Common examples of disaccharides include sucrose, lactose, and maltose.

Polysaccharides contain multiple monosaccharides connected by glycosidic bonds to form long polymers.

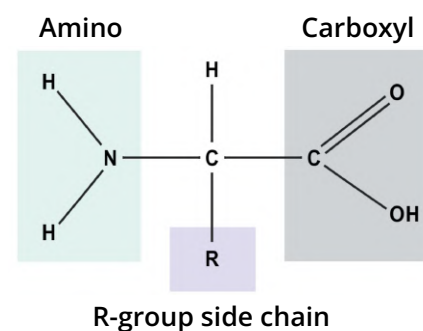
- **Starch:** Form of energy storage for plants.
- **Glycogen:** Form of energy storage in animals.
- **Cellulose:** Structural polysaccharide in plant cell walls.
- **Chitin:** Structural polysaccharide in exoskeletons and fungi cell walls.

Proteins

Proteins contain carbon, hydrogen, oxygen, and nitrogen atoms (CHON). These atoms combine to form **amino acids**, which link together to build **polypeptides** (or proteins).

Amino acids are the monomers of proteins and have the structure shown below. There are twenty different types of amino acids, characterized by a unique "**R-group**".

Amino Acid Structure

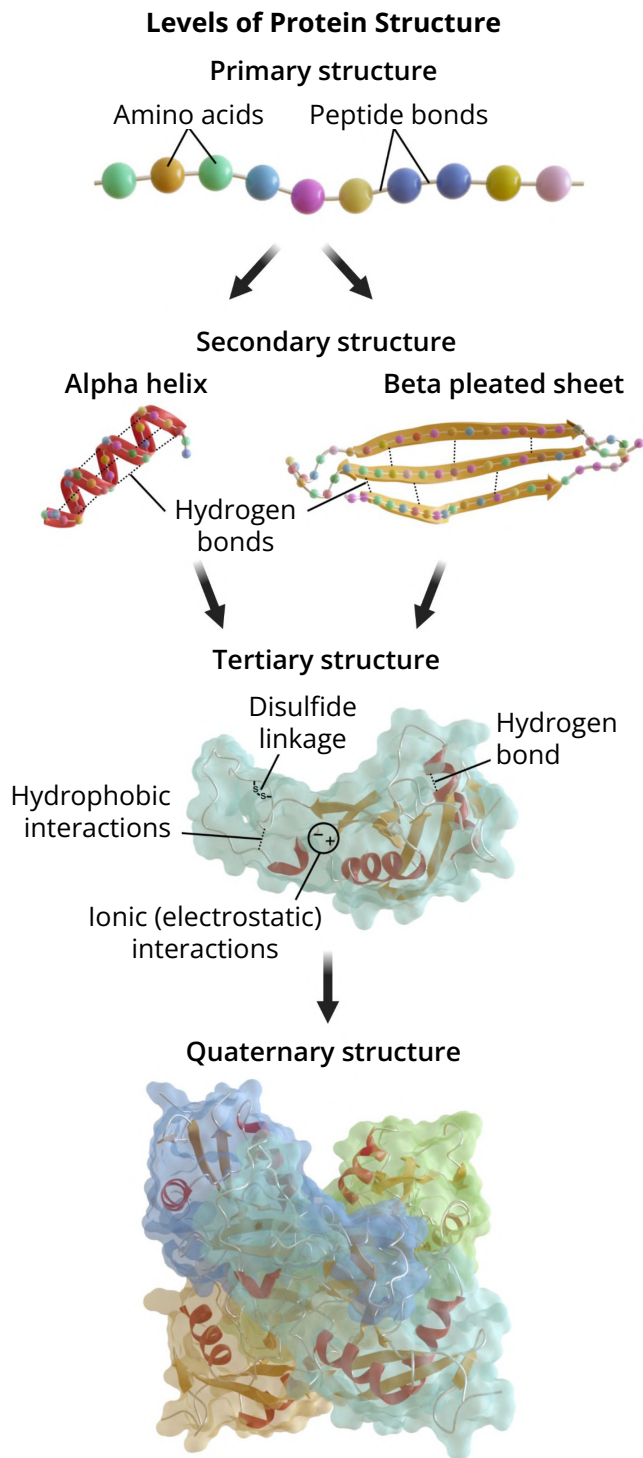


Polypeptides are polymers of amino acids and are joined by **peptide bonds** through **dehydration (condensation)** reactions. **Hydrolysis** reactions break peptide bonds.

Protein structure:

1. **Primary structure:** Sequence of amino acids connected through peptide bonds.
2. **Secondary structure:** Intermolecular forces between the polypeptide backbone (*not* R-groups) due to **hydrogen bonding**. Forms **α-helices** or **β-pleated sheets**.
3. **Tertiary structure:** Three-dimensional structure due to interactions between R-groups. Can create hydrophobic interactions based on the R-groups. **Disulfide bonds** are created by covalent bonding between the R-groups of two cysteine amino acids. Hydrogen bonding and ionic bonding between R groups also hold together the tertiary structure.
4. **Quaternary structure:** Multiple polypeptide chains come together to form one protein.

Protein denaturation describes the loss of protein function and higher order structures. Only the primary structure is unaffected. Proteins will denature as a result of high or low temperatures, pH changes, and salt concentrations. For example, cooking an egg in high heat will disrupt the intermolecular forces in the egg’s proteins, causing it to coagulate.



Protein function	Description
Storage	Reserve of amino acids
Hormones	Signaling molecules that regulate physiological processes
Receptors	Proteins in cell membranes which bind to signal molecules
Structure	Provide strength and support to tissues (hair, spider silk)
Immunity	Antibodies that protect against foreign substances
Enzymes	Regulate rate of chemical reactions

Catalysts *increase* reaction rates by lowering the **activation energy** of a reaction. The **transition state** is the unstable conformation between the reactants and the products. Catalysts *reduce* the energy of the transition state. Catalysts do *not* shift a chemical reaction or affect spontaneity.

Enzymes act as biological catalysts by binding to substrates (reactants) and converting them into products.

- Enzymes bind to substrates at an **active site**, which is specific for the substrate that it acts upon. Most enzymes are proteins.
- The **induced fit theory** describes how the active site molds itself and changes shape to fit the substrate when it binds.
- A **ribozyme** is an RNA molecule that can act as an enzyme (a non-protein enzyme).
- A **cofactor** is a non-protein molecule that helps enzymes perform reactions. A **coenzyme** is an organic cofactor (e.g., vitamins). Inorganic cofactors are usually metal ions.
- Protein enzymes are susceptible to **denaturation**. They require optimal temperatures and pH for function.

Enzymes catalyze reactions in the following ways:

- Conformational changes that bring reactive groups closer.
- The presence of acidic or basic groups.
- Induced fit of the enzyme-substrate complex.
- Electrostatic attractions between the enzyme and substrate.

High-yield biological enzymes:

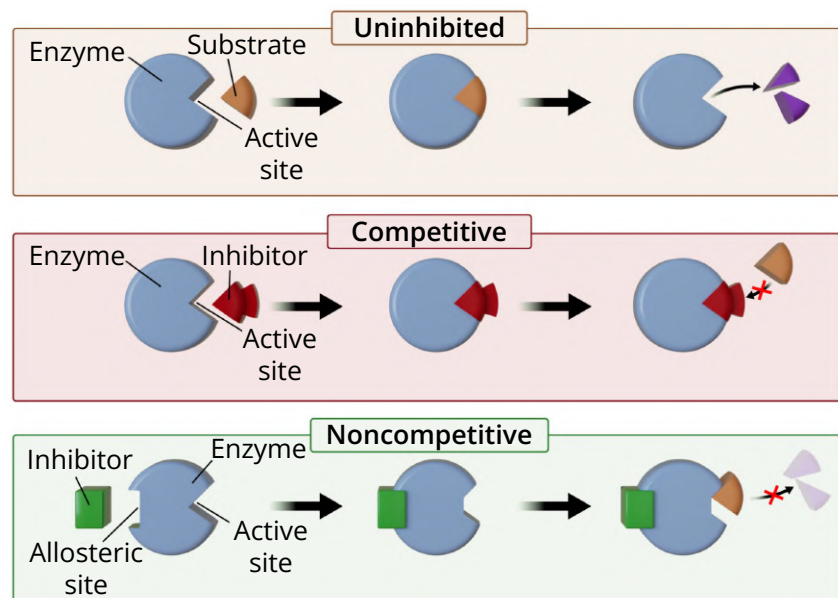
- Phosphatase:** Cleaves a phosphate group off of a substrate molecule
- Phosphorylase:** *Directly* adds a phosphate group to a substrate molecule by breaking bonds within a substrate molecule.
- Kinase:** *Indirectly* adds a phosphate group to a substrate molecule by transferring a phosphate group from an ATP molecule. These enzymes do *not* break bonds to add the phosphate group.

Feedback regulation of enzymes occurs when the end product of an enzyme-catalyzed reaction inhibits the enzyme's activity by binding to an **allosteric site** (a location on an enzyme that is different from the active site).

Competitive inhibition occurs when a **competitive inhibitor** competes *directly* with the substrate for active site binding. Competitive inhibitors *can* be outcompeted by adding more substrate.

Noncompetitive inhibition occurs when the **noncompetitive inhibitor** binds to an allosteric site that modifies the active site. Noncompetitive inhibitors *cannot* be outcompeted by adding more substrate.

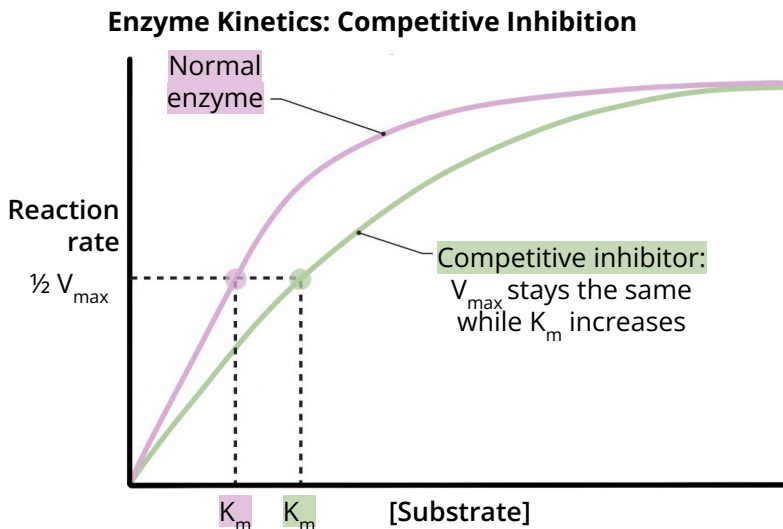
Types of Enzyme Inhibition



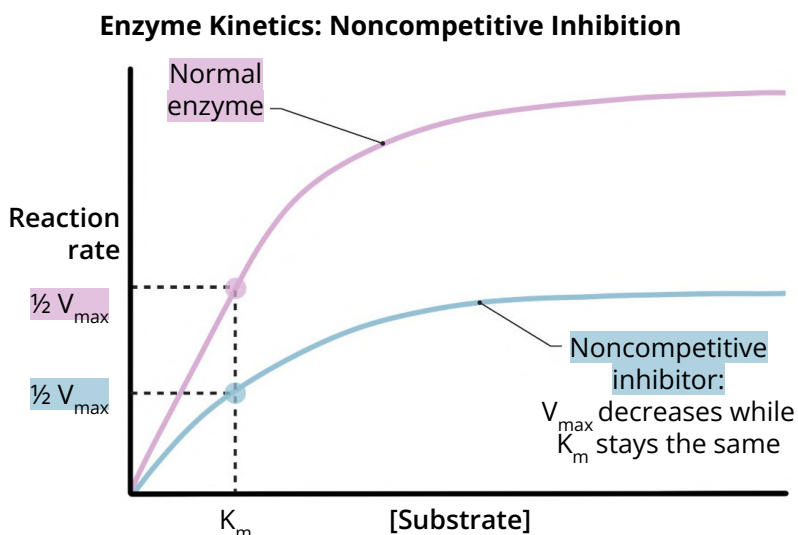
An **enzyme kinetics plot** can be used to visualize how inhibitors affect enzymes. Below are a few terms used to describe the plot:

1. The x-axis represents substrate concentration $[X]$ while the y-axis represents reaction rate or velocity (V).
2. $V_{\max}$ is the **maximum reaction velocity**.
3. **Michaelis Constant (K_M)** is the substrate concentration $[X]$ at which the velocity (V) is 50% of the maximum reaction velocity ($V_{\max}$).
4. **Saturation** occurs when all active sites are occupied, so the rate of reaction does *not* increase anymore despite increasing substrate concentration (causes graph plateaus).

Competitive inhibition: K_M increases, while $V_{\max}$ stays the same



Noncompetitive inhibition: K_M stays the same, while $V_{\max}$ decreases



Lipids

Lipids contain carbon, hydrogen, and oxygen atoms (CHO), like carbohydrates. They have long hydrocarbon tails that make them very **hydrophobic**.

Triacylglycerol (triglyceride) is a lipid molecule with a glycerol backbone (three carbons and three hydroxyl groups) and three fatty acids (long hydrocarbon tails). Glycerol and the three fatty acids are connected by **ester linkages**.

Saturated fatty acids have no double bonds and as a result pack tightly (solid at room temperature).

Unsaturated fatty acids have double bonds. Double bonds create kinks in the fatty acid chain, preventing tight packing and increasing membrane fluidity.

Phospholipids are lipid molecules that have a glycerol backbone, one phosphate group, and two fatty acid tails. The phosphate group is polar, while the fatty acids are nonpolar. As a result, they are **amphipathic** (both hydrophobic and hydrophilic). Furthermore, they spontaneously assemble to form **lipid bilayers**.

Cholesterol is an amphipathic lipid molecule that is a component of the cell membranes. It is the precursor to **steroid hormones** (lipids with four hydrocarbon rings). It is also the starting material for **vitamin D** and **bile acids**.

Lipoproteins allow the transport of lipid molecules in the bloodstream due to an outer coat of phospholipids, cholesterol, and proteins.

Waxes are simple lipids with long fatty acid chains connected to **alcohols**.

Carotenoids are lipid derivatives containing long carbon chains with double bonds and function mainly as **pigments**.

Sphingolipids have a backbone with **aliphatic** (non-aromatic) amino alcohols and have important functions in structural support, signal transduction, and cell recognition.

Glycolipids are lipids found in the cell membrane with a carbohydrate group attached instead of a phosphate group in phospholipids. Like phospholipids, they are amphipathic and contain a polar head and a fatty acid chain.

Nucleic Acids

Nucleic acids contain nucleotide monomers that build into **DNA (deoxyribonucleic acid)** and **RNA (ribonucleic acid)** polymers.

Nucleosides contain a five-carbon sugar and a nitrogenous base.

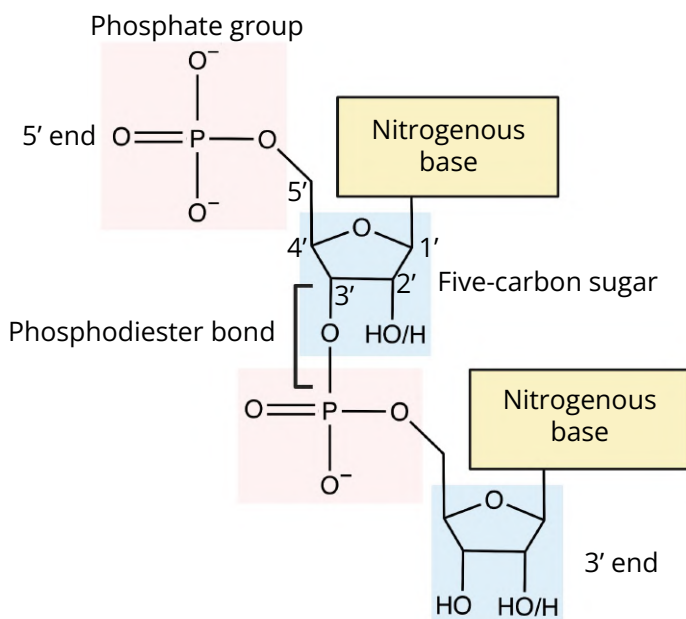
Nucleotides contain a five-carbon sugar, a nitrogenous base, and a phosphate group.

Deoxyribose sugars (in DNA) have a hydrogen at the 2' carbon while **ribose** five-carbon sugars (in RNA) have a hydroxyl group at the 2' carbon.

Phosphodiester bonds are formed through a condensation reaction where the phosphate group of one nucleotide (at the 5' carbon) connects to the hydroxyl group of another nucleotide (at the 3' carbon) and releases a water molecule as a by-product.

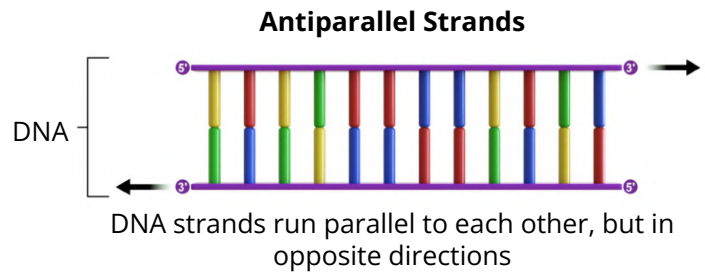
A series of phosphodiester bonds create the sugar-phosphate backbone, with a 5' end (free phosphate) and a 3' end (free hydroxyl).

Nucleic Acid Polymerization



Nucleic acid polymerization proceeds as nucleoside triphosphates are added to the 3' end of the sugar-phosphate backbone.

DNA is an antiparallel double helix, in which two complementary strands with opposite directionalities (positioning of 5' ends and 3' ends) twist around each other.



mRNA is single-stranded after being copied from DNA during transcription.

Other Types of RNA

miRNA (microRNA): Small RNA molecules that can silence gene expression by base pairing to complementary sequences in mRNA.

rRNA (ribosomal RNA): It is formed in the nucleolus of the cell and helps ribosomes translate mRNA.

dsRNA (double stranded RNA): Some viruses carry their code as double stranded RNA. **Pro-Tip:** dsRNA must pair its nucleotides, so it must have equal amounts of A/U and equal amounts of C/G.

tRNA (transfer RNA): Small RNA molecule that participates in protein synthesis.

Biological Hypotheses and Theories

The Universe is approximately 13.8 billion years old. The first cells appeared on Earth approximately 3.5 billion years ago.

Primordial Earth:

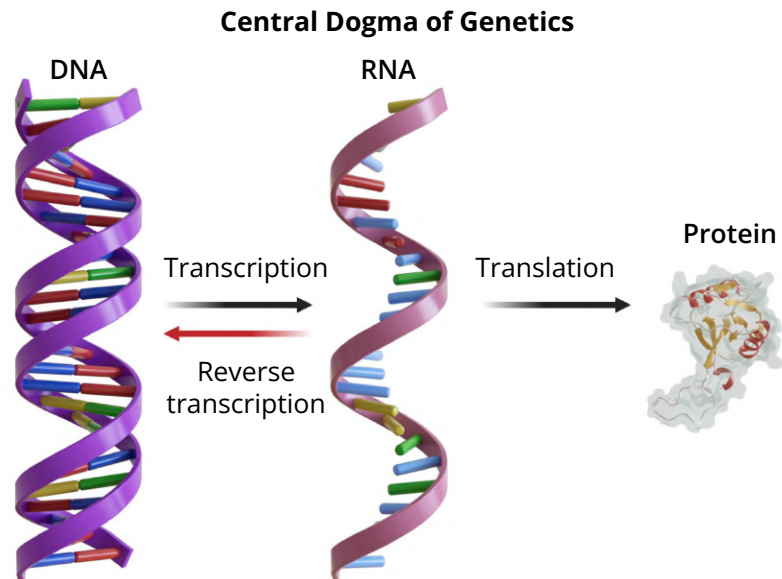
1. Earth's primordial atmosphere was comprised of inorganic compounds and was a reducing environment (little O₂ gas).
2. As Earth cooled, gases condensed, forming the primordial sea.
3. Simple compounds evolved into more complex organic compounds.
4. Organic monomers linked into polymers.
5. Protobionts emerged as precursors to cells.
6. Heterotrophic, obligate anaerobic prokaryotes developed.
7. Autotrophic prokaryotes, such as cyanobacteria capable of photosynthesis, formed. This led to oxygen production and accumulation, creating an oxidizing environment (high O₂ gas).
8. Primitive eukaryotes emerged, supporting the endosymbiotic theory where membrane bound organelles (mitochondria, chloroplasts), originally free-living, were engulfed by other prokaryotes, leading to a symbiotic relationship.
9. More complex eukaryotes and multicellular organisms began to evolve.

The **Miller-Urey** experiment simulated early Earth conditions by exposing a mixture of hydrogen gas, methane, ammonia, and water vapor to electrical sparks, resulting in the formation of amino acids and other simple organic compounds. This demonstrated that key building blocks of life can form abiotically.

Modern cell theory:

1. All lifeforms have one or more cells.
2. The cell is the basic structural, functional, and organizational unit of life.
3. All cells come from other cells (cell division).
4. Genetic information is stored and passed down through DNA.
5. An organism's activity is dependent on the total activity of its independent cells.
6. Metabolism and biochemistry (energy flow) occurs within cells.
7. All cells have the same chemical composition within organisms of similar species.

The **central dogma of genetics** states that information is passed from **DNA → RNA → proteins**. There are a few exceptions to this (e.g., reverse transcriptase and prions).



The **RNA world hypothesis** is the theory that early life forms relied on self-replicating RNA both to store genetic information and to catalyze chemical reactions before the evolution of DNA and proteins.

Because RNA is reactive and unstable:

- DNA replaced RNA in how genetic information is stored.
- Proteins largely replaced RNA in catalyzing reactions (ribozymes being a notable exception).

The **endosymbiotic theory** states that eukaryotes developed when aerobic bacteria were internalized as mitochondria while the photosynthetic bacteria became chloroplasts. Evidence for this theory includes the similarities between mitochondria and chloroplasts:

- They are similar in size.
- They possess their own circular DNA.
- They have ribosomes with a large and small subunit.
- They reproduce independently of the host cell.
- They contain a double membrane.

High-Yield Roots, Prefixes, and Suffixes

Prefix	Meaning	Example	Root	Meaning	Example	Suffix	Meaning	Example
Anti-	Against	Antibiotic	-adrena-	Adrenal	Adrenaline	-ase	Enzyme	Amylase
Ecto-	Outside	Ectoderm	-blast-	Formative cell	Osteoblast	-crine	Secrete	Exocrine
Endo-	Inside	Endocrine	-derma-	Skin	Endodermis	-cyte	Cell	Adipocyte
Epi-	Above	Epidermis	-enter-	Intestine	Enterocyte	-cytosis	Relating to cells	Transcytosis
Exo-	Outside	Exocytosis	-gastr-	Stomach	Gastrin	-gen	Producing, generating	Immunogen
Hypo-	Insufficient, low	Hypoglycemia	-glyc-	Sugar	Glycogen	-genesis	Formation of	Angiogenesis
Hyper-	Excessive, high	Hyperactive	-hem-	Blood	Hemoglobin	-lytic or -lysis	Destruction, breakdown	Hemolysis
Intra-	Within	Intracellular	-hydroxyl-	Hydroxyl group, alcohol	Hydroxylation	-oid	Resembles, related to	Nucleoid
Inter-	Between	Intermolecular	-leuk-	White	Leukocyte	-ose	Sugar	Glucose
Mono-	One	Monomer	-lip-	Fat or lipid	Lipid	-phobic	Fearing, repelling	Hydrophobic
Peri-	Around	Peripheral	-myo-	Muscle	Cardiomyocyte	-philic	Attracting	Hydrophilic
Poly-	Many	Polysaccharide	-nucle-	Nucleus, central part	Nucleus	-plasia	Development, formation	Hyperplasia
Sub-	Below, under	Subunit	-peptid-	Protein chain	Peptidase	-scopy	Examination, inspection	Microscopy
Supra-	Above	Supramolecular	-phag-	Eating, swallowing	Phagocytosis	-stasis	Stopping	Epistasis
Trans-	Across	Transduction	-ribos-	Ribose (five-carbon sugar)	Deoxyribose	-trophy	Growth	Hypertrophy

Ch. 2: Cells and Organelles

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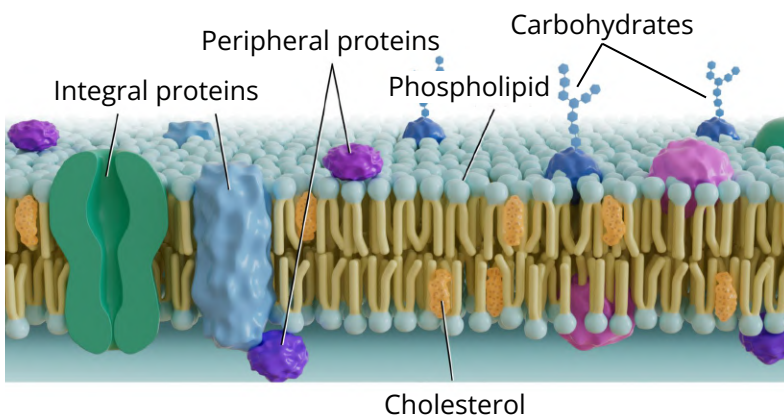
- Cell Membrane
- Crossing Cell Membranes
- Organelles
- Cytoskeleton
- Extracellular Matrix
- Cellular Tonicity and Cell Circulation
- Cellular Adaptations
- Cell Tissues

Cell Membrane

The **cell membrane** is a selectively permeable barrier composed of a phospholipid bilayer with embedded proteins. It regulates the movement of substances in and out of the cell. It also plays a role in cell signaling and communication.

1. **Phospholipid bilayer:** Possesses hydrophilic (water-attracting) heads that form the exterior surfaces and hydrophobic (water-repelling) tails that face inward, creating a barrier to most water-soluble substances.
2. **Membrane proteins:** Embedded within the phospholipid bilayer, they facilitate transport, act as receptors for signaling, and provide structural support.
3. **Carbohydrates:** Attached to proteins and lipids on the extracellular surface, they play key roles in cell recognition, signaling, and adhesion.
4. **Cholesterol:** Found within eukaryotic cell membranes (absent in prokaryotic membranes), it helps to stabilize membrane fluidity.

Cell Membrane Components



Integral proteins are embedded in the phospholipid bilayer. They are amphipathic (contain both hydrophilic and hydrophobic components).

Transmembrane proteins, a type of integral protein, traverse the entire bilayer. Integral proteins can function as receptors in cell signaling. They can also function as channels or carrier proteins for transport.

Peripheral membrane proteins are found on the surface of the bilayer (inside or outside of the cell) and are generally hydrophilic. They can function as receptors or assist with adhesion and cell recognition.

Membrane protein functions:

- **Receptor:** Transmit a signal to a cell. Trigger secondary responses within the cell.
 - **Agonists** are molecules that bind to receptors and functionally activate a target.
 - **Antagonists** bind and prevent other molecules from binding, inhibiting production of a response.
- **Adhesion:** Attach cells to other things (e.g., other cells) and act as anchors for the cytoskeleton.
- **Cellular recognition:** Proteins which have carbohydrate chains (**glycoproteins**). Used by cells to recognize other cells.

The **fluid mosaic model** describes how the components that make up the cell membrane can move freely within the membrane ("**fluid**"). Furthermore, the cell membrane contains many different kinds of structures ("**mosaic**").

The fluidity of the cell membrane can be affected by:

- **Temperature:** ↑ temperatures *increase* fluidity while ↓ temperatures *decrease* it.
- **Cholesterol:** Holds membrane together at high temperatures and keeps membrane fluid at low temperatures.
- **Degrees of unsaturation:** Saturated fatty acids pack more tightly than unsaturated fatty acids, which have double bonds that may introduce kinks.

Crossing Cell Membranes

Cells must regulate the travel of substances across the cell membrane. There are three types of transport across the cell membrane:

1. **Simple diffusion:** Diffusion of small uncharged molecules (e.g., O_2 , CO_2 , H_2O) or lipid soluble molecules (steroids) directly across the cell membrane, down their concentration gradient (high to low) without using energy.
2. **Facilitated transport:** Channel proteins allow the diffusion of large (e.g., glucose, sucrose) or charged molecules (e.g., Na^+ , K^+ , Cl^-) across the cell membrane, down their concentration gradient without using energy.
3. **Active transport:** Substances travel *against* their concentration gradient and require the consumption of energy by carrier proteins.
 - **Primary active transport** uses ATP hydrolysis to pump molecules *against* their concentration gradient. For example, the sodium-potassium (Na^+/K^+) pump sets a membrane potential.
 - **Secondary active transport** uses the energy from one molecule moving down its electrochemical gradient to drive the transport of another molecule against its concentration gradient. This requires an energy investment to establish the initial concentration gradient.

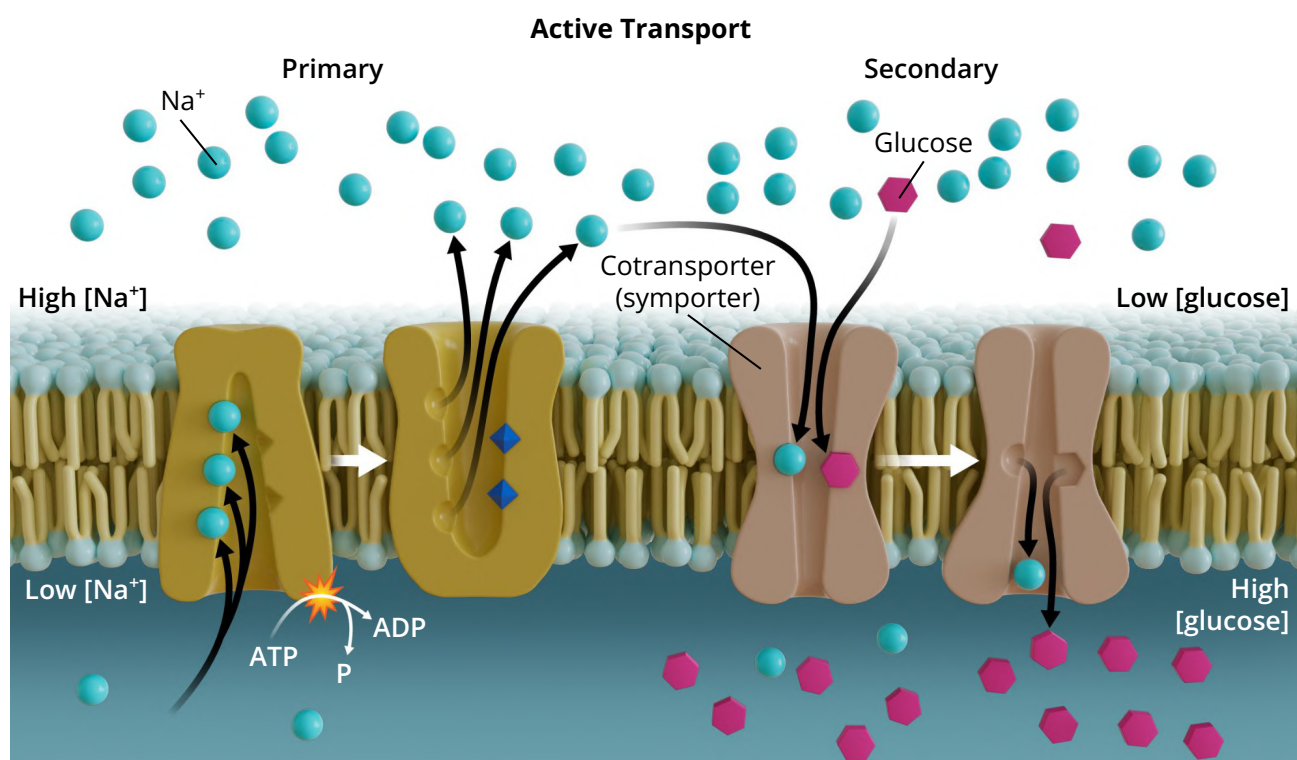
Cytosis refers to the bulk transport of molecules across the cell membrane.

Endocytosis involves the cell membrane wrapping around an extracellular substance, internalizing it into the cell via a **vesicle** or **vacuole**. Below are different forms of endocytosis:

- **Phagocytosis:** Cellular eating around solid objects.
- **Pinocytosis:** Cellular drinking around dissolved materials (liquids).
- **Receptor-mediated endocytosis:** Requires the binding of dissolved molecules to peripheral membrane receptor proteins, which initiates endocytosis.

Clathrin is a protein that aids in receptor-mediated endocytosis by forming a pit in the membrane that pinches off as a coated vesicle. This is known as **clathrin-mediated endocytosis**.

Exocytosis is the opposite of endocytosis, in which material is released to the extracellular environment through vesicle secretion.



Organelles

Organelles are cellular compartments enclosed by phospholipid bilayers (membrane bound). They are located within the **cytosol** (aqueous intracellular fluid) and help make up the **cytoplasm** (cytosol & organelles).

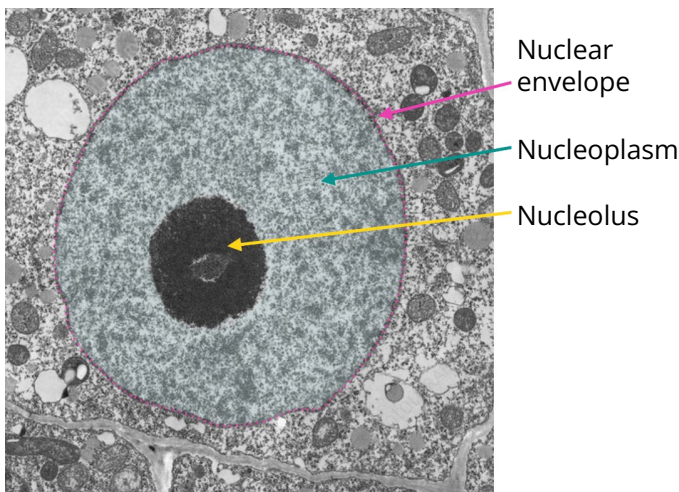
Only **eukaryotic** cells contain membrane-bound organelles. **Prokaryotes** do *not*, but they have other adaptations, such as keeping their genetic material in a region called the **nucleoid**.

The **nucleus** primarily functions to protect and house DNA. **DNA replication** and **transcription** (DNA → mRNA) occurs here.

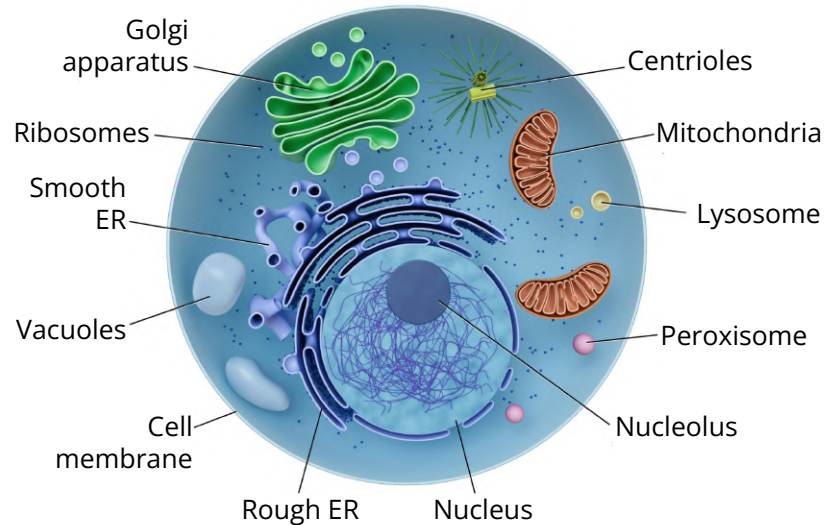
Parts of the nucleus:

- The **nucleoplasm** is the cytoplasm of the nucleus.
- The **nuclear envelope** is the membrane of the nucleus. It contains two phospholipid bilayers (one inner, one outer).
- **Nuclear pores** are holes in the nuclear envelope that allow molecules to travel in and out of the nucleus.
- The **nucleolus** is a dense area responsible for producing components of ribosomal subunits including rRNA (ribosomal RNA).

Nucleus



Cellular Structures



Ribosomes are *not* considered to be organelles; they work as small factories that carry out **translation** (mRNA → protein). They are composed of ribosomal subunits.

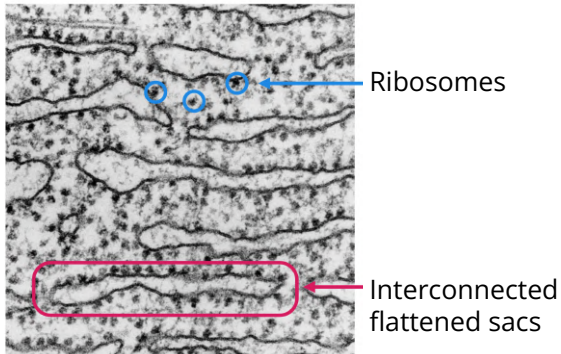
Eukaryotic ribosomal subunits (**60S** and **40S**) assemble in the nucleoplasm and are then exported from the nucleus to form the complete ribosome in the cytosol (**80S**).

Prokaryotic ribosomal subunits (**50S** and **30S**) assemble in the cytosol and also form complete ribosomes (**70S**) there.

Free-floating ribosomes make proteins that function in the cytosol while ribosomes embedded in the rough endoplasmic reticulum (rough ER) make proteins that are sent out of the cell or to the cell membrane.

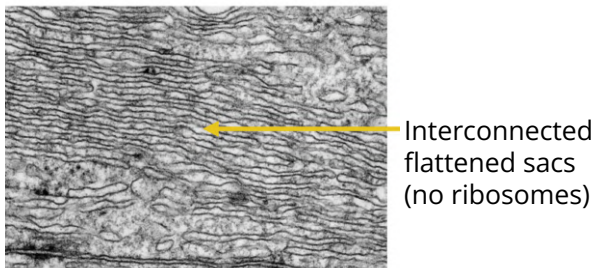
The **rough endoplasmic reticulum (rough ER)** is continuous with the outer membrane of the nuclear envelope and is “rough” because it has ribosomes embedded in it. Proteins synthesized by the embedded ribosomes are sent into the lumen (inside of the rough ER) for modifications (e.g., glycosylation). Afterwards, they are either sent out of the cell or become part of the cell membrane.

Rough ER



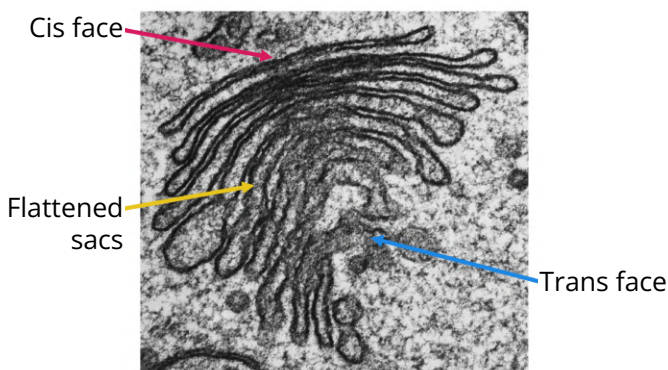
The **smooth endoplasmic reticulum (smooth ER)** is an extension of the rough ER. Its main function is to synthesize lipids, produce steroid hormones, and detoxify cells.

Smooth ER



The **Golgi apparatus** stores, modifies, and exports proteins that will be secreted from the cell. It is made up of **cisternae** (flattened sacs) that modify and package substances. Vesicles come from the ER and reach the **cis face** (side closest to ER) of the Golgi apparatus. Vesicles leave the Golgi apparatus from the **trans face** (side closest to cell membrane).

Golgi Apparatus



Lysosomes, found in animal cells, are membrane-bound organelles that contain digestive (**hydrolytic**) enzymes which break down cellular waste. They also play a role in **apoptosis** (programmed cell death).

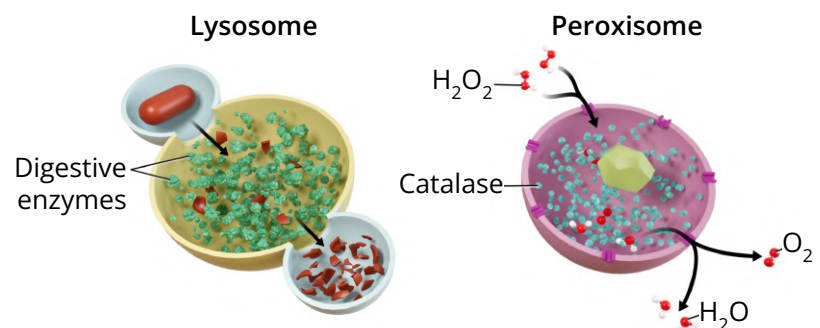
Vacuoles are membrane-bound vesicles typically used for storage (water, nutrients, waste) or transport. Plant cells have large, fluid-filled **central** vacuoles that help maintain cell rigidity (**turgor**). Plant cell vacuoles also function similarly to lysosomes in animal cells. **Contractile** vacuoles actively pump out excess water and are found mostly in single-celled protists.

The **endomembrane system** is composed of the different membranes that are suspended in the cytoplasm within a eukaryotic cell. It is a group of organelles and membranes that work together to modify, package, and transport proteins and lipids that are entering or exiting a cell. The components of the endomembrane system include the nucleus, rough and smooth ERs, Golgi apparatus, lysosomes, vacuoles, vesicles, and cell membrane.

The Golgi apparatus has a significant role in the endomembrane system: it receives vesicles from the ER on the cis face that empty proteins and lipids into the lumen of the Golgi. These proteins/lipids undergo modifications and are then sorted, tagged, packaged, and distributed as secretory products.

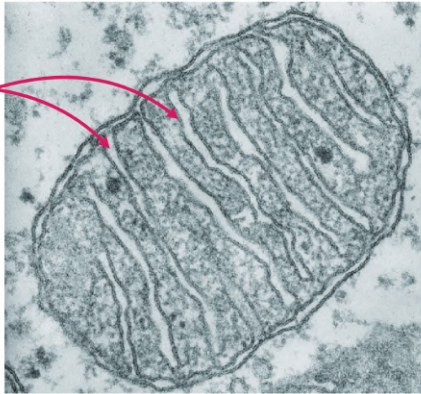
Peroxisomes perform hydrolysis, break down stored fatty acids, and help with **detoxification**. These processes generate hydrogen peroxide, which is toxic since it can produce **reactive oxygen species (ROS)**. ROS damage cells through **free radicals**. Peroxisomes contain an enzyme called **catalase**, which quickly breaks down hydrogen peroxide into water and oxygen.

Lysosomes vs. Peroxisomes



Mitochondria are the powerhouses of the cell, producing ATP for energy use through **cellular respiration**. Mitochondrial inheritance is **maternal**, meaning all mitochondrial DNA in humans originates from the mother.

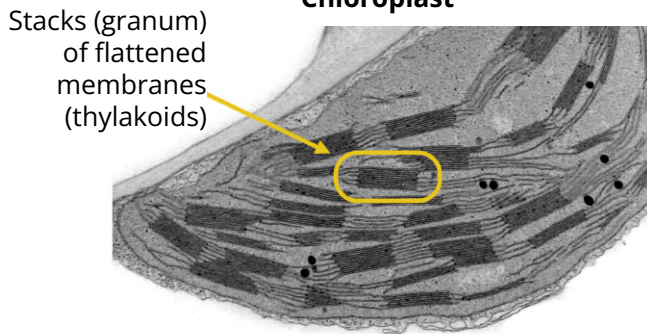
Mitochondria



Chloroplasts are found in plants and some protists. They carry out photosynthesis.

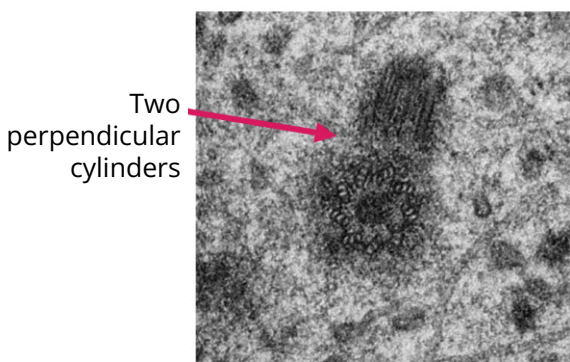
Chloroplasts are a type of **plastid**. Plastids are double-membraned organelles found exclusively within plant cells and algae, that function in photosynthesis and storage of metabolites.

Chloroplast



Centrosomes are organelles found in animal cells containing a pair of **centrioles**. They act as **microtubule organizing centers (MTOCs)** during cell division.

Centrosomes



Cytoskeleton

The **cytoskeleton** provides structure and function within the cytoplasm.

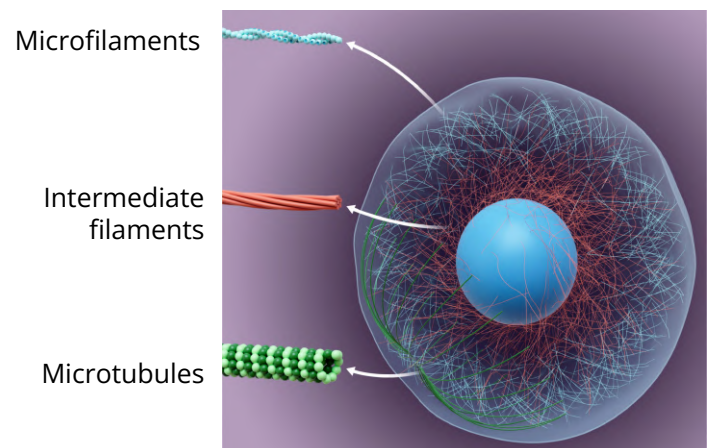
Microfilaments are the smallest structure of the cytoskeleton and are composed of a double helix made of actin filaments. They are mainly involved in cell movement and can quickly assemble and disassemble. Below are some of their functions:

1. **Cleavage furrow:** During cell division, myosin motors and actin microfilaments form contractile rings that split the cell.
2. **Cyclosis (cytoplasmic streaming):** The flow (or stirring) of the cytoplasm inside the cell. It is driven by forces via actin (microfilaments) and myosin movement, in a manner similar to muscle contraction.
3. **Muscle contraction:** Actin microfilaments have directionality, allowing **myosin motor proteins** to pull on them for muscle contraction.

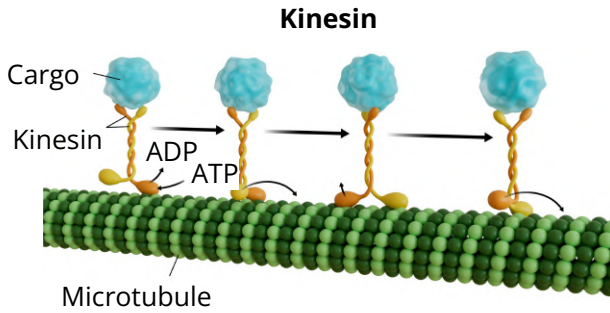
Intermediate filaments are between microfilaments and microtubules in size. They are more stable than microfilaments and mainly help with structural support. For example, **keratin** is an important intermediate filament protein in skin, hair, and nails.

Microtubules are the largest in size and give structural integrity to cells. They are made of tubulin protein. Microtubules also form centrioles (used in cell division) and are found in cilia and flagella (beating appendages).

Cytoskeletal Proteins



Kinesin and **dynein** are motor proteins that transport cargo along microtubules.



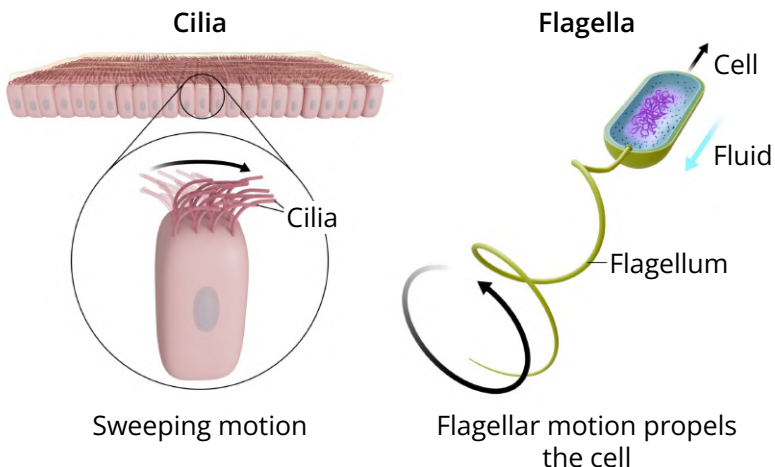
Microtubule organizing centers (MTOCs) are present in eukaryotic cells and help organize microtubule extension. **Centrioles** are hollow cylinders made of microtubules. **Centrosomes** contain a pair of centrioles oriented at 90 degree angles to one-another.

Cilia and Flagella

Cilia are small hair-like projections made of tubulin found *only* in eukaryotes. They line the outside of eukaryotic cells and function in locomotion of either the cell itself or fluids (e.g., cilia remove debris in the lungs).

Flagella are longer hair-like structures found in *both* prokaryotes and eukaryotes. Like cilia, flagella also function in locomotion of the cell or fluids (e.g., flagella propel sperm). Eukaryotic flagella are composed of polymers of **tubulin**. Prokaryotic flagella are composed of polymers of **flagellin** (they are *not* microtubules)

Cilia vs. Flagella



Extracellular Matrix

The **extracellular matrix (ECM)** provides extracellular mechanical support for cells.

ECM components:

- **Proteoglycan:** A type of glycoprotein that has a high proportion of carbohydrates.
- **Collagen:** The most common structural protein, secreted by fibroblasts; organized into **collagen fibrils**.
- **Integrin:** A transmembrane protein that facilitates **ECM adhesion** and signals to cells how to respond to the extracellular environment (growth, apoptosis, etc.).
- **Fibronectin:** A protein that connects integrin to ECM and helps with signal transduction.
- **Laminin:** Behaves similarly to fibronectin. Influences cell differentiation, adhesion, and movement.

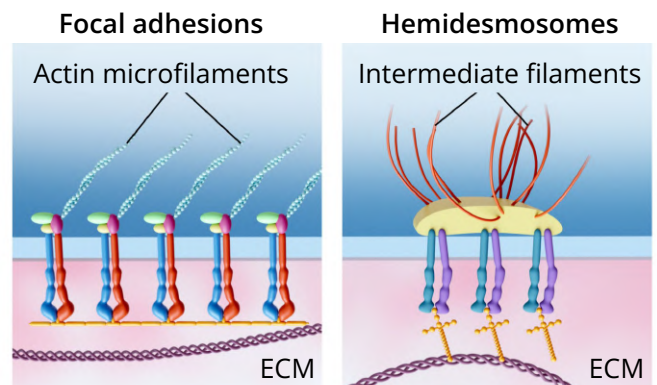
Cell walls are carbohydrate-based structures that act like a substitute ECM because they provide structural support to cells that either do *not* have ECM, or have a minimal ECM. They are present in plants (**cellulose**), fungi (**chitin**), bacteria (**peptidoglycan**), and archaea (**non-peptidoglycan polysaccharides**).

Peptidoglycan is a polysaccharide with peptide chains. This is the primary component of bacterial cell walls. The cell wall of archaea is also made of polysaccharides, but does *not* contain peptidoglycan.

Cell-matrix junctions (connect ECM → cytoskeleton):

1. **Focal adhesions:** ECM connects via integrins to actin microfilaments inside the cell.
2. **Hemidesmosomes:** ECM connects via integrins to intermediate filaments inside the cell.

Cell-Matrix Junctions



Cell-cell junctions (connect adjacent cells):

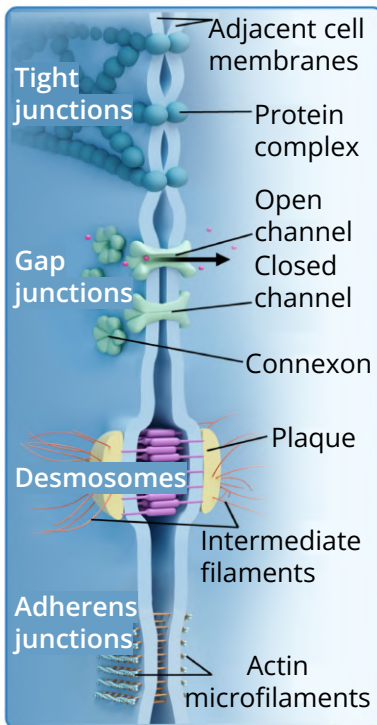
1. **Tight junctions:** Form water-tight seals between cells to ensure substances pass through cells and *not* between them.
2. **Desmosomes:** Provide support against mechanical stress. Connects neighboring cells via intermediate filaments.
3. **Adherens junctions:** Similar in structure and function to desmosomes, but connects neighboring cells via actin microfilaments.
4. **Gap junctions:** Allow passage of ions and small molecules between cells. Formed from transmembrane proteins known as **connexons**. Gap junctions are *only* present in animal cells.

Plant cells contain a few unique cell junctions:

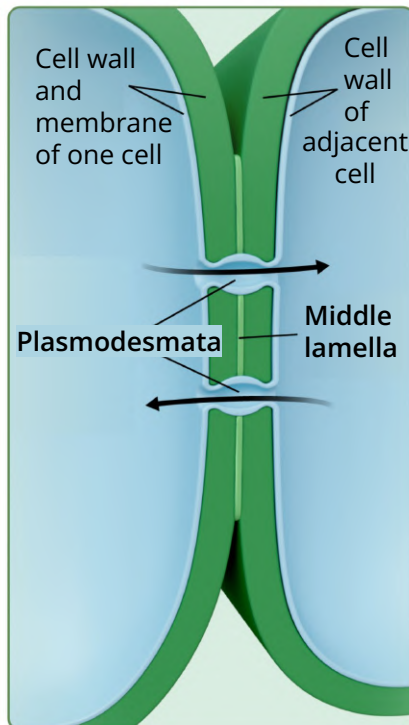
1. **Middle lamella:** Sticky cement similar in function to tight junctions.
2. **Plasmodesmata:** Tunnels with tubes between plant cells. Allows cytosol fluids to freely travel between plant cells.

Intercellular Junctions

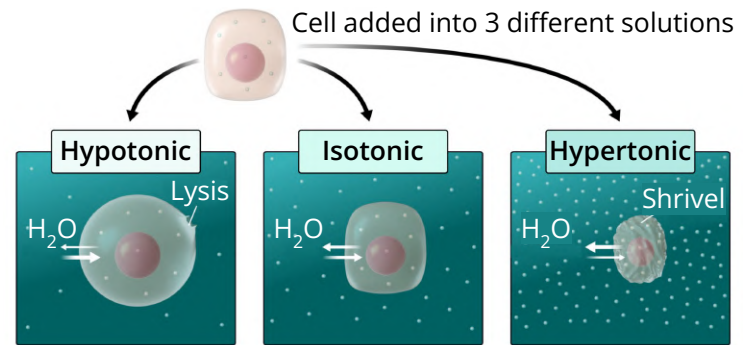
Animal cell junctions



Plant cell junctions



Cellular Tonicity and Cell Circulation



Osmosis: Passive movement of water (solvent) across a semipermeable membrane from low to high solute concentration.

Isotonic solutions have the *same* solute concentration as the cells placed in them.

Hypertonic solutions have a *higher* solute concentration than the cells placed in them, causing water to leave the cell.

- Animal cells shrivel.
- Cells that have a cell wall (plants, bacteria, fungi, etc.) undergo **plasmolysis**, the process of the cell's cytoplasm shrinking away from the cell wall.

Hypotonic solutions have a *lower* solute concentration than the cells placed in them, causing water to enter the cell.

- Animal cells swell and can burst (**lysis**).
- Plant, bacterial, and fungal cells become **turgid** (swollen and firm).

Cell Tissues

There are four basic types of **tissues**:

1. **Epithelial** tissues are cohesive sheets of cells that line internal organs and cover the body.
2. **Connective** tissues support the structure of the organism. Sparse connective tissue cells are scattered within an extracellular matrix. Cartilage, bone, blood, and adipose (fat) are all types of connective tissue.
3. **Muscle** tissues function in locomotion (skeletal muscle) and support organ systems (smooth muscle and cardiac muscle).
4. **Nervous** tissues process and transmit information within the body. Nervous tissue is composed of neurons to send information and various support cells called glial cells.

Ch. 3: Cellular Energy

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- Biothermodynamics
- Adenosine Triphosphate (ATP)
- Mitochondria
- Aerobic Cellular Respiration
- ATP Yield of Aerobic Cellular Respiration
- Anaerobic Energy Pathways
- Fermentation
- Alternative Sources of Energy Generation

Biothermodynamics

Laws of thermodynamics:

1. Energy *cannot* be created nor destroyed, but can be transformed from one form to another.
2. The **entropy** (disorder) of the universe is always *increasing*. The combined change in entropy (system and surroundings) must be positive.
3. The entropy of a substance at absolute zero is 0.

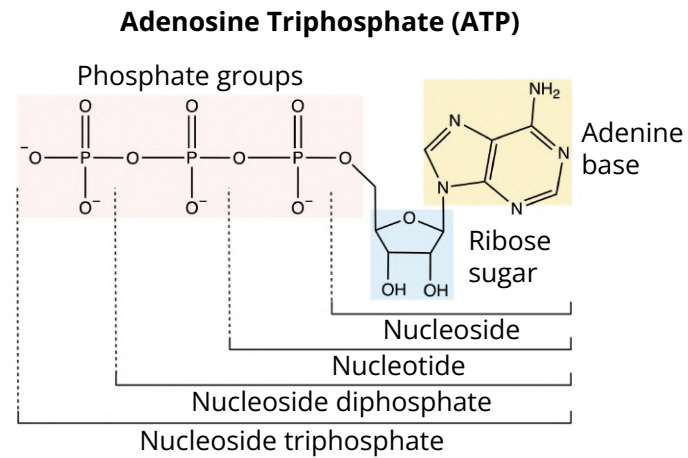
Metabolism refers to all the metabolic pathways (series of chemical reactions) that are happening in a given organism. **Catabolic** processes involve breaking down larger molecules for energy (*less* ordered state, *increased* entropy), while **anabolic** processes involve using energy to build larger macromolecules (*more* ordered state, *decreased* entropy).

Q₁₀ (temperature coefficient) is a factor that measures the change in reaction rates for every 10 °C change in temperature. For example, in animals, if Q₁₀ = 1.5, an increase of 10 °C in body temperature would increase the metabolic rate of the body by 50%.

To break down carbohydrates for energy, cells either utilize **aerobic cellular respiration** (consumes oxygen, more energy produced) or **anaerobic** pathways (no oxygen needed, but less energy produced).

Adenosine Triphosphate (ATP)

Adenosine triphosphate (ATP) is an RNA nucleoside triphosphate. It contains an adenine nitrogenous base linked to a ribose sugar (RNA nucleoside part), and three phosphate groups connected to the sugar (triphosphate part).

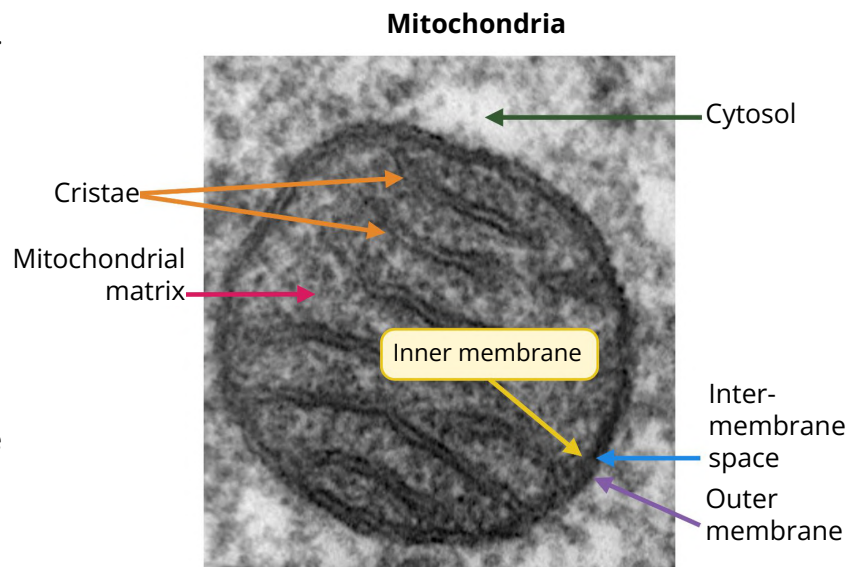


ATP is used as cellular energy currency. The bond between phosphate groups releases energy upon hydrolysis, resulting in ATP becoming **adenosine diphosphate (ADP)**. Because of the additional negatively-charged phosphate group, ATP is *less* stable than ADP.

Reaction coupling is the process of powering an energy-requiring reaction with an energy-releasing one. It allows an unfavorable reaction to be powered by a favorable reaction, making the net Gibbs free energy negative ($-\Delta G$ = exergonic = releases energy + spontaneous).

Mitochondria

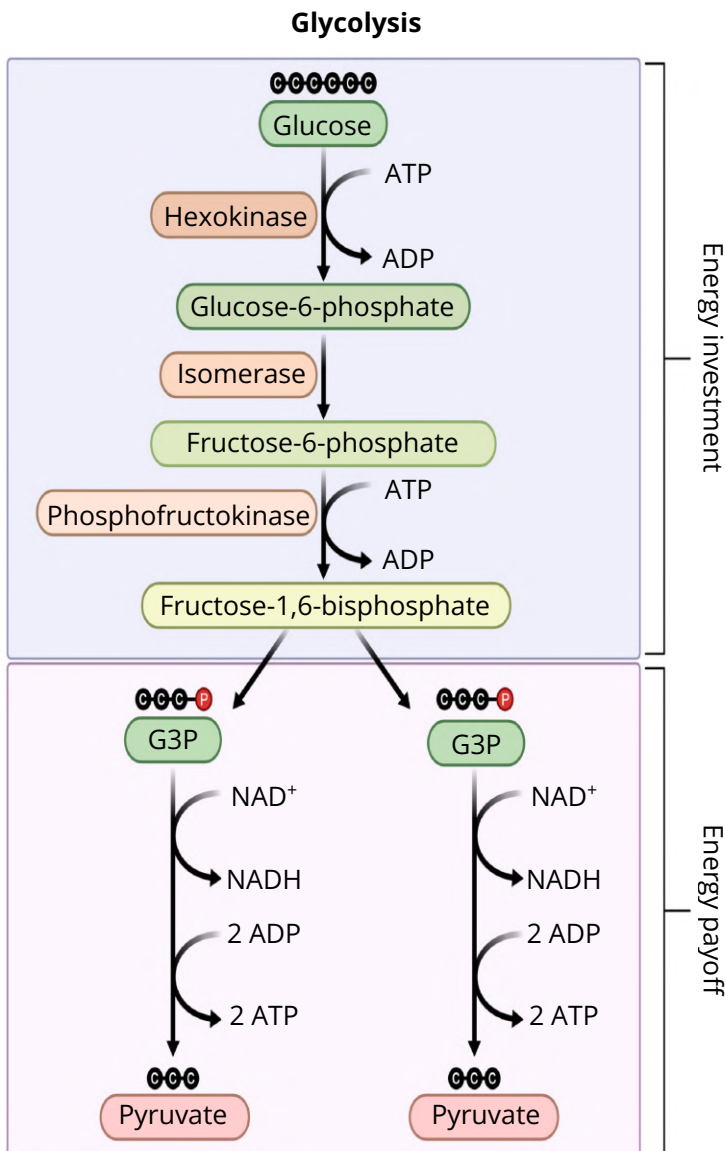
Mitochondria are organelles that produce ATP through cellular respiration (catabolic process). They have an **outer membrane** and an **inner membrane** with many infoldings called **cristae**. The **intermembrane space** is located between the outer and inner membranes while the **mitochondrial matrix** is located inside the inner membrane.



Aerobic Cellular Respiration

Aerobic cellular respiration is performed to phosphorylate ADP into ATP by breaking down glucose and moving electrons around (oxidation and reduction reactions). Aerobic cellular respiration involves four **catabolic** processes:

1. Glycolysis
2. Pyruvate oxidation
3. Krebs cycle
4. Oxidative phosphorylation



Glycolysis



Glycolysis takes place in the cytosol and does *not* require oxygen, so it is also used in **fermentation**.

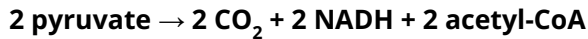
Substrate-level phosphorylation is the process used to generate ATP in glycolysis by transferring a phosphate group to ADP directly from a phosphorylated compound.

Glycolysis has an **energy investment phase** and an **energy payoff phase**:

1. **Hexokinase** uses one ATP to phosphorylate glucose into **glucose-6-phosphate**, which *cannot* leave the cell (it becomes trapped by the phosphorylation).
2. **Isomerase** modifies glucose-6-phosphate into **fructose-6-phosphate**.
3. **Phosphofructokinase** uses a second ATP to phosphorylate fructose-6-phosphate into fructose-1,6-bisphosphate. This is the key regulatory step in glycolysis.
4. Fructose-1,6-bisphosphate is broken into dihydroxyacetone phosphate (DHAP) and **glyceraldehyde-3-phosphate (G3P)**, which are in equilibrium with one another.
5. G3P proceeds to the energy payoff phase so DHAP is constantly converted into G3P to maintain equilibrium. Thus, one glucose molecule will produce two G3P that continue into the next steps.
6. G3P undergoes a series of redox reactions to produce four ATP through substrate-level-phosphorylation, two pyruvate and two NADH.

Since two ATP are used up in the energy investment phase and four ATP are produced in the energy payoff phase, a net of two ATP is produced per glucose molecule within glycolysis.

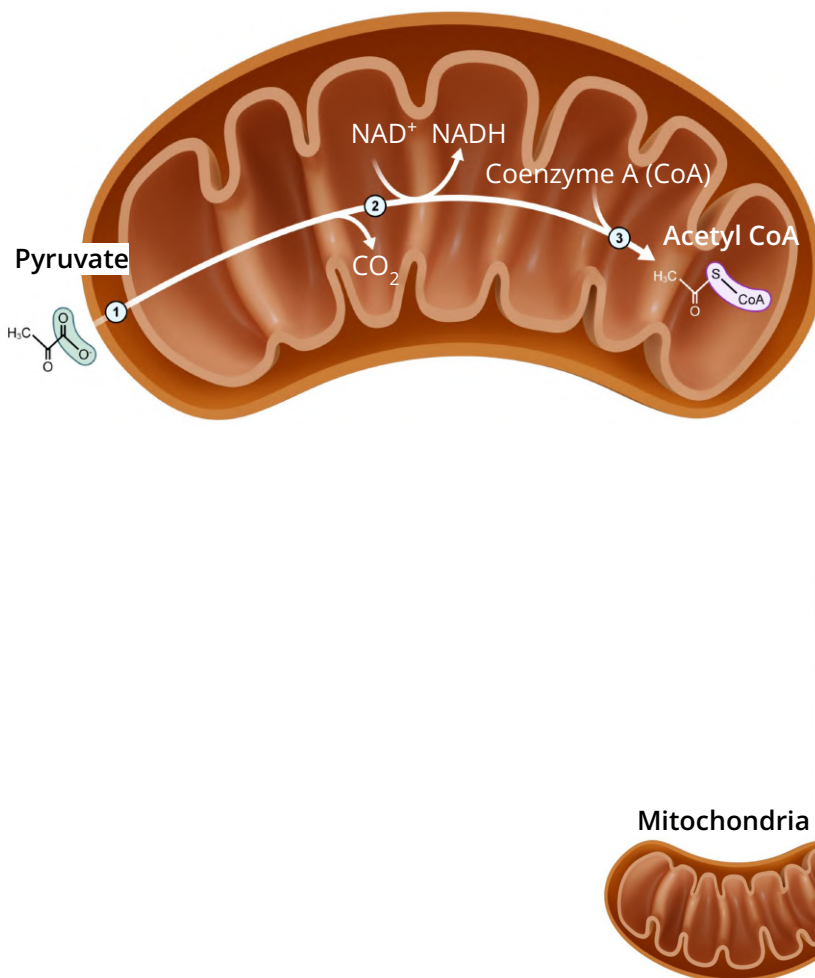
Pyruvate oxidation



Pyruvate dehydrogenase is an enzyme that carries out the pyruvate oxidation steps below:

1. **Decarboxylation:** Pyruvate molecules (three-carbon molecule) move from the cytosol into the mitochondrial matrix (stays in the cytosol for prokaryotes), where they undergo decarboxylation, producing one CO_2 and one two-carbon molecule per pyruvate.
2. **Oxidation:** The two-carbon molecule is converted into an **acetyl group**, giving electrons to NAD^+ to convert it into NADH .
3. **Coenzyme A (CoA):** CoA binds to the acetyl group, producing **acetyl-CoA**.

Pyruvate Oxidation



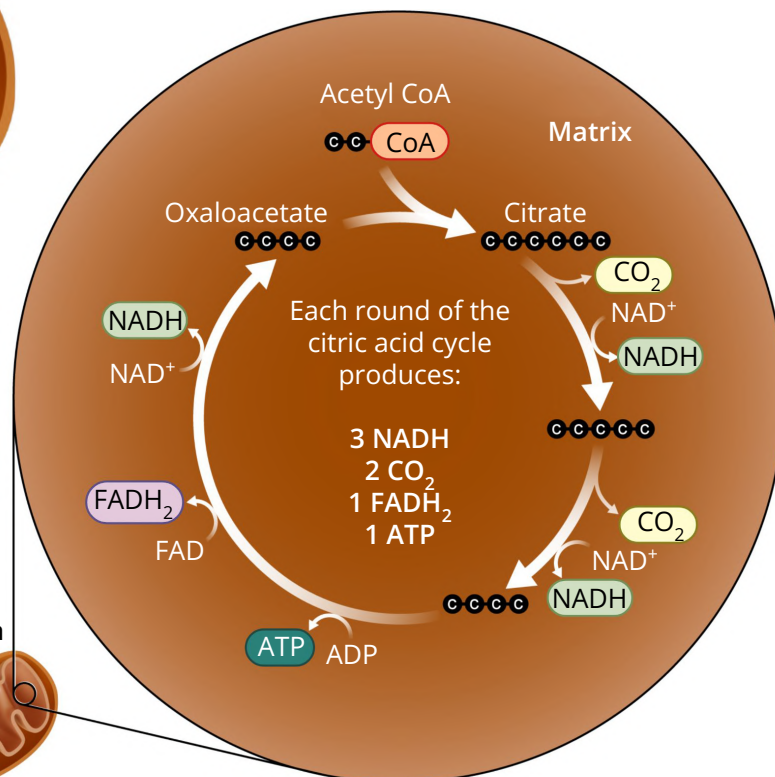
Krebs cycle



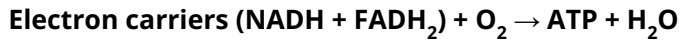
The Krebs cycle is also known as the **citric acid cycle** or the **tricarboxylic acid (TCA) cycle**. Like pyruvate oxidation, it also occurs in the mitochondrial matrix and the cytosol for prokaryotes.

1. Acetyl-CoA joins **oxaloacetate** (four-carbon) to form **citrate** (six-carbon).
2. Citrate undergoes rearrangements that produce two CO_2 and two NADH .
3. After the loss of two CO_2 , the resulting four-carbon molecule produces one ATP through substrate-level phosphorylation.
4. The molecule will now transfer electrons to one FAD , which is reduced into one FADH_2 .
5. Lastly, the molecule is converted back into oxaloacetate and also gives electrons to produce one NADH .
6. Two acetyl-CoA molecules produce four CO_2 , six NADH , two FADH_2 , two ATP.

Krebs Cycle



Oxidative phosphorylation



The **electron transport chain (ETC)** and **chemiosmosis** (ions moving down electrochemical gradients) work together to produce ATP in oxidative phosphorylation. Oxygen acts as a **final electron acceptor** and gets reduced to form water.

ETC goal: Regenerate electron carriers and create an electrochemical gradient to power ATP production.

The mitochondrial **inner membrane** is the location of the ETC for eukaryotes while the **cell membrane** is the location of the ETC for prokaryotes.

Four **protein complexes (I-IV)** are responsible for moving electrons through a series of **oxidation-reduction (redox) reactions** in the ETC. As the series of redox reactions occurs, protons are pumped from the mitochondrial matrix to the intermembrane space, forming an **electrochemical gradient**. This is the reason the intermembrane space is highly acidic.

NADH donates electrons to complex-I, regenerating NAD^+ . FADH_2 donates electrons to complex-II, regenerating FAD. NADH generates more proton pumping than FADH_2 .

Chemiosmosis goal: Use the proton electrochemical gradient (**proton-motive force**) to synthesize ATP.

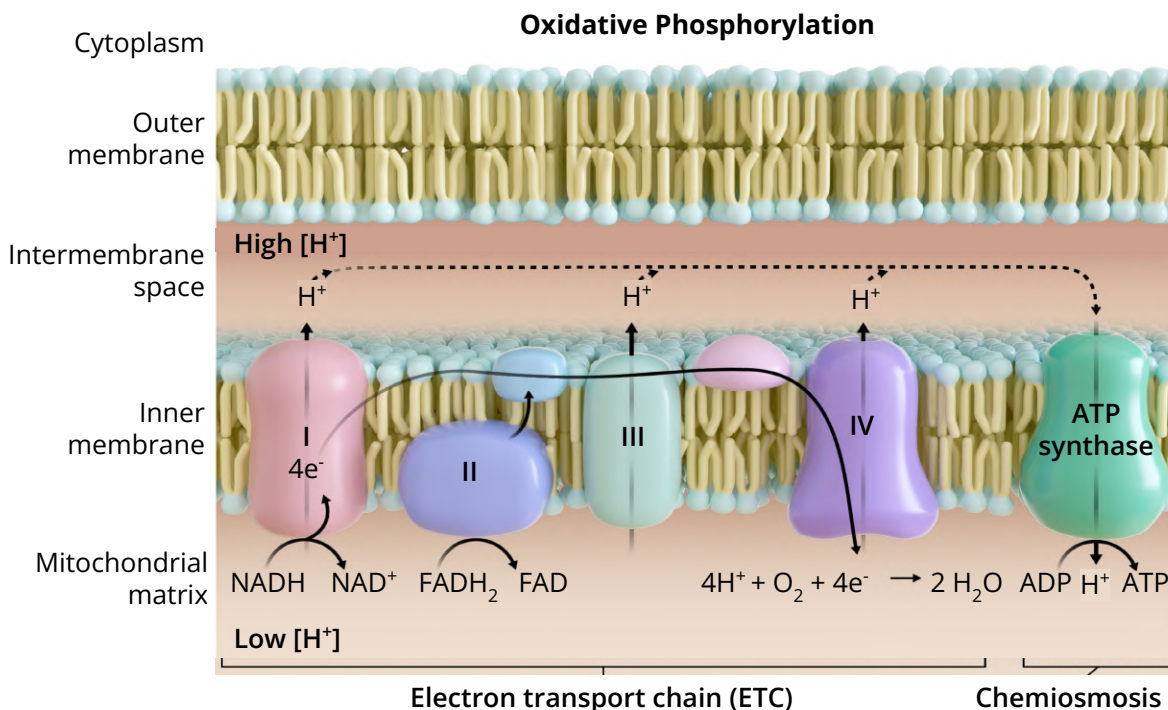
ATP synthase is a channel protein that allows protons to flow *down* their electrochemical gradient (from the intermembrane space back to the mitochondrial matrix). The spontaneous movement of protons generates energy that is used to convert ADP into ATP, a condensation reaction that is **endergonic** (requires energy + nonspontaneous = $+\Delta G$).

ATP Yield of Aerobic Cellular Respiration

Aerobic respiration is **exergonic** ($-\Delta G$).

NADH produces three ATP (NADH from glycolysis produces less). FADH_2 produces two ATP.

Stage	Net products	Net ATP yield
Glycolysis	2 ATP (substrate level)	2 ATP
	2 NADH	4-6 ATP
Pyruvate decarboxylation	2 NADH	6 ATP
Krebs cycle	2 ATP	2 ATP
	6 NADH	18 ATP
	2 FADH_2	4 ATP
Total		36-38 ATP



Anaerobic Energy Pathways

While aerobic respiration is the most efficient method of energy generation, there are two anaerobic alternatives:

- **Anaerobic respiration:** Common in many microorganisms. Uses alternative electron acceptors (e.g., sulfate, nitrate) instead of oxygen at the end of an electron transport chain.
- **Fermentation:** Common in yeast and human muscle cells. Creates byproducts while regenerating reactants needed for glycolysis.

Fermentation

Fermentation is an anaerobic pathway (no oxygen) that only relies on glycolysis by converting the produced pyruvate into different molecules in order to oxidize NADH back to NAD^+ . Regenerating NAD^+ means glycolysis can continue to make ATP. Fermentation occurs within the cytosol. The two most common types of fermentation are lactic acid fermentation and alcohol fermentation.

1. Lactic acid fermentation

Lactic acid fermentation uses the two NADH from glycolysis to reduce the two pyruvate into two lactic acid molecules. Thus, NADH is oxidized back to NAD^+ so that glycolysis may continue.

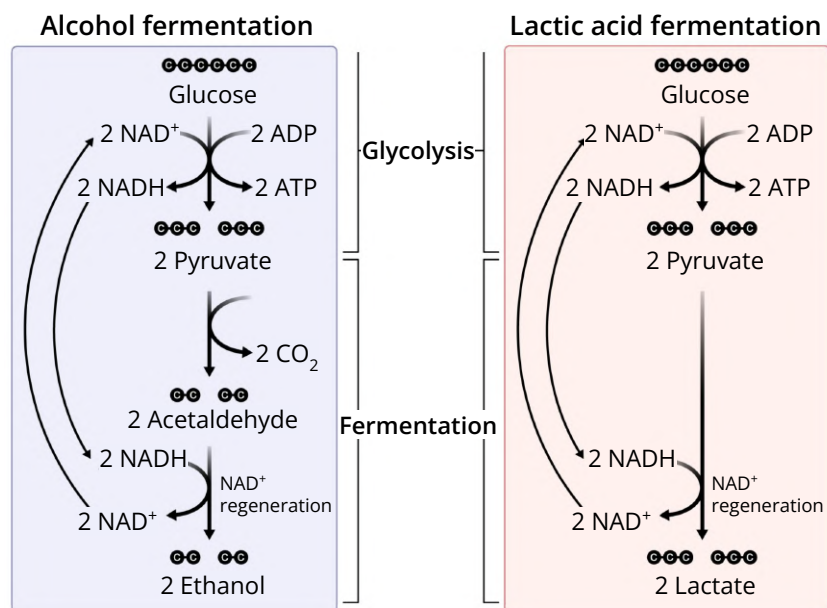
Lactic acid fermentation is used by muscle cells during periods of intense exercise, when aerobically-produced ATP is quickly depleted and there is low oxygen availability. Muscle cells will then default to lactic acid fermentation to produce energy anaerobically to continue to meet demand.

Additionally, lactic acid fermentation occurs continuously in **red blood cells**, which lack mitochondria needed for aerobic respiration.

2. Alcohol fermentation

Alcohol fermentation, conducted by yeast, uses the two NADH from glycolysis to convert the two pyruvate into two ethanol. Thus, NADH is oxidized back to NAD^+ so that glycolysis may continue. However, this process has an extra step that first involves the decarboxylation of pyruvate into acetaldehyde, which is only then reduced by NADH into ethanol.

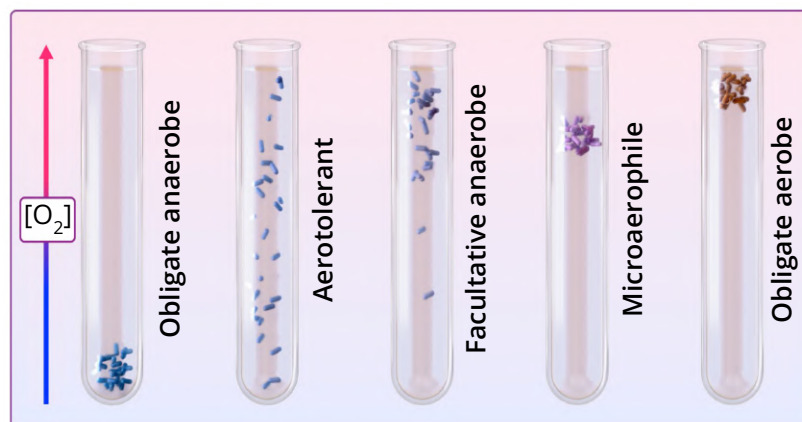
Types of Fermentation



Types of organisms based on ability to grow in oxygen:

- **Obligate aerobes:** Only perform aerobic respiration, so they need the presence of oxygen to survive.
- **Obligate anaerobes:** Only undergo anaerobic respiration or fermentation; oxygen is poison to them.
- **Facultative anaerobes:** Can do aerobic respiration, anaerobic respiration, or fermentation, but prefer aerobic respiration because it generates the most ATP.
- **Microaerophiles:** Only perform aerobic respiration, but high amounts of oxygen are harmful to them.
- **Aerotolerant organisms:** Only undergo anaerobic respiration or fermentation, but oxygen is *not* poisonous to them.

Microorganism Oxygen Preferences

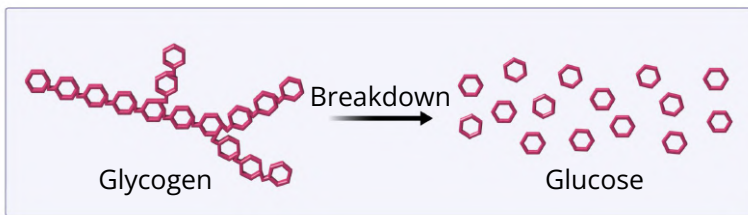


Alternative Sources of Energy Generation

Molecules other than glucose, such as other types of carbohydrates, fats, and proteins can be modified to enter cellular respiration at various stages for energy generation.

1. **Other carbohydrates** mostly enter during glycolysis. **Glycogenolysis** describes the release of glucose-6-phosphate from **glycogen**, a highly branched polysaccharide of glucose. Disaccharides can undergo hydrolysis to release two carbohydrate monomers, which can enter glycolysis.

Glycogenolysis



Carbohydrates are the preferred energy source since they are easily catabolized and are high yield.

Glycogenesis refers to the reverse process, the conversion of glucose into glycogen to be stored in the liver when energy and fuel is sufficient. Glycogen is stored in the liver and muscle cells.

2. **Fats** are mostly present in the body as triglycerides. **Lipases** are required to first digest fats into free fatty acids and alcohols.

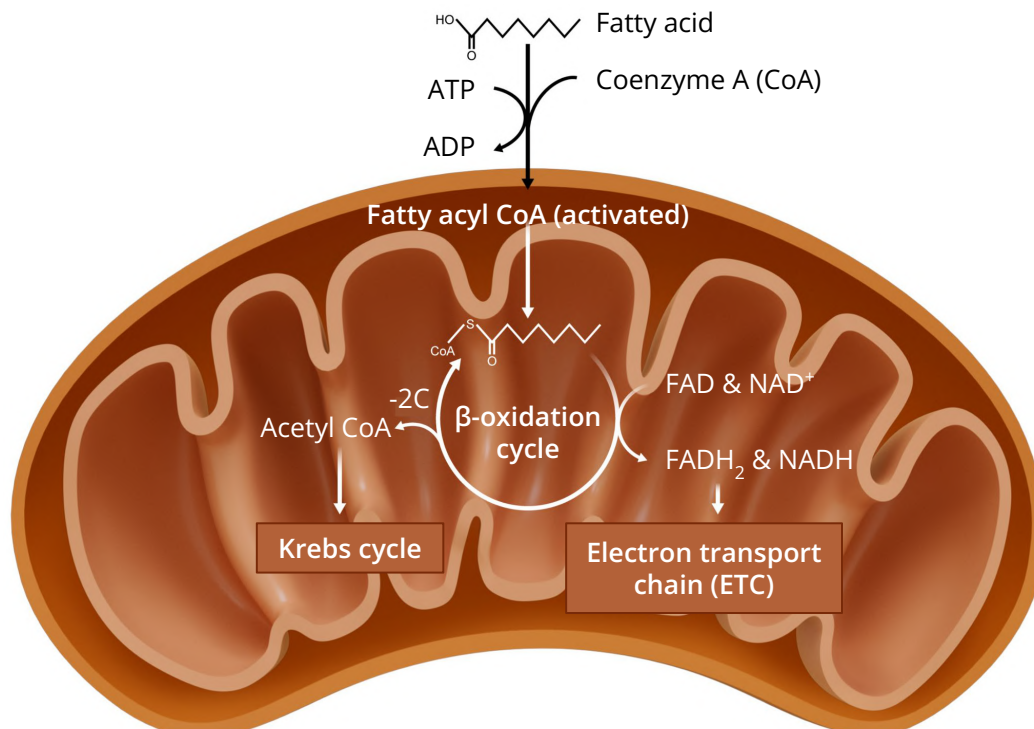
Adipocytes are cells that store fat (triglycerides) and have hormone-sensitive lipase enzymes to help release triglycerides back into circulation as **lipoproteins** or as free fatty acids bound by a protein called **albumin**.

Free fatty acids undergo **beta-oxidation** to be converted into acetyl-CoA. Beta-oxidation occurs in the mitochondrial matrix of eukaryotic cells and requires an initial investment of ATP. The fatty acid chain is then continuously cleaved to yield two-carbon acetyl-CoA molecules and electron carriers NADH and FADH_2 . Acetyl-CoA enters the Krebs cycle while NADH/ FADH_2 go directly into the ETC for ATP production.

Fats are harder to catabolize than carbohydrates as they must undergo beta-oxidation and transport away from fat cells. However, per carbon molecule, fats contain the most energy.

3. **Proteins** are the least desirable energy source because the processes to get them into cellular respiration take considerable energy and proteins are needed for many essential functions in the body.

Beta-Oxidation



Ch. 4: Photosynthesis

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- Objective of Photosynthesis
- Photosynthesis and Cellular Respiration
- Leaf Structures of Photosynthesis
- Light-Dependent Reactions of Photosynthesis
- The Calvin Cycle
- Photorespiration
- Alternative Photosynthetic Pathways

Objective of Photosynthesis

While **heterotrophs** must get energy from the food they eat, **autotrophs** can make their own food. Photoautotrophs take light energy and convert it to chemical energy using **photosynthesis**. This process originally developed in **cyanobacteria**.

Photosynthesis reduces atmospheric carbon dioxide, releases oxygen, and creates chemical energy that can be transferred through food chains. **Photons** (light energy) are used to synthesize sugars (glucose) in photosynthesis.

Carbon fixation is the process by which inorganic carbon (CO_2) is converted into an organic molecule (glucose). Photosynthesis takes electrons released from **photolysis** (the process of splitting water molecules) and excites them using solar energy. These excited electrons are then used to power carbon fixation.

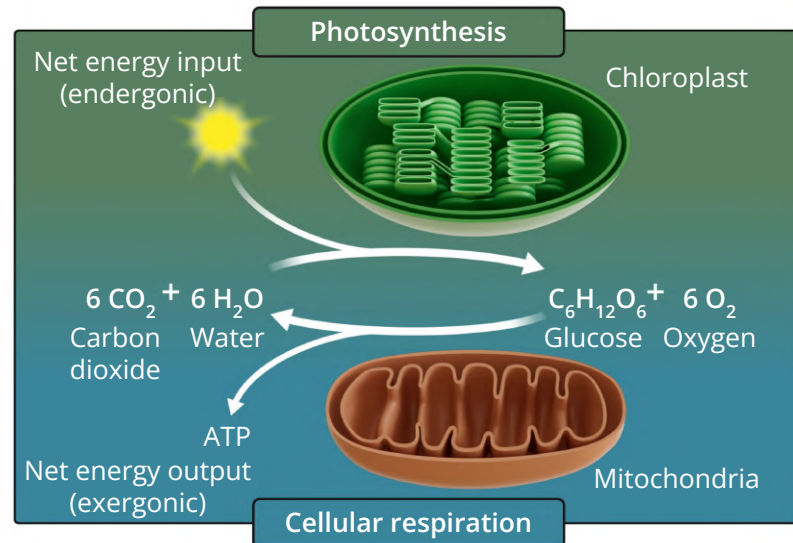
Photosynthesis and Cellular Respiration

Photosynthesis and cellular respiration are reverse processes in terms of their overall reactions:

Photosynthesis is non-spontaneous and endergonic, producing glucose after an input of solar energy.

Cellular respiration is spontaneous and exergonic, breaking down glucose to generate energy in the form of ATP.

Photosynthesis vs. Cellular Respiration



Leaf Structures of Photosynthesis

Mesophyll cells: Located between the upper and lower epidermis of leaves. Facilitate gas movement within the leaf. Contain chloroplasts.

Chloroplasts: Organelle that carries out photosynthesis. Found in plants and photosynthetic algae, but *not* in cyanobacteria. They are similar to mitochondria and contain the following structures (outermost to innermost):

Structure	Description
Outer membrane	Outer plasma membrane made of phospholipid bilayer.
Intermembrane space	Space between the outer and inner membranes
Inner membrane	Inner plasma membrane made of phospholipid bilayer.
Stroma	Fluid material that fills the area inside the inner membrane. The Calvin cycle occurs here.
Thylakoids	A membrane structure within the stroma. Multiple stack up to form a granum . They are the site of light-dependent reactions .
Thylakoid lumen	Interior of the thylakoid. H^+ ions accumulate here, making it acidic.

Light-Dependent Reactions of Photosynthesis

The **light-dependent reactions** occur in the **thylakoid membrane** and harness light energy to produce ATP and **NADPH** (an electron carrier) for later use in the **Calvin cycle**. ATP generated in the light-dependent reactions is *not* used to power the cell; it is consumed in the Calvin cycle.

Photosystems contain special pigments, such as **chlorophyll**, that absorb photons. Chlorophyll absorbs red and blue light and reflects green light, giving plants their green color.

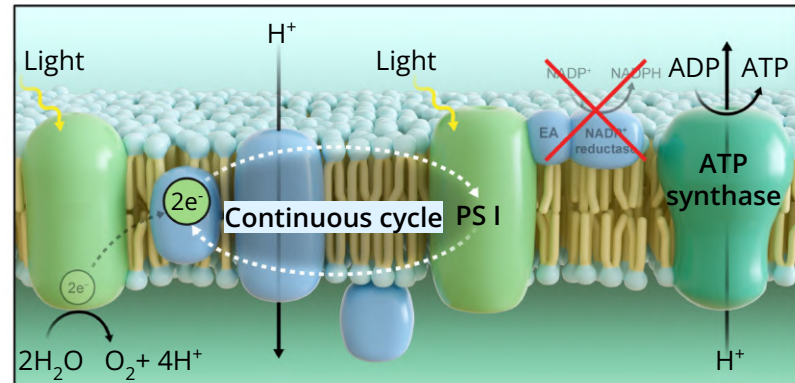
The **reaction center** is a special pair of chlorophyll molecules in the center of these proteins. Chlorophyll has a **porphyrin ring** structure with a magnesium atom bound in its center. **Photosystem II (P_{680})** and **photosystem I (P_{700})** are used in photosynthesis.

Non-cyclic photophosphorylation is carried out by the light-dependent reactions. Below are the important steps of this process:

1. Water is split (**photolysis**), passing electrons to photosystem II (PS II) and releasing protons into the thylakoid lumen.
2. Photons excite electrons in the **reaction center** of PS II, passing the electrons to a **primary electron acceptor**.

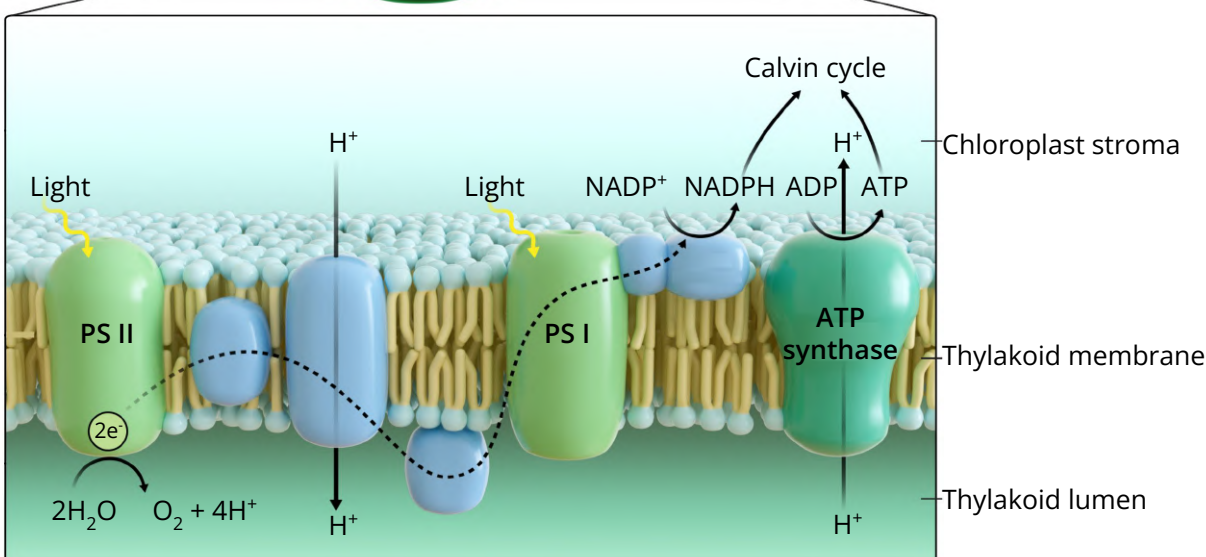
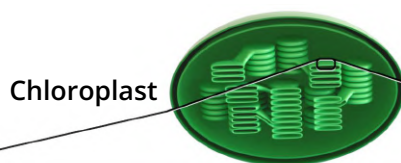
3. The primary electron acceptor sends the excited electrons to the **electron transport chain (ETC)**. During the redox reactions within the ETC, protons are pumped from the stroma to the thylakoid lumen. The electrons are then deposited into photosystem I (PS I).
4. Photons excite pigments in PS I, energizing the electrons in the reaction center to be passed to another primary electron acceptor.
5. The electrons are sent to a short electron transport chain that terminates with **NADP⁺ reductase**, an enzyme then reduces NADP⁺ into NADPH using electrons and protons.
6. The accumulation of protons in the thylakoid lumen generates an **electrochemical gradient** that is used to produce ATP using an ATP synthase, as H⁺ moves from the thylakoid lumen back into the stroma.

Cyclic Photophosphorylation



Cyclic photophosphorylation: Electrons from PS I are cycled back to the first ETC. This process increases proton pumping across the thylakoid membrane, enhancing ATP production. Electrons bypass NADP⁺ reductase, so no NADPH is generated.

Non-Cyclic Photophosphorylation

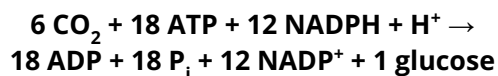


The Calvin Cycle

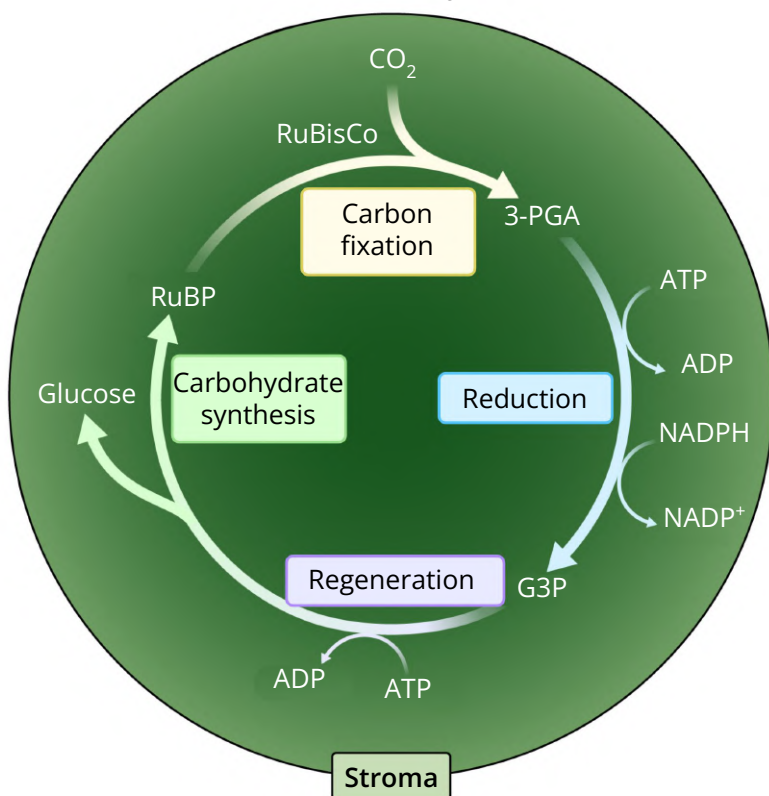
The **Calvin cycle** is made up of reactions known as **light-independent reactions** because they do *not* directly use light energy, but can only occur if the light-dependent reactions are providing ATP and NADPH.

The Calvin cycle fixes carbon dioxide molecules that enter plant leaves through **stomata** (pores in the epidermis). It takes place in the chloroplast stroma of plant mesophyll cells.

1. **Carbon fixation:** Carbon dioxide combines with five-carbon ribulose-1,5-bisphosphate (**RuBP**) to form six-carbon molecules, which quickly break down into three-carbon phosphoglycerates (**PGA**). This reaction is catalyzed by **RuBisCo**.
2. **Reduction:** **PGA** is phosphorylated by ATP and subsequently reduced by NADPH to form glyceraldehyde-3-phosphate (**G3P**).
3. **Regeneration:** Most of the G3P is converted back to RuBP.
4. **Carbohydrate synthesis:** Some of the G3P is used to make glucose.



The Calvin Cycle



Photorespiration

RuBisCo, in addition to fixing carbon dioxide into RuBP, can also cause oxygen to bind to RuBP in a process called **photorespiration**.

Photorespiration produces two-carbon **phosphoglycolate** and eventually converts it into PGA. There is a net loss of fixed carbon in the process and no new glucose produced.

Photorespiration is also called **C₂ photosynthesis**, since a two-carbon molecule is produced.

Hot and dry: Stomata are closed to minimize water loss; oxygen accumulates inside the leaf while carbon dioxide is used up. RuBisCo binds oxygen and photorespiration occurs.

Alternative Photosynthetic Pathways

C₃ photosynthesis: Normal photosynthesis, where three-carbon PGA is produced.

C₄ photosynthesis: Produces a four-carbon intermediate. Uses **spatial isolation** of carbon dioxide to prevent photorespiration in plants living in hot environments. Below are the important steps:

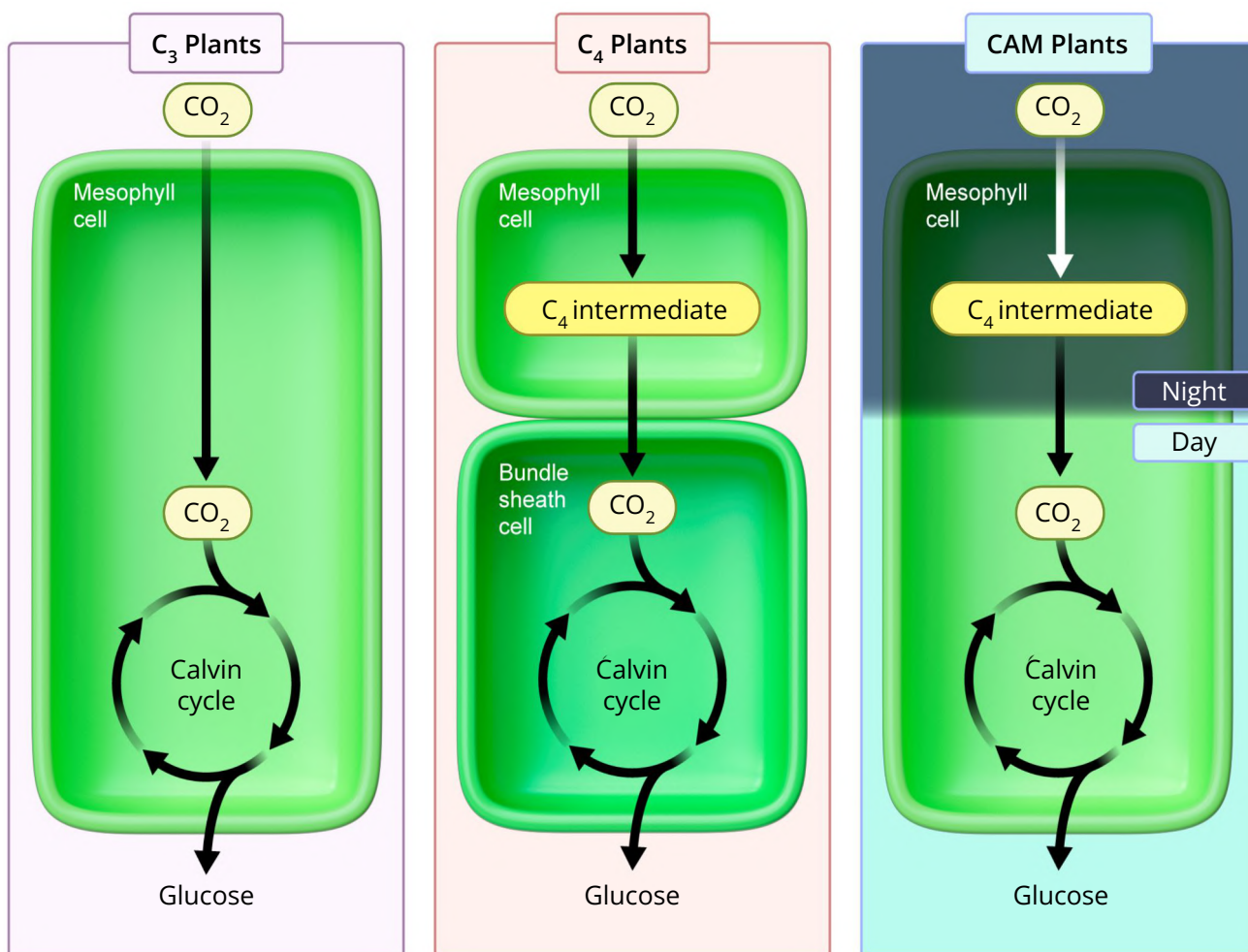
1. CO₂ is fixed and produces a four-carbon **oxaloacetate** molecule (first intermediate), which is then converted into **malate** (second intermediate) in the mesophyll cells.
2. Malate is transferred to **bundle sheath cells**, which have less oxygen.
3. Malate is decarboxylated, spatially isolating CO₂. It is then fixed by RuBisCo and enters the Calvin cycle.

Crassulacean acid metabolism (CAM)

photosynthesis: Uses **temporal isolation** of carbon dioxide to prevent photorespiration in hot and dry environments. Below are the important steps:

1. During the day, stomata are closed to prevent **transpiration** (evaporation of water from plants).
2. During the night, stomata are open to let carbon dioxide in. Just as in C₄ photosynthesis, CO₂ is fixed, producing oxaloacetate and afterwards malate. However, malate is not shuttled to bundle sheath cells.
3. During the next day, the stomata are closed again and malate is decarboxylated. CO₂ is temporally isolated from O₂, which cannot enter during the day. CO₂ is fixed by RuBisCo and enters the Calvin cycle.

Types of Photosynthesis in Plants



Ch. 5: Cell Division

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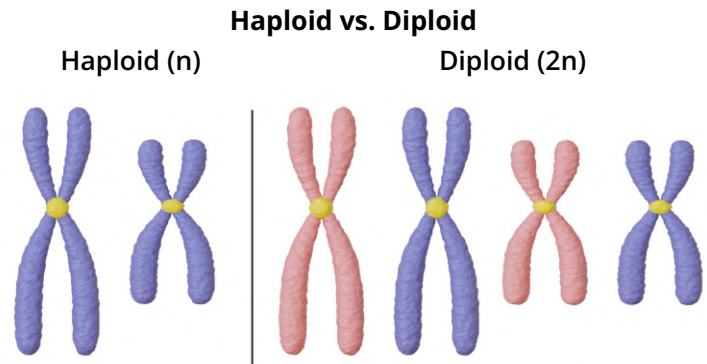
- Key Terms
- The Cell Cycle
- Components of Interphase
- Cell Cycle Regulation
- Microtubule Organizing Centers
- Components of the M Phase
- Binary Fission
- Meiosis
- Chromosome and Chromatid Numbers During Mitosis and Meiosis

A part of **cell theory** states that all cells arise from pre-existing cells through **cell division**.

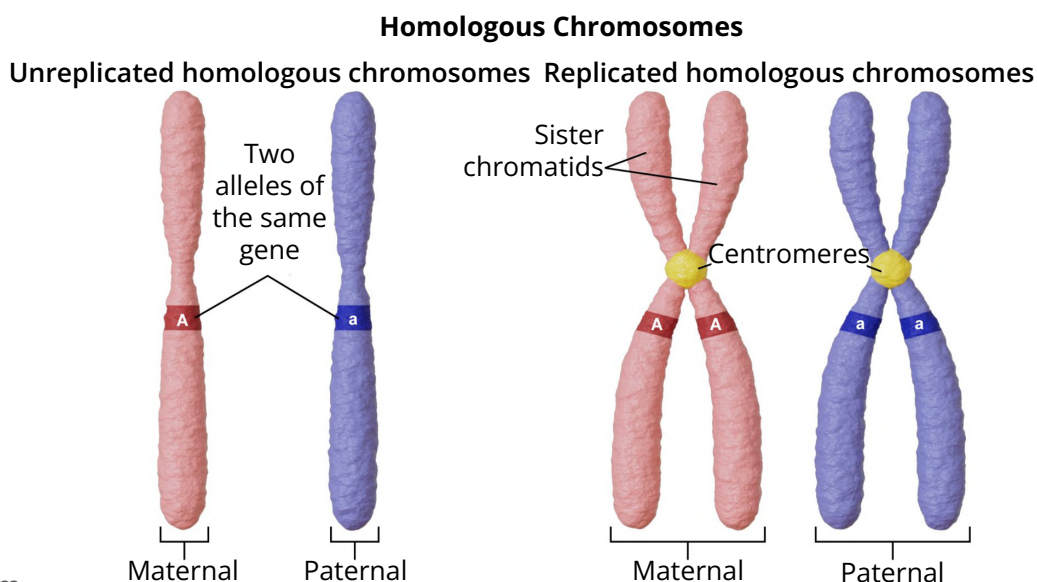
Key Terms

- **Genome:** All the DNA in a cell.
- **Chromosomes:** Individual molecules of DNA that each contain a portion of the entire genome.
- **Homologous chromosome pairs:** Pairs of chromosomes that contain the same genes in the same order, with one inherited from each parent.
- **Sister chromatids:** Identical, attached copies of a single chromosome that form dyads.
- **Dyads:** Replicated chromosomes containing two sister chromatids that look like an "X".
- **Centromeres:** Regions of DNA that connect sister chromatids in a dyad.
- **Kinetochores:** Proteins on the sides of centromeres that help microtubules pull sister chromatids apart during cell division.

- **Karyokinesis:** Division of the nucleus.
- **Cytokinesis:** Physical division of the cytoplasm and cell membrane.
- **Parent cell:** Original cell that undergoes cell division to produce two **daughter cells**.
- **Ploidy:** Describes the number of chromosome sets found in the body. Humans are **diploid** because they contain two sets of chromosomes (46 chromosomes, 23 pairs), one from each parent. However, they also have **haploid** cells (gametes) that only contain one chromosome set (23 chromosomes).



- **Sex chromosomes:** One pair in the human body; they determine sex.
- **Autosomes:** 22 pairs in the human body; they are nonsex chromosomes.
- **Gametes:** Haploid cells (sperm and eggs).
- **Germ cells:** Diploid cells that divide and increase by mitosis and differentiate into gametocytes.
- **Gametocyte:** Eukaryotic differentiated germ cells that can undergo meiosis to produce gametes.
- **Somatic cells:** All body cells excluding the gametes; diploid in humans.



The Cell Cycle

The **cell cycle** is divided into **interphase** (**G₁**, **G₀**, **S**, and **G₂**) and the **M phase**. Interphase precedes mitosis and meiosis, and 90% of the cell cycle happens during interphase. M phase is where **karyokinesis** and **cytokinesis** occur.

Mnemonic:

- **Go**: Gap phase 1 (G₁) of interphase
- **Sam**: Synthesis phase (S) of interphase
- **Go**: Gap phase 2 (G₂) of interphase
- **Make**: Mitosis of the M phase
- **Cake**: Cytokinesis of the M phase

Components of Interphase

1. **Gap phase 1 (G₁)**: Cell grows in preparation for cell division. Also checks for favorable conditions. If favorable, the cell will enter the S phase. If unfavorable, the cell will enter G₀ phase.
 - **G₀ phase**: Cells still carry out their functions but halt in the cell cycle. Cells that do *not* divide are stuck here.
2. **Synthesis phase (S)**: Cell replicates its genome here and moves to G₂ phase when completed. Centrosome duplicates.
3. **Gap phase 2 (G₂)**: Cell continues to grow and prepare for cell division by checking DNA for any errors after replication. Also checks for **mitosis promoting factor (MPF)**, aka **maturation promoting factor**, which needs to be present in adequate amounts for cell cycle continuation. Organelles are replicated here.

Cell Cycle Regulation

The cell cycle influences cell division through limitations to growth and regulation to prevent cancerous growth.

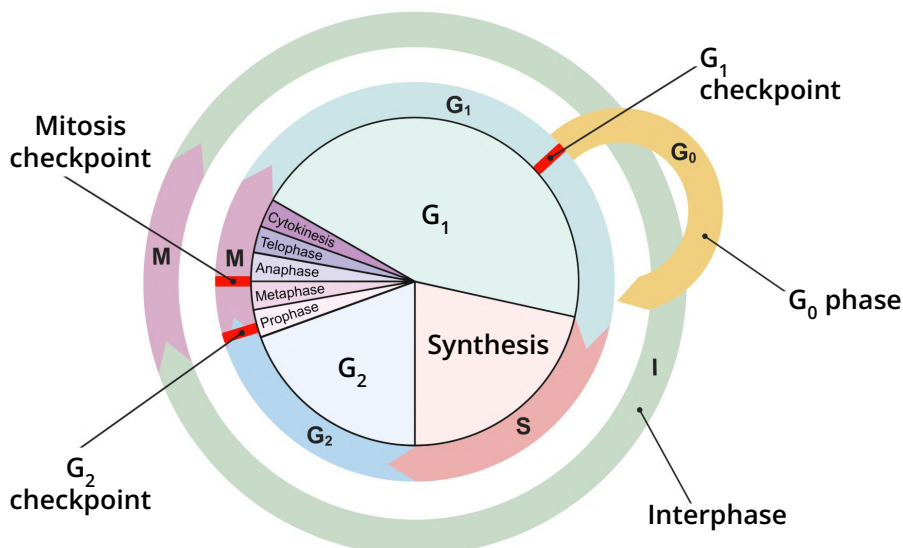
Functional limitations:

- **Surface to volume ratio (S/V)**: As a cell grows, its volume increases *more* than its surface area. Cells rely on the surface area of their cell membrane for transport of material in and out of the cell. Cell division occurs if the S/V ratio *decreases* (i.e., surface area is too small for the volume).

Cell specific regulations:

- **Cell specific checkpoints**:
 - G₁ restriction point checks for favorable conditions to grow; enters G₀ phase if unfavorable.
 - G₂ restriction point checks accuracy of DNA replication and MPF levels.
 - M checkpoint (during metaphase) checks for chromosomal attachment to spindle fibers.
- **Cyclin-dependent kinases (CDKs)**: Phosphorylate certain substrates to signal cell cycle progression. Activated by **cyclin**, a protein that cycles through stages of synthesis and degradation.
- **Growth factors**: Bind to receptors in the plasma membrane to signal for cell division.
- **Density dependent inhibition**: Halting of cell division when density of cells is high.
- **Anchorage dependence**: Cells divide only when attached to an external surface.

The Cell Cycle



Microtubule Organizing Centers

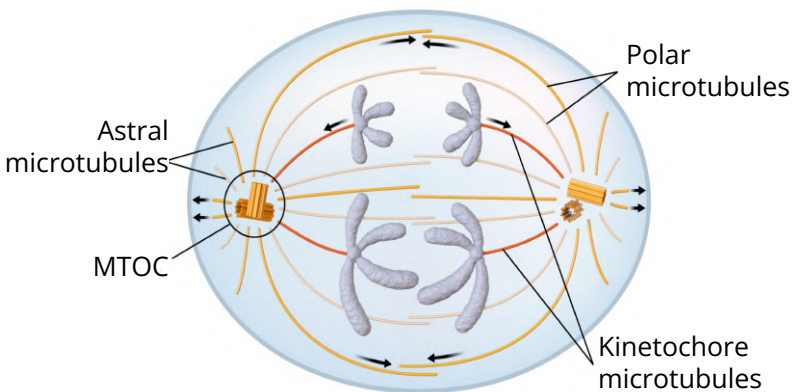
Microtubule organizing centers (MTOCs) are present in eukaryotic cells; they organize extension of microtubules, which are made of the protein **tubulin**. MTOCs are responsible for forming the spindle apparatus, which guides chromosomes during **karyokinesis**.

Centrosomes are organelles found in animal cells that contain a pair of **centrioles**. They act as microtubule organizing centers (MTOCs).

Microtubules in the **spindle apparatus**:

1. **Kinetochores microtubules:** Extend from centrosomes and attach to kinetochores on chromosomes.
2. **Astral microtubules:** Extend from centrosomes to the cell membrane to orient the spindle apparatus.
3. **Polar microtubules:** Extend from the two centrosomes and connect with each other. They push centrosomes to opposite ends of the cell.

Spindle Apparatus



Centrioles are hollow cylinders made of microtubules. **Centrosomes** are located near the nucleus and contain a pair of centrioles oriented at 90 degree angles to one another. They replicate during the S phase of the cell cycle so that each daughter cell after cell division has one centrosome.

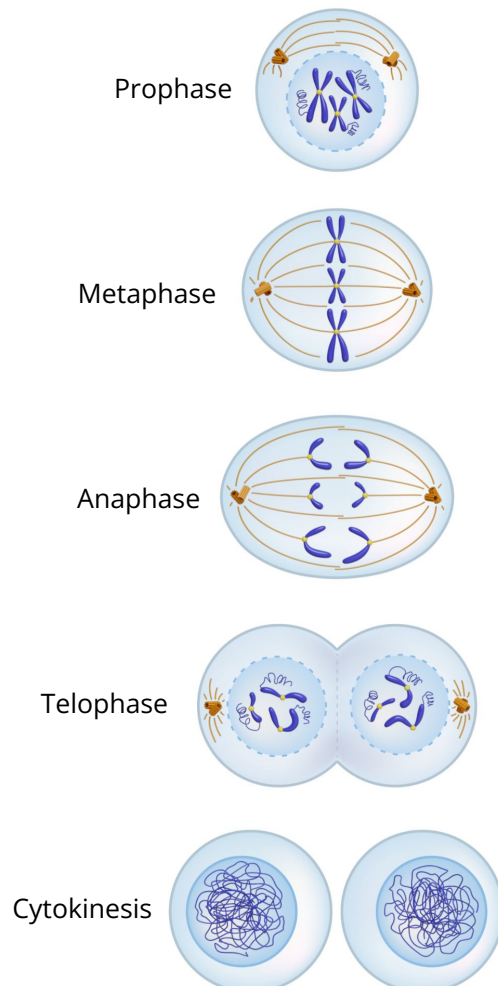
Components of the M Phase

The **M phase** is the stage of the cell cycle where **karyokinesis** (nuclear division) occurs and is then followed by **cytokinesis** (cell division).

Four phases of **mitosis**:

1. **Prophase:** Chromatin DNA condenses into chromosomes. The nucleolus and nuclear envelope disappear. Spindle apparatus forms.
2. **Metaphase:** The chromosomes line up single file along the **metaphase plate** (midpoint of the cell), guided by the spindle apparatus.
3. **Anaphase:** Kinetochores microtubules shorten to pull sister chromatids apart. The sister chromatids are now considered separate chromosomes. The chromosome number doubles.
4. **Telophase:** Chromosomes have segregated and nuclear membranes reform. In addition, nucleoli reappear and chromosomes decondense into chromatin.

Mitosis

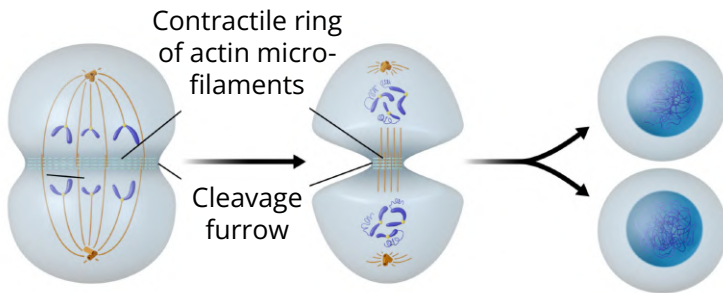


Binary Fission

Cytokinesis is the physical separation of the cytoplasm and cell membrane into two daughter cells.

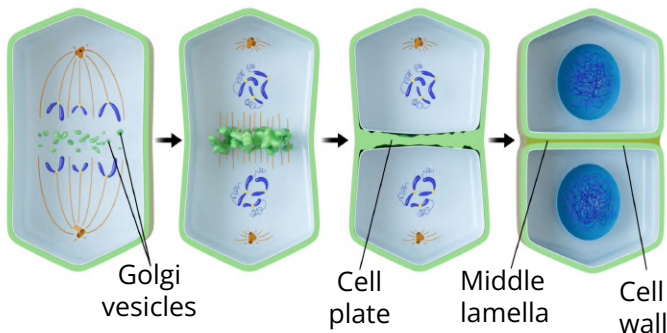
In animal cells, cytokinesis begins in late anaphase with the formation of a **cleavage furrow**. The cleavage furrow is a contractile ring of actin microfilaments and myosin motors that pinches the cell into two.

Animal Cell Cytokinesis



In plant cells, cytokinesis begins in telophase with the formation of a **cell plate**. The cell plate is created by vesicles from the Golgi apparatus and ends up producing the **middle lamella** (cements plant cells together).

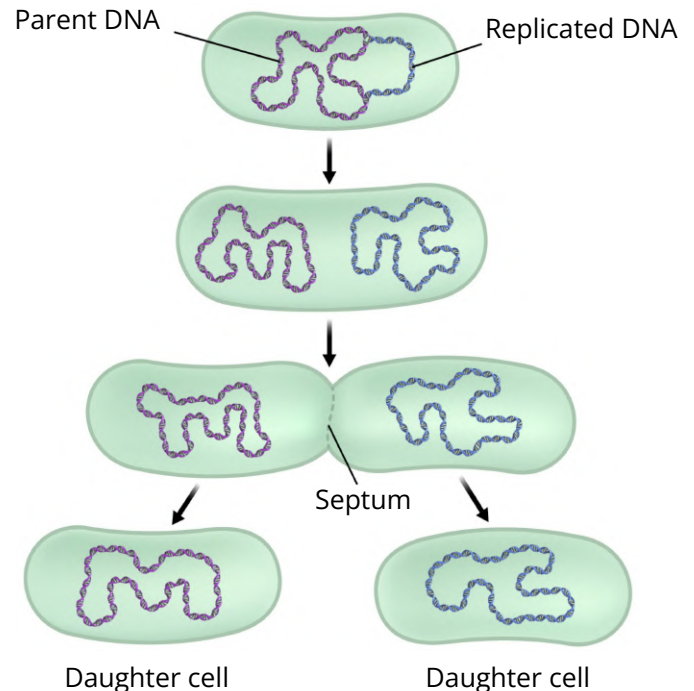
Plant Cell Cytokinesis



Binary fission is used by archaea, bacteria, and certain organelles (mitochondria, chloroplasts) to reproduce.

During binary fission, organisms replicate their genome while cell division is happening. Replicated DNA segregate to opposite ends of the cell, and a septum forms to separate the cell into two daughter cells.

Binary Fission



Meiosis

Meiosis produces four haploid daughter cells from one diploid parent cell. It does this by repeating the steps of karyokinesis twice. Meiosis can be divided into meiosis I (homologous chromosomes separate) and meiosis II (sister chromatids separate).

Meiosis I (reductional division) produces two haploid daughter cells through separation of **homologous chromosomes**.

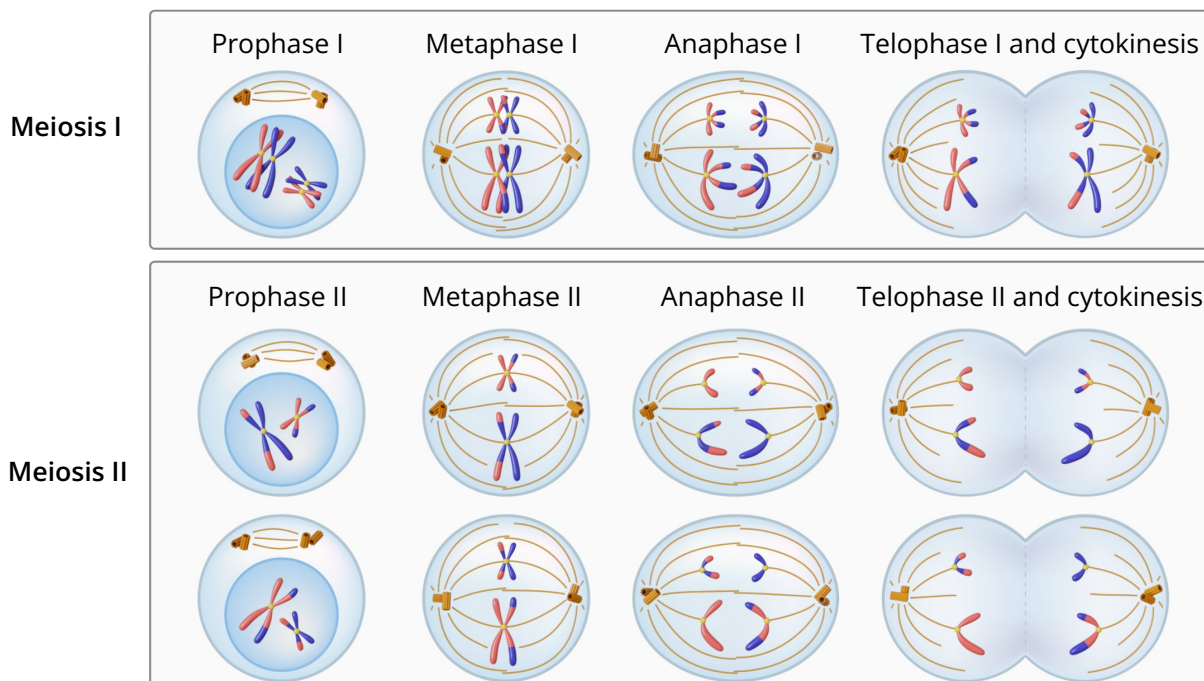
1. **Prophase I:** Chromatin condenses into chromosomes. The nucleolus and nuclear envelope will disappear. Homologous chromosomes pair up and crossing over occurs.
 - **Synapsis:** The pairing up of homologous chromosomes to form **tetrads**.
 - **Tetrads (bivalents):** Pair of two homologous chromosomes each with two sister chromatids.
 - **Chiasmata:** Where two chromosomes of a homologous pair crossover during synapsis, causing **genetic recombination**.
 - **Genetic recombination:** Exchange of DNA between chromosomes to produce genetically diverse offspring.

2. **Metaphase I:** Tetrads randomly line up double-file on the metaphase plate; this contributes to genetic diversity.
3. **Anaphase I:** Kinetochore microtubules shorten to separate homologous chromosomes from each other.
4. **Telophase and Cytokinesis I:** After tetrads have been pulled to opposite poles, nuclear membranes reform. In addition, nucleoli reappear and chromosomes decondense into chromatin. A **cleavage furrow** forms in animal cells and a **cell plate** forms in plant cells.

Meiosis II is very similar to mitosis because sister chromatids are separated. Two haploid cells divide into four haploid daughter cells.

1. **Prophase II:** Chromatin condenses into chromosomes. The nucleolus and nuclear envelope will disappear. Spindle apparatus forms. No crossing over occurs.
2. **Metaphase II:** Chromosomes line up single-file at the metaphase plate just like in mitosis.
3. **Anaphase II:** Kinetochore microtubules shorten to pull sister chromatids apart. Sister chromatids become separate chromosomes and the chromosome number doubles.
4. **Telophase and Cytokinesis II:** Nuclear membranes reform, nucleoli reappear, and chromosomes decondense into chromatin. Four haploid daughter cells are produced in total.

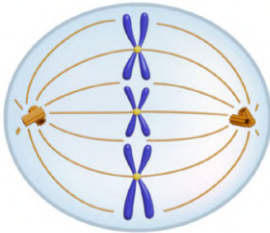
Meiosis



Chromosome and Chromatid Numbers During Mitosis and Meiosis

Mitosis:

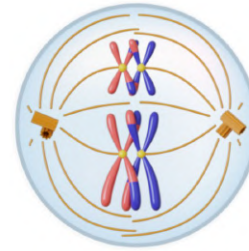
During the **S phase** of the cell cycle, a human's 46 chromosomes are duplicated. Afterwards, there are still 46 chromosomes but also 92 chromatids. They line up in **metaphase** individually as shown below:



During **anaphase** of mitosis, sister chromatids split. This produces 92 separate chromosomes, which are also counted as 92 chromatids. Each separated cell will have 46 chromosomes (46 chromatids). These cells are **diploid**.

Meiosis:

For **meiosis I**, a human goes through the same DNA replication in **S phase** as mitosis that results in 46 chromosomes and 92 chromatids. However, during **metaphase** the chromosomes double up as shown below:



During **anaphase** of meiosis I, homologous chromosomes split up. This results in the same total numbers: 46 chromosomes and 92 chromatids. Each daughter cell will have 23 chromosomes and 46 chromatids.

Meiosis II is very similar to mitosis and involves chromosomes lining up individually in **metaphase**. During **anaphase**, sister chromatids are separated, resulting in 23 chromosomes (23 chromatids) in each daughter cell. These cells are **haploid**.

Phase	# of chromosomes	# of chromatids
Mitosis		
Prophase	46	92
Metaphase	46	92
Anaphase	92	92
Telophase	92	92
End of mitosis (separated cells)	46	46
Meiosis I		
Prophase I	46	92
Metaphase I	46	92
Anaphase I	46	92
Telophase I	46	92
End of meiosis I (separated cells)	23	46
Meiosis II		
Prophase II	23	46
Metaphase II	23	46
Anaphase II	46	46
Telophase II	46	46
End of meiosis II (separated cells)	23	23

Ch. 6: Molecular Genetics

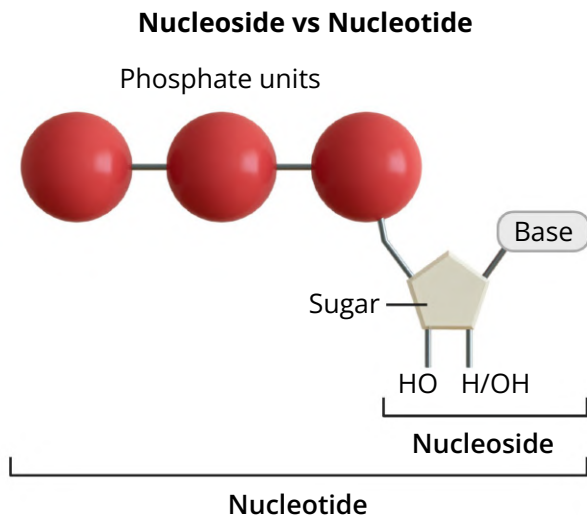
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Genetic Building Blocks

Nucleoside: Ribose sugar and nitrogenous base.

Nucleotide: Ribose sugar, nitrogenous base, and phosphate group.



Nucleic acids are polymers of nucleotides:

- **RNA** (ribonucleic acid) nucleotides have ribose sugars with hydroxyl groups (-OH) on the 2' and 3' carbons.
- **DNA** (deoxyribonucleic acid) nucleotides have deoxyribose sugars; their 3' carbon has a hydroxyl group, the 2' carbon does *not*.

Purines are the double-ringed nitrogenous bases **adenine (A)** and **guanine (G)**.

Pyrimidines are the single-ringed nitrogenous bases **cytosine (C)**, **thymine (T)**, and **uracil (U)**.

Mnemonic:

PUR As **Gold**: **PUR**ines are **A**denine and **G**uanine.

CUT the **PY**: **C**ytosine, **U**racil, and **T**hymine are **PY**rimidines.

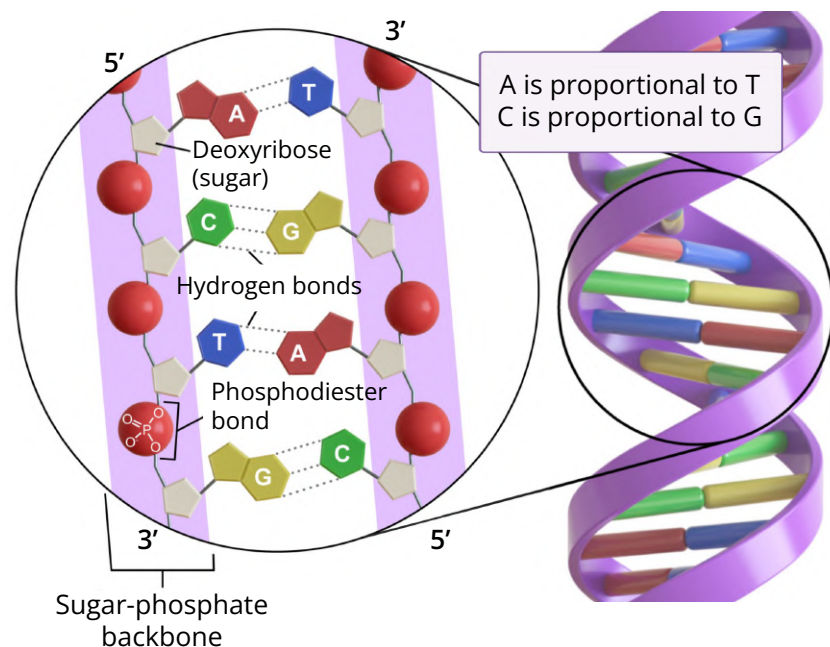
In **DNA**:

A binds to **T** (with two hydrogen bonds).

G binds to **C** (with three hydrogen bonds).

Since **G-C bonds** have more hydrogen bonds, a *higher* temperature is needed to break DNA strands with a larger proportion of G-C bonds.

DNA Structure



In **RNA**:

Uracil replaces thymine.

A binds to **U** (with two hydrogen bonds).

G binds to **C** (with three hydrogen bonds).

DNA Replication

An **origin of replication** is where the DNA strands first separate to initiate DNA replication. Organisms with circular DNA (bacteria) have a single origin of replication while organisms with linear DNA (humans) have multiple.

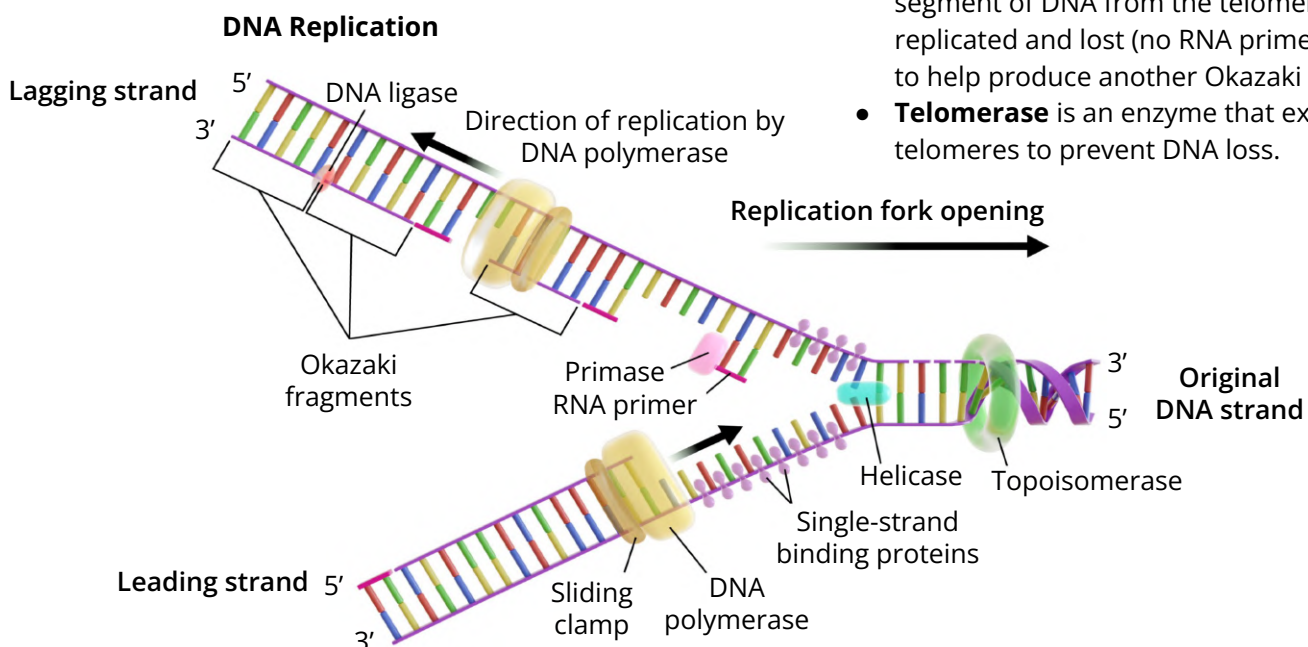
DNA undergoes **semiconservative replication**, where each new double helix produced by replication has one "new" strand and one "old" strand.

DNA is antiparallel, meaning that the 5' end (terminal phosphate group) of one strand is always next to the 3' end (terminal hydroxyl group) of the other strand and vice versa.

The G_1/S checkpoint regulates cell cycle transition from the G_1 phase into the S phase, checking for favorable conditions to grow. If unfavorable, the cell will stay in G_0 , halting DNA replication.

Steps of DNA replication:

1. **Initiation:** Origins of replication form at A-T rich DNA segments. A-T pairs have less hydrogen bonds than G-C pairs and are easier to split.
 - **Helicase** unzips DNA by breaking hydrogen bonds between strands, creating a **replication fork**. As it unzips the strands, helicase leads to **supercoiling** (tension ahead of the replication fork).
2. **Elongation:** Producing new DNA strands using different types of enzymes.
 - **Single-strand binding proteins** bind to uncoiled DNA strands, preventing reattachment of the strands to each other.
 - **Topoisomerase** nicks the DNA double helix ahead of helicase to relieve built-up tension and supercoiling.
 - **Primase** places **RNA primers** along the template strands to create 3' ends for nucleotide addition.
 - **Sliding clamp proteins** hold DNA polymerase onto the template strand.
 - **DNA polymerase** adds free nucleoside triphosphates to 3' ends. DNA polymerase can only add nucleotides onto a preexisting 3' hydroxyl group with primase.
 - The **leading strand** is produced continuously because it has a 3' end that faces the replication fork.
 - The **lagging strand** is produced discontinuously because its 3' end is facing away from the replication fork. Thus, many RNA primers are needed to produce short DNA fragments called **Okazaki fragments**.
 - A different **DNA polymerase** replaces RNA primers with DNA.
 - **DNA ligase** glues separated fragments of DNA together.
3. **Termination:** Replication fork cannot continue, ending DNA replication.
 - **Telomeres** are noncoding, repeated nucleotide sequences at the ends of linear chromosomes. They are necessary in **eukaryotes** because when the replication fork reaches the end of a chromosome, a small segment of DNA from the telomere is *not* replicated and lost (no RNA primer is present to help produce another Okazaki fragment).
 - **Telomerase** is an enzyme that extends telomeres to prevent DNA loss.



DNA replication proteins	Function
DNA helicase	Separates complementary strands at the replication fork.
Single-strand binding proteins	Proteins that prevent the two strands from coming back together after separating.
Topoisomerase (DNA gyrase)	Relaxes the DNA double helix from the tension and supercoiling the opening helix is creating.
Primase	Provides a 3' hydroxyl group for DNA polymerase to attach new nucleotides to.
DNA polymerases	The class of enzymes that extends DNA in the 5' to the 3' direction. Several have proofreading capabilities that allow them to catch synthesis errors.
DNA sliding clamp	Helps to hold DNA polymerase to the template strand.
DNA ligase	Glues together separate pieces of DNA.
Telomerase	Adds repetitive DNA to the ends of eukaryotic chromosomes, preventing critical information from being lost.

DNA Organization

Nucleosomes are complexes of DNA wrapped around **histone** proteins.

Histones are positively charged while DNA is negatively charged, allowing proper binding.

Chromatin refers to the overall packaging of DNA and histones. Below are two types of chromatin:

1. **Euchromatin:** Nucleosomes are "loosely packed", so DNA is readily accessible for transcription.
2. **Heterochromatin:** Nucleosomes are "tightly packed", so DNA is mostly inactive.

Acetylation of histones removes positive charges, relaxing DNA-histone attractions and allowing for *more* transcription to happen.

Deacetylation of histones *increases* positive charges, tightening DNA-histone attractions and *decreasing* transcription.

Methylation of histones adds methyl groups, either *increasing* or *decreasing* transcription.

Transcription

Genes are instructions within DNA that code for proteins. However, they must first be transcribed into RNA before being translated into proteins. In a gene, the promoter region comes first, then the gene operator, then the gene.

Specifically, DNA undergoes transcription to produce single-stranded **messenger RNA (mRNA)**.

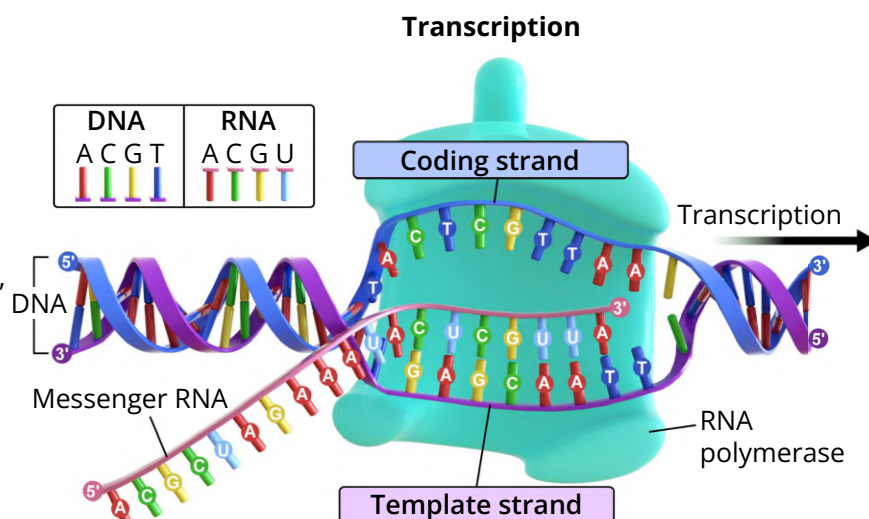
Because prokaryotes do *not* have membrane-enclosed nuclei, both transcription and translation occur simultaneously in the cytosol.

Steps of **transcription**:

1. **Initiation:** A promoter sequence (aka **promoter**) next to the gene attracts RNA polymerase to transcribe the gene.
2. **Elongation:** **Transcription bubble** forms and RNA polymerase travels in the 3' → 5' direction on the **template strand**. However, it extends RNA in the 5' → 3' direction.
3. **Termination:** A termination sequence (aka **terminator**) signals to RNA polymerase to stop transcribing the gene.

Because there are two complementary strands of DNA:

- The template strand used to transcribe the mRNA is called the **template/antisense/noncoding strand**.
- The other DNA strand is called the **coding/sense strand** because it is almost equivalent to the transcribed mRNA. However, the T in DNA is replaced with U in mRNA once transcribed.



Prokaryotic Transcriptional Control

In prokaryotes, transcription occurs in the **cytosol**. RNA polymerase opens up DNA, forming a **transcription bubble**.

An **operon** is a group of genes that function as a single unit that is controlled by one promoter. The **operator region** is present near the operon promoter. To regulate the promoter, **repressors** bind to the operator regions, while **activators** bind to the promoter sites.

The ***lac* operon** is an **inducible operon** (it must be induced to become active). ***LacZ***, ***lacY***, and ***lacA*** are the three genes contained within the *lac* operon that encode proteins required for **lactose metabolism**. The *lac* operon will only be induced when glucose is not available as an energy source and lactose must be metabolized.

The ***lac* repressor protein** is the first way that the *lac* operon is controlled. This protein is **constitutively expressed** (always on). Thus, the *lac* repressor protein is always bound to the **operator**, blocking transcription. However, when lactose is present, it is converted into **allolactose**. Allolactose binds directly to the repressor and removes it from the operator, allowing transcription to occur.

cAMP levels and **catabolite activator protein (CAP)** are the second level of *lac* operon regulation. cAMP levels are *inversely* related to glucose levels, so when glucose is low, cAMP is high.

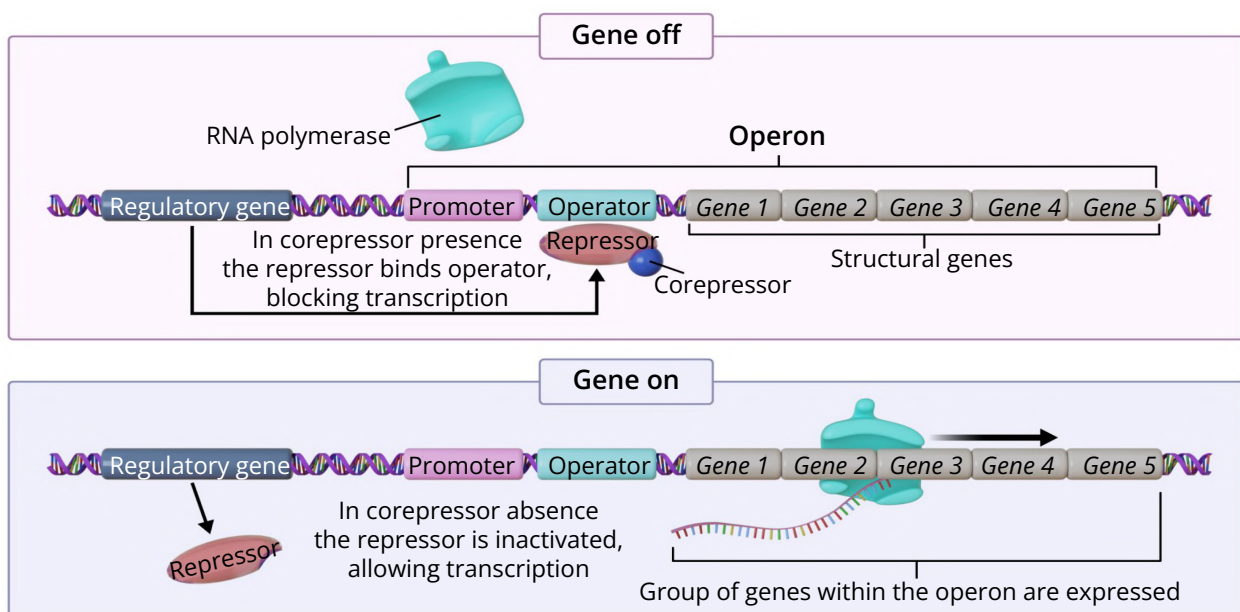
cAMP binds to **catabolite activator protein (CAP)**, which then attaches near the *lac* operon promoter to help attract RNA polymerase, promoting transcription.

	Lactose present	Lactose absent
Glucose present	Moderate transcription. Repressor not bound and CAP not bound.	No transcription. Repressor is bound and CAP is not bound.
Glucose absent	High transcription. Repressor not bound and CAP is bound.	No transcription. Repressor is bound and CAP is bound.

Another operon employed by prokaryotes is the ***trp* operon**, which is responsible for producing the amino acid **tryptophan**. It is known as a **repressible operon** because it codes for **tryptophan synthetase** and is always active unless the presence of tryptophan in the environment represses the operon.

Tryptophan binds to the ***trp* repressor protein**, which then attaches to the **operator** on the *trp* operon to prevent tryptophan production. Thus, this is the first level of *trp* operon regulation. When tryptophan is *not* present in the environment, the *trp* operon will undergo transcription because the *trp* repressor protein will be inactive.

Prokaryotic Transcriptional Control



Eukaryotic Transcriptional Control

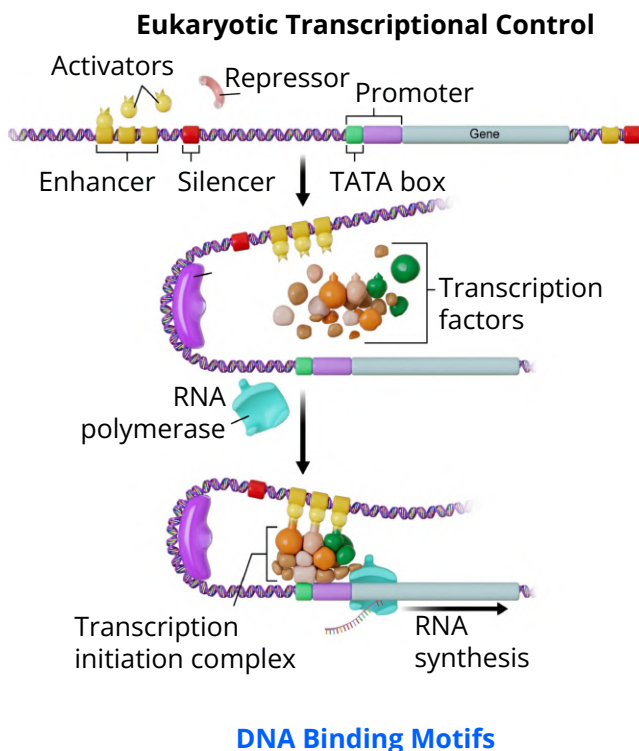
Unlike in prokaryotes, eukaryotic transcription occurs in the nucleus and uses **RNA polymerase** to transcribe most genes.

Transcription factors are needed in eukaryotes to help RNA polymerase bind to promoters. The **TATA box** is an AT-rich sequence in many promoters that transcription factors can recognize and bind to.

- **Enhancers** are DNA sites that **activator proteins** can bind to; they help *increase* transcription of a gene.
- **Silencers** are DNA sites that **repressor proteins** can bind to; they *decrease* transcription of a gene.

Enhancers and silencers can be far upstream or downstream from the gene, so DNA from these sites are thought to loop around to colocalize with RNA polymerase.

The terminator sequence ends transcription. It includes a **poly-A signal** in eukaryotes.



Regulatory proteins in eukaryotes and prokaryotes contain **DNA-binding motifs** that recognize and bind the **major groove** of regulatory sequences without unwinding the double helix. The four main types of binding motifs are zinc finger, helix-turn-helix, homeodomain, and leucine zipper.

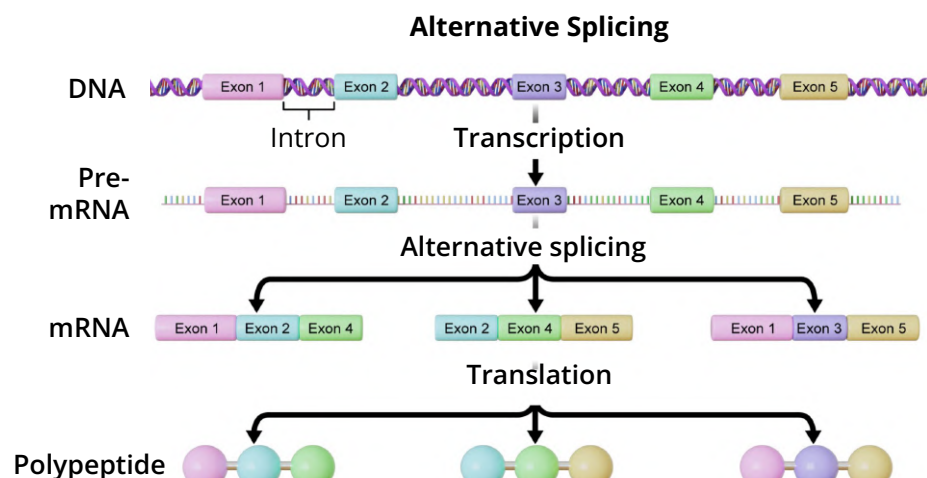
Eukaryotic Post-Transcriptional Modifications

Post-transcriptional modification describes the conversion of pre-mRNA into processed mRNA, which leaves the nucleus. Below are the three main types of post-transcriptional modifications:

1. **5' capping:** A guanine cap is added to the 5' end of the mRNA during elongation, protecting the mRNA from degradation.
2. **3' polyadenylation:** Addition of a poly-A tail (adenine nucleotides) to the 3' end to prevent mRNA degradation.
3. **Splicing out introns:** Introns are stretches of noncoding DNA that lie between regions of coding DNA (exons). **Splicing** refers to the removal of introns from pre-mRNA by **spliceosomes**.
 - The human genome contains approximately 20,000 protein coding genes.
 - The human genome contains more *non-coding* DNA than coding DNA.

Alternative splicing describes a single pre-mRNA having multiple possible spliced mRNA products. Thus, the same pre-mRNA can produce many different proteins.

miRNAs (micro RNA) are small RNA molecules that silence mRNA expression as a post-transcriptional gene regulation. miRNAs base-pair with segments of mRNA, inhibiting translation.



Translation

Ribosomes and **tRNA** (transfer RNA) are important players in **translation**, the process of converting mRNA into protein products.

Ribosomes are made up of one small subunit and one large subunit as described below:

- **Eukaryotes:** Small (**40S**) and large (**60S**) subunits form a **80S ribosome**. They are composed of rRNA (ribosomal RNA) and proteins.
- **Prokaryotes:** Small (**30S**) and large (**50S**) subunits form a **70S ribosome**. They are also composed of rRNA and proteins.

A **codon** is a group of three mRNA bases (A, U, G, or C) that code for an amino acid or terminate translation. There are 64 codon combinations total but only 20 amino acids, so degeneracy is present (multiple codons code for the same amino acid).

Memorize the following codons:

Start codon: **AUG** (methionine)

Stop codons: **UAA, UAG, UGA** (end translation; do not code for any amino acid)

The stretch of DNA between the start and stop codons is known as the **open reading frame**.

An **anticodon** is a group of three tRNA bases (A, U, G, or C) that base pairs with a codon. Each tRNA carries an amino acid to be added to the growing protein.

Aminoacyl-tRNA refers to a tRNA bound to an amino acid.

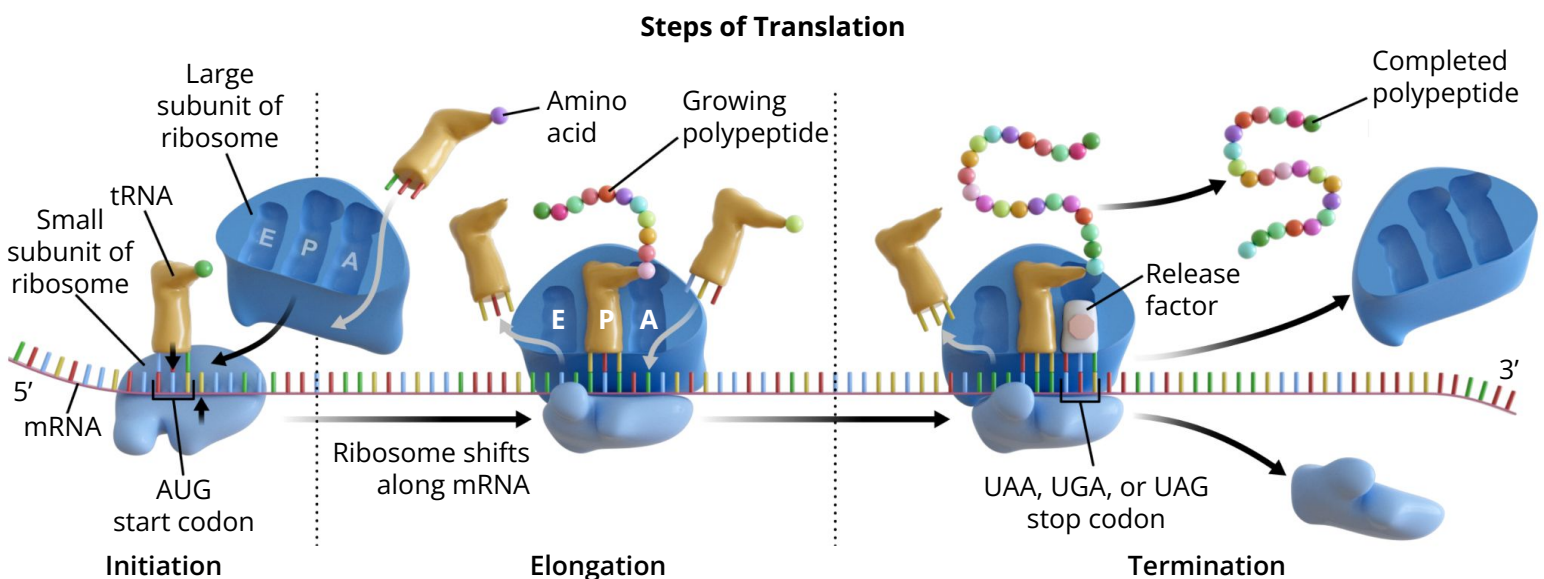
Ribosomal binding sites for tRNA:

1. **A site:** A for **aminoacyl-tRNA**, which first enters at this site.
2. **P site:** P for **peptidyl-tRNA**, which carries the growing polypeptide. The polypeptide chain moves from the P site to the tRNA on the A site during peptide bond formation.
3. **E site:** E for **exit site**. The tRNA from the P site is sent here and released from the ribosome.

Steps of translation:

1. **Initiation:** AUG (start codon) pairs with a methionine-tRNA at the P site.
2. **Elongation:** Subsequent aminoacyl-tRNAs enter at the A site. **Peptide bond** formation between the polypeptide of the P site and the amino acid at the A site is catalyzed by the ribosome. The resultant polypeptide is on the A site tRNA. Ribosome **translocation** occurs, shifting the tRNA at the A site to the P site and the tRNA at the P site to the E site (A→P→E).
3. **Termination:** A stop codon pairs with release factors and the polypeptide is released from the ribosome.

Specialized proteins known as **chaperonins** are found in both eukaryotic and prokaryotic organisms and function in assisting newly synthesized polypeptides to fold into their correct shape.



Mutations

A **DNA mutation** is a change in the DNA nucleotide sequence. A mutation that occurs in a **germ cell** (cells that give rise to gametes) can be inherited by offspring, whereas a mutation that occurs in a **somatic cell** (all other cells) only affects the individual.

Two main types of DNA mutations:

1. **Base substitutions (point mutations):** One nucleotide is replaced by another.
 - **Silent mutations:** No change in amino acid sequence. Due to “**third base wobble**”, mutations in the DNA sequence that affect the third base of a codon can still result in the same amino acid. Relies on the **degeneracy** (redundancy) of translation.
 - **Missense mutations:** Single change in amino acid sequence. Can either be **conservative** (similar amino acid) or **non-conservative** (different amino acid).
 - **Nonsense mutations:** Single change in amino acid sequence that results in a **stop codon**. Results in early termination of protein.
2. **Frameshift:** Mutations that result in a shift in the reading frame, changing the way the mRNA transcript is read. There are two types:
 - **Insertions:** Adding nucleotides into the DNA sequence.
 - **Deletions:** Removing nucleotides from the DNA sequence.

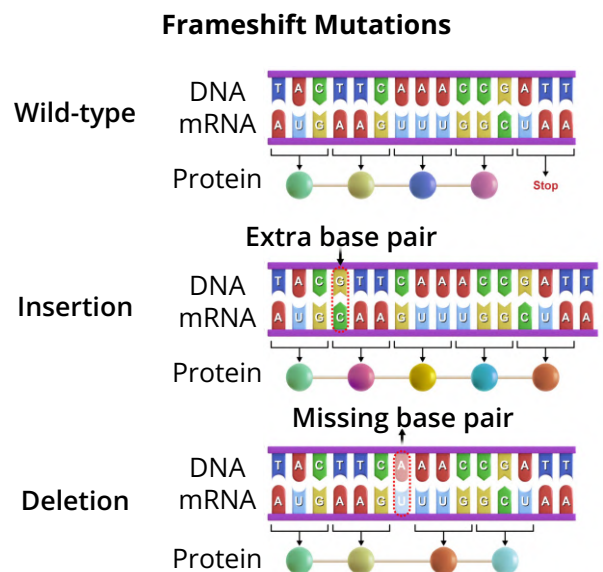
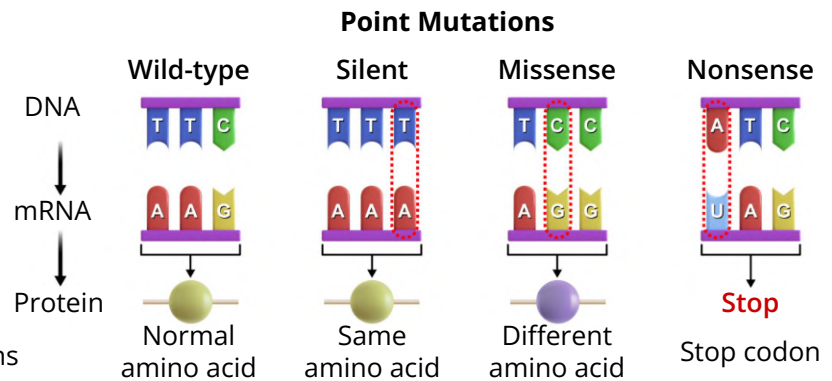
Factors that contribute to DNA mutations:

- **DNA polymerase errors** during replication.
- Loss of DNA during meiosis **crossing over**.
- **Chemical damage** from drugs.
- **Radiation**
- **Transposons (jumping genes):** DNA sequences in prokaryotes and eukaryotes that can move and integrate into different places in the genome.

Factors that prevent DNA mutations:

- **DNA polymerase proofreading** by DNA polymerase.
- **Mismatch repair** machinery that checks uncaught errors.
- **Nucleotide excision repair** that cuts out damaged DNA and replaces it with correct DNA using complementary base pairing.

A **null mutation** can also occur, a null (non-functional) allele is produced that lacks the function of the normal, wild-type allele.



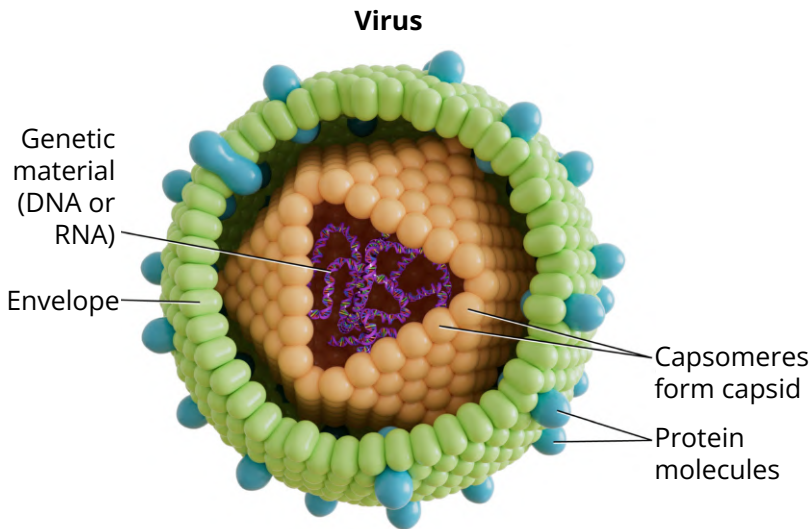
Chromosomal mutations occur and affect the entire chromosome rather than individual nucleotides. There are four types:

- **Deletion:** A portion of the chromosome is deleted, resulting in a shorter chromosomal arm.
- **Duplication:** A piece of the chromosome is duplicated, resulting in a larger chromosomal arm and an atypical banding pattern.
- **Inversion:** A portion of the chromosome becomes inverted on the arm of the chromosome. Results in an abnormal banding pattern, but does *not* affect the length of the chromosome.
- **Translocation:** A piece of one chromosome breaks off and attaches to another chromosome. Translocation increases chromosomal arm length and results in an abnormal banding pattern. This is the only mutation that affects both chromosomes.

Molecular Genetics of Viruses

Viruses are *not* living because they must infect living cells to multiply.

The **capsid** is a viral protein coat that is made of subunits called **capsomeres**. Some viruses also have a **phospholipid envelope** that they pick up from the host cell membrane.



In order for an infection to continue to spread, viruses undergo **viral replication** to create new viruses that can further infect other cells/hosts. The **viral replication cycle** is as follows:

1. **Attachment:** Binding of a virus to host cell
2. **Penetration:** Virus crosses through the host's cell membrane.
3. **Uncoating:** Viral capsid is removed and degraded by host enzymes.
4. **Replication:** Viral genome replicated and components of viral capsid are manufactured.
5. **Assembly:** Viral capsid components assemble to form the viral capsid.
6. **Release:** Last step of viral replication, fully assembled viruses are released.

Mnemonic: APURAR

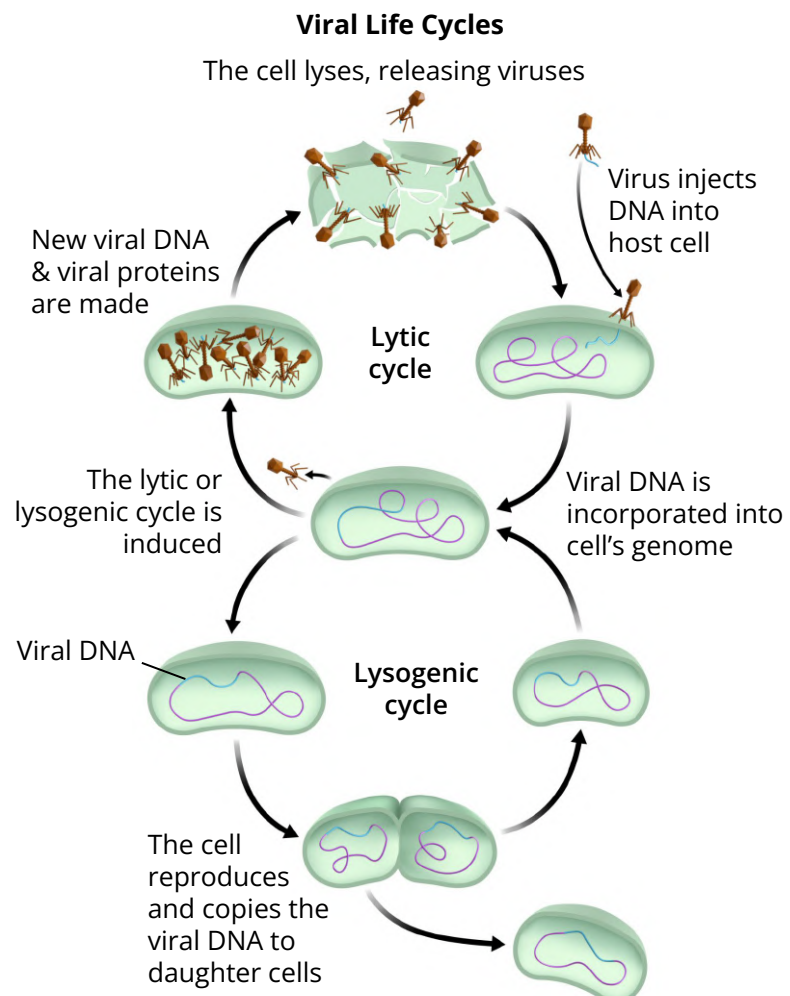
- While there are two **As** and **Rs**, remember the first step is always attachment and the last step is always release.

Two viral life cycle types:

1. **Lysogenic cycle:** Virus is considered dormant because it inserts its own genome into the host's genome and does *not* harm the host. Each time the host genome undergoes replication, so does the viral genome.
 - E.g., **Prophages** are bacteriophage genomes that have been integrated into the bacterial host genome.
2. **Lytic cycle:** Virus takes over host to replicate and does cause harm to the host. The viral particles produced can lyse the host cell to find other hosts to infect.

Viruses can switch between the lysogenic and lytic cycles. For example, favorable conditions can stimulate a virus in the lysogenic cycle to replicate and enter the lytic cycle.

Retroviruses (e.g., HIV) have an RNA genome that infects host cells. They contain an enzyme called **reverse transcriptase**, which converts their RNA into cDNA (complementary DNA). The cDNA can integrate into the host genome and enter the lysogenic cycle.



Molecular Genetics of Bacteria

Bacteria are asexual and divide by **binary fission**; they only receive genes from one parent cell and do *not* increase genetic diversity through reproduction.

Bacteria must *increase* genetic diversity through **horizontal gene transfer**, which describes the transfer of genes between individual organisms. Below are the three methods of horizontal gene transfer:

1. **Conjugation:** Bacteria use a cytoplasmic bridge called a **pili** to copy and transfer a special plasmid known as the **F plasmid (fertility factor)**. If a bacteria contains an F plasmid, it is referred to as **F+**. If not, it is referred to as **F-**. To review, **plasmids** are circular DNA pieces that are independent of a bacterium's single circular chromosome.

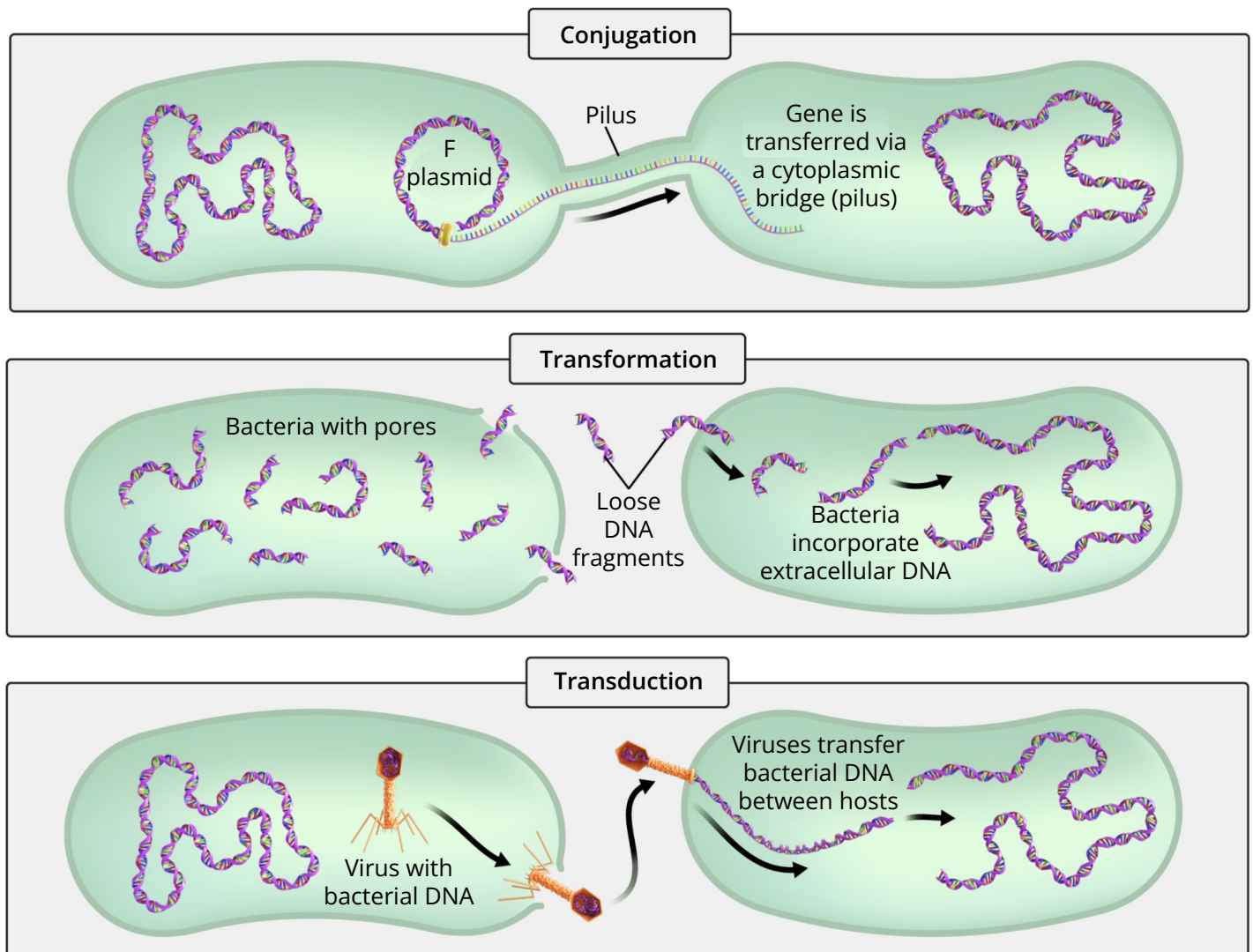
2. **Transformation:** Bacteria take up extracellular DNA. Bacteria are referred to as **competent** if they can perform transformation.

Electroporation is the process of using electrical impulses to force bacteria to become competent.

3. **Transduction:** Viruses transfer bacterial DNA between different bacterial hosts. This occurs when a bacteriophage enters the lysogenic cycle in its host and carries bacterial DNA along with its own genome upon re-entering the lytic cycle.

Bacteria can also contain **R-factors**, which are extrachromosomal pieces of DNA in the form of a plasmid that contains antibiotic resistance genes.

Horizontal Gene Transfer



Ch. 7: Heredity

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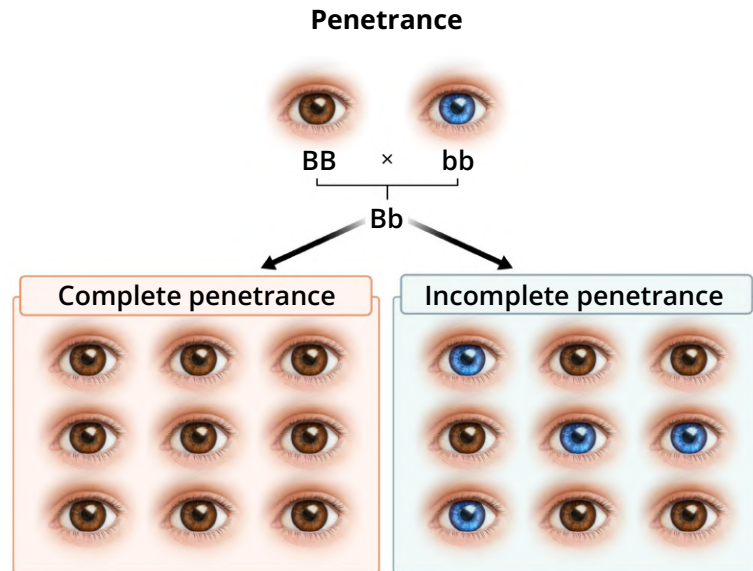
- Key Terms
- Patterns of Inheritance
- Gene Defects
- Mendel's Laws
- Nondisjunction and Aneuploidies
- Crosses
- Genetic Diversity
- Gene Linkage
- Pedigree Analysis
- Epigenetics

Heredity is the passing of traits from parents to offspring. These traits can be passed down **sexually** (mating in animals) or **asexually** (binary fission in bacteria).

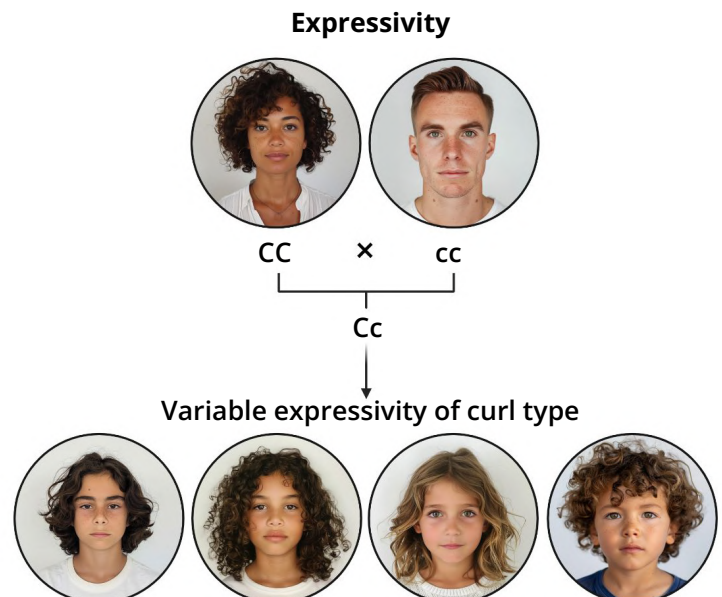
Key Terms

- **Genome:** All the DNA within a cell.
- **Gene:** Sequence of DNA that codes for a trait.
- **Locus:** Location of a gene on a chromosome. Plural is **gene loci**.
- **Allele:** One variation of a gene. Alleles are found at the same loci on both chromosomes in a homologous pair.
- **Wild-type allele:** Normal allele that is most common in nature. Can turn into a **mutant allele**.
- **Mutation:** A change in DNA sequence.
- **Genotype:** Genetic composition of an organism.
- **Phenotype:** Observable traits that result from a genotype.
- **Dominant alleles:** Mask the expression of recessive alleles. Typically represented by uppercase letters ("A").
- **Recessive alleles:** Only show up in a phenotype if dominant alleles are not present. Typically represented by lowercase letters ("a").
- **Homologous pairs:** Two different copies of the same chromosome in a diploid organism. One from each parent. Each copy is very similar, with minor nucleotide variations that generate unique alleles.
- **Heterozygous:** One dominant allele and one recessive allele in its homologous pair.
- **Homozygous:** Same allele in both homologs. Can be **homozygous dominant** or **homozygous recessive**.

- **Hemizygous:** Only one allele is present. For example, men have one X and one Y chromosome (not homologous), which contain hemizygous genes.
- **Penetrance:** Proportion of individuals who have the phenotype associated with a specific allele. Can be **complete penetrance** or **incomplete penetrance**. In the example below, Bb individuals all have brown eyes only when there is complete penetrance.



- **Expressivity:** Describes the degree of a certain phenotype for a given genotype. All of the children of the parents in the example below have genotype Cc (curly hair). Due to variable expressivity, the level of hair curliness varies among the children.

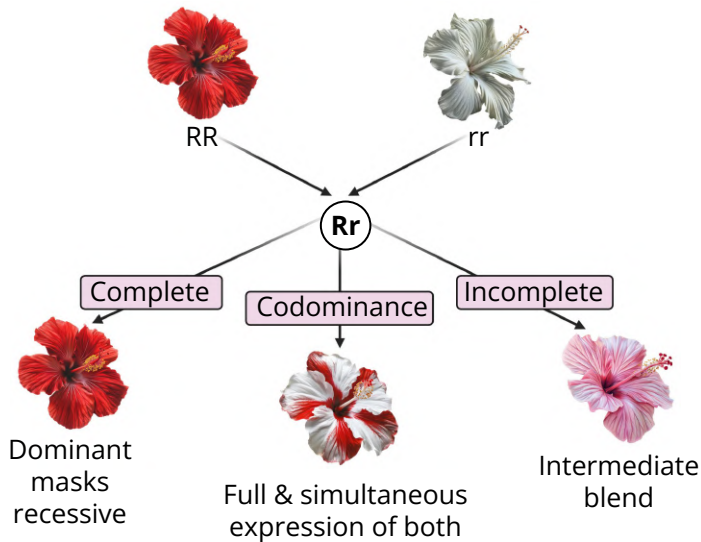


Patterns of Inheritance

Incomplete dominance is when one allele is *not* completely expressed over its paired allele. The heterozygous will have an intermediate state (e.g., red x white = pink).

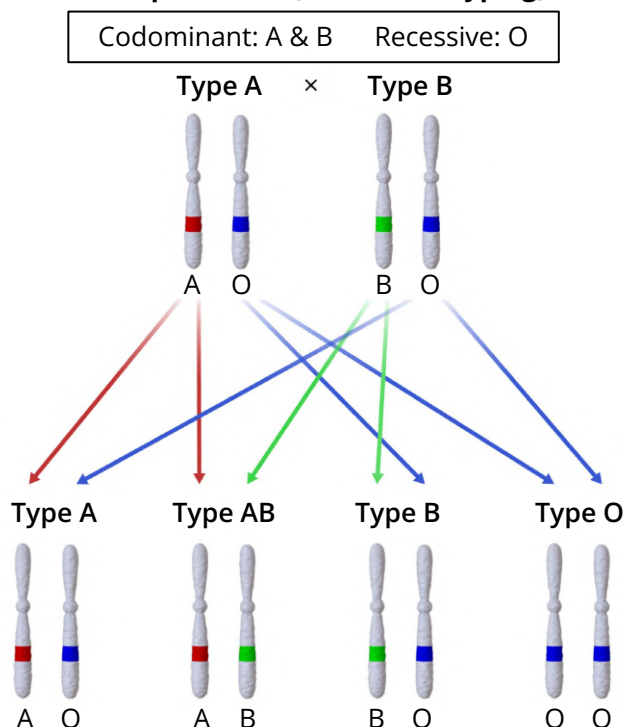
Codominance is when the heterozygous genotype expresses both alleles (e.g., red x white = red + white spots).

Dominance Inheritance Patterns



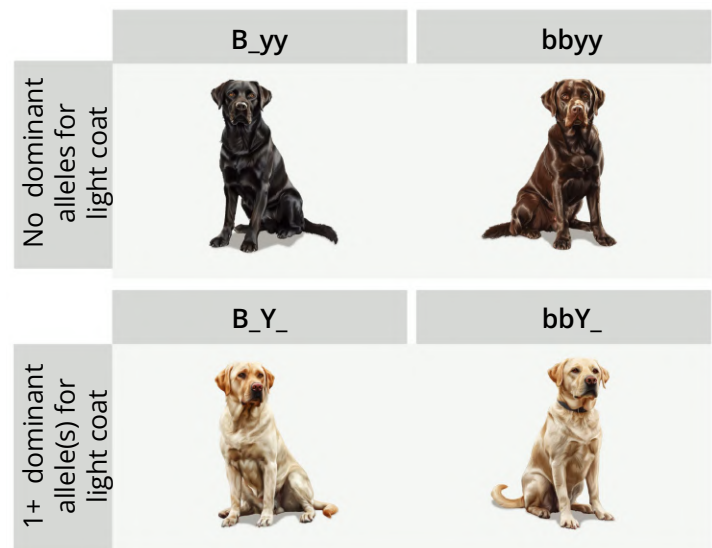
Multiple alleles describe when there are more than just two allele options (e.g., ABO blood typing - A, B, O alleles).

Multiple Alleles (ABO Blood Typing)



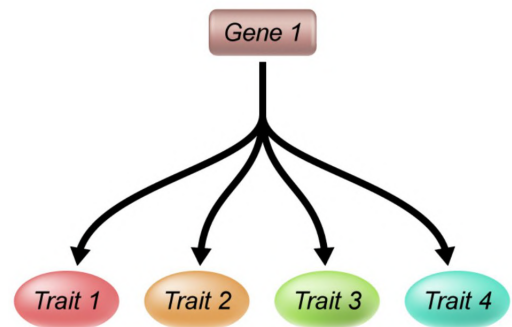
Epistasis occurs when one gene affects the expression of a different gene (e.g., light coat gene covers up the gene for black/chocolate coat in labradors).

Epistasis



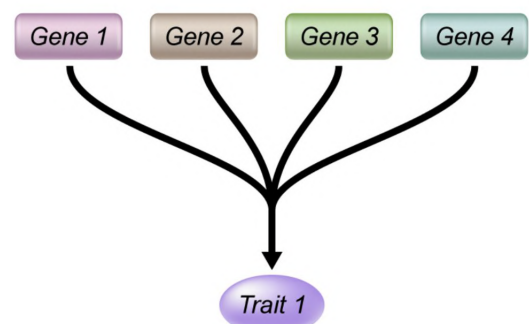
Pleiotropy describes when one gene is responsible for many traits (e.g., cystic fibrosis is a disease with many symptoms caused by a single gene).

Pleiotropy



Polygenic inheritance is when **many genes** are responsible for **one trait**. This gives the trait **continuous variation** (e.g., height is a single trait affected by many genes).

Polygenic Inheritance



Gene Defects

Haploinsufficiency occurs when one copy of the gene is lost or nonfunctional and the expression of the remaining copy is *not* sufficient enough to result in a normal phenotype. It can result in an intermediate phenotype.

Haplosufficiency describes when the remaining copy of the gene is sufficient enough to result in a normal phenotype.

Proto-oncogenes are genes that can become **oncogenes** (cancer-causing genes) due to **gain-of-function** mutations. Gain-of-function mutations can cause too much protein to be made or production of an over-active protein; cancerous growth occurs as a result. Proto-oncogenes are normally involved in cell cycle control.

Proto-oncogenes follow the **one hit hypothesis**, which states that a gain-of-function mutation in one copy of the gene turns it into an oncogene.

Tumor-suppressor genes are genes that become cancerous as a result of **loss-of-function** mutations, because they are normally needed to suppress cancerous growth.

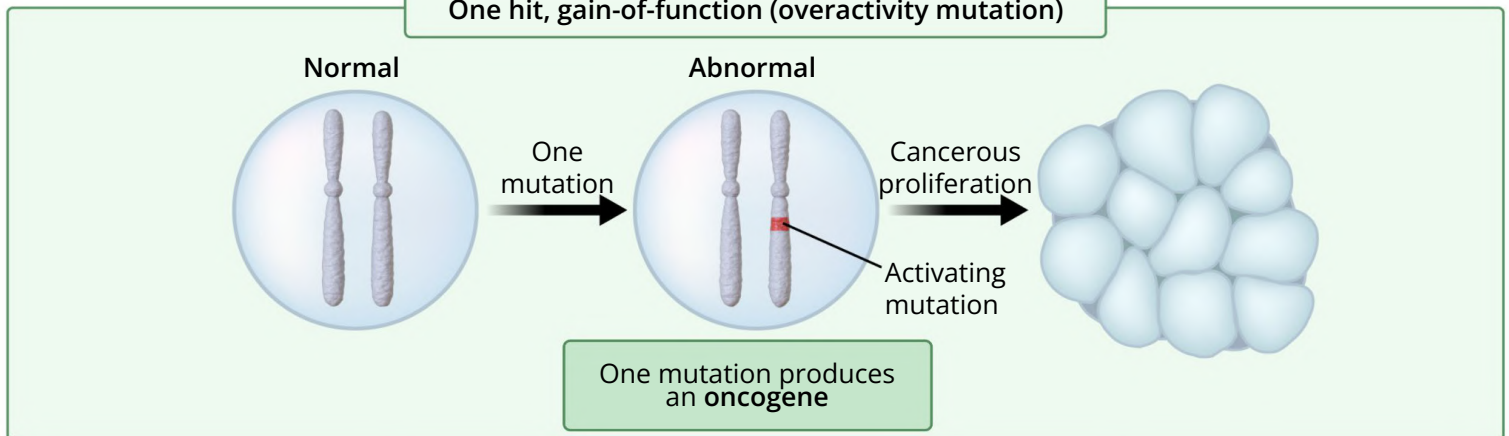
Tumor-suppressor genes follow the **two hit hypothesis**, which states that a loss-of-function mutation in both copies of the gene are needed to make it cause cancer. Thus, tumor-suppressor genes are haplosufficient.

Null alleles come from mutations that cause the alleles to lack normal function. Tumor-suppressor genes have null alleles when they become cancer-causing.

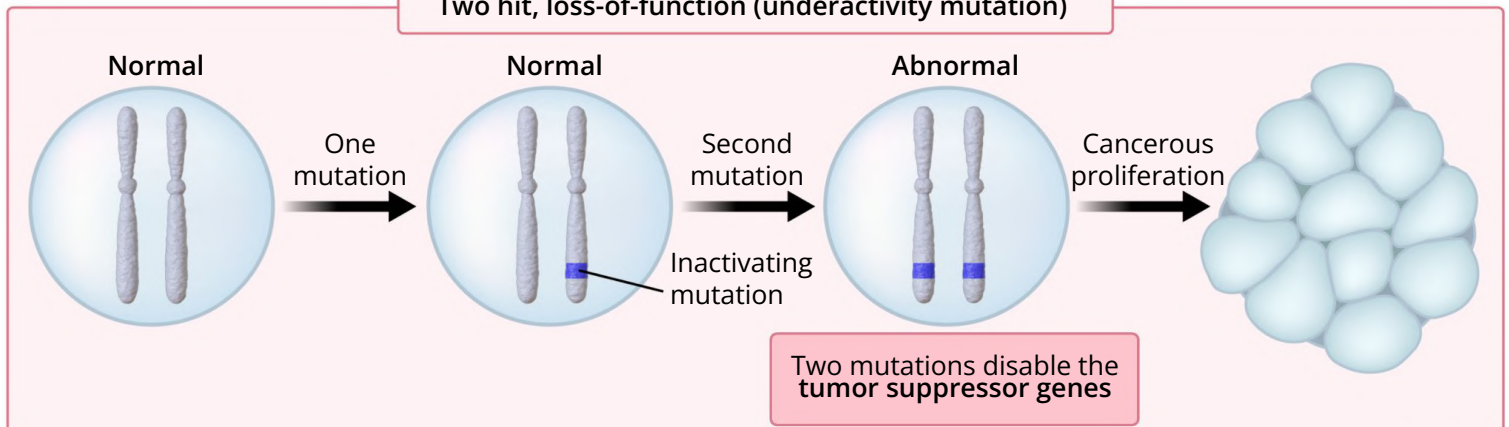
- **p53** is an important tumor-suppressor gene that is known as the guardian of the cell. It is upregulated to prevent cells from becoming cancerous.
- **p21** is another tumor-suppressor gene that inhibits phosphorylation activity in order to decrease rampant cell division.
- **Retinoblastoma gene (RB)** is a tumor-suppressor gene that codes for a retinoblastoma protein, which prevents excessive cell growth during interphase.

Cancer Causing Mutations

One hit, gain-of-function (overactivity mutation)



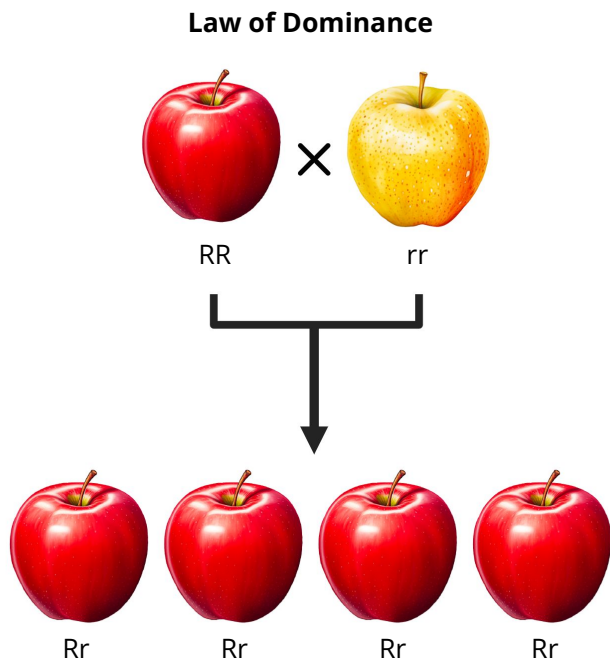
Two hit, loss-of-function (underactivity mutation)



Mendel's Laws

Gregor Mendel studied genetics and proposed three laws:

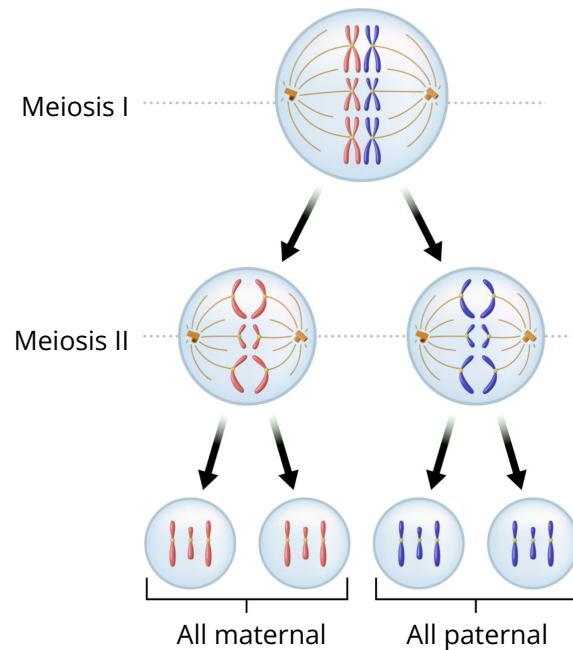
1. **Law of dominance:** Dominant alleles mask the expression of recessive alleles.



2. **Law of segregation:** Homologous chromosomes separate during meiosis (anaphase I) so that one allele is passed down. Thus, Aa individuals will produce gametes with "A" or "a" alleles.
3. **Law of independent assortment:** Homologous chromosomes line up independently during metaphase I of meiosis so that alleles separate randomly (this *increases* genetic variability). Metaphase II is different, during which sister chromatids are pulled apart instead.

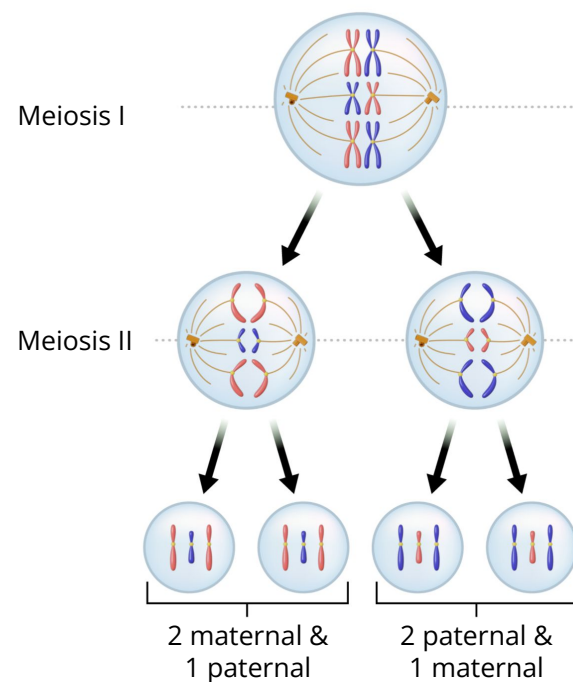
Under the law of independent assortment, a six chromosome diploid organism (haploid number is 3) could assort with:

Trial 1: All paternal on one side, all maternal on the other.



However, they also could randomly align like this:

Trial 2: Two maternal with one paternal on one side and two paternal with one maternal on the other.

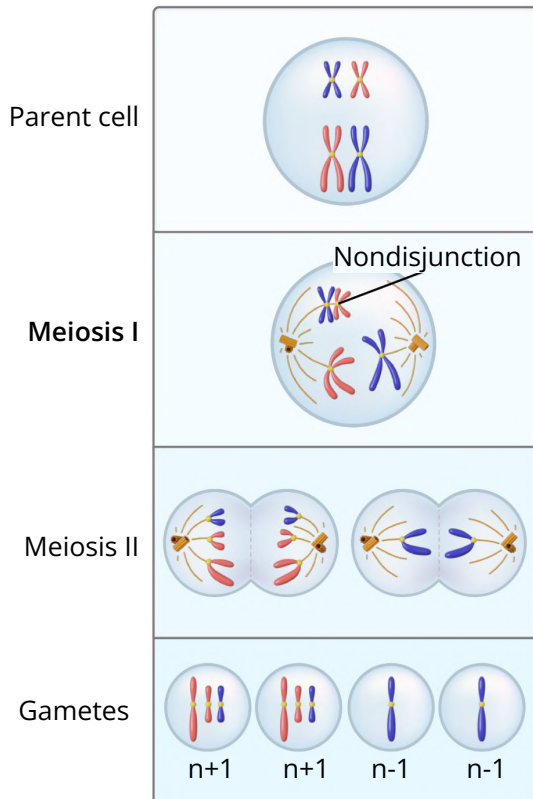


Nondisjunction and Aneuploidies

Nondisjunction is the improper segregation of chromosome pairs during anaphase; it produces daughter cells with an incorrect number of chromosomes.

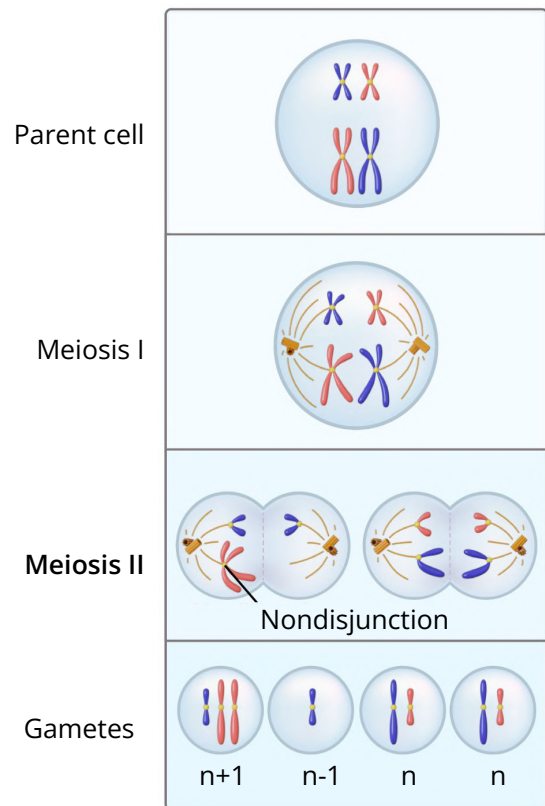
1. **Single nondisjunction** of homologous chromosomes during **meiosis I**

46 chromosomes in diploid parent cell →
24, 24, 22, 22 chromosomes in haploid daughter cells



2. **Single nondisjunction** of sister chromatids during **meiosis II**

46 chromosomes in diploid parent cell →
24, 22, 23, 23 chromosomes in haploid daughter cells



3. **Single nondisjunction** of sister chromatids during **mitosis**

46 chromosomes in diploid parent cell →
47, 45 chromosomes in diploid daughter cells

Aneuploidy refers to an abnormal number of chromosomes in the daughter cells. After fertilization, **trisomy** (three chromosome copies) or **monosomy** (one chromosome copy) can occur. **Disomy** refers to a normal diploid cell.

Crosses

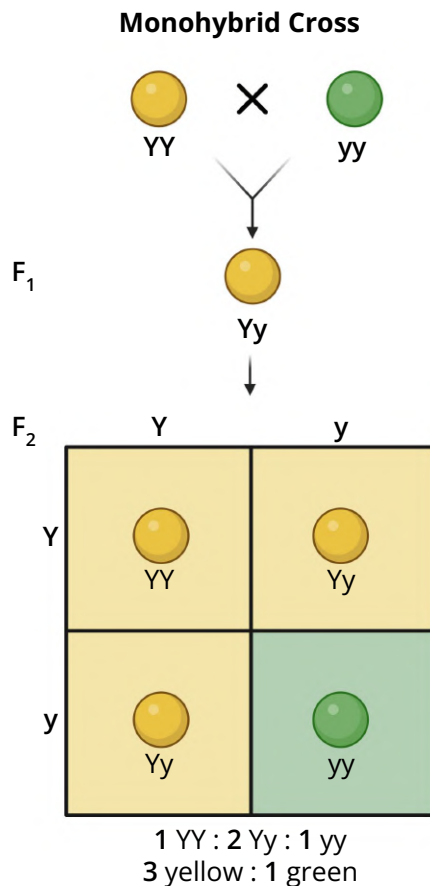
A **cross** refers to when two organisms are mated to produce offspring.

- The **F1 generation** (aka **filial 1 hybrid**) is the first generation of offspring from a cross.
- The **F2 generation** (aka **filial 2 hybrid**) is the second generation of offspring from a cross.

A **test cross** pairs an individual of unknown genotype with one that is homozygous recessive. By looking at the offspring from a test cross, the unknown genotype can be determined.

True-breeding organisms are homozygous for all the traits of interest.

If the F1 and F2 generations of true-breeding organisms are studied under **monohybrid crosses**, then only a single gene is examined. In the F2 generation, the **genotype ratio** (AA:Aa:aa) should be (1:2:1) and the **phenotype ratio** (dominant:recessive) should be (3:1).

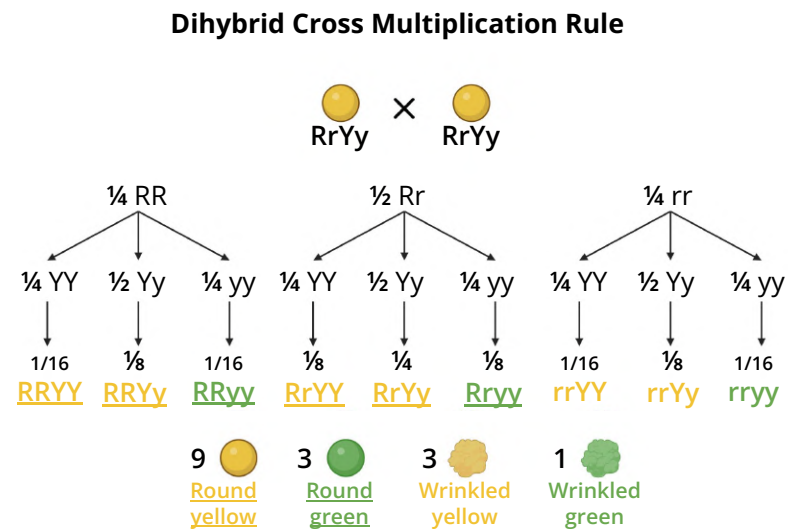


On the other hand, a **dihybrid cross** examines the inheritance of two genes on separate chromosomes. Although the genotype ratio is complex in the F2 generation, just remember that the phenotype ratio (both dominant:one dominant and one recessive:one dominant and one recessive:both recessive) should be (9:3:3:1).

Punnett squares are used to visualize these crosses but are too complex for dihybrid crosses. Thus, **one-gene cross ratios** can be used to solve these questions faster. Below are the single allele crosses you should memorize:

1. **Homozygous x homozygous** = 1/1 AA or 1/1 Aa or 1/1 aa
2. **Homozygous x heterozygous** = ½ AA (or aa) and ½ Aa
3. **Heterozygous x heterozygous** = ¼ AA, ½ Aa, ¼ aa

Multiple-locus crosses can then be solved using these single allele crosses. As shown below, RrYy individuals cross with each other. The Rr single cross probabilities can be multiplied with the Yy single cross probabilities to get the **dihybrid offspring probabilities**.

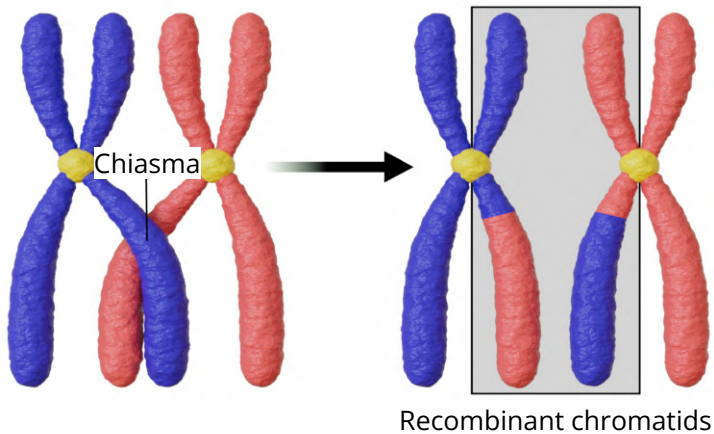


Genetic Diversity

Crossing over creates genetic diversity and occurs during prophase I of meiosis. Homologous chromosomes join together to form **tetrads** (aka **bivalents**) and exchange genetic material at points referred to as **chiasmata**. Afterwards, genetically unique chromatids are produced as a result of crossing over.

Recombinant gametes describe the gametes that receive the genetically unique chromatids (new combination of alleles), while **non-recombinant gametes** refer to the gametes that receive parental chromatids (alleles match parent's alleles).

Crossing Over During Prophase I



Gene Linkage

Linked genes are found close together on the same chromosome. By looking at **recombination frequencies**, the relative distance between these genes can be deduced.

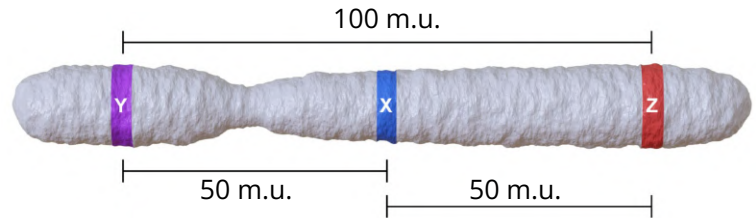
1 **map unit** (m.u.) equals a 1% chance of recombination between two genes, equivalent to 1 centimorgan (cM) or 0.01 probability. For example, 20 m.u. or 20 cM means there is a 20% chance (0.20 probability) of recombination between the two genes per generation.

Recombination frequencies of less than 50% mean that the two genes are **linked**. A recombination frequency of 0.5 indicates that the genes are **unlinked**, assort independently, and have a 50% chance of recombining.

Linkage maps are tables that are used to determine the probability of inheritance. Linkage maps use map units to infer the distance between genes on a chromosome.

Linkage Map

1% recombination frequency = 1 map unit (m.u.)

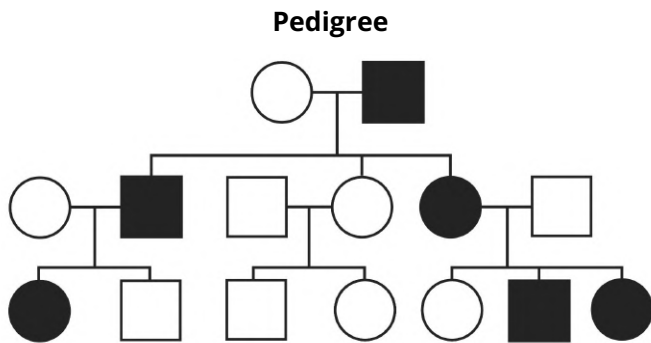


Sex-linked traits come from genes located on the sex chromosomes. Most sex-linked disorders have **X-chromosome linkage**. Below are three types of sex-linked traits:

1. **X-linked dominant:** Dominant inheritance on the X chromosome. Any offspring (male or female) that receive the affected allele will end up with the disorder.
2. **X-linked recessive:** Recessive inheritance on the X chromosome. For males, only one affected allele is needed to cause the disorder. For females, two affected alleles are needed to cause the disorder because females have two X chromosomes. Hemophilia and color-blindness are examples of X-linked recessive conditions.
3. **Y-linked:** Inheritance on the Y chromosome. Can only be passed from father to son. Will always be expressed whether it is dominant or recessive because males only have one Y chromosome.

Pedigree Analysis

Pedigree charts are used to track inherited traits over many generations to see inheritance patterns. Females are represented by circles, and males are represented by squares. Individuals affected by the trait in question are shaded; unaffected individuals are *not* shaded.



Example: Given that the “affected” trait is autosomal dominant, this chart can be used to solve for the genotype of the affected male in the third generation.

Since the male is affected, he can be heterozygous or homozygous dominant. However, because his father is unaffected, the male could not have received a dominant (affected) allele from his father. Thus, this individual must be heterozygous. The single dominant allele came from his mother.

Epigenetics

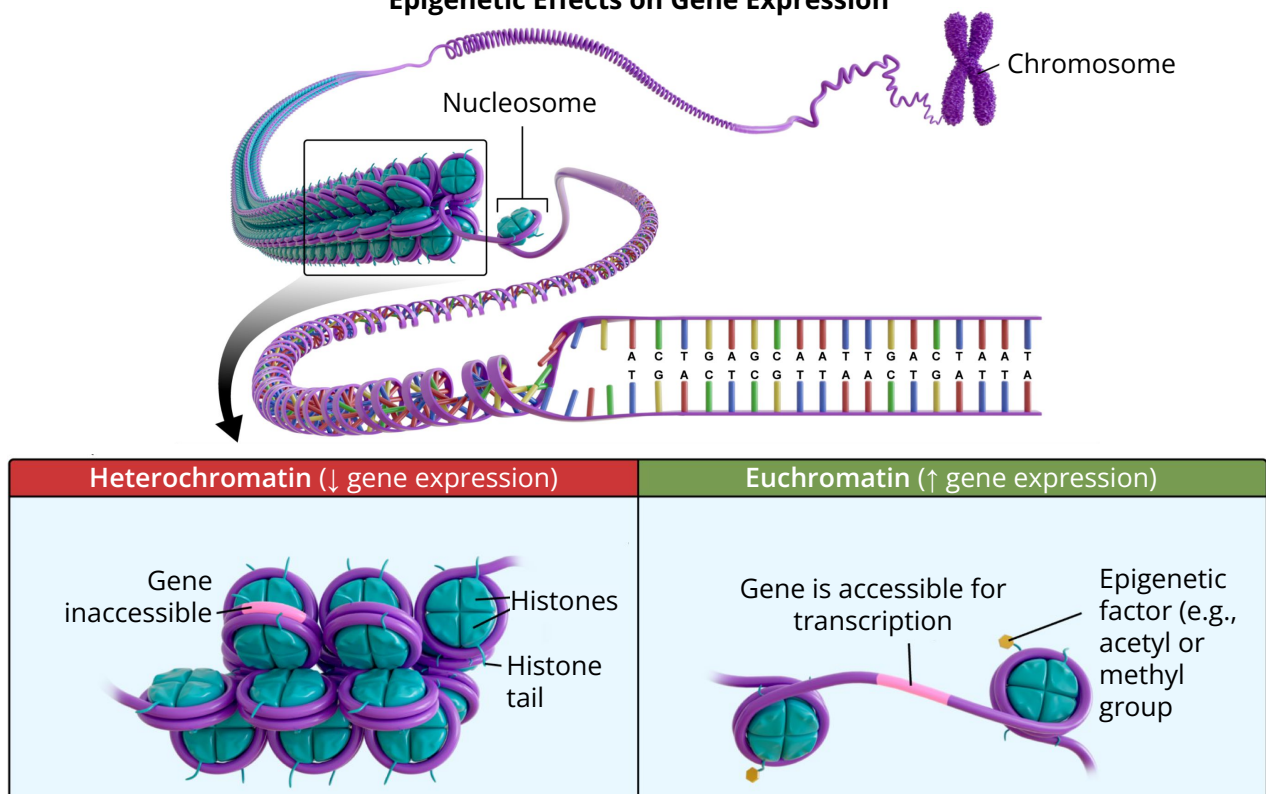
Epigenetics does not involve modifying the genetic code, but instead the regulation of when genes are expressed. By modifying the DNA with methylation or acetylation, the same genetic code can be expressed differently.

Below are some examples of **epigenetic changes**:

- **DNA methylation:** Regulates gene expression by suppressing genes. This is done through the addition of methyl groups, which prevent transcription factors from binding.
- **Histone acetylation:** Causes formation of euchromatin (easily accessible DNA), resulting in gene activation.
- **Histone deacetylation:** Causes formation of heterochromatin (hard to access DNA), resulting in gene suppression (think ‘de-activation’).
- **Histone methylation:** Can upregulate or downregulate gene expression depending on methyl group location and number.

Pro-Tip: Epigenetic changes can cause monozygotic twins to have different susceptibilities to the same disease.

Epigenetic Effects on Gene Expression



Ch. 8: Microscopy & Lab Techniques

Table of Contents

- Overview of Microscopy
- Types of Optical Microscopes
- Types of Electron Microscopes
- Cellular Biological Lab Techniques
- Biological Lab Techniques for Nucleic Acids and Proteins
- Genomics
- Miscellaneous Biological Laboratory Techniques

Overview of Microscopy

The **objective** is the part of the microscope that magnifies the image of the specimen.

Before beginning microscopy, the cells must first be fixed and stained:

1. **Fixation:** 'Sticks' cells to the slide, preserving them in their most life-like state.
2. **Staining** adds color or contrast to cells, making cell structures easier to visualize. Staining often kills the cells.

General types of **microscopy**:

1. **Optical microscopy:** Cells are viewed directly. Light shines on a sample which is magnified via lenses. Can be used to observe living cells.
2. **Electron microscopy:** Cells are viewed indirectly. Electrons are fired on a sample, then bounce off and pass through magnetic fields. Can be used to view *smaller* objects, but cells must be fixed, stained (metal coated) and killed.

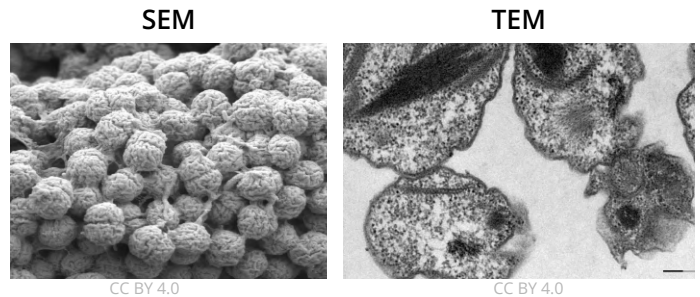
Types of Optical Microscopes

1. **Stereo microscopes (dissection microscopes):** Use low magnification to view the surface of an object.
2. **Compound microscopes:** Have multiple lenses to view simple, one-cell thick samples. Poor contrast. Usually need fixation and staining.
3. **Bright field microscopes:** Compound microscopes with a bright light.

4. **Phase contrast microscopes:** View thin samples of live, unstained cells with high contrast.
5. **Fluorescence microscopy:** Live cells tagged with fluorophores (fluorescent chemicals). Can visualize cellular components.

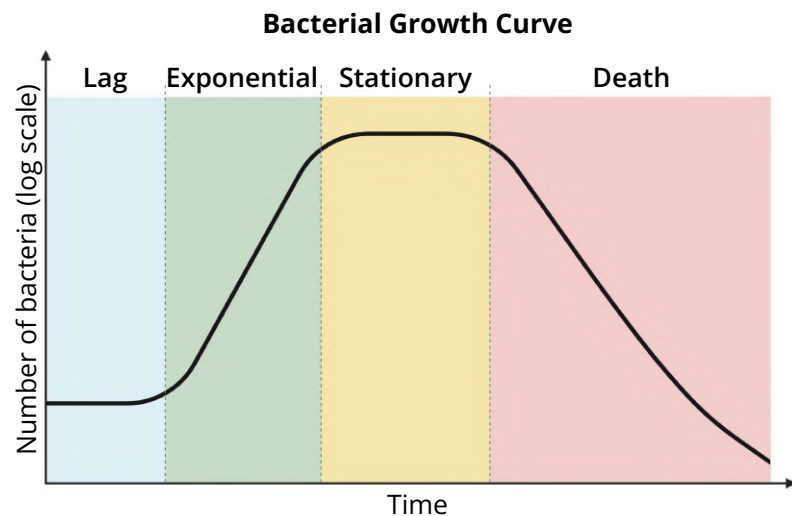
Types of Electron Microscopes

1. **Scanning electron microscopy (SEM):** High resolution 3D images of the surface of a dehydrated sample.
2. **Transmission electron microscopy (TEM):** High resolution 2D images of the sample's internal structures.



Cellular Biological Lab Techniques

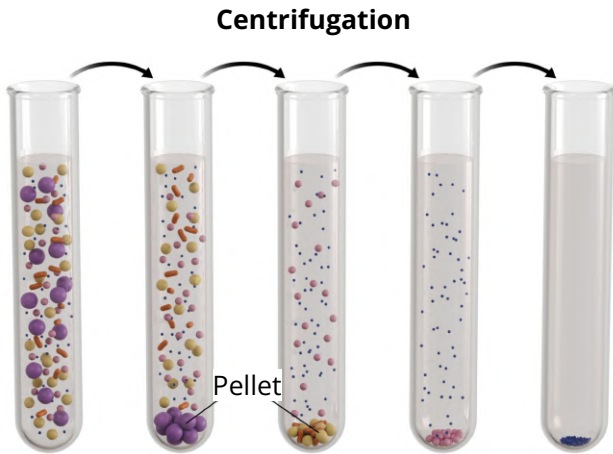
A **bacterial growth curve** can be used to show the phases of a microbial culture:



1. **Lag phase:** Adaptation prior to cell division.
2. **Exponential (log) phase:** Rapid doubling.
3. **Stationary phase:** Growth rate = death rate.
4. **Death phase:** Decline due to lack of food/other variable.

Differential centrifugation is used to separate cellular components based on their size and density by subjecting them to varying centrifugal forces in a series of steps. Each step selectively pellets components of a specific density range, allowing for the fractionation of cell organelles and macromolecules.

Most dense → least dense: Nuclei > mitochondria, chloroplasts, & lysosomes > ER & Golgi fragments > ribosomes & large macromolecules



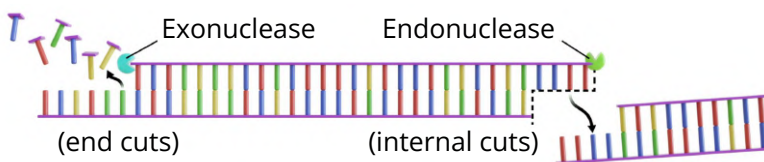
Biological Lab Techniques for Nucleic Acids and Proteins

Exonucleases are enzymes that cleave nucleotides from the polynucleotide chain at the ends of the chain. Exonuclease activity only results in sticky ends.

Endonucleases are enzymes that cleave nucleotides from the polynucleotide chain from the inside of the polynucleotide chain. Endonuclease activity can result in either sticky or blunt ends.

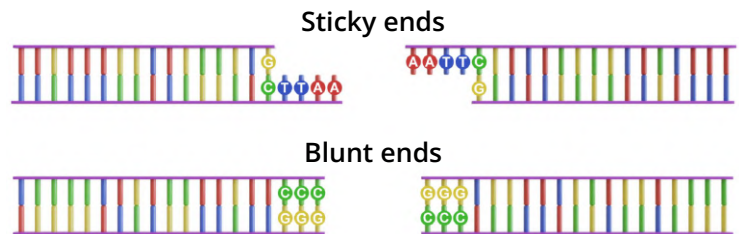
These enzymes can be used in nucleic acid sequencing, analyzing, editing, or cloning techniques.

Exonucleases vs. Endonucleases



Restriction enzymes are specialized endonucleases that mostly cut DNA at specific palindromic sequences (short regions where the sequence reads the same in the 5' to 3' direction on both strands). This cut can produce either **sticky ends**, with overhanging unpaired bases, or **blunt ends**, with no overhangs. DNA fragments from different sources can be joined at these ends to create **recombinant DNA**.

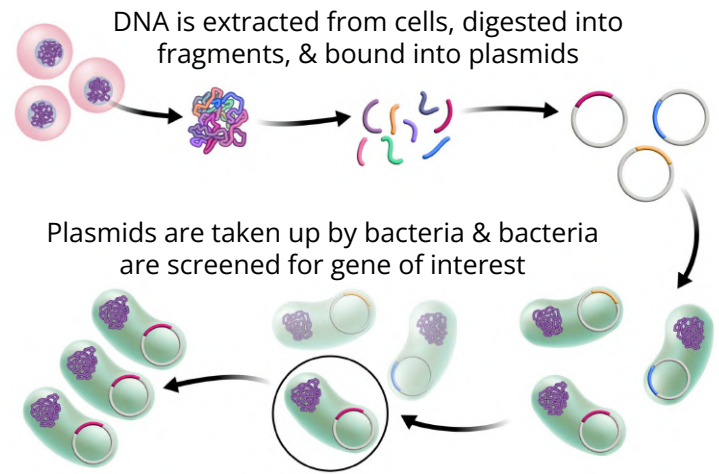
Sticky vs. Blunt Ends



1. **Karyotyping:** Observing chromosomes under a light microscope during metaphase. Used to identify abnormalities in chromosomes (e.g., Down syndrome/trisomy 21).
2. **CRISPR:** Used to edit specific genomic regions of interest by adding or deleting targeted sequences of DNA. Used in gene therapy.
3. **DNA fingerprinting:** Identifies individuals through analyzing specific regions of non-coding DNA, fragmented with restriction enzymes. Used in paternity and forensic cases.
4. **DNA sequencing:** Determines the order of nucleotides in a DNA strand.
 - Can be done with **Sanger sequencing** method. DNA copies are made by PCR, using deoxynucleotides (dNTPs) and **dideoxynucleotides (ddNTPs)**. ddNTPs lack a 3' OH necessary to create phosphodiester bonds, terminating elongation. Using a mix of dNTPs and ddNTPs ensures that many chains of different lengths exist, allowing the sequence to be determined.

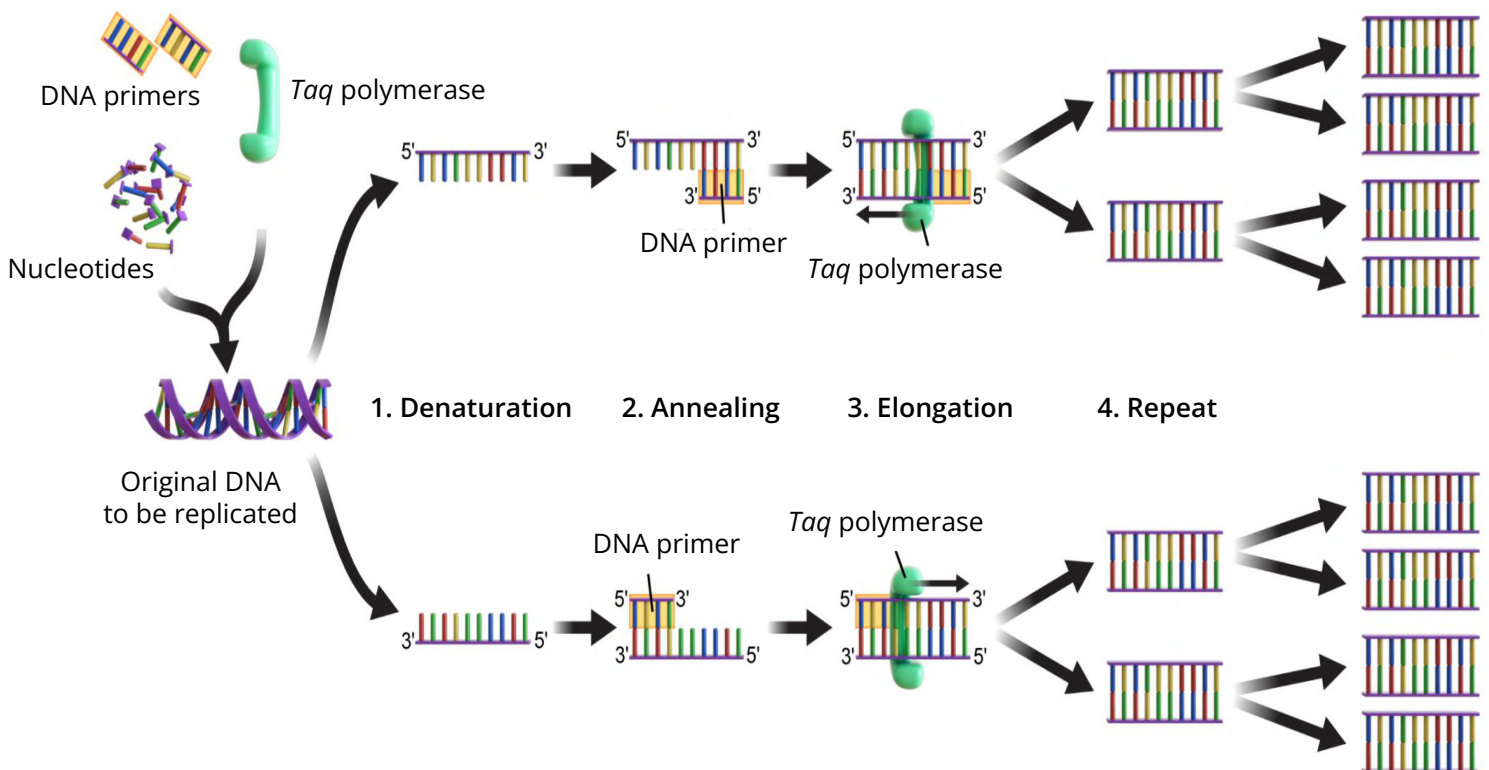
5. **Bacterial cloning:** Cloning eukaryotic gene products in prokaryotic cells. Used to produce medicine.
 - Protocol: Processed mRNA for eukaryotic gene is isolated and treated with **reverse transcriptase** to make **cDNA** → cDNA is incorporated into a **plasmid** (transfer vector) using restriction enzymes and **DNA ligase** → vector taken up by **competent** bacterial cells, which undergo **transformation** → gene of interest is found using **antibiotic resistance** (antibiotic resistant gene attached to target gene) or **color change** methods.
6. **Polymerase chain reaction (PCR):** Automated process creating 1+ billion copies of DNA in three steps:
 - **Denaturation (~95 °C):** Heating separates DNA into single strands.
 - **Primer annealing (~65 °C):** DNA primers hybridize with single strands.
 - **Elongation (~70 °C):** Nucleotides are added to the 3' end of DNA using **Taq polymerase**.

Bacterial Cloning



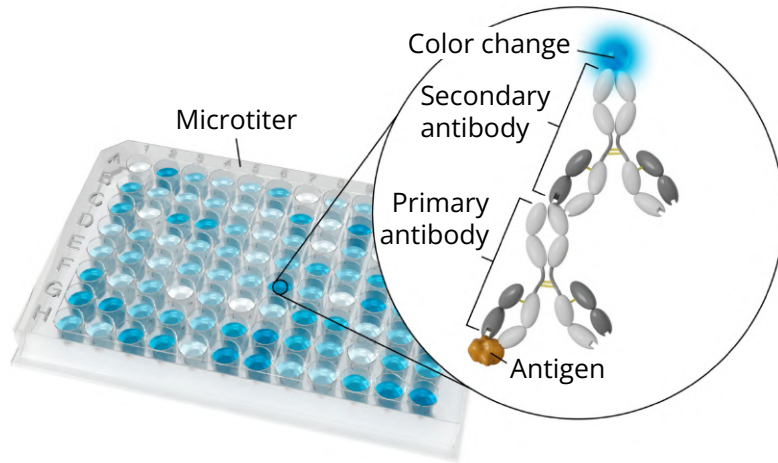
7. **Gene therapy:** The process of inserting genes into cells to treat disease. Viruses are the preferred vectors because they have the highest transduction efficiency. However, viruses may cause an immune response. Non-viral vectors, such as **naked plasmid DNA** and **CRISPR**, do not cause an immune response but are less efficient at inserting genes.

Polymerase Chain Reaction (PCR)



8. **Enzyme-linked immunosorbent assay (ELISA):**
 Determines if a person has a specific antigen.
 Important to diagnose diseases (e.g., HIV).
 Antibodies are placed on a microtiter plate with a sample and change color if antigens are present.

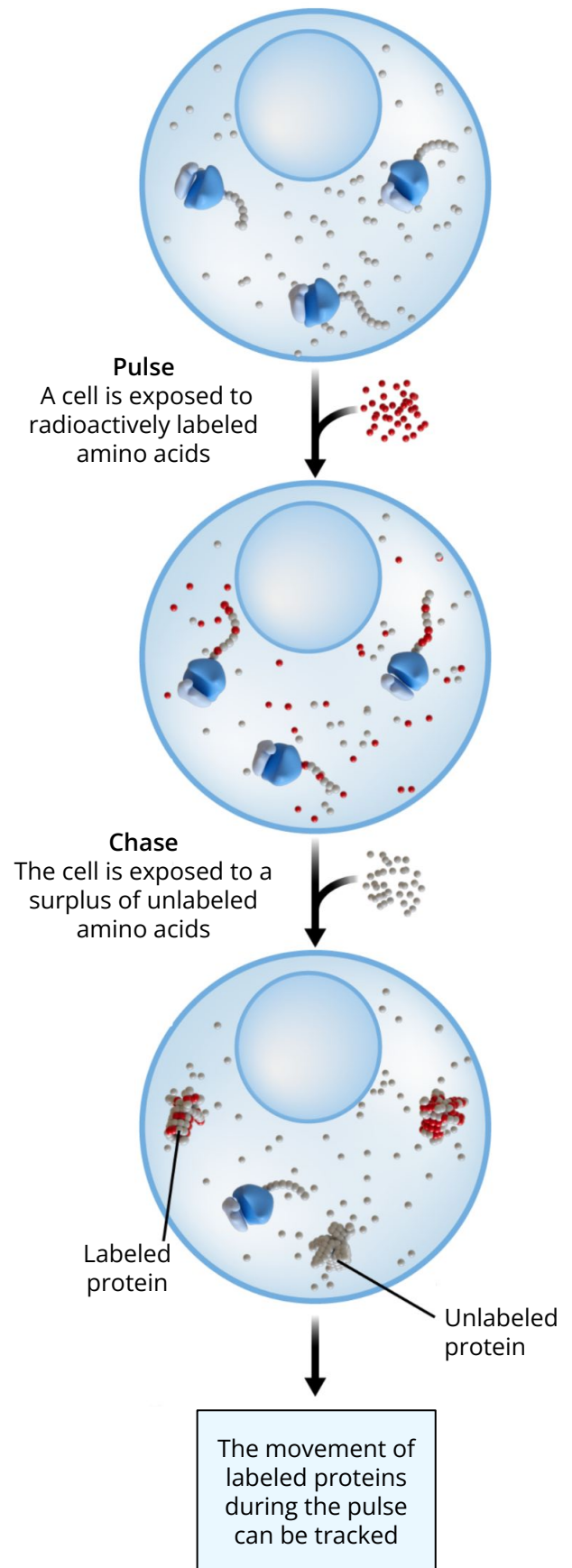
ELISA



A color is produced when both antibodies are bound

9. **Pulse chase experiments:** Useful for studying gene expression and the fate of proteins by viewing how a protein moves through a cell. During the **pulse phase** amino acids are radioactively labeled and then incorporated into proteins. The **chase phase** prevents radioactively labeled protein production. Using simple staining, the radioactive proteins can be tracked.

Pulse Chase Experiment

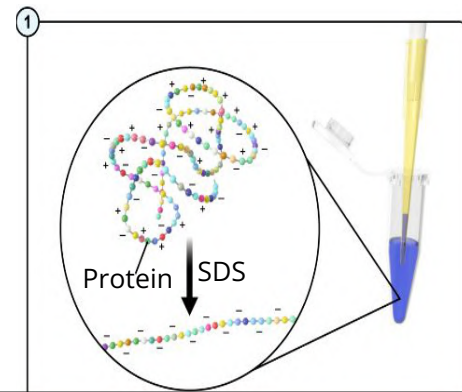


10. **Gel electrophoresis:** Separates fragments of nucleic acids or proteins by charge and size. An electric field is applied to **agarose gel** (top: negative cathode, bottom: positive anode). Smaller fragments travel further from top of gel.

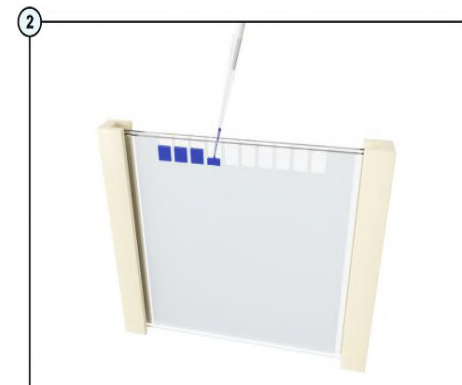
Sodium dodecyl sulfate (SDS): A strong detergent used in protein gel electrophoresis. SDS denatures, linearizes, and adds a negative charge to proteins to separate them by size and charge.

11. **Southern blotting:** Identifies fragments of known DNA sequence in a large population of DNA. Electrophoresed DNA is separated into single strands and identified via **complementary DNA probes**.
12. **Northern blotting:** Identifying fragments of known RNA using an **RNA probe**.
13. **Western blotting:** Quantifies amount of target protein in a sample using sodium dodecyl sulfate polyacrylamide gel electrophoresis or **SDS PAGE** (proteins denatured and given negative charge proportional to their mass). Treated with **primary antibody** (binds to target protein) and **secondary antibody** (attached to indicator and binds to primary antibody).

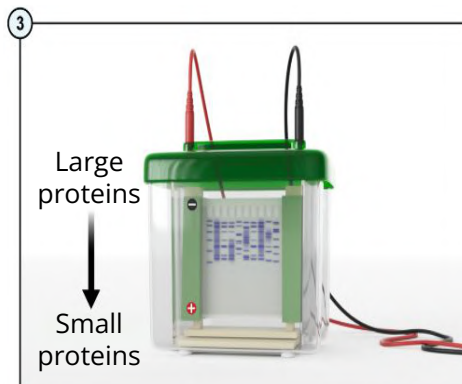
SDS-PAGE (Polyacrylamide Gel Electrophoresis)



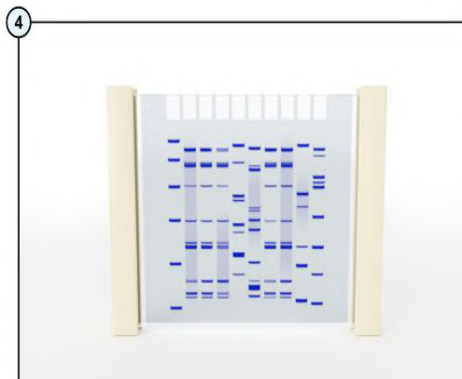
SDS denatures proteins & imparts a negative charge



Mixture is pipetted into polyacrylamide gel walls



Proteins are separated by size, smaller fragments travel faster and further



The sample can be analyzed or prepared for more analyses, such as Western blotting

Mnemonic: SNOW DROP

S outhern blotting	D N
N orthern blotting	R NA
O nothing	O nothing
W estern blotting	P rotein

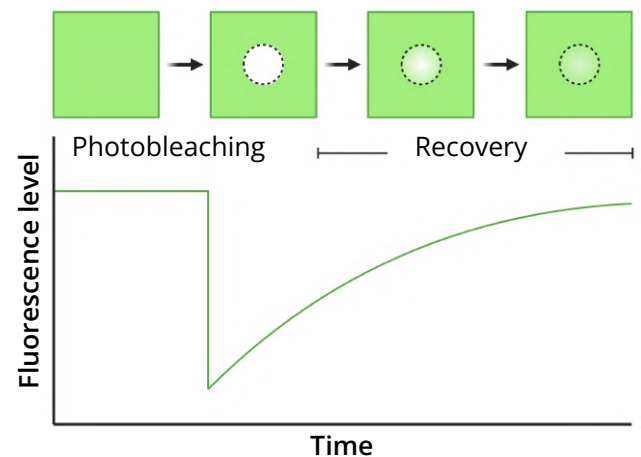
Genomics is the study of all genes present in an organism's genome and how they interact.

After a gene has been sequenced, it must be annotated via **genome annotation**. This process identifies the location of genes and coding regions in a genome, and determines each of their functions.

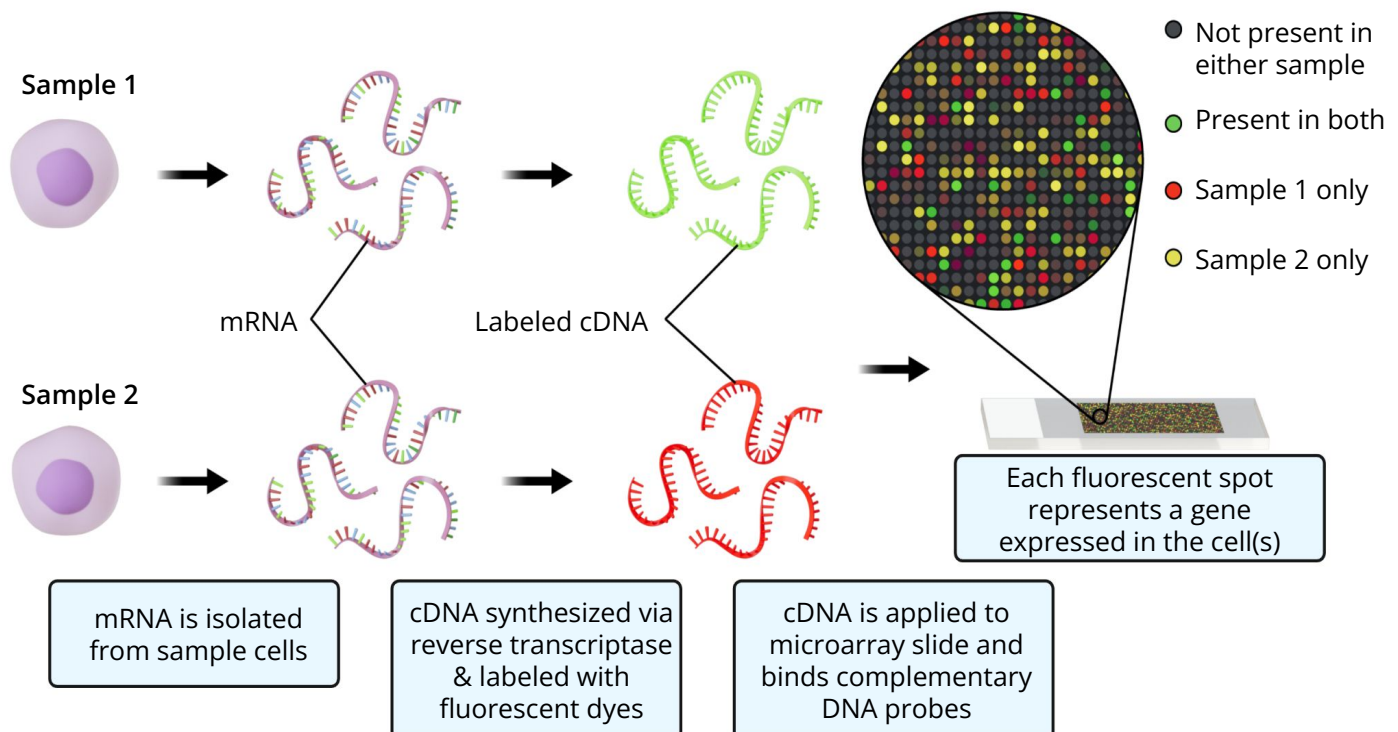
1. A **genomic library** stores the DNA of an organism's genome. DNA fragments are incorporated into plasmids and can be screened for by using antibiotic resistance and color changing techniques. They can then be cloned via bacterial cloning.
2. **DNA microarrays** contain thousands of DNA probes that bind to complementary DNA fragments, allowing researchers to see which genes are expressed.
 - Protocol: Isolate a cell and remove mRNA (active transcription) → synthesize cDNA from mRNA using reverse transcriptase → hybridize cDNA with DNA probes → examine microarray for fluorescence → compare microarray with the sequenced genome.

1. **Immunofluorescence microscopy**: A technique that identifies the localization of proteins of interest, using fluorophores.
2. **Fluorescence recovery after photobleaching (FRAP)**: Quantitative measure of how and where biomolecules move in a live cell.
 - Protocol: Baseline fluorescence is measured → area of the sample is photobleached → photobleached molecules are replaced by unbleached molecules over time due to cell dynamics → area gradually recovers fluorescence.

Fluorescence Recovery After Photobleaching (FRAP)



DNA Microarray



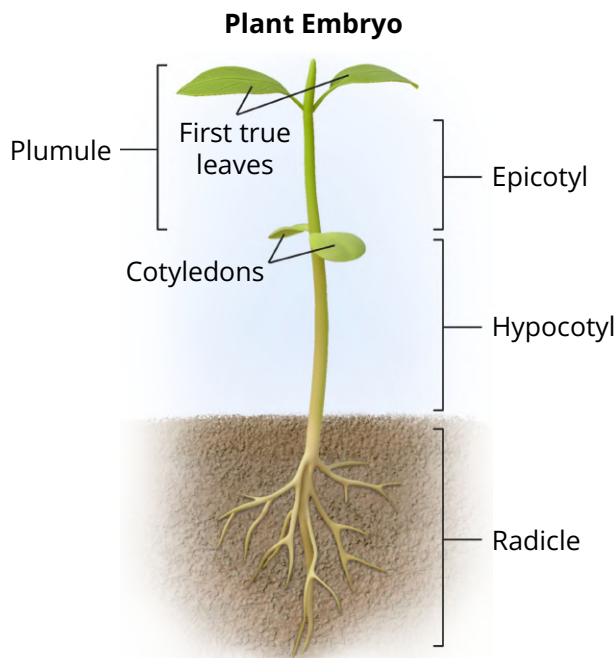
Ch. 9: Plants

Table of Contents

- The Seed and Germination
- Primary vs. Secondary Growth
- Plant Tissues
- Leaf Anatomy
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- Movement of Food
- Plant Hormones
- Alternation of Generations
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- Bryophytes
- Tracheophytes
- Flower Structures
- Angiosperms: Monocots vs. Dicots
- Double Fertilization
- Nitrogen Fixation

The Seed and Germination

1. **Seed coat:** Hard outer layer that covers and protects the seed.
2. **Endosperm:** Storage material, provides the embryo with nutrients.
3. **Embryo:** Consists of four parts:
 - **Radicle:** First to emerge, develops into root, anchors the plant into soil.
 - **Hypocotyl:** Bottom region of young shoot.
 - **Plumule:** Develops into leaves.
 - **Epicotyl:** Top region (shoot tip).



Germination: The process of a seed sprouting into a seedling. Water is the most important condition to initiate growth.

Primary vs. Secondary Growth

Plant growth takes place via mitosis at **meristems**.

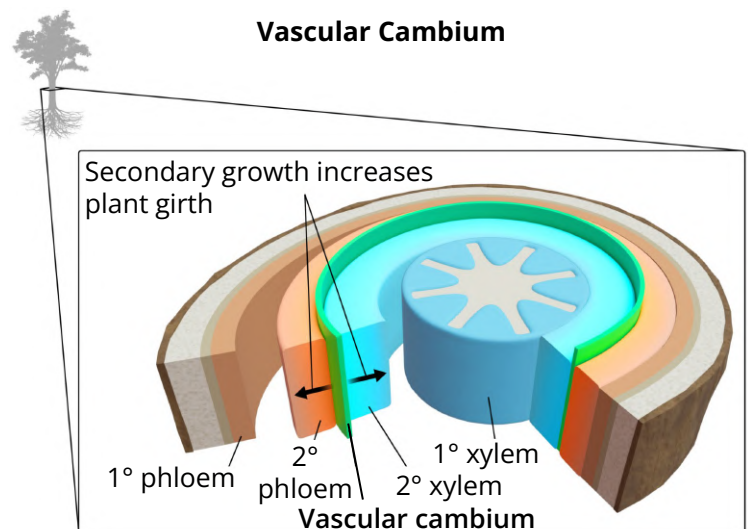
Primary growth is vertical growth occurring at apical meristems (located at tips of roots and shoots). Occurs before secondary growth.

Root growth: Root cap covers roots, protecting the apical meristem.

Secondary growth is horizontal growth occurring at lateral meristems (vascular cambium and cork cambium). Mainly occurs in woody plants.

Vascular cambium is a ring of meristematic tissue located between **primary xylem** (closer to center) and **primary phloem** (closer to outer edge). Cells produced inside the ring of vascular cambium become **secondary xylem** (forms wood) and cells outside become **secondary phloem** (contributes to inner bark). New xylem accumulates every year, forming growth rings, whereas new phloem replaces old phloem and does not accumulate.

Cork cambium is a ring of meristematic tissue located outside the phloem. It produces cork, the outermost protective layer. Bark is formed by secondary phloem, cork cambium, and cork.



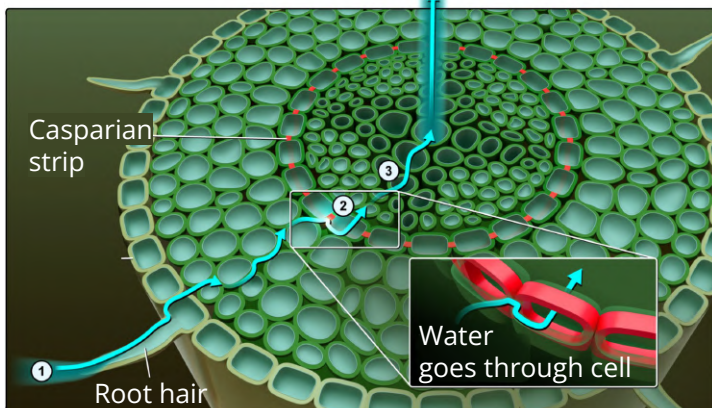
Plant Tissues

1. **Ground tissue:** Provides structural support, makes up most of the plant's mass.
2. **Vascular tissue:** Transports materials from a source to a sink (**source to sink theory**).
 - **Phloem:** Transports sugars from leaves (source) to roots and other areas (sink). Made of **sieve cells** (long cells, lacking organelles, connected to form a tunnel for transport) and **companion cells** (connected to sieve cells, contain organelles for metabolic functions).
 - **Xylem:** Transports water and dissolved minerals from roots (source) to leaves (sink) and provides structural support. Made up of **tracheids** (long and thin, water travels through pits in their tapered ends) and **vessel elements** (short and stout, water travels via perforations in cell walls).
3. **Dermal tissue:** Outer layer of the plant, provides protection and regulation. A waxy **cuticle** layer limits water evaporation.

Casparian strip (made of fat and wax) is an impenetrable substance in the cell walls of the root endoderm. In order for water to pass from the roots into the vascular tissue, water is forced out of the cell walls into the endodermal cell cytoplasm to be filtered.

Casparian Strip

Water travels up through xylem



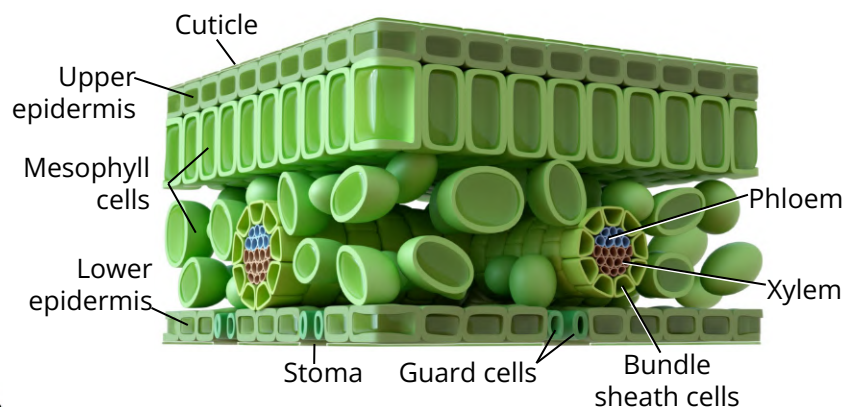
Leaf Anatomy

Leaves are covered by an **epidermal layer** and a waxy **cuticle**. **Stomata** are pores in the lower epidermis that open and close for gas exchange. Water influx makes **guard cells** turgid, opening the stomata. Stomata also open if CO_2 concentration is low (allows for CO_2 intake and photosynthesis). Stomata close if CO_2 concentration is high or when temperatures are high (prevents transpiration). There must be balance between opening stomata for food production and closing them to prevent water loss.

Between the upper and lower epidermis are the **mesophyll cells**, which are involved in photosynthesis and gas exchange.

Bundle sheath cells surround and protect the vascular bundles.

Leaf Structures



Movement of Water

1. **Cohesion-tension theory:** Transpiration, the driving force, causes water to evaporate from the stomata and leads to a transpirational pull. This **cohesive force** (between similar substances, e.g., the water molecules) pulls the water column upward.
2. **Capillary action:** An **adhesive force** (between dissimilar substances) due to attraction between water and xylem vessels that causes water to climb upwards.
3. **Root pressure:** Water osmotically travels into roots from soil. An **osmotic gradient** in the roots then drives water into the xylem.

Movement of Food

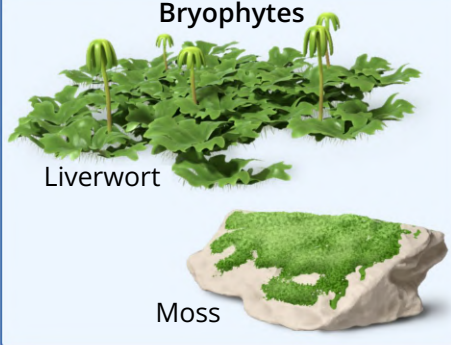
Pressure flow hypothesis: Source cells produce sugar and load it into phloem → increased sugar concentration creates a gradient that pulls water into phloem → turgor pressure in phloem *increases*, resulting in bulk flow movement of sugar from leaves down to roots.

Plant Hormones

1. **Ethylene:** Gas that *increases* fruit ripening.
2. **Auxins:** Cause cell growth. Work with cytokinins. Responsible for plant **tropisms** (growth in certain directions). Auxin concentrated on one side of a stem leads to asymmetric growth.
 - **Phototropism:** Growth towards light.
 - **Gravitropism:** Growth away from pull of gravity.
 - **Thigmotropism:** Growth in response to contact (e.g., vine growing up a wall).
3. **Cytokinins:** Regulate cell differentiation and division with auxins. Can prevent aging.
4. **Gibberellins:** Responsible for stem and shoot elongation, elimination of dormancy of a seed, flowering, fruit production, leaf and fruit death.
5. **Abscisic acid:** Functions during stress. Promotes dormant seeds, closes stomata (drought), inhibits growth.

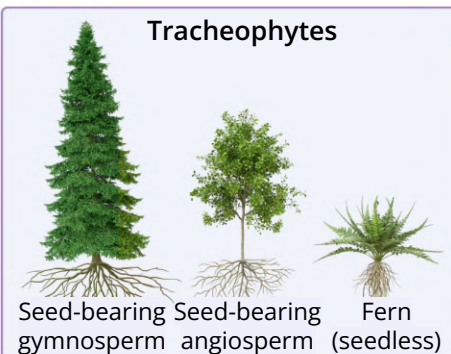
Bryophytes vs. Tracheophytes

Bryophytes



- Nonvascular (no xylem/phloem)
- Seedless (no flowers/fruit)
- Low-growing and spread horizontally
- Contain rhizoids, not roots

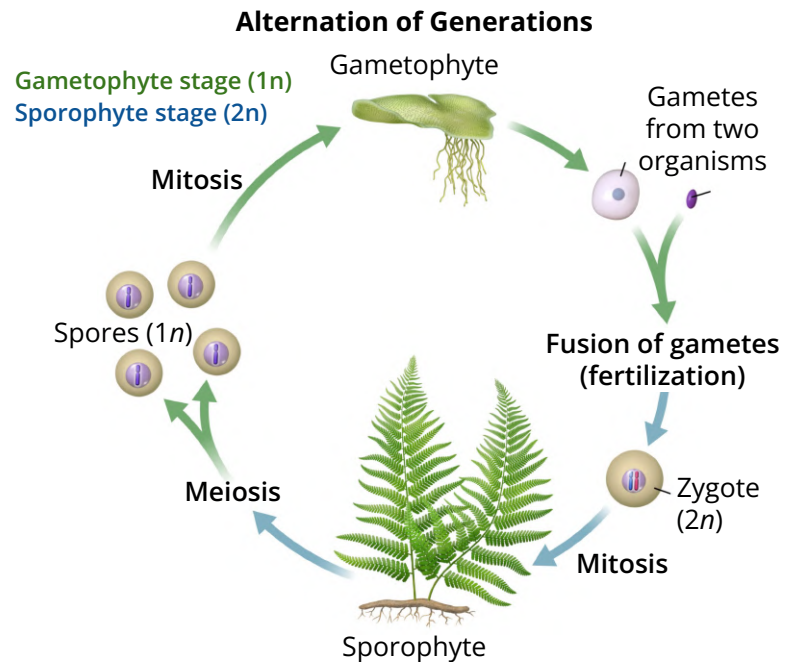
Tracheophytes



- Vascular (contain xylem/phloem)
- Can be seed-bearing or seedless
- Grow vertically and horizontally
- Contain root systems for anchorage

Alternation of Generations

Alternation between **diploid** and **haploid**.



Two haploid **gametes** fuse producing diploid **zygote** → zygote becomes **sporophyte** via mitosis → in their sporangia, sporophyte undergoes meiosis to produce haploid **spores** → spore becomes **gametophyte** via mitosis → gametophyte produces **gametes** → cycle repeats.

Homosporous vs. Heterosporous Plants

Homosporous plants: Produce one type of spore that develops into a bisexual gametophyte.

Heterosporous plants: Produce two types of spores. **Megaspores** develop into the embryo sac (female gametophyte) and **microspores** develop into pollen grains (male gametophyte).

Bryophytes

Nonvascular plants (e.g., mosses, hornworts, liverworts) are small and short. Found in moist habitats and grow horizontally to remain close to water and nutrients. Contain **rhizoids** (hair-like projections) which aid in water absorption and minor anchorage. Majority of their life cycle is spent in the gametophyte stage; they have a reduced sporophyte which depends on and is attached to the gametophyte.

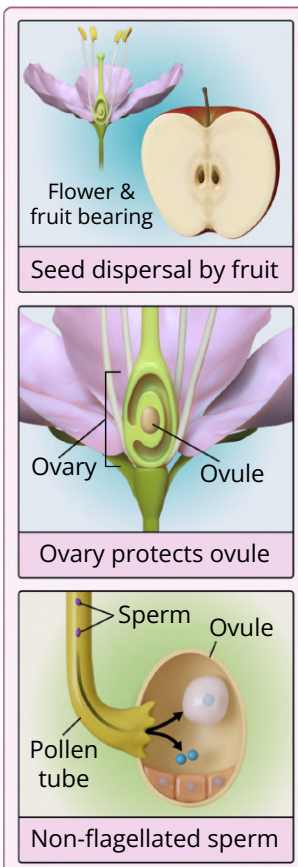
Tracheophytes

Tracheophytes are **vascular** (contain xylem and phloem), allowing them to grow vertically and tall and have a root system for anchorage. Most of the life cycle is spent in the sporophyte stage.

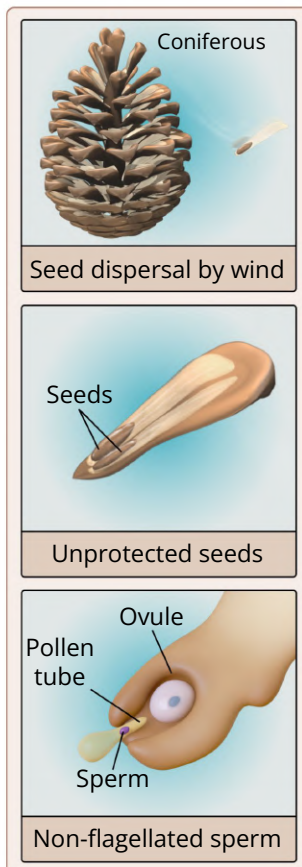
1. **Seedless tracheophytes:** (Lycophytes and pteridophytes; e.g., club moss, quillworts, fern, horsetail). Most are homosporous with **flagellated sperm** (can move on their own). Also, seedless tracheophytes have independent gametophyte and sporophyte life-cycles.
2. **Seed-bearing tracheophytes** (all heterosporous)
 - **Gymnosperms:** The first seeded plants. Seed are not protected (e.g., **conifers** such as firs, spruce, pine, redwood). Most gymnosperms have non-flagellated sperm that is dispersed in seeds by wind.
 - **Angiosperms:** Most abundant plant. Flower-bearing and fruit-producing. Plant ovary protects seeds. Non-flagellated sperm is dispersed by wind or animals often as **pollen**. Can exhibit **double fertilization** (female gamete fertilized by two male sperm).

Seed-Bearing Tracheophytes

Angiosperm



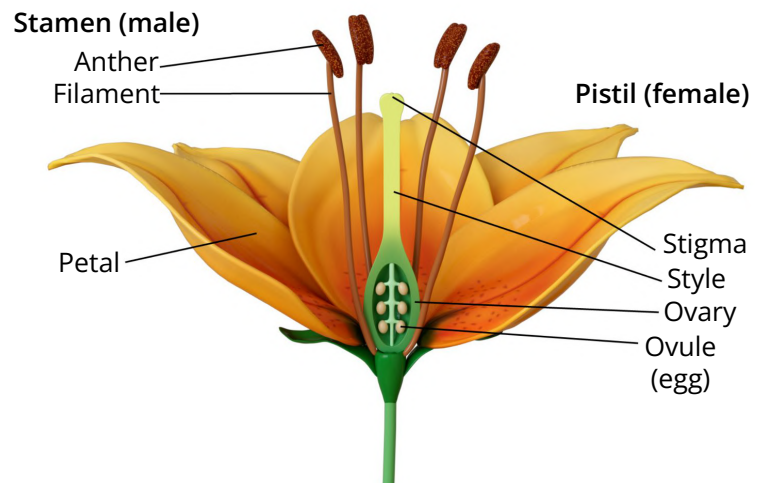
Gymnosperm



Flower Structures

1. **Petals:** Attract animals to achieve pollination.
2. **Stamen:** Male plant sex organ. Composed of **anther** (site of microspore formation) and **filament** (supports anther).
 - Microspore undergoes mitosis to form two cells that combine to form **pollen**.
3. **Pistil:** Female plant sex organ. Composed of three parts:
 - **Stigma:** The top of the pistil and the site where pollen lands for germination.
 - **Style:** Tube that leads to the ovary.
 - **Ovary:** Stores ovule. The female gametophyte is known as the **embryo sac** and develops within the ovule. Eggs are in the embryo sac.

Flower Anatomy



Double Fertilization

1. Pollen lands on the stigma and forms a pollen tube down the style.
2. Two sperm cells are released:
 - One sperm nucleus fuses with an egg in the ovule to form zygote, which develops into the **embryo**.
 - The other sperm nucleus combines with the ovule's polar nuclei to form the **endosperm**, which provides nutrients for the embryo within seed.

Angiosperms: Monocots vs. Dicots

Cotyledons: First leaves to appear on seedlings. Contain nutrients from seed to feed the growing seedling.

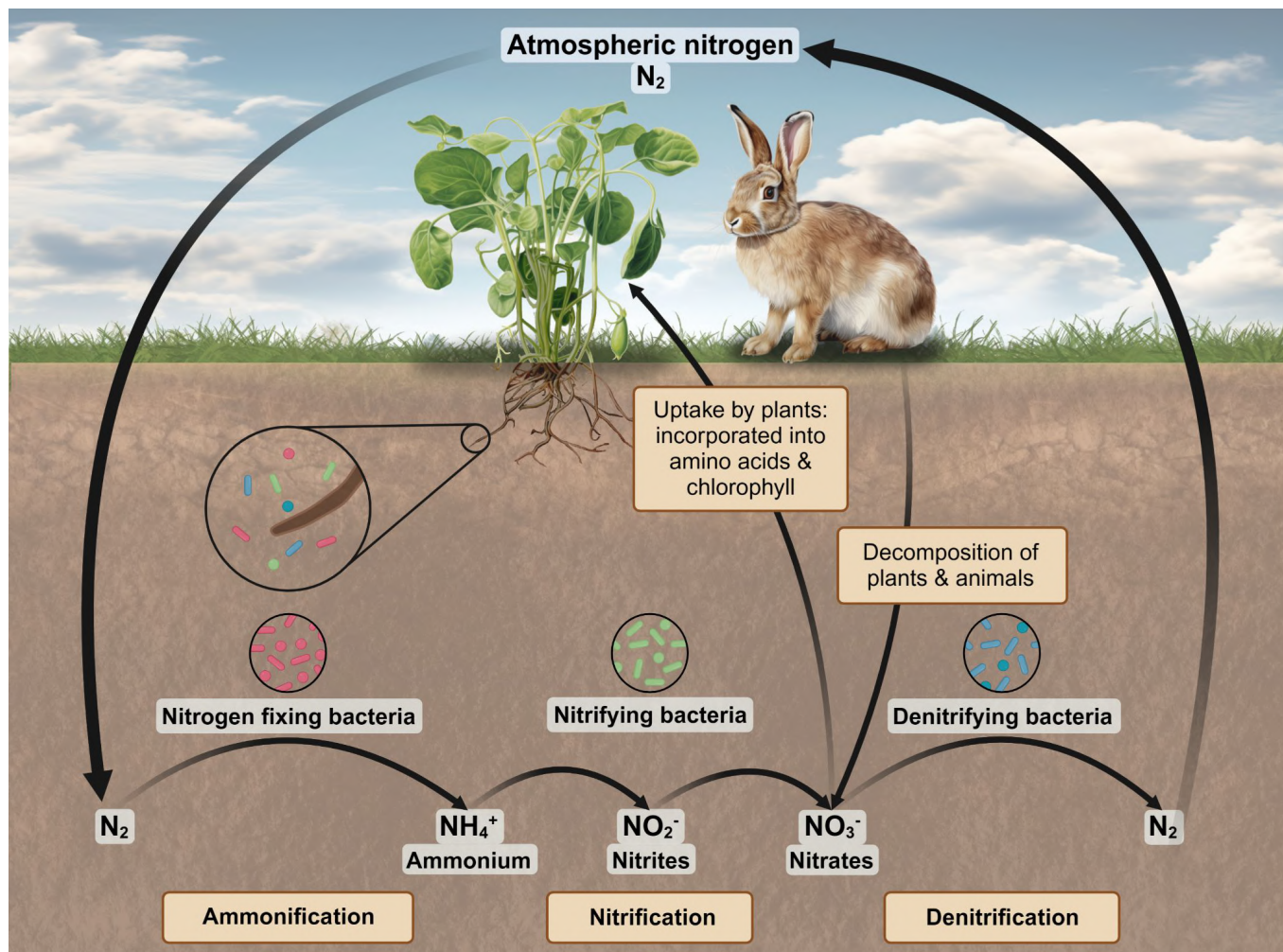
Monocotyledons (Monocots)	Dicotyledons (Dicots)
Single cotyledon	Two cotyledon
Long narrow leaf & parallel veins	Broad leaf & a network of veins
Vascular bundles scattered	Vascular bundles in a ring
Floral parts in multiples of three	Floral parts in multiples of four or five
Fibrous fine root system near surface	Single taproot with branching

Nitrogen Fixation

Plants have a **symbiotic relationship** with nitrogen-fixing bacteria. Bacteria fix atmospheric nitrogen into a usable form for plants; in return, plants produce food for bacteria via photosynthesis.

1. **Nitrogen-fixing bacteria** in root nodules of legumes fix atmospheric nitrogen (N_2) to ammonia (NH_3) and ammonium (NH_4^+).
2. **Nitrifying bacteria** convert ammonia and ammonium into nitrites (NO_2^-) and then into nitrates (NO_3^-).
3. **Nitrates** are taken up by plants (assimilation of nitrogen) and incorporated into amino acids and chlorophyll. Animals (consumers) acquire nitrogen by eating plants (producers).
4. **Detritus** of dead decaying plants and animals provides soil with nitrates.
5. **Denitrifying bacteria** convert nitrates back to atmospheric nitrogen.

Nitrogen Cycle



Ch. 10.1: Circulatory System

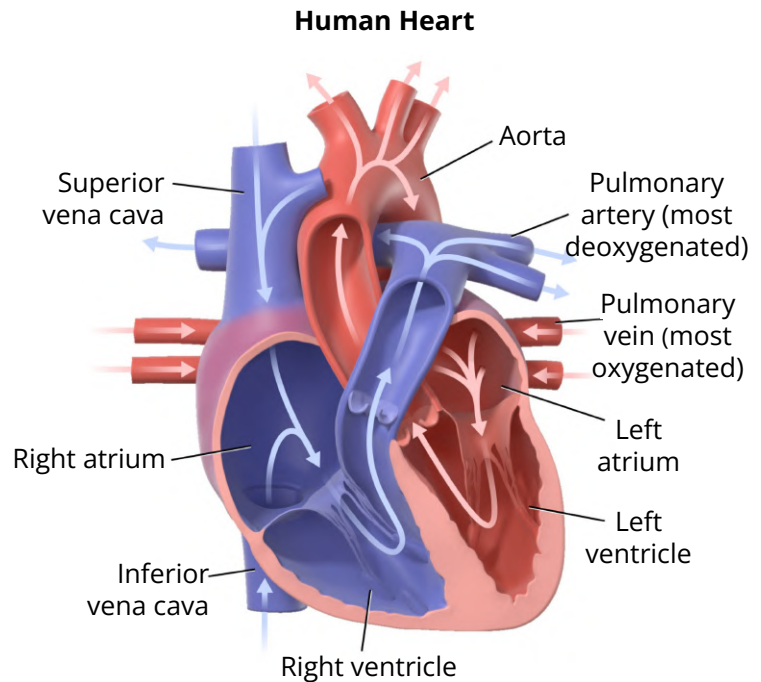
Table of Contents

- Human Heart
- Human Cardiac Cycle
- Heart Function Measurements
- Blood Vessels
- Components of Blood
- Blood Clotting
- Blood Types
- Fetal Circulation
- The Lymphatic System

Human Heart

Flow of blood through the heart:

1. **Right atrium:** Deoxygenated blood flows in through the **superior** (upper) **vena cava** and the lower **inferior** (lower) **vena cava**. Blood passes from the right atrium to the right ventricle through the right atrioventricular valve (AV or **tricuspid valve**).
2. **Right ventricle:** Pumps deoxygenated blood into the pulmonary artery for pulmonary circulation through the **pulmonary semilunar valve**. When the ventricle contracts, the AV valve closes and the pulmonary semilunar valve opens. When the ventricle is relaxed, the AV valve opens to refill the ventricle, and the pulmonary semilunar valve closes to prevent backflow from the artery.
3. **Left atrium:** Oxygenated blood from the lungs flows in through the **pulmonary vein**. Blood passes from the left atrium into the left ventricle through the left AV valve (**bicuspid** or **mitral valve**).
4. **Left ventricle:** Most muscular chamber of the heart. Pumps oxygenated blood into the **aorta** for systemic circulation through the aortic semilunar valve.



Pulmonary circulation moves deoxygenated blood from heart to the lungs and back in order for it to become oxygenated. Pathway:

Right atrium → tricuspid valve → right ventricle → pulmonary semilunar valve → pulmonary arteries → lung → pulmonary veins → left atrium

Systemic circulation moves oxygenated blood from the heart throughout the body. Pathway:

Left atrium → bicuspid / mitral valve → left ventricle → aortic semilunar valve → aorta → body → vena cava → right atrium

Coronary circulation supplies oxygenated blood to the heart itself for function.

Human Cardiac Cycle

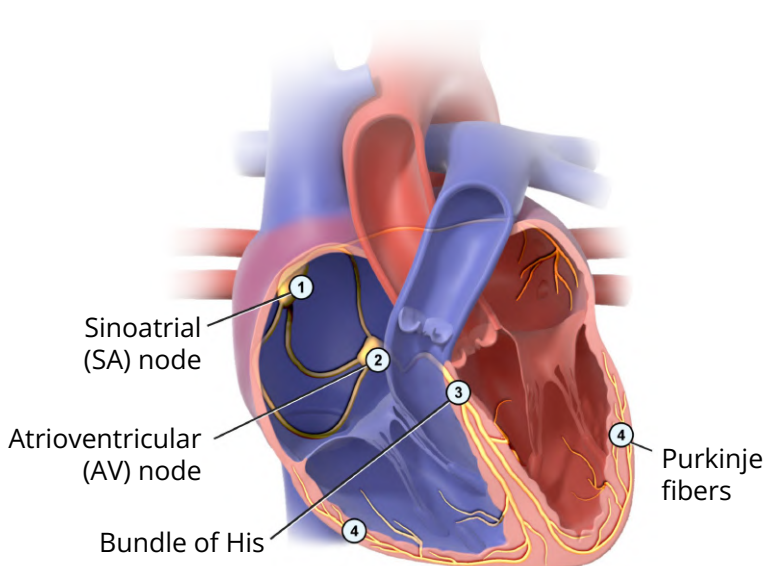
The heart needs to contract and relax rhythmically in order to pump blood throughout the body.

Cardiomyocytes (heart muscle cells) have **automaticity**, which means they are self-excitabile and able to initiate an action potential without an external nerve.

The cardiac cycle:

1. The **SA node (pacemaker)** is located in the upper wall of the right atrium and usually initiates the cardiac cycle. It has the greatest automaticity and is most likely to reach threshold to stimulate a heartbeat. It sends a signal to contract both atria to send blood to the ventricles. It also sends a signal to the AV node to initiate contraction.
2. The **AV node** is located in the lower wall of the right atrium. The function of the AV node is to add a brief delay between the contraction of the atria and the contraction of the ventricles. It also sends a signal to the **bundle of His**, located in the interventricular septum between the ventricles. The bundle of His carries the signal to the **Purkinje fibers**, which contract the ventricles.

Innervation of the Heart



Signal transduction: The heart has **intercalated discs** that connect adjacent heart cells (cardiomyocytes). Intercalated discs are made of desmosomes and gap junctions and function to transmit the signal to contract in a coordinated, rhythmic fashion.

Systole occurs right after the **ventricles** eject their blood into the **arteries** they connect to. Therefore, it is the phase of the cardiac cycle where blood pressure is highest in the arteries. The **aorta** is the blood vessel that experiences the highest blood pressure.

Diastole occurs right after the **atria** contract to fill the **ventricles**. The **myocardium** is completely relaxed at this point. Diastole is the phase of the cardiac cycle where blood pressure is lowest in the arteries.

Heart Function Measurements

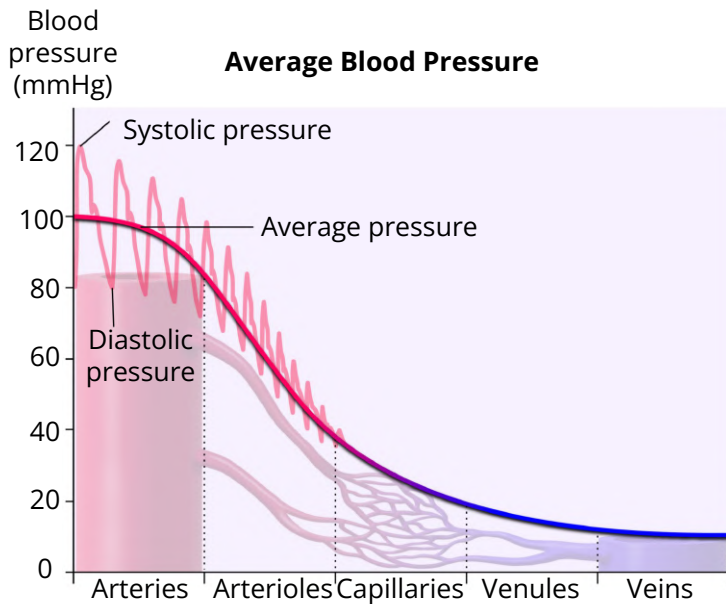
Heart rate (HR) is how fast the heart beats.

Stroke volume (SV) is the volume of blood pumped from the heart with each beat.

Cardiac output (CO) is the stroke volume multiplied by the heart rate. This provides the volume of blood being pumped by the heart in one minute.

$$CO = HR \times SV$$

Blood Vessels



Vessels transport blood to and from the heart in a closed circulatory system.

- **Arteries:** Move blood away from the heart. Blood pressure is highest in arteries due to hydrostatic pressure of the heart.
- **Arterioles:** Branch off of arteries (smaller). Greatest drop of blood pressure is seen from the arteries to the arterioles.
- **Capillaries:** One-cell-thick vessels that branch off of arterioles. Diffuse gas and nutrients into interstitial fluid. Collect waste and CO_2 .
- **Venules:** Intermediate vessel connecting capillaries to veins.
- **Veins:** Move blood toward the heart. Veins contain more blood by volume than arteries, but blood pressure is lowest in veins. Veins contain valves (aka **pocket** or **venous valves**) that help blood move towards the heart by preventing backflow. Blood is pushed forward when skeletal muscles squeeze the veins, *not* when the heart pumps.

Blood pressure in peripheral blood vessels can be regulated by vasoconstriction or vasodilation through endocrine, paracrine, or neural signaling.

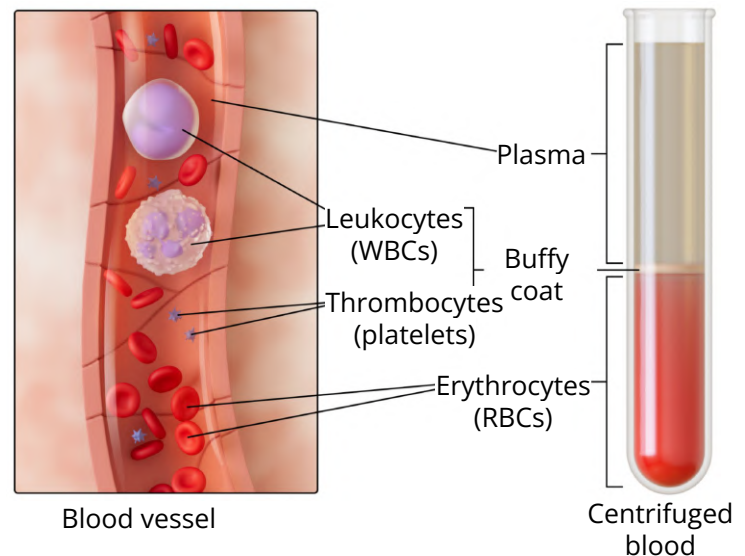
Vasoconstriction decreases vessel diameter, causing resistance to blood flow (*higher* blood pressure).

Vasodilation increases vessel diameter, reducing resistance to blood flow (*lower* blood pressure).

Components of Blood

1. **Plasma** contains water, proteins, nutrients, hormones, and makes up most of the blood volume. Makes up ~55% of blood volume.
2. **White blood cells (leukocytes)** are our immune cells and defend against infection. The most common white blood cell is the **neutrophil**.
3. **Platelets (thrombocytes)** are cytoplasmic cell fragments that do not have a nucleus; they are responsible for clotting. Large bone marrow cells called **megakaryocytes** are the precursor to platelets. Platelets release factors that help convert **fibrinogen** into **fibrin**, which creates a 'net' to stop bleeding. Many of the clotting factors are synthesized with **Vitamin K**, a deficiency in Vitamin K will lead to increased bleeding. Platelets are also immune cells that function in the innate immunity. Leukocytes and thrombocytes make up <1% of blood volume.
4. **Red blood cells (erythrocytes)** are responsible for transporting oxygen attached to hemoglobin. Mature red blood cells are **anucleate** (they don't have a nucleus) in order to maximize the amount of space they have to carry hemoglobin and oxygen, making them very flexible. Makes up ~45% of blood volume.

Components of Blood



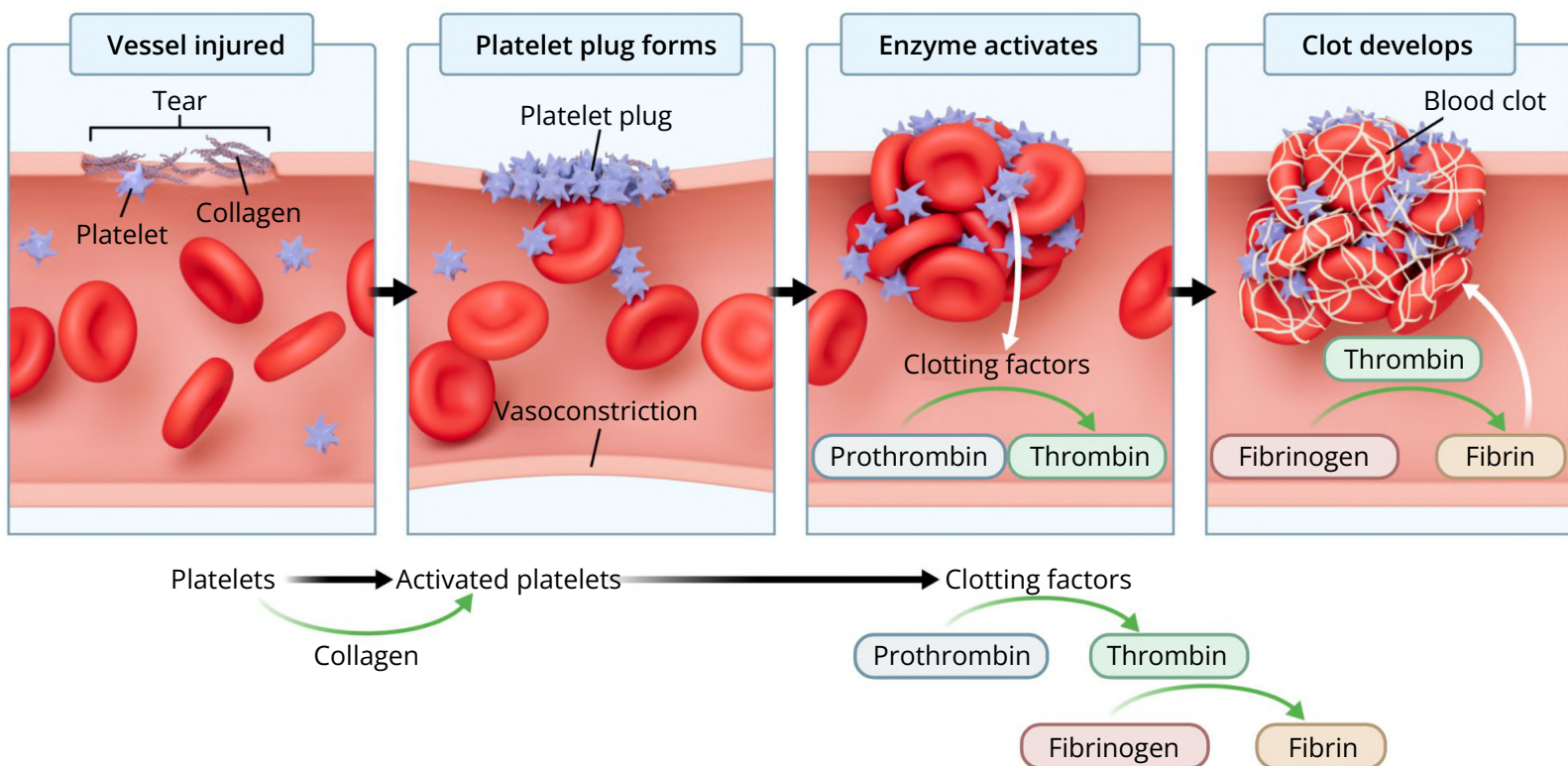
Blood Clotting

Blood clotting: A mechanism that protects our blood vessels from damage. When a tear in our blood vessels occurs, the **blood clotting cascade** then 'plugs' the tear, sealing any vessel leaks.

The blood clotting cascade is an example of a **positive feedback mechanism**. Damaged tissues release a signal that attracts platelets to the site → each platelet will then release a signal to attract *more* platelets → a clot is formed. The steps to this process are as follows:

1. **Tissue damage:** Damaged tissue tears blood vessel walls, exposing their **collagen**.
2. **Platelet activation:** Exposed collagen triggers platelet activation. Platelets will adhere and aggregate at the site of the tear, forming a **platelet plug**.
3. **Clotting factors release:** Activated platelets release clotting factors that convert **prothrombin** (inactive precursor) → **thrombin** (active form).
4. **Formation of clot:** Activated thrombin converts **fibrinogen** (inactive precursor) → **fibrin** (active form). Fibrin strands polymerize with other fibrin strands, and attach platelets to form a **blood clot** (**hemostatic plug**).

Clotting Cascade



Blood Types

Red blood cells (erythrocytes) have **antigens** on their surfaces. These antigens are little sugars and proteins that mark our blood as a certain type. Blood types are described as follows.

- **Type A blood:** Has 'A' antigen
- **Type B blood:** Has 'B' antigen
- **Type AB blood:** Has both 'A' and 'B' antigens
- **Type O blood:** Has neither 'A' or 'B' antigens

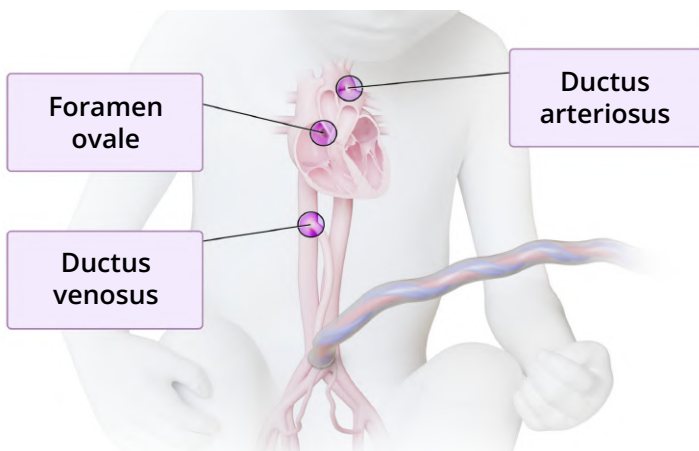
If a person receives a blood transfusion with the incorrect blood type, their immune system will cause the **agglutination** (clumping together) of antibodies of that blood type.

In addition to blood type A and B, your body also has another surface protein called the **Rhesus factor (Rh)**. You either have the factor (Rh+) or you do not (Rh-). If a donor is Rh+, they cannot donate to someone who is Rh-, because the donor has antigens on the surface of the blood cell.

A **universal donor** (blood donor who can donate to anyone) is O-. O blood type has neither A nor B surface antigens, and O- blood also does not have an Rh surface antigen. This means there are no blood cell surface antigens that would stimulate immune clearance by someone receiving the O- blood.

A **universal acceptor** is AB+. Because an AB+ person has both A and B cell surface antigens, as well as an Rh surface antigen, they can receive any blood type and not mount an immune response. Any blood cell surface antigen they receive would be something their blood cells already have.

Fetal Shunts



Blood Types

A	B	AB	O
 Anti-B	 Anti-A	No antibodies	 Anti-A & Anti-B
 A antigen	 B antigen	 A & B antigens	No antigens
A, O compatible	B, O compatible	A, B, AB, O compatible	O compatible

Fetal Circulation

A fetus receives oxygen and nutrients from the **placenta** (mother) via the **umbilical cord**. The **umbilical vein** carries oxygenated blood from the placenta to the fetal heart, while the **umbilical artery** carries deoxygenated blood from the fetal heart back to the placenta. Fetal lungs and liver are nonfunctional and filled with fluid. To compensate, fetal circulation has:

1. **Ductus venosus:** A duct (tunnel) that connects the **umbilical vein** to the **inferior vena cava**. Blood that passes through the ductus venosus bypasses the nonfunctional fetal liver.
2. **Foramen ovale:** A hole that allows blood in the **right atrium** to flow into the **left atrium**. Recall that the right ventricle pumps to the lungs, which are nonfunctional in the fetus. Blood that passes through the foramen ovale bypasses pulmonary circulation.
3. **Ductus arteriosus:** A duct that connects the **pulmonary artery** to the **aorta**. Blood pumped by the right ventricle into the pulmonary artery can bypass the lungs by moving through the ductus arteriosus.

Mnemonic: The ductus **arteriosus** connects two **arteries** (pulmonary artery to aorta). The ductus **venosus** connects two **veins** (umbilical vein to vena cava). The foramen **ovale** is the **other one** (right to left atria).

Wastes and CO_2 from the fetus are removed via the umbilical artery in the umbilical cord. Nutrients and gas exchange across a barrier to prevent mixing of fetal and maternal blood in the placenta.

After birth, these structures are no longer needed. The umbilical cord dries up and falls off (if not cut at birth) and the ducts and foramen ovale close in order for normal circulation to the lungs and liver to occur.

Erythroblastosis fetalis: If the mother has Rh- blood type and the fetus has Rh+ blood, during labor, the fetal Rh+ blood will enter the mother's system, and she will develop **anti-Rh antibodies**. This will not pose a problem in the first pregnancy, but if the mother becomes pregnant again with another Rh+ fetus, the mother's anti-Rh antibodies will attack the fetus, because antibodies are small enough to cross the placental barrier.

Lymphatic System

The **lymphatic system** is a subsystem of the circulatory system that regulates fluid levels and produces immune cells. Its components are **lymph nodes**, **lymph vessels**, **adenoids** (lymphatic tissue), the **spleen**, and the **thymus**. The thymus is the maturation site of T cell lymphocytes, immune cells that circulate through blood and lymph. Lymphocytes are also found in the spleen, which filters blood for pathogens and old or damaged RBCs.

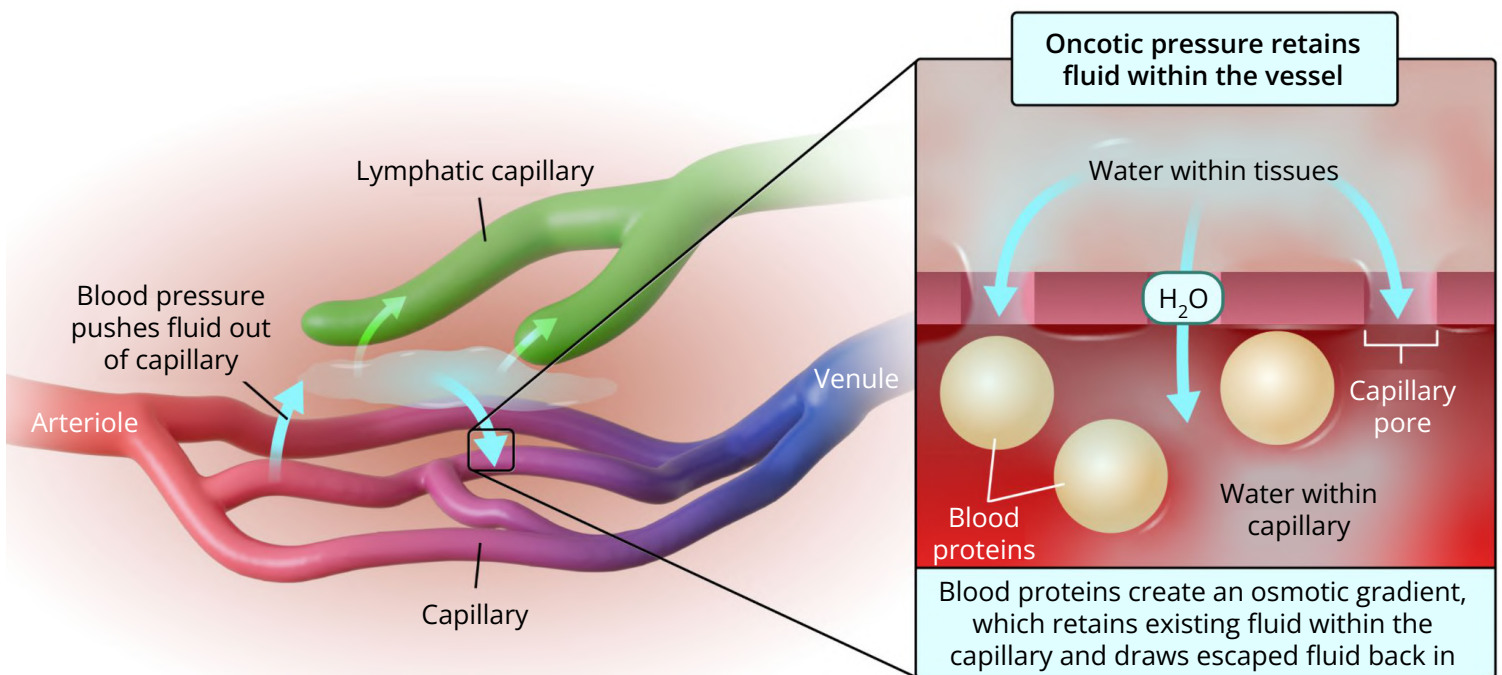
Nutrient and gas exchange occur at the level of the capillaries. **Hydrostatic pressure** pushes fluid out of the capillaries on the arterial end into interstitial space. **Oncotic pressure**, a type of osmotic pressure, brings fluid back into the capillaries at the venule end. However, not all the fluid is reabsorbed from the interstitial space into the venule. Lymphatic capillaries collect the remaining fluid, called **lymph**, which consists of interstitial fluid, bacteria, fats, and proteins.

The lymphatic capillaries merge together to form larger vessels that travel to the heart. Along the way, the lymph is filtered through **lymph nodes**, which are centers for the immune response system to eliminate infections.

Lymph vessels have no pressure (like veins), but the constriction of skeletal muscle, in conjunction with the smooth muscle present in the lymphatic vessel walls allows for the propulsion of lymph via **peristalsis**. This allows fluid to move towards the heart, and backflow of fluid is prevented with a system of valves, similar to veins.

Solute concentration influences lymph volume. If there is an increased amount of proteins (e.g., albumin) within the blood vessels, water will flow into these vessels, reducing the amount of water left in the interstitial fluid and *decreasing* lymph volume.

Blood Plasma Proteins and Oncotic Pressure



Ch. 10.2: Respiratory System

Pathways of Air

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- Human Lungs
- Pathways of Air
- Oxygen Dissociation Curve
- Gas Exchange
- Breathing Pace

Respiration: The exchange of gases between the outside environment and the inside of an organism.

Human Lungs

Lungs are located in the thoracic cavity and are covered by the rib cage. The left lung has two lobes and is smaller than the right lung, which has three lobes.

The **pleura** covers the lungs and is a dual-layered membrane.

The **diaphragm** is a large skeletal muscle at the bottom of the lungs and is involved in inspiration and expiration. This is the only organ unique to all mammals.

The **pleural space** is a fluid-filled space in between the two layers of the pleura. This space is at a lower pressure than the atmosphere, and creates the **intrapleural pressure**.

Inspiration or **inhalation** involves the contraction of the **diaphragm** (pulls lungs downwards) and the **external intercostal muscles** (expands the rib cage). These contractions cause the pressure of the intrapleural space to *decrease* and the volume of the lungs to *increase*, bringing air into the lungs.

Expiration or **exhalation** involves the relaxation of the diaphragm and the external intercostal muscles, bringing the lungs back up and closing up the rib cage through **elastic recoil**. This causes the pressure of the intrapleural space to *increase* and the volume of the lungs to *decrease*, driving air out of the lungs. The **internal intercostal muscles** can also contract during a more forced expiration, closing the rib cage even more.

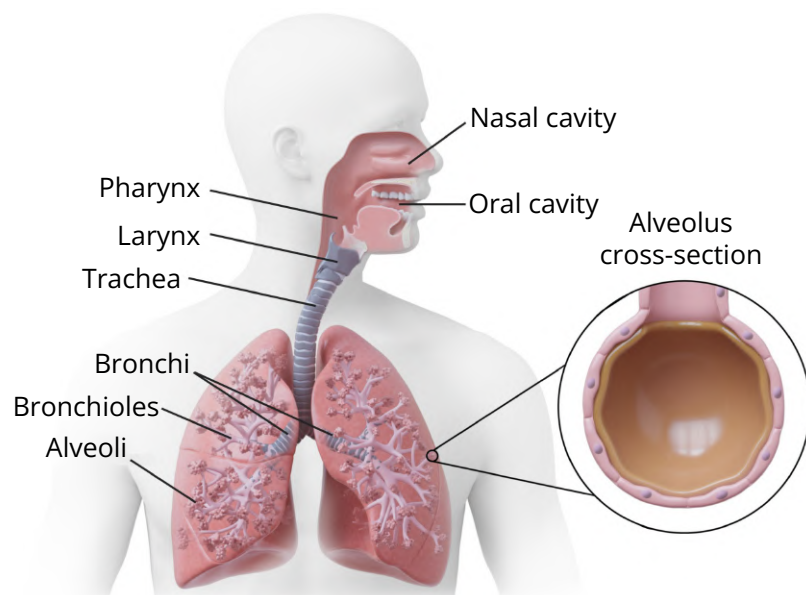
1. Air is inhaled through the nose and mouth.
2. The **pharynx** is the beginning of the throat, after the nasal cavity and mouth. Both food and air enter the pharynx.
3. The **larynx** receives air and contains the voice box. The **epiglottis** diverts food and liquids into the esophagus, and prevents anything but air from entering the larynx and lungs.
4. The **trachea** is below the larynx and splits into the bronchi.
5. The left and right **bronchi** (main lung tubes) split into smaller **bronchioles** (branched, smaller lung tubes), and eventually into alveoli. **Alveoli** contain specialized cells that produce surfactant. **Surfactant** is a substance that prevents the alveoli from collapsing by reducing the surface tension within them.

Within the respiratory tract, the nasal cavity, bronchi, and bronchioles contain **goblet cells** (secrete mucus) and **ciliated epithelial cells** (possess beating cilia) to trap and remove debris that enters the respiratory tract.

Overall Pathway of Air

Nasal cavity → pharynx → larynx → trachea → bronchi
→ bronchioles → alveoli

Respiratory Tract



Differences in **partial pressure** allow gases to flow from areas of high pressure to low pressure by simple diffusion. This is required for **external respiration** (gas exchange between inspired air and lung alveolar capillaries) and **internal respiration** (gas exchange between blood and tissues).

Oxygen: Air → blood → tissues

Carbon dioxide: Tissues → blood → air

Erythrocytes (red blood cells) contain **hemoglobin**. Hemoglobin is tetrameric and has a **heme** cofactor in each of its four subunits. Heme cofactors are organic molecules that contain iron atoms, which bind oxygen. Thus, each hemoglobin can carry up to four oxygen molecules. **Carbon monoxide** (CO) poisoning happens because hemoglobin has a higher binding affinity for CO than for oxygen.

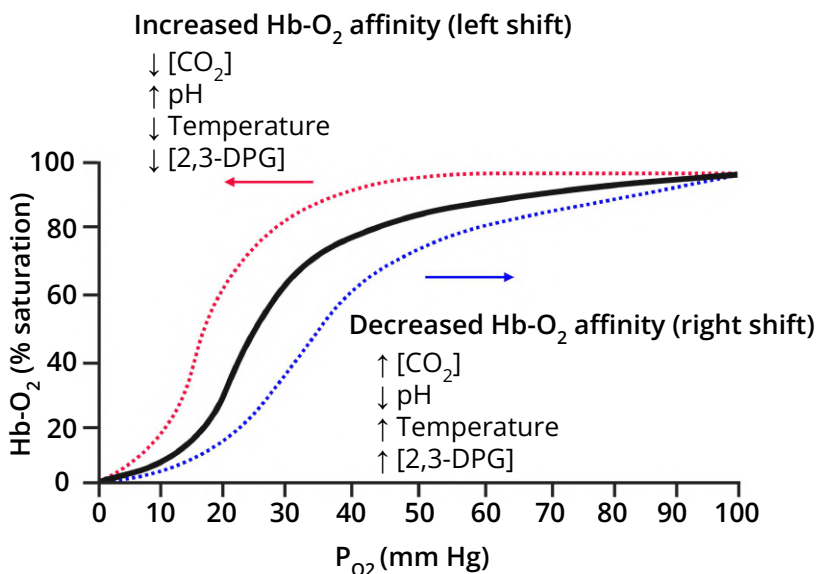
Oxyhemoglobin (HbO₂) transports most of the oxygen traveling in the blood.

Cooperativity describes the process by which the binding of one oxygen molecule to hemoglobin makes it easier for others to bind due to changes in the shape of the hemoglobin polypeptide. This also works in reverse, allowing efficient unloading of oxygen in body tissues.

Oxygen Dissociation Curve

The **oxygen dissociation curve** reveals the relationship between the saturation of hemoglobin with oxygen in the blood and the partial pressure of oxygen. Certain conditions shift this curve left or right.

Oxygen Dissociation Curve



A **right-shifted curve** corresponds to a lowered affinity for oxygen in hemoglobin. Below are the main reasons for a right-shifted curve:

- **Decreased pH:** A lower pH means there is a higher concentration of protons (H⁺), which produces reduced hemoglobin. Reduced hemoglobin (H⁺Hb) has a lowered affinity for binding oxygen, resulting in less HbO₂.
- **High partial pressure of carbon dioxide:** More carbon dioxide is converted to bicarbonate anions (HCO₃⁻) and protons (H⁺), which lower oxygen binding affinity through decreased pH.
- **2,3-diphosphoglycerate (2,3-DPG) aka 2,3-bisphosphoglycerate (2,3-BPG):** Accumulates in cells that undergo anaerobic respiration as a result of the loss of oxygen. This compound *decreases* oxygen binding affinity so more oxygen is released from hemoglobin to fuel aerobic respiration.
- **Increased body temperature:** Correlates to more cellular respiration, which uses up oxygen and produces more carbon dioxide. Thus, hemoglobin will need to unload more oxygen for tissues to use and have *decreased* oxygen binding affinity.

A **left-shifted curve** corresponds to an *increased* affinity for oxygen in hemoglobin. Below are the main reasons for a left-shifted curve:

- **Increased pH (more basic):** Fewer protons (H⁺) to produce reduced hemoglobin (H⁺Hb), so more oxyhemoglobin (HbO₂) remains.
- **Low partial pressure of carbon dioxide:** Less carbon dioxide is converted to bicarbonate anions (HCO₃⁻) and protons (H⁺), leading to *increased* oxygen binding affinity through *increased* pH.
- **Fetal hemoglobin:** Binds oxygen better than adult hemoglobin to help give oxygen to the fetus.
- **Decreased body temperature:** Less cellular respiration, so hemoglobin isn't influenced to unload more oxygen and has an *increased* oxygen binding affinity.

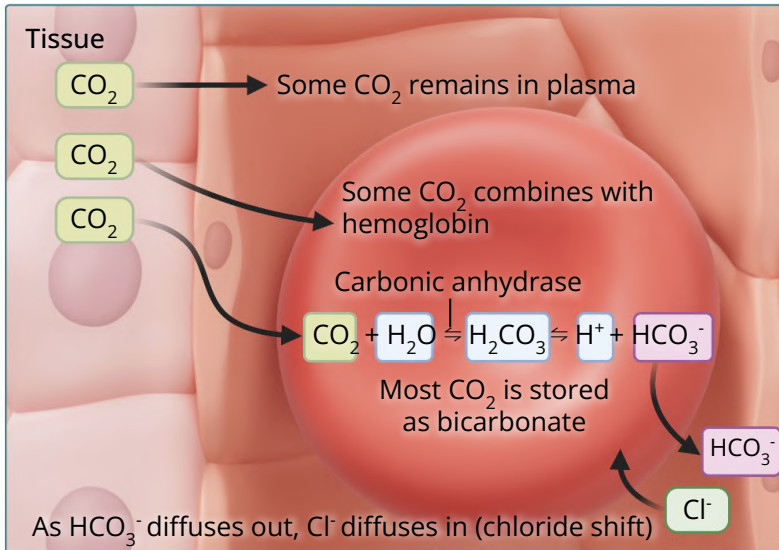
Mnemonic: CADET, face Right!

CADET = Carbon dioxide, Acid, 2,3-Diphosphoglycerate, Exercise and Temperature.

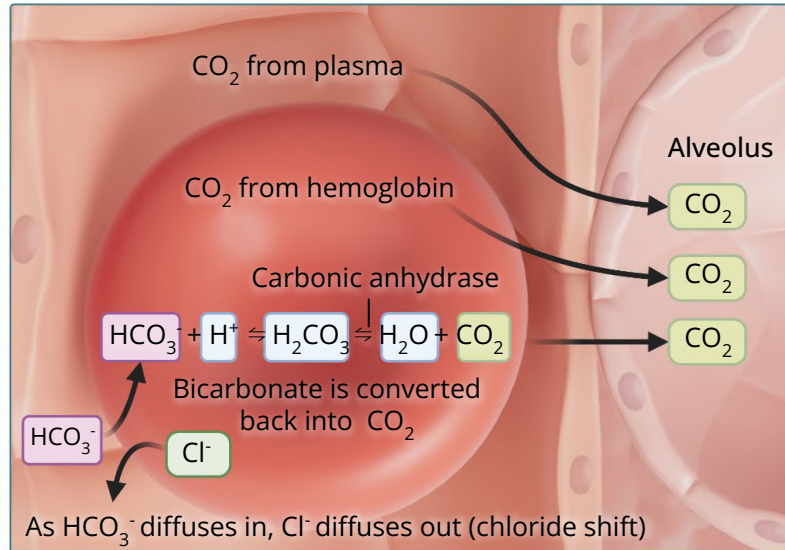
CADET Increase → **Right** shifted curve

Carbon Dioxide Transport

CO₂ transport from tissues



CO₂ transport to lungs



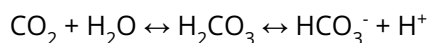
Changes in Hemoglobin Affinity

Bohr effect: Hemoglobin has *decreased* oxygen affinity when carbon dioxide is high. Carbon dioxide is converted to bicarbonate anions and protons, which produce reduced hemoglobin (H⁺Hb).

Haldane effect: Hemoglobin has *increased* carbon dioxide affinity when oxygen is low. As a result of low oxygen, reduced hemoglobin (H⁺Hb) levels are higher and have a greater affinity for carbon dioxide.

Gas Exchange in Tissues

The **bicarbonate buffering system** is the main extracellular buffering system in the body. It maintains our blood pH of 7.4 and can be described by the equation below:



Carbonic acid (H₂CO₃)

Bicarbonate anion (HCO₃⁻)

This system is catalyzed by **carbonic anhydrase** in both directions based on concentrations. Carbonic anhydrase is an enzyme present in red blood cells.

1. In **erythrocytes** (red blood cells) in the systemic circulation, the partial pressure of carbon dioxide is low. As a result, carbon dioxide continuously diffuses in from the tissues, and is converted into bicarbonate and protons. Bicarbonate is able to diffuse out of the cell, however, protons (H⁺) cannot leave. As some bicarbonate diffuses out, this creates a positive charge within the erythrocyte, and chloride ions (Cl⁻) must diffuse into the blood cell to cancel out the positive charge of the protons. This process is known as the **chloride shift**.
2. Influx of protons causes the pH to decrease within the erythrocyte, resulting in the conversion of oxyhemoglobin into reduced hemoglobin. Reduced hemoglobin has lower affinity for O₂, leading to release of oxygen which diffuses to the tissues.

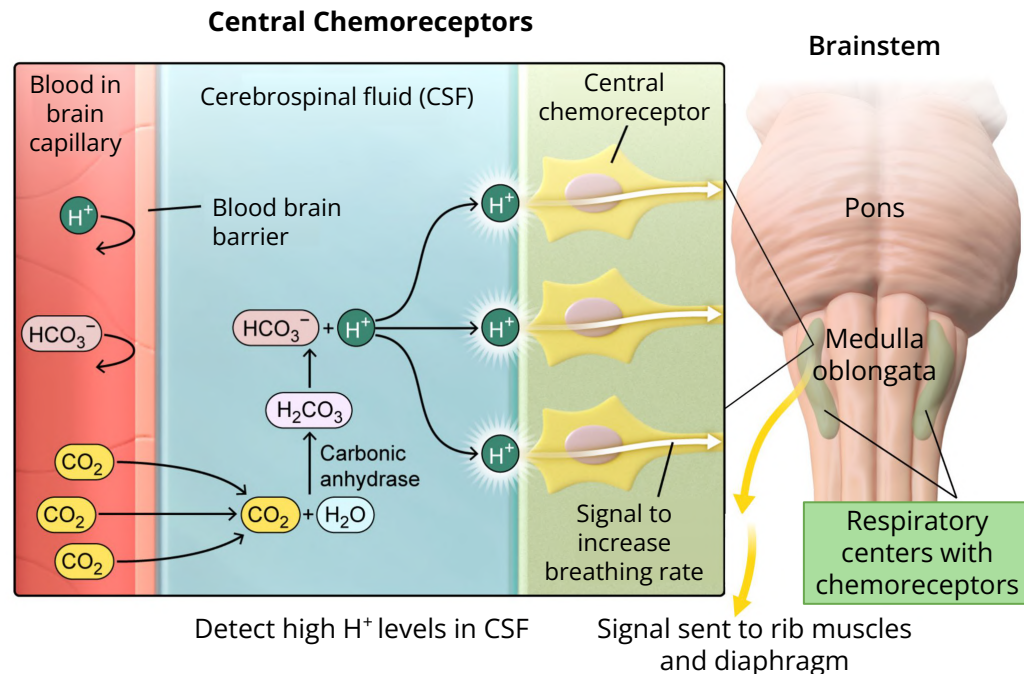
Gas Exchange in Lungs

1. Blood travels to the lungs through **bulk flow**.
2. Since most of the carbon dioxide is present in the blood plasma as bicarbonate ions (HCO₃⁻), the bicarbonate ions re-enter erythrocytes at the lungs and chloride ions leave through the **reverse chloride shift**.
3. The **bicarbonate buffer system equation** proceeds in the reverse direction, producing carbon dioxide and water. The carbon dioxide exits into the alveoli as gas while oxygen enters the blood, forming oxyhemoglobin.

Breathing Pace

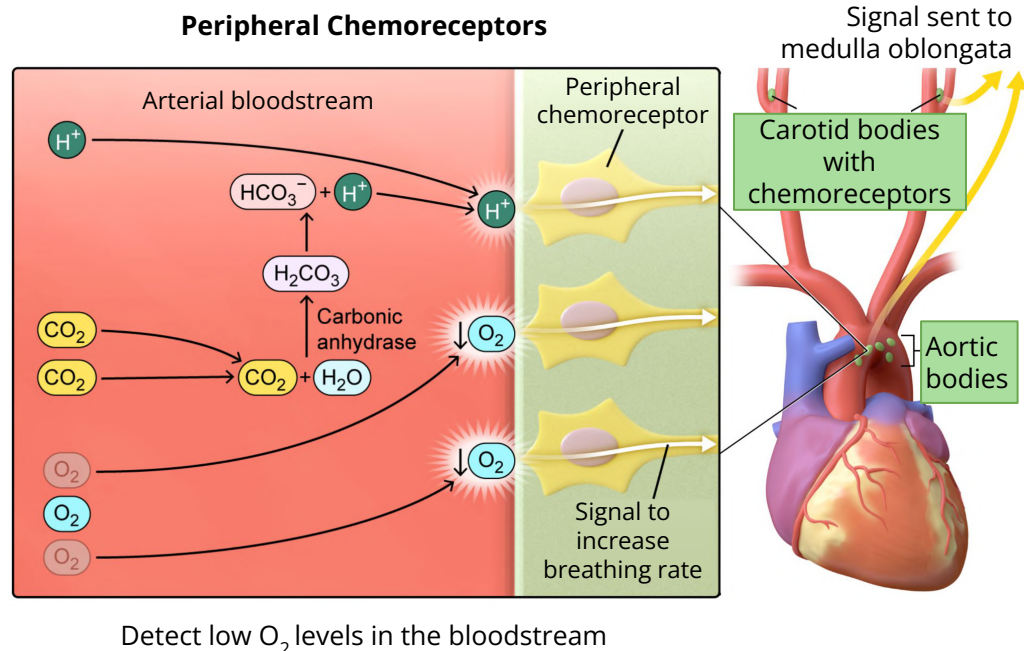
The **medulla oblongata** is located in the brain and controls the diaphragm to regulate respiratory rate. Central chemoreceptors and peripheral chemoreceptors signal to the medulla.

Central chemoreceptors are located in the medulla oblongata and are contained within the **blood-brain barrier**. Since carbonic anhydrase is present in the cerebrospinal fluid, carbon dioxide is converted into **bicarbonate ions** and **protons** here. However, protons cannot exit through the blood-brain barrier. As carbon dioxide accumulates, acidity *increases* and is directly sensed by central chemoreceptors, which signal to the medulla oblongata to *increase* breathing rate.



Peripheral chemoreceptors surround the aortic arch and carotid arteries. These peripheral chemoreceptors directly sense oxygen, carbon dioxide, and proton levels to signal to the medulla oblongata. When carbon dioxide is high and oxygen is low, peripheral chemoreceptors signal to the medulla oblongata to *increase* breathing rate.

Blood oxygen levels have significantly less of an effect on the respiratory center than carbon dioxide levels. Carbon dioxide in the blood is the primary stimulator for breathing.



Rate of Breathing

Respiratory acidosis: Lowered blood pH occurs due to inadequate breathing (**hypoventilation**).

Respiratory alkalosis: Increased blood pH occurs due to rapid breathing (**hyperventilation**).

Metabolic acidosis (lowered blood pH) and **metabolic alkalosis** (increased blood pH) occur as a result of imbalances in carbon dioxide, oxygen, or proton levels.

Ch. 10.3: Immune System

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- Innate Immunity: Overview
- Innate Immunity: Inflammatory Responses
- Innate Immunity: Immune Cells and Molecules
- Innate Immunity: Complement System
- Adaptive Immunity: Overview
- Adaptive Immunity: B cells
- Adaptive Immunity: T cells
- Passive vs. Active Immunity
- Classification of Common Diseases

Pathogens: Harmful microorganisms that cause disease.

Leukocytes: White blood cells.

Lymphocytes: Include T cells, B cells, and natural killer (NK) cells. These leukocytes originate from bone marrow and are primarily found in lymphatic organs.

T cells mature in the thymus, while **B cells** mature in the bone marrow.

Innate Immunity: Overview

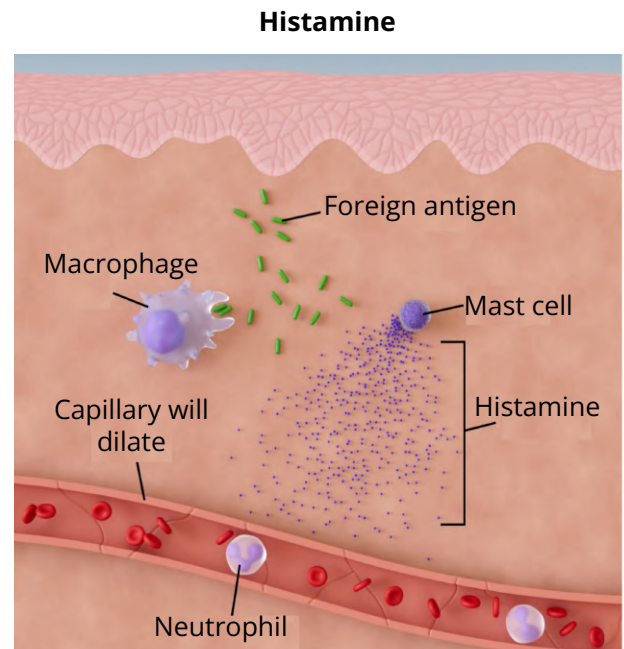
The **innate immune system** is the first line of defense and generates a nonspecific immune response (generalized). There are two parts:

- **External immunity:** Physical/physiological barriers preventing pathogen entry. These barriers include skin, hair, cilia, mucous membranes, chemical secretions, and symbiotic bacteria.
- **Internal immunity:** Internal defenses activated by the innate immune system to neutralize pathogens that have entered. The body's internal immunity is composed of the inflammatory response, complement proteins, and phagocytic and natural killer cells.

Innate Immunity: Inflammatory Responses

Mast cells are a type of leukocyte responsible for the first part of the inflammatory response, known as **rally signaling**:

1. Mast cells sit in the tissue in preparation for injury.
2. If there is an injury, mast cells will release **histamine**, which dilates blood vessels. This *increases* blood flow and makes vessels more permeable to let immune cells into the tissues.
3. Mast cells also release **heparin**, an anticoagulant that prevents blood clotting.



Mnemonic: Inflammatory response → SLIPR

- **Swelling:** Permeable capillaries result in fluids leaking into tissues.
- **Loss of function:** Body part with inflammation becomes less usable.
- **Increased heat:** Increased blood flow results in higher temperature.
- **Pain:** Throbbing pain caused by swelling, which puts continuous pressure on nerve endings.
- **Redness:** Increased blood flow causes redness of skin.

A **fever** can also occur due to the inflammatory response; this is controlled by the brain and causes a **systemic response** to kill pathogens with higher temperatures.

Innate Immunity: Immune Cells and Molecules

Diapedesis is the process by which cells move from the capillaries to the tissues in order to fight pathogens.

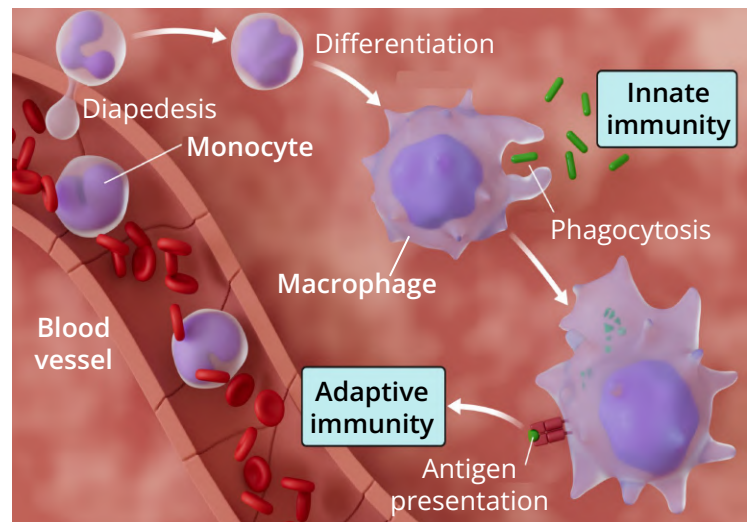
Granulocytes are cells in the innate immune system with specific granules in their cytoplasm. The four types of granulocytes include neutrophils, eosinophils, basophils, and mast cells.

Mnemonic: Five main types of **leukocytes** from highest to lowest in quantity →
Never **L**et **M**onkeys **E**at **B**ananas

1. **Neutrophils:** Phagocytes in innate immunity that make up over half of all leukocytes. Neutrophils are the most common type of leukocyte found in blood and are one of the first cells to be recruited to a site of inflammation.
2. **Lymphocytes:** B cells, T cells, and natural killer cells. B and T cells are part of adaptive immunity and must be activated. **Natural killer (NK) cells** are part of innate immunity and attack virally-infected cells and cancerous cells. NK cells use **perforin** (create holes) and **granzyme** (stimulate apoptosis) to lyse cells. B and T cells are the most common type of leukocyte found in lymph.
3. **Monocytes & macrophages:** Phagocytes in innate immunity. Monocytes are immature form found in blood vessels and macrophages are mature form after diapedesis. Can also act as **antigen-presenting cells** to activate adaptive immunity.
4. **Eosinophils:** Part of innate immunity. Have **granules** that can be released to kill pathogens, especially parasites.
5. **Basophils:** Least numerous leukocyte; contain granules with **histamine** (vasodilation) and **heparin** (an anticoagulant to prevent blood clotting). Very similar to mast cells, except basophils circulate as mature cells while mast cells circulate as immature cells.

Dendritic cells are also part of innate immunity and scan tissues using **pinocytosis** (cell "drinking" fluids & solutes) and **phagocytosis** (cell "eating" solid particles). They act as **antigen-presenting cells** like macrophages, migrating to the lymph nodes to activate adaptive immunity.

Macrophage Roles in Immunity



Macrophages and dendritic cells use **toll-like receptors (TLRs)** to recognize conserved parts of microbes. Binding to these receptors triggers phagocytosis and activates the innate immune system.

Interferons are secreted by **virally-** or **bacterially-infected cells** and bind to non-infected cells to prepare them for an attack.

Platelets are anucleate cell fragments that are involved in blood clotting and in activating the innate immune system. They help regulate macrophages and dendritic cells.

Innate Immunity: Complement System

The **complement system** is a group of approximately 30 proteins that aid immune cells in fighting pathogens. While small, these proteins turn each other on through the activation of a **complement cascade**, producing a large effect. Upon recognizing a pathogen, a chain reaction of protease activity is triggered for the proteins to activate each other.

Complement protein actions:

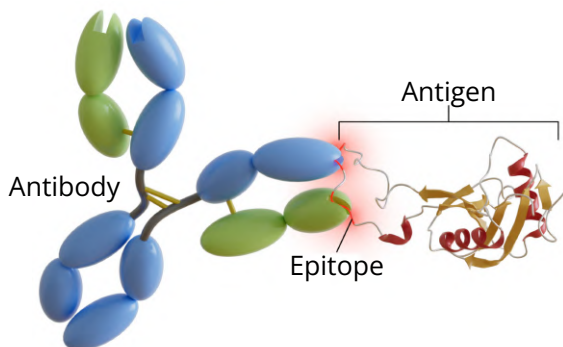
- Tags antigens for phagocytosis in a process called **opsonization**.
- Amplifies **inflammatory response** (e.g., binds to mast cells for increased histamine release).
- Forms a **membrane attack complex (MAC)**, which pokes holes in pathogens and lyses them.

Adaptive Immunity: Overview

The **adaptive immune system** is a specific immune response (targets specific antigens).

An **antigen** is an immunogenic foreign molecule and is the target of the immune response. The **epitope** is the important part of the antigen that is recognized by the immune cell.

Antibody Epitope Interaction



The immune system recognizes foreign cells and antigens using **major histocompatibility complex (MHC)** molecules. Once recognized, the immune system destroys the foreign or infected cell.

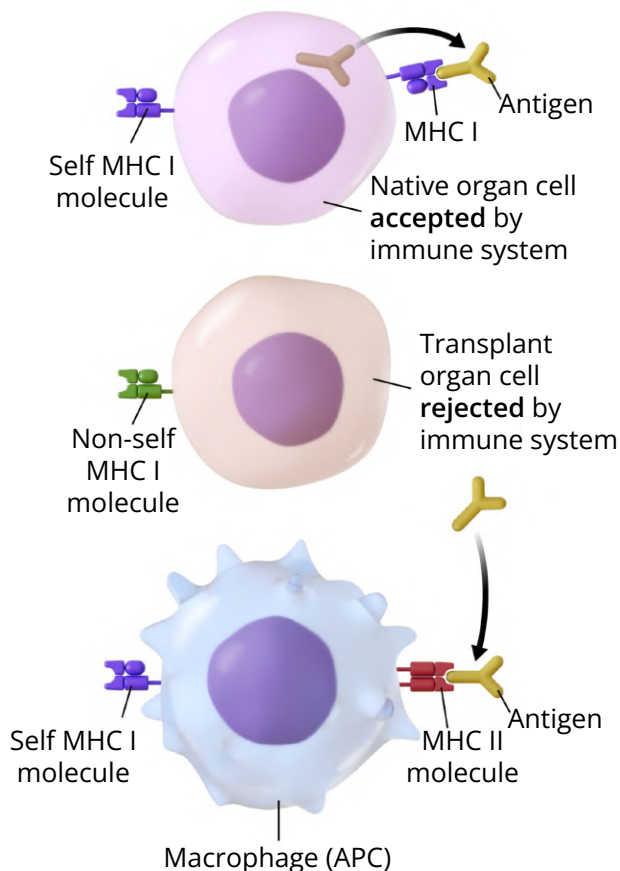
MHC class I is a surface molecule on all nucleated cells that present intracellular antigens. Each genetically different individual will have a unique MHC I molecule, thus the immune system can use this to distinguish between self and non-self cells.

MHC class II is a surface molecule on antigen-presenting cells (dendritic cells, macrophages) that present extracellular antigens.

Organ transplants that have different MHC I may lead to failure and rejection, so immunosuppressants are given to transplant patients. **Autoimmune diseases** occur when the immune system attacks self MHC I.

Identical twins have identical major histocompatibility complex molecules. This allows identical twins to donate organs to each other without the need for immunosuppression (the donated organ cells won't be marked as foreign).

Major Histocompatibility (MHC) Molecules



Adaptive Immunity: B cells

B cells control **antibody-mediated immunity (humoral immunity)** by managing the production and release of antibodies. They can also act as antigen-presenting cells.

B cell receptors (BCRs) are located on B cells and bind to antigen epitopes either free-floating or on APCs. Each B cell has a unique BCR.

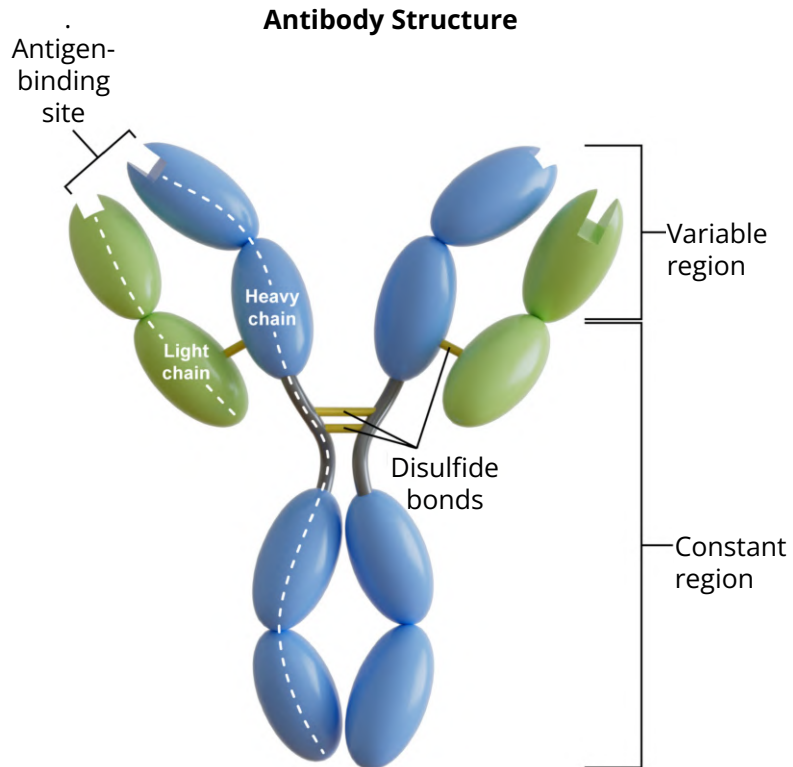
The **clonal selection model** describes the development of one type of BCR for every B cell. Through **clonal expansion**, these B cells divide into either **plasma cells** (antibody-secreting cells) or **memory B cells** (to be activated later in case of another attack).

Memory B cells survive for a long time and lay dormant until reactivated by the same antigen that triggered the original clonal expansion. They are the key to **vaccinations** because vaccines cause memory B cell production for later reactivation. After reactivation, memory B cells cause massive antibody production.

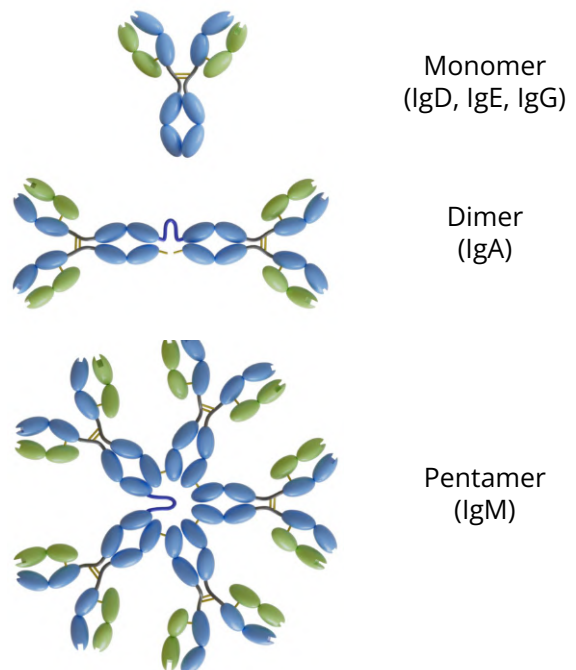
Antibodies (immunoglobulins) are structurally identical to BCRs but freely circulate in blood and lymph. They can tag antigens for phagocytosis, neutralize the antigen by coating it, or activate the complement system. Antibodies contain **light chains** and **heavy chains** that are linked together by **disulfide bonds**. In addition, the **variable region** recognizes different antigens while the **constant region** is the same for antibodies within the same class.

Five classes of antibodies:

1. **IgM**: Present in a pentameric form and is the largest antibody. The first antibody to be produced; activates the complement system.
2. **IgA**: Present in a dimeric form and found most abundantly in bodily secretions. Newborns receive **passive immunity** through breast milk containing IgA.
3. **IgE**: Monomer that is present on **basophils** and **mast cells** as antigen receptors. When bound to an **allergen**, it triggers **histamine** release and an allergic reaction. Think Ig sneEze.
4. **IgD**: Monomer with very little information known. Only small amounts are produced.
5. **IgG**: Monomer that is the most abundant antibody in circulation. Also the only antibody that can cross the **placenta** to give fetus passive immunity. Helps the complement system to cause **opsonization** (tags antigens and subsequent phagocytosis). Helps IgM activate the complement system.



Immunoglobulins



Adaptive Immunity: T cells

T cells control **cell-mediated immunity** by directly acting on cells instead of sending antibodies out.

T cell receptors (TCRs) are unique just like BCRs, binding only to one type of antigen per T cell. Thus, T cells also undergo **clonal selection** just like B cells.

T cells must bind to antigens presented on **APCs (antigen-presenting cells)** to be activated. There are two ways antigens may be presented to T cells:

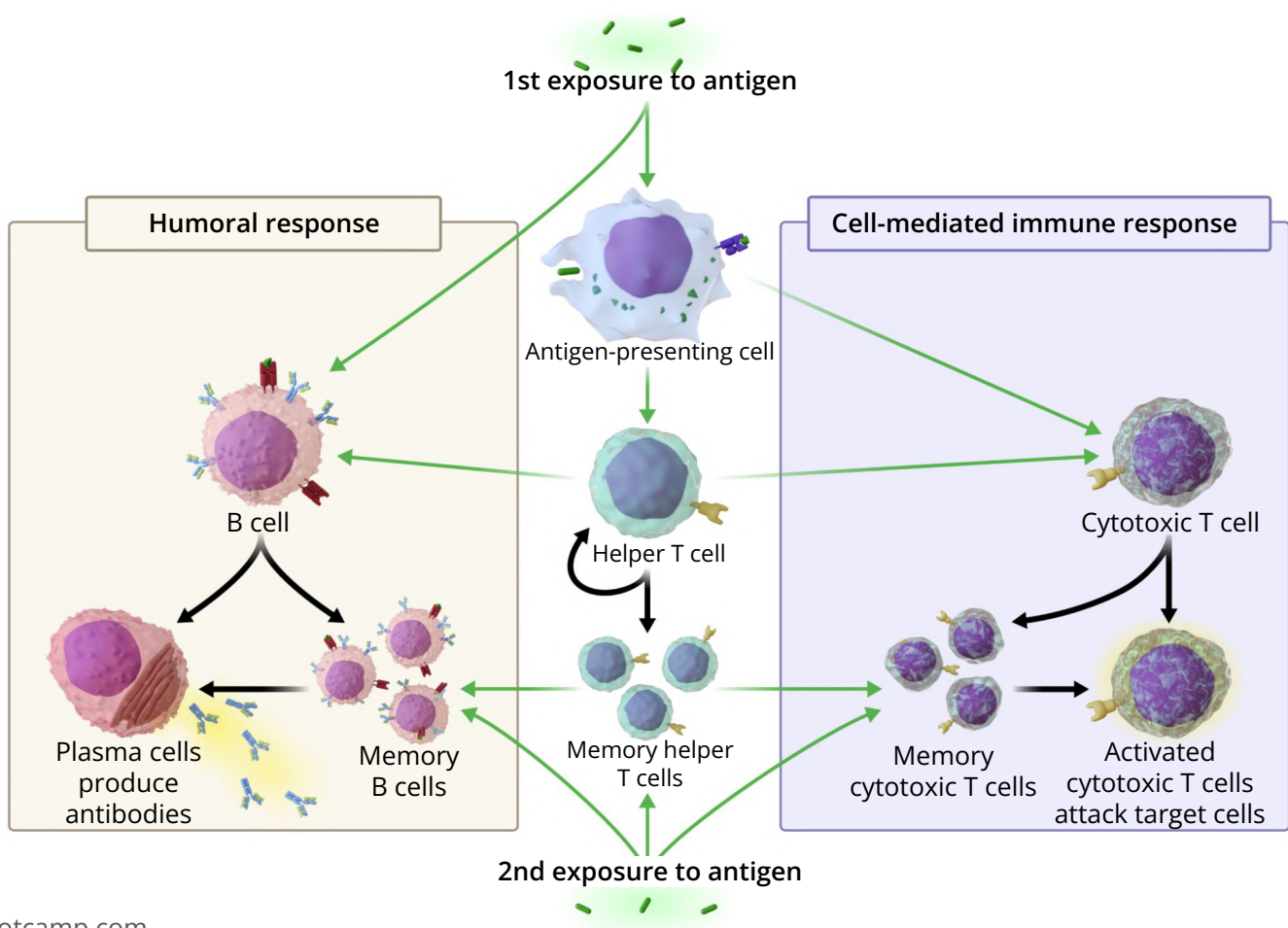
1. **MHC I presentation:** T cells differentiate into **cytotoxic T cells (CD8+)**, which directly kill infected cells through **perforin** (poke holes) and **granzymes** (cause apoptosis). However, T cells are different from **natural killer cells** because they are more specific and require antigen presentation.
2. **MHC II presentation:** T cells differentiate into **helper T cells (CD4+)**, which release **interleukins** to boost both innate immunity and adaptive immunity. These interleukins help attract innate immune cells and increase proliferation of other T and B cells.

Active vs. Passive Immunity

Passive immunity refers to the immunity one organism gains from receiving the **antibodies** from another organism that already has that immunity. For example, a fetus gains passive immunity through the **placenta (IgG)** while a newborn gains passive immunity through **breast milk (IgA)**. The fetus and newborn are referred to as **immuno-naïve** because they do not yet have their own **active immunity**.

Active immunity refers to the immunity an organism gains from being infected once already by a pathogen. A **vaccination** introduces the antigen or pathogen in a deactivated state to stimulate active immunity, which is referred to as **artificial immunity** in this case and induces memory B and T cell formation.

Adaptive Immune System



Classification of Common Diseases

Disease type	Common examples
Bacterial	• Tuberculosis • Gonorrhea • Leprosy • Syphilis • <i>E. coli</i> • Streptococcus, bacillus, staphylococcus, mycoplasma, & spirochete infections
Viral	• Influenza • Hepatitis • Herpes • Chickenpox • Human papillomavirus (HPV) • Human immunodeficiency virus (HIV) • Measles • Polio
Genetic	• Down syndrome • Cystic fibrosis • Huntington's • Sickle cell • Tay-Sachs
Parasitic	• Malaria • Tapeworms
Fungal	• Yeast infections • Athlete's foot

Ch. 10.4: Nervous System

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- Synaptic Transmission
- Neurotransmitters
- Central vs. Peripheral Nervous System
- Central Nervous System
- Peripheral Nervous System: Somatic vs. Autonomic Nervous System
- Autonomic Nervous System: Sympathetic vs. Parasympathetic Nervous System
- Special Senses

The Neuron

The **neuron** is the most basic unit of the nervous system. It has three parts: the **soma** (cell body), **dendrites** (extensions that receive signals), and the **axon** (sends signals out).

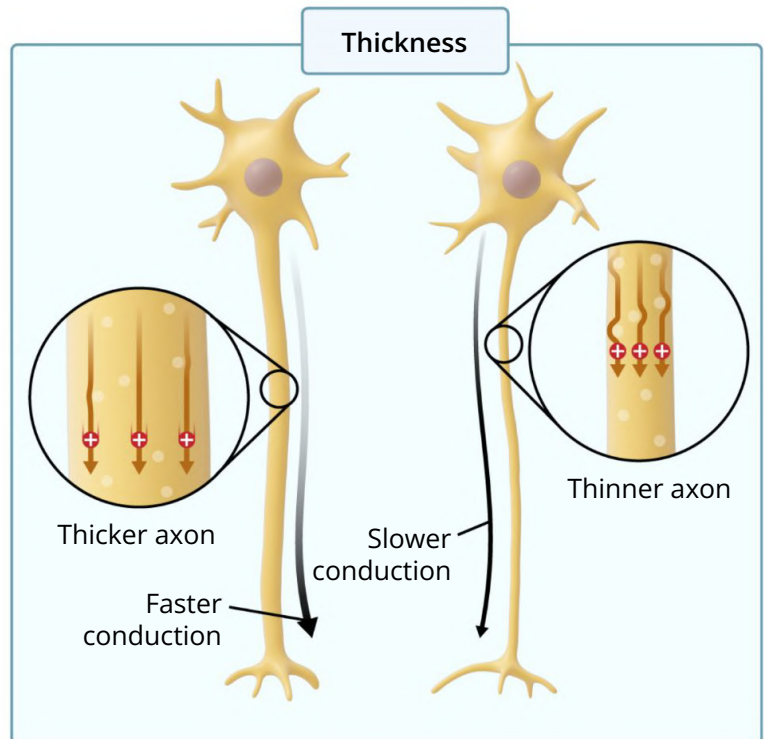
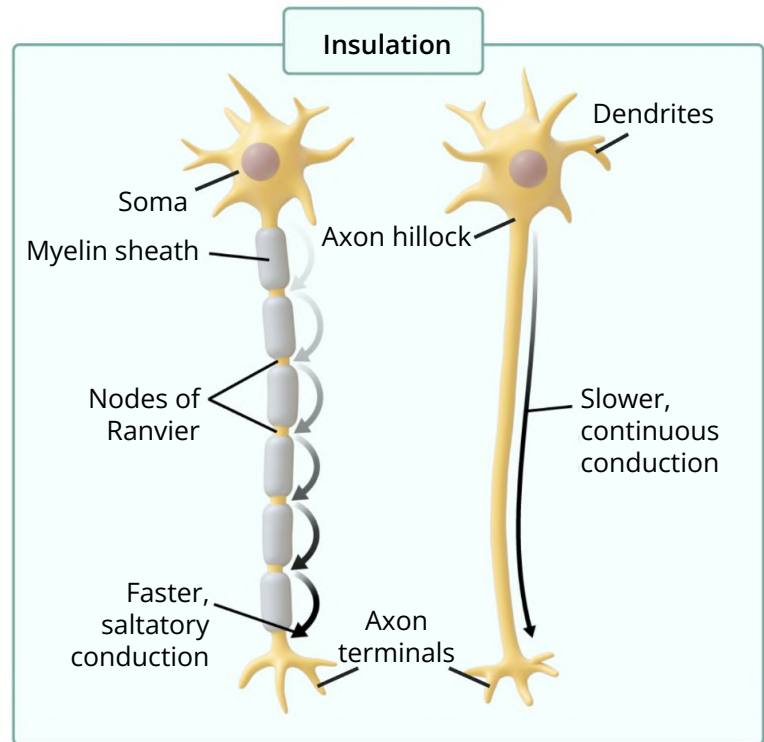
The **axon**:

- **Axon hillock:** Area where the axon is connected to the cell body. Responsible for the **summation** of graded potentials.
- **Myelin sheath:** Fatty insulation of the axon that speeds up action potential propagation by limiting ion exchange along the axon. The myelin sheath is formed by **oligodendrocytes** (in the central nervous system) and **Schwann cells** (in the peripheral nervous system). Thicker myelinated neurons fire signals faster.
- **Nodes of Ranvier:** Gaps between myelin sheaths where ion exchange occurs. Propagation of the action potential occurs here, jumping from gap to gap (node to node) in a process called **saltatory conduction**.

Mnemonic: COPS

- **CNS:** Oligodendrocytes
- **PNS:** Schwann cells

Neuron: Conduction Speed Factors



Action Potentials

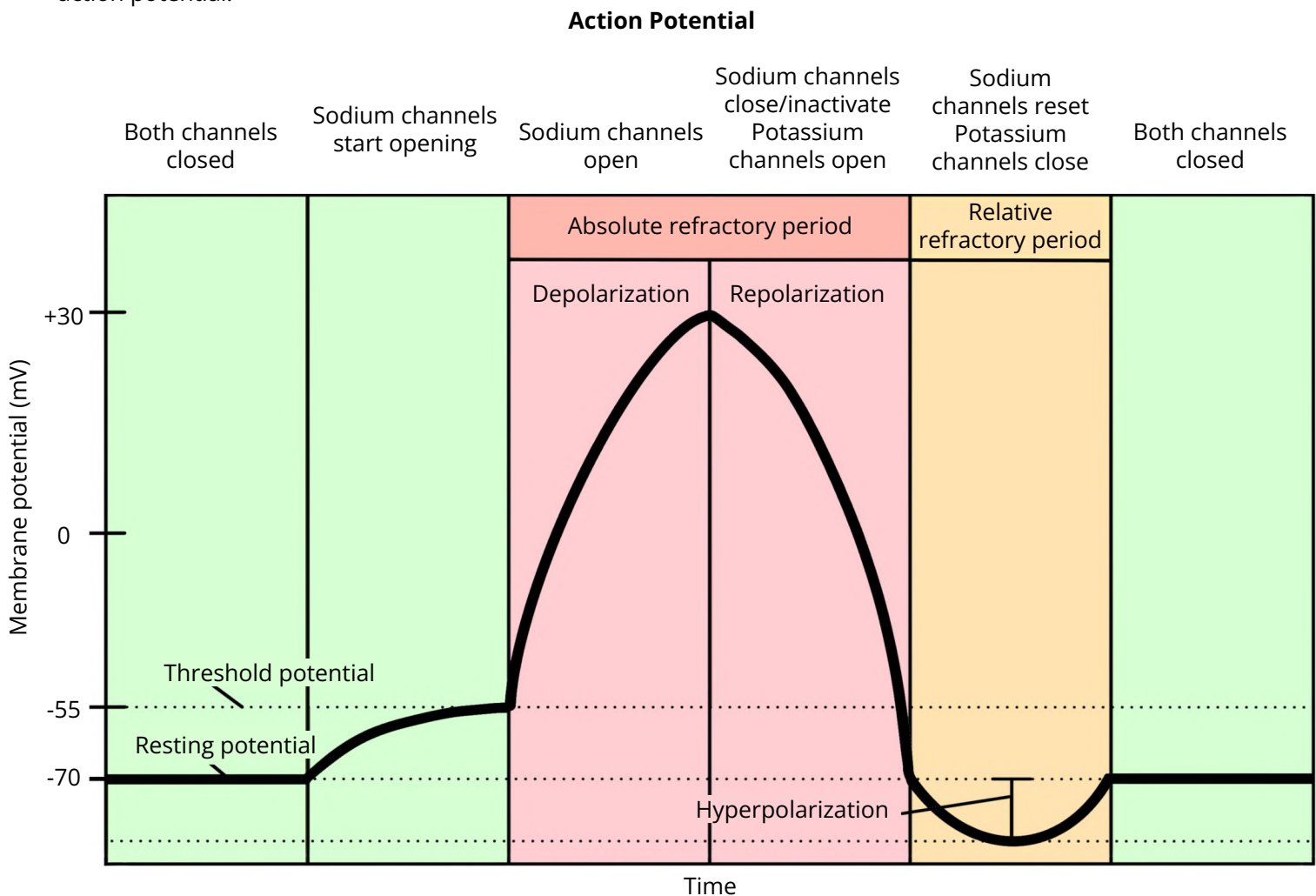
Steps of an action potential:

1. At **resting potential**, the membrane potential of the neuron is around -70mV and is maintained by **Na⁺/K⁺ ATPases**, which pump three Na⁺ ions out and two K⁺ ions in, powered by hydrolysis of one ATP. **K⁺ leak channels** are also present and help maintain resting potential through passive K⁺ leakage.
2. When a stimulus causes **threshold potential** to be reached (around -55mV in neurons), **voltage-gated Na⁺ channels** open up, letting Na⁺ in, resulting in **depolarization** of the neuron. K⁺ channels are closed.
3. Next is **repolarization** of the neuron due to the opening of **voltage-gated K⁺ channels**, letting K⁺ out, and the closing of Na channels. This causes the membrane potential to become *less* positive since positive ions are leaving. This is the **absolute refractory period**: no stimulus can cause an action potential.

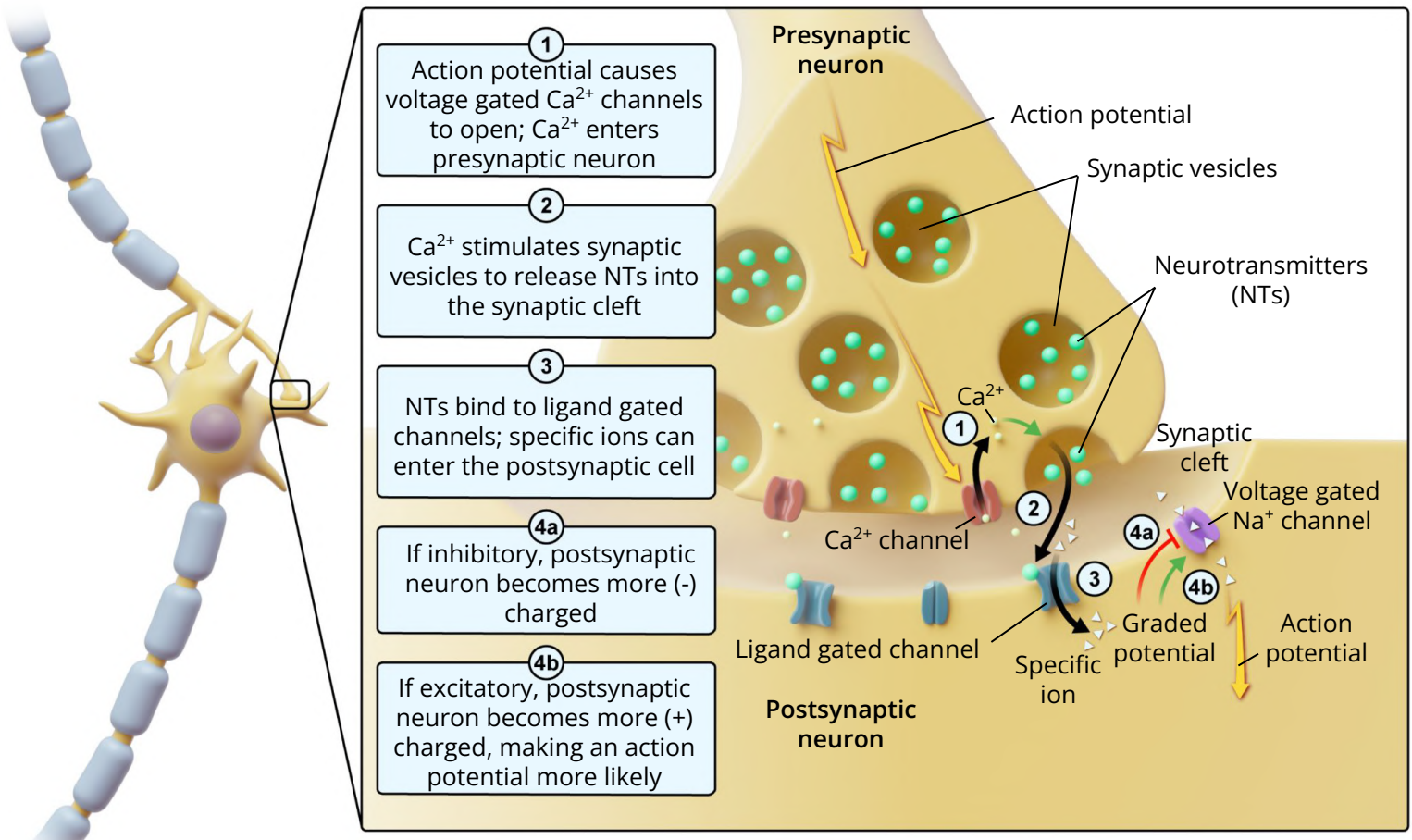
4. When the membrane potential becomes even more **negative** than the normal resting potential, this is known as **hyperpolarization**. This results in a **relative refractory period** being established, during which another action potential can be fired, but it requires a much *stronger* stimulus.
5. The membrane potential returns to **normal resting potential** through the pumping of **Na⁺/K⁺ ATPases** and **K⁺ leak channels**.

The **absolute refractory period** refers to the period after the initiation of the action potential during which another action potential *cannot* be fired no matter how powerful the stimulus is. It is due to the inactivation of voltage-gated Na⁺ channels after they open.

The **relative refractory period** refers to the period after the action potential fires during which a stronger than normal stimulus could cause another action potential to be fired.



Synaptic Transmission



Synaptic Transmission

The **synapse** is the space between two neurons. The **presynaptic** neuron sends the signal and releases neurotransmitters into the synapse, while the **postsynaptic** neuron receives the signal by interacting with the released neurotransmitters.

Steps of synaptic transmission:

1. Action potential reaches the end of the presynaptic axon, causing **voltage-gated calcium channels** to open and letting Ca^{2+} ions into the neuron.
2. The **Ca^{2+} ions** cause **synaptic vesicles** to fuse and undergo exocytosis, releasing neurotransmitters into the synapse.
3. The neurotransmitters (described in the table on the next page) bind to **ligand-gated ion channels** on the postsynaptic neuron, producing **graded potentials** (depolarizations or hyperpolarizations of the membrane).

4. These graded potentials **summate** at the axon hillock and an action potential will fire if the summation of graded potentials is higher than the threshold potential of neurons.

An **excitatory postsynaptic potential (EPSP)** is a graded potential that **depolarizes** the membrane.

An **inhibitory postsynaptic potential (IPSP)** is a graded potential that **hyperpolarizes** the membrane.

Neurotransmitters and Their Functions

	Neurotransmitter	Function
Excitatory neurotransmitter (ENT)	Glutamate	• Main ENT of the central nervous system
	Dopamine	• Involved in reward motivated behavior
	Epinephrine	• Postsynaptic NT of the sympathetic nervous system • Constricts blood vessels and dilates bronchioles • Increases blood flow to skeletal muscles • Decreases blood flow to smooth muscles
	Norepinephrine	• Postsynaptic NT of the sympathetic nervous system • Aids epinephrine in the "fight or flight" response
	Acetylcholine	• Most common NT in the muscular system • Released from presynaptic motor neurons to signal muscle fibers • Signals muscle fibers by binding to ligand-gated sodium channels
Inhibitory neurotransmitter (INT)	Gamma-aminobutyric acid (GABA)	• Main INT of the central nervous system
	Glycine	• INT of the central nervous system
	Serotonin	• INT of the brain

Central vs. Peripheral Nervous System

The **central nervous system (CNS)** is composed of the brain and spinal cord.

The **peripheral nervous system (PNS)** is composed of nerves branching off the CNS.

Central Nervous System

During **embryonic development**, the brain can be divided into three areas:

The...	Develops into..	Which gives rise to...
Forebrain	Telencephalon	Cerebrum
	Diencephalon	Thalamus, hypothalamus, and pineal gland
Midbrain	Mesencephalon	Midbrain
Hindbrain	Metencephalon	Pons, cerebellum
	Myelencephalon	Medulla oblongata

The developed brain cortex is divided into four main lobes:

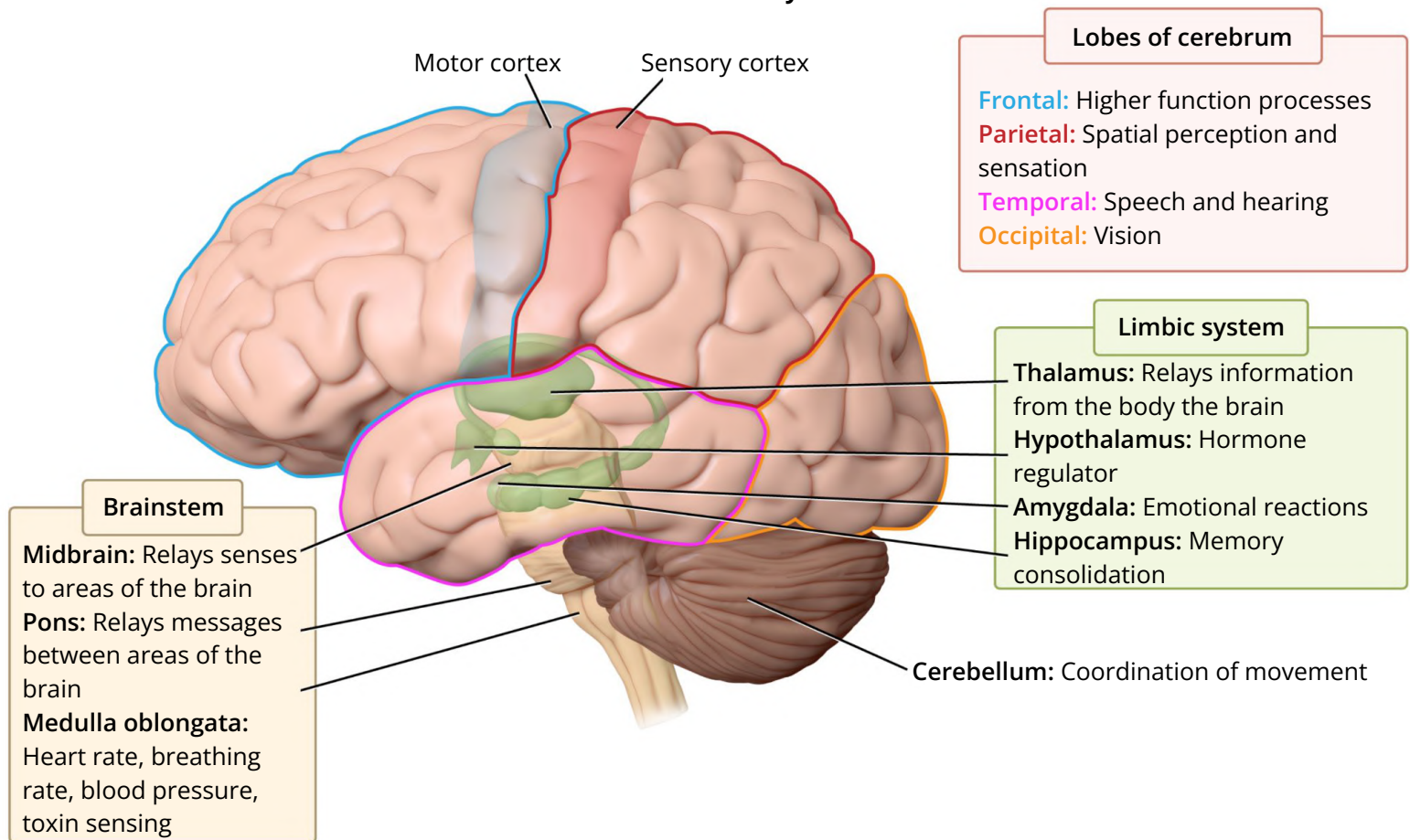
- **Frontal lobe:** Known for higher function processes such as decision making, problem solving, attention and concentration.
- **Temporal lobe:** Known for speech and hearing.
- **Occipital lobe:** Known for visual perception.
- **Parietal lobe:** Known for spatial perception and sensation.

The **cerebellum** is located underneath the occipital lobe and is responsible for the coordination of movement.

The **brainstem** is composed of the following components:

- **Midbrain:** Relays senses to other parts of brain.
- **Pons:** Relays messages between the forebrain, cerebellum, and medulla.
- **Medulla oblongata:** Heart and breathing rate, blood pressure, toxin sensing. Connects the cerebrum/cerebellum to the spinal cord.
- **Reticular formations:** Neurons throughout the brainstem that are involved in cortical arousal, and consciousness.

Brain Anatomy



The **limbic system** is composed of the **thalamus**, **hypothalamus**, **hippocampus**, and **amygdala**. It is responsible for emotion, memory, learning, and motivation.

- **Thalamus:** The “relay center” of the brain and is located between the cerebrum and the midbrain. Relays sensory and motor signals from the body to the brain.
- **Hypothalamus:** Regulates hormone secretion in the body.
- **Hippocampus:** Responsible for memory consolidation.
- **Amygdala:** Responsible for the emotional reaction to certain scents.

Finally, the **spinal cord** is nervous tissue in part of the central nervous system; it connects the brain to the body.

Somatic vs. Autonomic Nervous System

The **peripheral nervous system** is divided into the **somatic nervous system** (voluntary motor action and sensory input) and the **autonomic nervous system** (involuntary).

Sensory (afferent) neurons in the peripheral nervous system receive stimuli and send signals into the brain through the spinal cord. **Motor (efferent)** neurons send signals from the brain to the muscles.

Autonomic Nervous System: Sympathetic vs. Parasympathetic Nervous System

The **autonomic nervous system** can be further divided into the **sympathetic nervous system** (fight or flight) and the **parasympathetic nervous system** (rest and digest).

Sympathetic nervous system effects:

- Release of sugar into blood for energy.
- Increase in heart rate for oxygen delivery to the brain and muscles.
- Vasodilation of skeletal blood vessels, and vasoconstriction of the digestive system.
- Dilation of bronchi and bronchioles to allow more oxygen into lungs.
- Dilation of the pupil to give the brain more visual information.

Parasympathetic nervous system effects (through vagus nerve):

- Relaxation of muscles.
- Decrease in heart rate.
- Maintenance of homeostasis.
- Increase in gastrointestinal activity.

A **ganglion** is defined as a cluster of nerve bodies in the peripheral nervous system. The autonomic nervous system neurons are either **preganglionic** or **postganglionic**. The preganglionic neuron comes from the central nervous system and synapses with the postganglionic neuron at the ganglion.

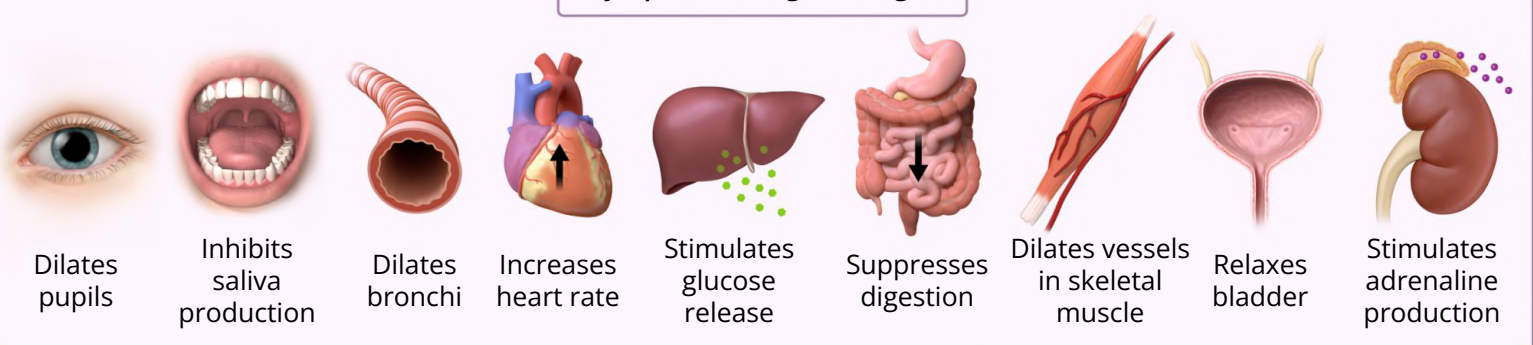
Sympathetic nervous system → uses **acetylcholine (ACh)** for preganglionic nerves and **norepinephrine (NE)/epinephrine (E)** for postganglionic nerves. The sympathetic nervous system also can stimulate the adrenal medulla to release **NE/E** into the blood.

Parasympathetic nervous system → uses **acetylcholine (ACh)** for both preganglionic and postganglionic nerves.

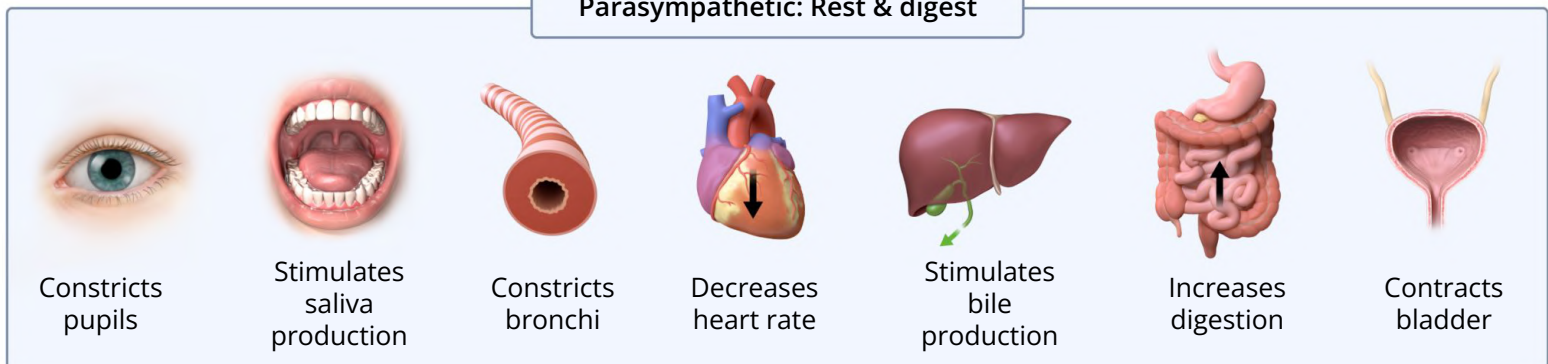
Acetylcholinesterase is an enzyme that is responsible for the breakdown of acetylcholine via hydrolysis.

Autonomic Nervous System

Sympathetic: Fight or flight



Parasympathetic: Rest & digest

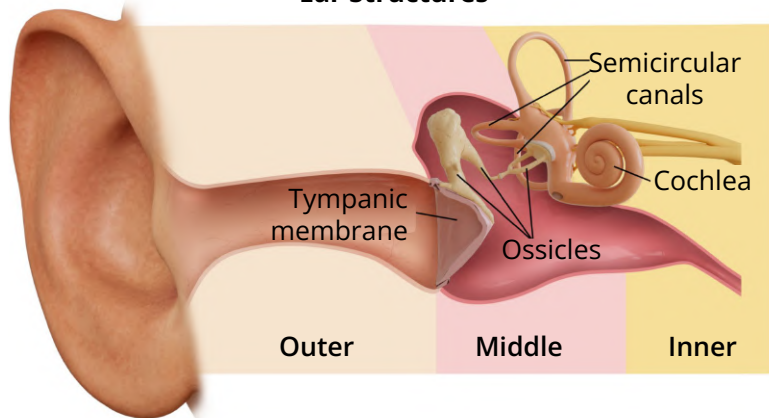


Special Senses

Ear:

- The **outer ear** takes in sound waves, and the **tympanic membrane** transfers the sound from outer ear to middle ear.
- The **middle ear** is composed of three bony **ossicles**. The ossicles transfer vibrations through the middle ear and amplify the signal. The vibrations get transmitted from the middle to the inner ear.
- The **cochlea** uses fluid and hairs to convert the mechanical signal into a neuronal signal, known as **transduction**.
- The **semicircular canal** has fluid and hairs just like the cochlea but gives information about the person's **movement**. It is also the reason people get dizzy.

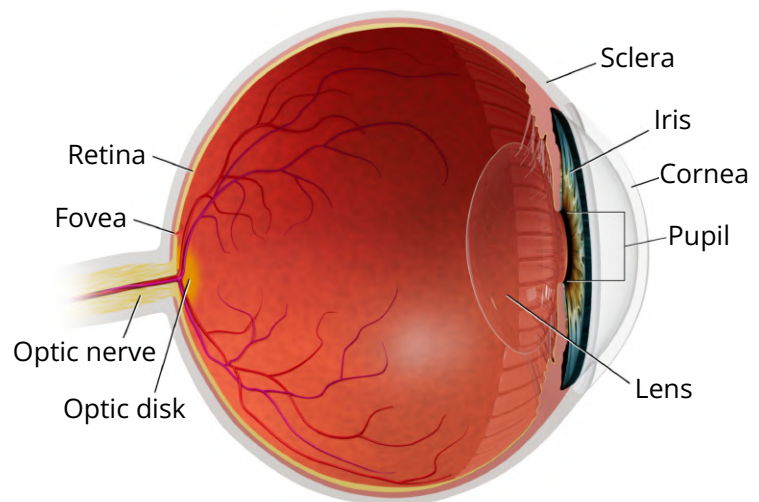
Ear Structures



Eye:

- **Cornea:** Transparent; focuses light and protects the eye.
- **Iris:** Controls the size of the pupil.
- **Pupil:** Controls how much light enters the eye.
- **Lens:** Focuses images on retina.
- **Retina:** Back of the eye that has photoreceptors (rods + cones).
 - **Rods** function at low levels of light and are responsible for low-light perception.
 - **Cones** function at high levels of light and are responsible for color perception.
 - **Fovea:** Highest concentration of photoreceptors in the retina. Responsible for high acuity vision.
- **Optic nerve:** Bundle of axons that transmits visual information to the brain.
- **Optic disk:** The blind spot of the eye, where the optic nerve passes through to reach the brain.
- **Sclera:** Protective layer of tissue; the “white part” of the eye.

Anatomy of the Eye



Ch. 10.5: Muscular System

Table of Contents

- Types of Muscle
- Skeletal Muscle Anatomy
- Sliding Filament Theory of Muscle Contraction
- The Sarcomere
- Motor Units

Types of Muscle

There are three types of muscles: **smooth muscle**, **cardiac muscle**, and **skeletal muscle**.

Smooth muscle	Cardiac muscle	Skeletal muscle
Present in organs, airways, blood vessels	Present in heart	Present around bone
Involuntary	Involuntary	Voluntary
1 nucleus per cell	1 nucleus per cell	Many nuclei per cell
Not striated	Striated	Striated

Striated means the muscle contains **sarcomeres**. Smooth muscle therefore lacks sarcomeres, whereas cardiac and skeletal muscle contain them.

Cardiac muscle contains **intercalated discs**, made of **desmosomes** (hold cells together) and **gap junctions** (channels for rapid ion exchange) that enable coordinated contractions of the heart.

Skeletal Muscle Anatomy

Skeletal muscle is composed of many bundles within bundles.

Muscle → **muscle fascicles** → **muscle fibers** (muscle cells) → **myofibrils** (composed of contractile protein filaments)

The **sarcolemma** is the muscle fiber's cellular membrane, and it protects each muscle fiber.

Sliding Filament Theory of Muscle Contraction

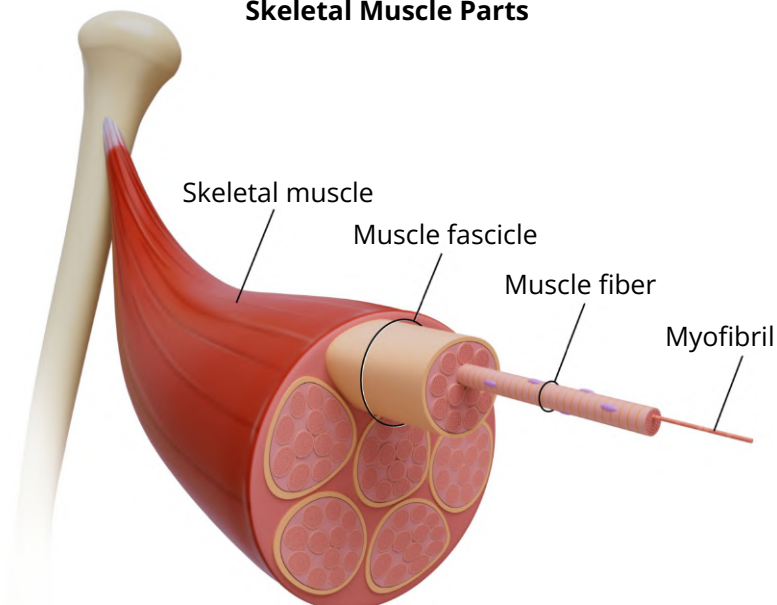
All muscles always contract (pull) across a joint to move body parts, they never push.

Myofibrils are made of repeating units of **sarcomeres**, the functional unit of skeletal muscle that shortens to cause muscle contraction. Sarcomeres are made of thin **actin** filaments and thick **myosin** filaments. These filaments slide past each other to shorten sarcomeres through the **sliding filament model** of muscle contraction.

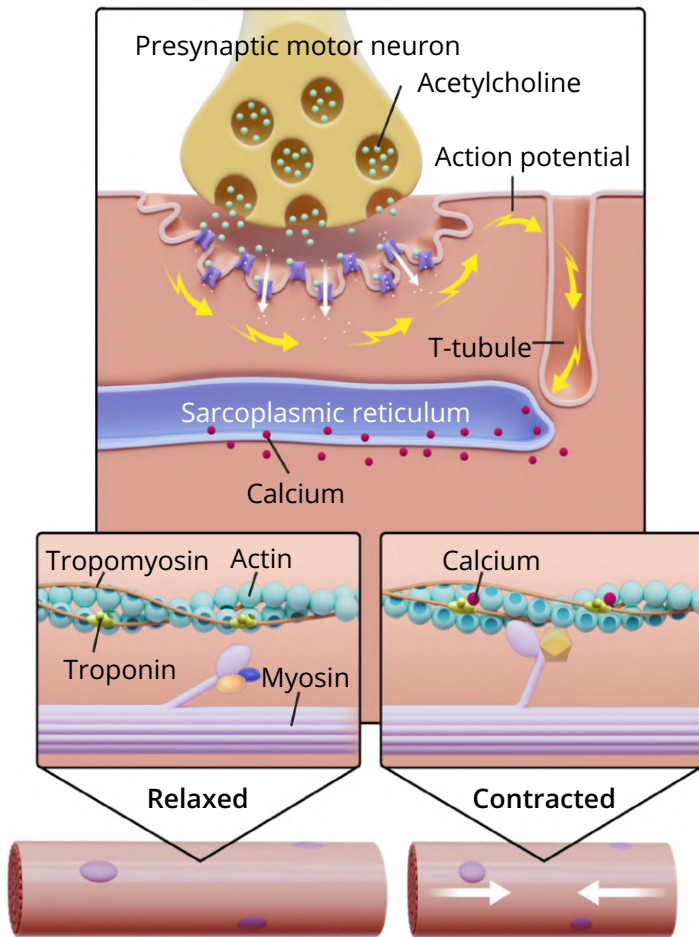
Stimulation of **skeletal muscle contraction**:

1. Action potential propagation reaches the motor neuron axon terminal, causing **voltage-gated calcium channels** to open, which allows Ca^{2+} ions to enter the neuron.
2. The influx of Ca^{2+} ions causes synaptic vesicles containing **acetylcholine** to fuse with the presynaptic membrane, releasing acetylcholine into the **neuromuscular junction**.
3. Acetylcholine binds to and opens **ligand-gated sodium channels** on the postsynaptic muscle fiber (**motor end plate**), causing sodium to enter the postsynaptic cell. This creates **graded potentials** on the muscle fibers.
4. The graded potentials open **voltage-gated sodium channels**, further depolarizing the postsynaptic cell membrane.
5. If the membrane reaches **threshold potential**, an action potential is triggered that spreads along the muscle fiber membrane, initiating a process that leads to **muscle contraction**.

Skeletal Muscle Parts



Stimulation of a Skeletal Muscle Contraction



The **sarcolemma** contains **T-tubules**, invaginations that propagate action potentials throughout the muscle, ensuring coordinated contraction.

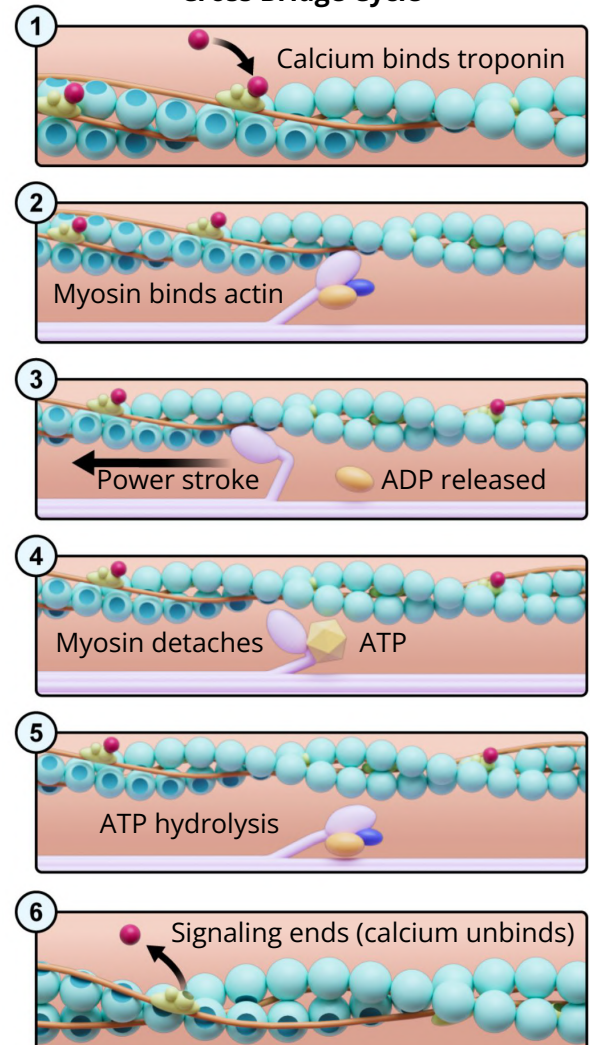
The **sarcoplasmic reticulum** is the endoplasmic reticulum of muscle fibers that releases stored calcium ions into the muscle cell through **voltage-gated calcium channels** when triggered by the depolarization of the muscle cell.

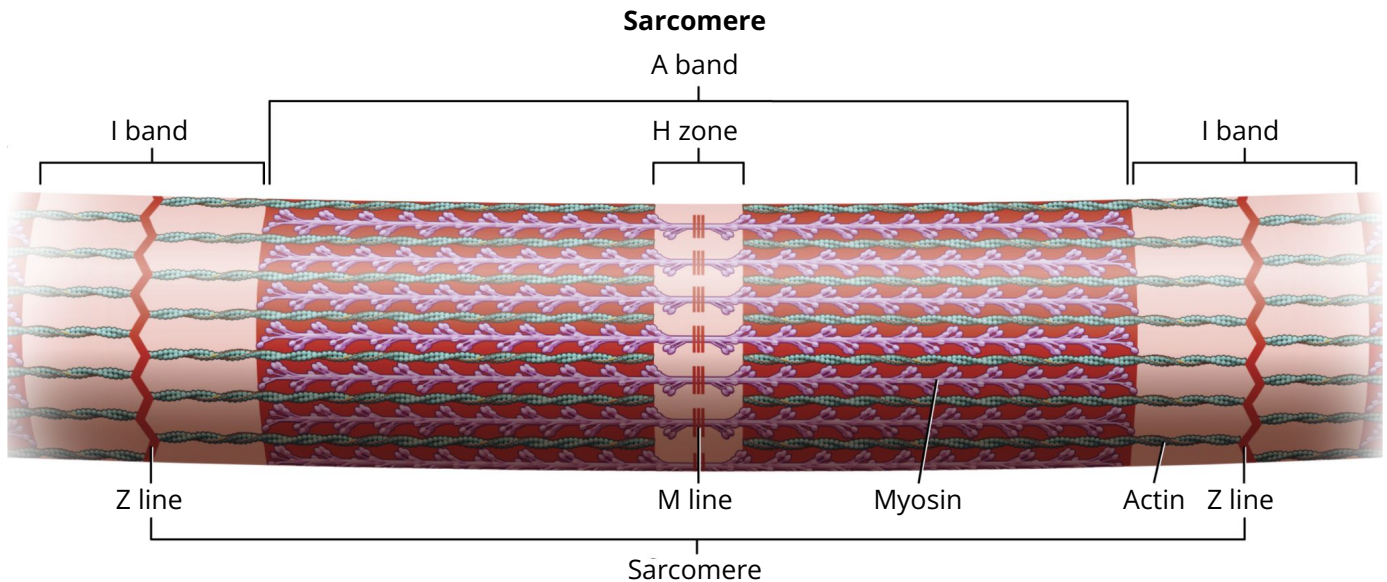
The **calcium ions** then bind to **troponin**, which removes **tropomyosin** from the myosin-binding sites on actin, allowing myosin to interact with actin and cause sarcomere shortening, via sliding filaments.

Cross bridge cycling:

1. **Initiation:** Calcium ions expose the myosin-binding sites on actin.
2. A cocked back, high energy myosin head (containing ADP) binds to actin, forming a **cross bridge**.
3. The myosin head contracts (known as a **power stroke**) and releases ADP, bringing the myosin head back to a **low energy state**. The power stroke causes the sarcomere to shorten.
4. A new ATP molecule binds to myosin, causing detachment of the myosin head from the actin filament.
5. The myosin head hydrolyzes the ATP into ADP and re-enters a cocked back, **high energy state**. If calcium is still present, the cross-bridge cycle continues.
6. **Termination:** Motor neuron signaling ends. The sarcoplasmic reticulum pumps calcium back into itself, and troponin brings tropomyosin back to cover myosin-binding sites on actin.

Cross Bridge Cycle





The Sarcomere

The **Z lines** are the ends of the sarcomeres. **Thin actin filaments** branch from the Z lines towards the middle of the sarcomere.

The **M lines** are the midpoints of the sarcomeres. **Thick myosin filaments** branch from the M lines towards the ends of the sarcomere.

The **I band** is the area in the sarcomere where only actin filaments are present.

Mnemonic: "I" is a *thin* letter, representing thin actin filaments.

The **A band** is the entire length of the myosin filament, including where myosin overlaps actin.

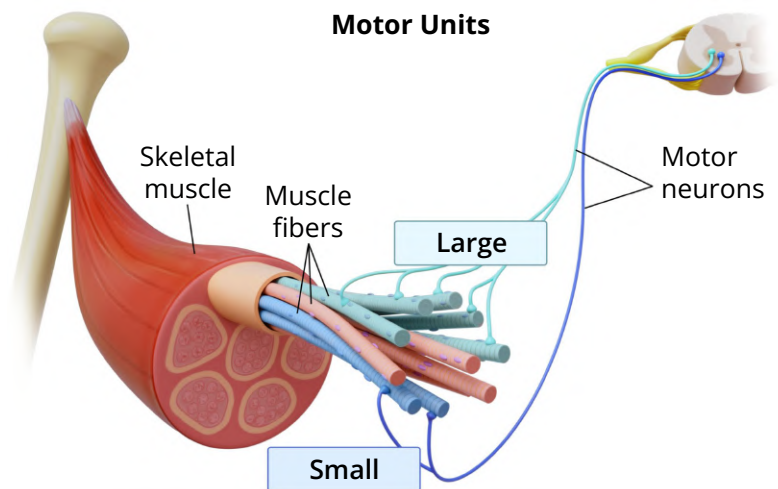
The **H zone** is the area in the sarcomere where only myosin is present.

Mnemonic: "H" is a *thick* letter, representing thick myosin filaments.

Motor Units

Motor units make up muscles; a motor unit refers to all the muscle fibers innervated by a single neuron.

Small motor units include only a few muscle fibers and are used in precision movement. **Large motor units** include many muscle fibers that are innervated by a single neuron and are used in powerful movements.



Ch. 10.6: Skeletal System

Table of Contents

- Types of Skeletons
- Long Bones and Their Structure
- Bone Remodeling
- Embryonic Ossification
- Connective Tissues and Joints

Types of Skeletons

An **exoskeleton** is an external skeleton. Many **invertebrates** and all arthropods possess exoskeletons.

Vertebrates contain an **endoskeleton** on the inside. An endoskeleton can be divided into the **axial skeleton** (skull, spinal column, and ribcage) and the **appendicular skeleton** (appendages).

Long Bones and Their Structure

Long bones, cylindrical bones in the appendicular skeleton that are longer than wide, are important for support, movement, and hematopoiesis.

Made of **cortical bone** (compact) and pockets of **cancellous bone** (spongy). Important features include:

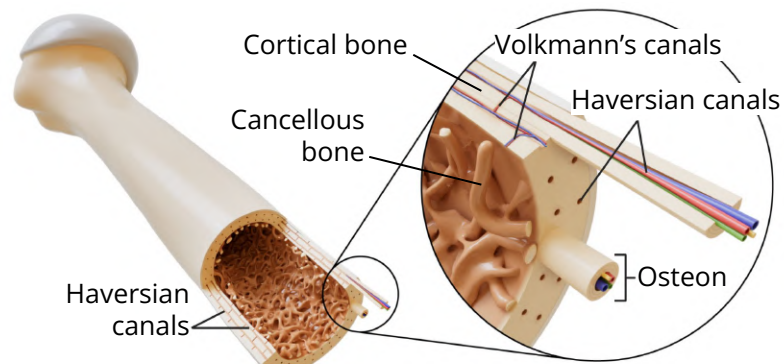
- **Epiphysis:** End of a long bone that forms joints with other bones and contains **red bone marrow** for **hematopoiesis** (blood cell synthesis).
- **Diaphysis:** Long hollow shaft in center of bone.
- **Red bone marrow:** Contains stem cells that generate red and white blood cells.
- **Yellow bone marrow:** Contains stem cells that generate fat, bone, cartilage, and muscle. Also stores fat.
- **Metaphysis:** Similar to epiphyses. Found between the diaphysis and epiphyseal plates.
- **Epiphyseal plate:** "Growth plate" located between epiphysis and metaphysis. Made out of **cartilage** and works to lengthen the diaphysis through growth and **ossification**.

Cancellous bone is the spongy inner layer of bone that soaks up red bone marrow via a web of **trabeculae** (connective tissue that supports cancellous bone).

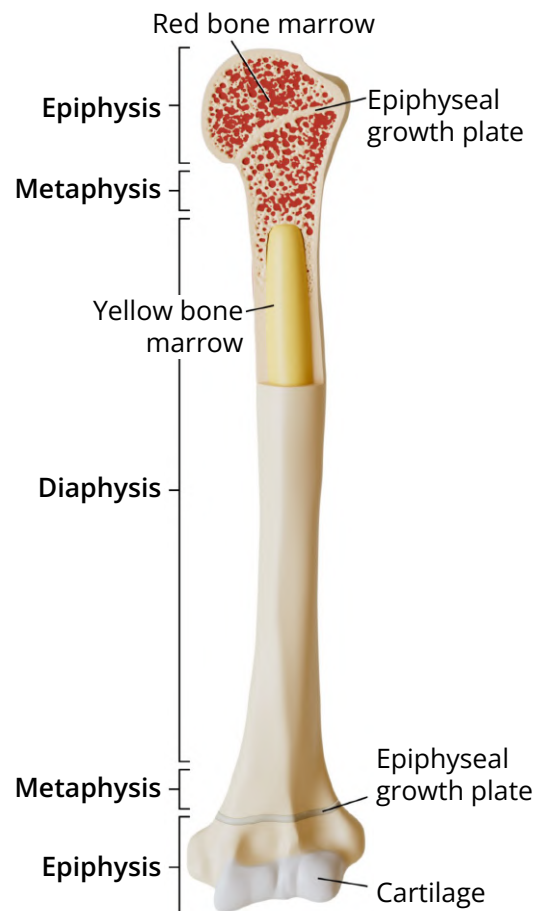
Cortical bone is the dense outer layer of bone that supports the weight of our bodies. It is composed of many microstructures:

- **Osteons:** Cortical bone functional unit, composed of tiny multi-layered cylinders. Also known as **haversian systems** because they contain a haversian canal in their center.
- **Haversian canals:** Tubes that contain blood vessels for nutrient supply.
- **Lamellae:** Layers of the osteon.
- **Volkman's canals:** Connect Haversian systems to the periosteum.

Cortical vs. Cancellous Bone



Long Bone Anatomy



Bone Remodeling

Bone remodeling is the process of alternating between **ossification** (bone formation) and **resorption** (bone loss).

Types of cells involved in bone remodeling:

- **Osteoblasts:** Build bone by secreting proteins and utilizing blood calcium. They mature into osteocytes after getting trapped inside the bone matrix they create.
- **Osteocytes:** Live in osteons to maintain bone.
- **Osteoclasts:** Eat and resorb bone, releasing calcium and phosphate back into the blood. Derived from **monocytes**.

Mechanisms involved in bone remodeling:

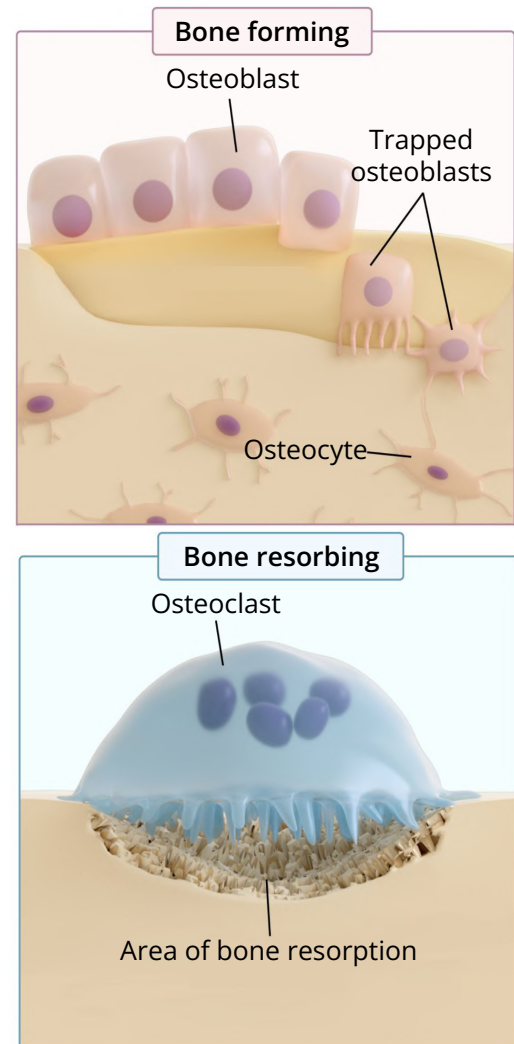
- **Parathyroid hormone (PTH):**
 - Stimulates osteoclasts to *increase* bone resorption, which releases calcium into the bloodstream.
 - Increases reabsorption of calcium in the kidneys, leading to *decreased* excretion in the urine.
- **Calcitonin:**
 - Inhibits osteoclast activity, reducing bone resorption and leading to *decreased* blood calcium levels.
 - Decreases reabsorption of calcium in the kidneys, leading to *increased* excretion in the urine

Mnemonic: Calcitonin “tones down calcium.”

Osteoid is the organic component of bone containing many proteins such as collagen (gives bone tensile strength).

Hydroxyapatite is the inorganic mineral component of bone that gives the bone density and strength.

Bone Remodeling Cells



Embryonic Ossification

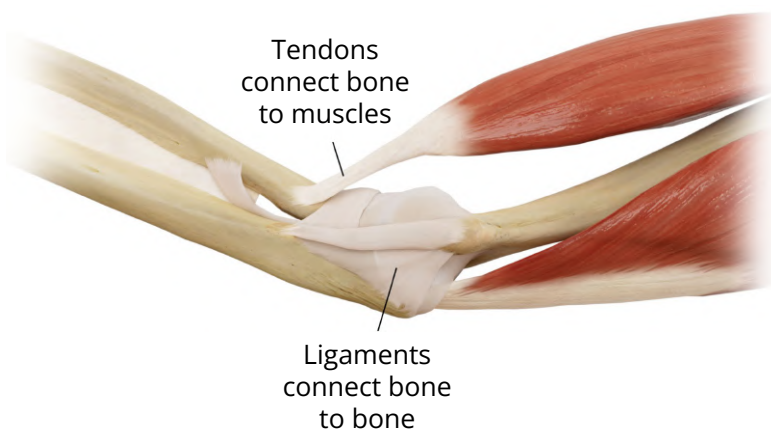
Embryonic ossification is the process of bone formation during fetal development. Two types of **embryonic ossification**:

1. **Intramembranous ossification:** Bone forms directly in sheets of embryonic connective tissue. Osteoblasts deposit osteoid, which calcifies and forms cortical bone. Occurs primarily for flat bones (e.g., skull, ribs).
2. **Endochondral ossification:** An indirect process of bone formation. A cartilage model is initially formed. Cartilage is gradually replaced by bone as osteoid is deposited and calcifies. Occurs for most bones in the body, including long bones (e.g., femur).

Connective Tissue and Joints

- Connective tissue containing a high degree of collagenous fibers is called **fibrous connective tissue**. It is secreted and maintained by cells called **fibroblasts**, with several important functions in the body.
 - **Tendons:** Attach muscles to bones.
 - **Ligaments:** Attach bones to other bones.
 - **Periosteum:** Layer that surrounds cortical (dense) bone. Provides nutrients and nervous innervation to bone.
 - **Endosteum:** Layer between cortical and spongy bone.

Tendon vs. Ligament



- **Cartilage** is a type of fibrous connective tissue that is avascular and is *not* innervated by nerves. This is distinguished from bone, which is heavily innervated by nerves and vasculature.
- **Joints** are the meeting points between two or more bones. Muscles run across joints and pull the bones on either side, producing body movement. Joints are vascularized, innervated by nerves, and contain connective tissues.

Ch. 10.7: Endocrine System

Table of Contents

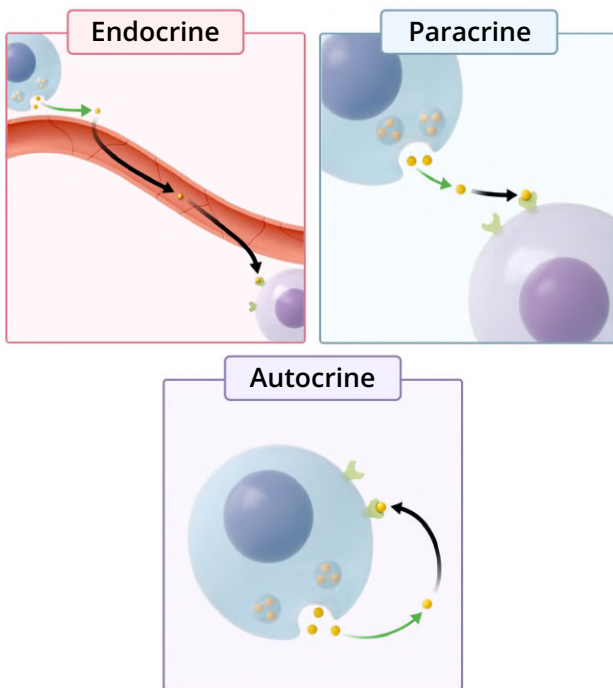
- Hormones
- Hypothalamus and Pituitary
- Thyroid and Parathyroid
- Pancreas
- Adrenal Gland
- Testes and Ovaries
- Feedback Loops

Hormones

Hormones can be secreted in the following ways:

- **Endocrine:** Through the bloodstream.
- **Paracrine:** To neighboring cells.
- **Autocrine:** Onto the same cell that is secreting the hormone.

Cell Signaling



There are three types of hormones: peptide hormones, steroid hormones, and amino acid-derived hormones.

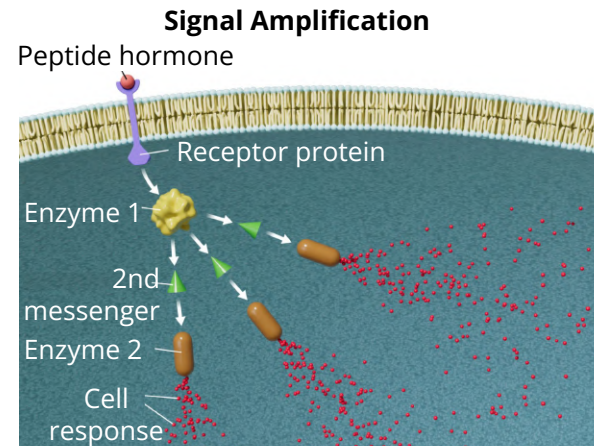
Peptide hormones (protein hormones):

- **Synthesis:** Produced in the **rough ER** and made of amino acids connected by peptide bonds.

- **Action:** Bind to **cell surface receptors** because they are **water-soluble** (not lipid-soluble) and cannot pass freely through the cell membrane. Therefore, peptide hormones function by **indirect stimulation**. Their signal can be transmitted through ligand-gated ion channels or by intracellular second messengers.

Ligand-gated ion channels change shape upon binding to peptide hormones, allowing ions to flow across the cell membrane. No second messengers are involved.

The **second messenger system** of peptide hormone signaling allows for quick and immediate physiological changes. This process is known as **signal amplification**.



A single signaling molecule can lead to a robust cellular response due to **signal amplification**. For example, as second messengers become activated and, in turn, activate multiple enzymes that catalyze multiple reactions, each step in the **signaling cascade** results in the progressive *increase* in the intensity of the initial signal.

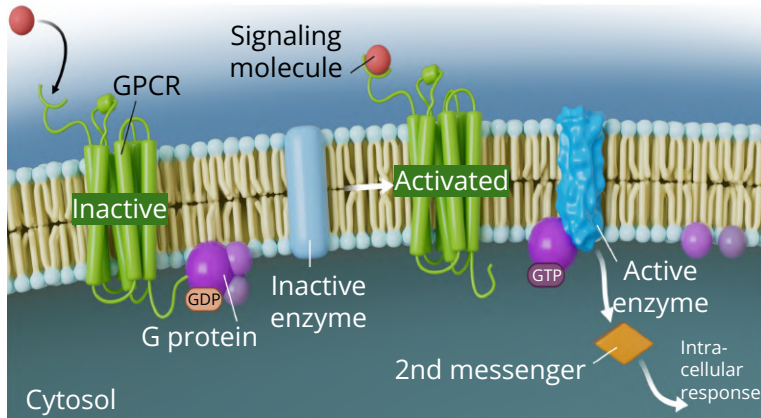
Common second messengers include:

- cAMP (cyclic AMP)
- IP₃ (inositol triphosphate)
- DAG (diacylglycerol)
- Calcium ions (Ca²⁺)

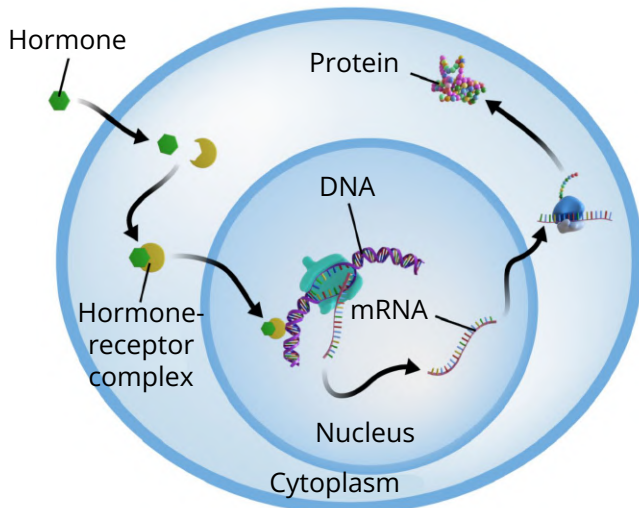
The **IP₃/DAG** pathway uses second messengers. It regulates calcium levels in the cell. A membrane signaling molecule activates IP₃ and DAG. IP₃ binds to calcium channels on the endoplasmic reticulum, triggering these channels to open and release calcium ions into the cytosol. Calcium can then act as another signaling molecule to continue the signaling cascade.

G protein-coupled receptors (GPCRs) are cell surface receptors that can initiate a second messenger response after binding to a peptide hormone extracellularly. A **G protein** is coupled to the receptor and dissociates into subunits (alpha, beta, and gamma) after activation. These subunits then act upon intracellular second messengers to propagate the signal.

G Protein-Coupled Receptor (GPCR)

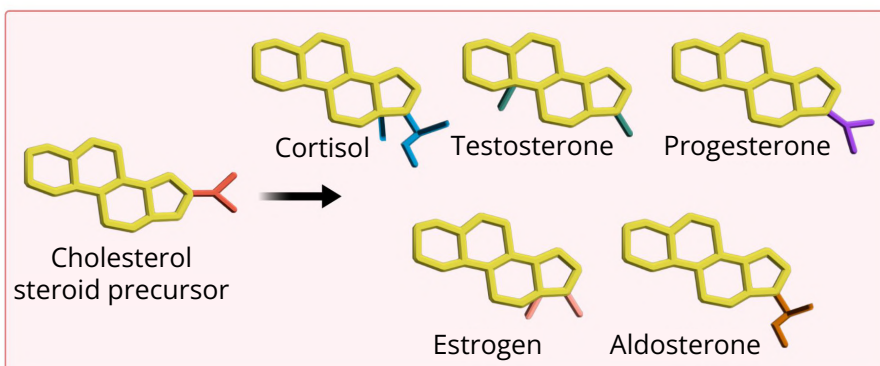


Steroid Hormone Mechanism

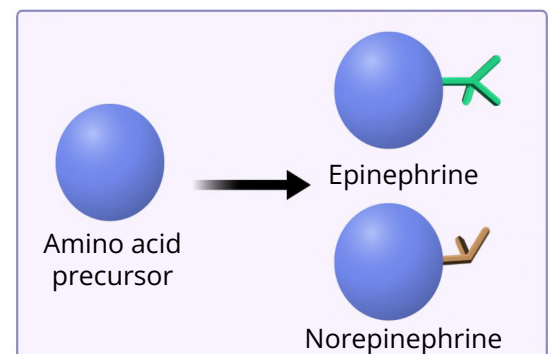


Types of Hormones

Steroid derived



Amino acid-derived



Steroid hormones:

- **Synthesis:** Produced in the **smooth ER** and made up of a fused four-ring structure. All steroid hormones are derived from **cholesterol**.
- **Examples:** All hormones produced by the **adrenal cortex** (glucocorticoids, mineralocorticoids, androgenic steroids) and **reproductive organs** (progesterone, testosterone, estrogen).
- **Action:** Lipid-soluble (**lipophilic**). Therefore, require a protein carrier to travel through the bloodstream, but can freely cross the cell membrane. Bind to receptors either in the cytoplasm or the nucleus to form **hormone-receptor complexes** that bind to DNA, and influence gene transcription. This process is known as **direct stimulation**. Steroid hormones cause slow and gradual physiological changes.

Amino acid-derived hormones:

Can have properties that are similar to both peptide hormones and steroid hormones.

- **Synthesis:** Produced in **rough ER** and cytosol. Mainly derived from the amino acid **tyrosine**.
- **Examples:** All hormones produced by the **adrenal medulla** (epinephrine and norepinephrine, which are water-soluble). Also includes charged amino acid derivatives **T₃** and **T₄** (lipid-soluble).

Hypothalamus and Pituitary

The **hypothalamus** coordinates the body's internal environment and maintains homeostasis.

The **pituitary gland (hypophysis)** is under the hypothalamus and is composed of two lobes: the **anterior pituitary** and **posterior pituitary**.

1. **Posterior pituitary:** It is a direct neuronal extension of the **hypothalamus**.

Two hormones (produced by the hypothalamus) are stored and released by the posterior pituitary:

- **Antidiuretic hormone (ADH aka vasopressin):** Decreases urination by increasing water retention. Targets **nephrons**, increasing the number of **aquaporins** for water reuptake.
- **Oxytocin:** Causes uterine contractions during child labor and the release of milk during breastfeeding (mammary gland). Oxytocin also plays an important role in facilitating maternal behavior (drive to be a good parent).

2. **Anterior pituitary:** Produces its own hormones. It is connected to the hypothalamus through a **hypophyseal portal system**, which allows for quick diffusion of hormones through a **portal vein**.

Hypothalamic-releasing hormones are released by the hypothalamus to stimulate the **anterior pituitary** to release other hormones.

- **GnRH (gonadotropin-releasing hormone):** Causes release of luteinizing hormone (LH) and follicle stimulating hormone (FSH).
- **TRH (thyrotropin-releasing hormone):** Causes release of thyroid stimulating hormone (TSH).
- **CRH (corticotropin-releasing hormone):** Causes release of adrenocorticotropic hormone (ACTH).
- **GHRH (growth hormone-releasing hormone):** Causes release of growth hormone (GH).

The **anterior pituitary** then produces its own hormones, classified as **tropic hormones** and **direct hormones**.

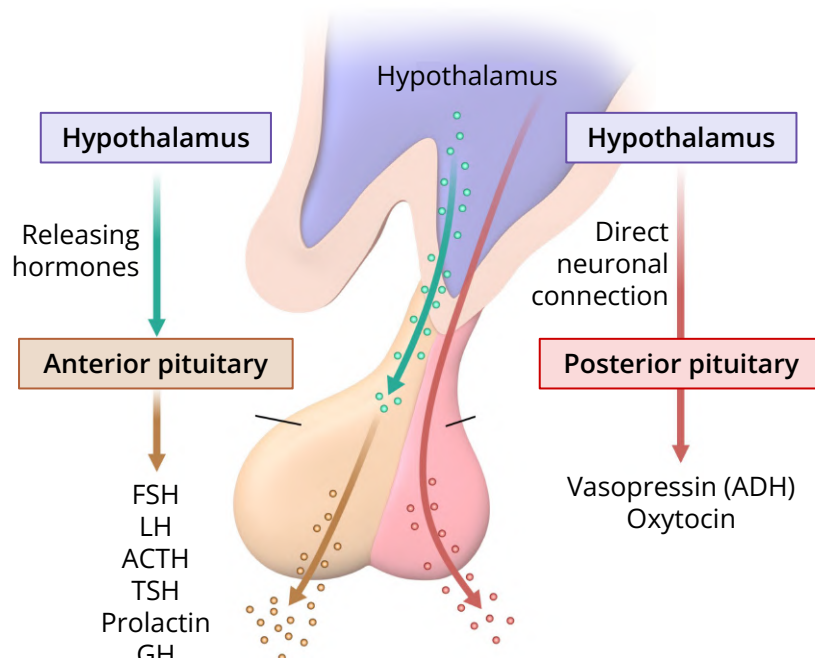
Tropic hormones target other endocrine glands for further hormone release. Important examples released from the **anterior pituitary**:

- **FSH (follicle stimulating hormone):** Follicle growth (females) and sperm maturation (males) in the gonads.
- **LH (luteinizing hormone):** Stimulates ovulation, corpus luteum formation (females), and testosterone production (males) in the gonads.
- **ACTH (adrenocorticotropic hormone):** Stimulates release of glucocorticoids from the adrenal gland to fight stress. This also leads to an increase in glucose levels in the body.
- **TSH (thyroid stimulating hormone):** Stimulates T_3 and T_4 production by the thyroid gland to increase metabolism.

Direct hormones target organs directly for effects. Important examples released from the **anterior pituitary**:

- **Prolactin:** Stimulates mammary gland development and increases **milk production** after childbirth.
- **Growth hormone (GH, somatotropin):** Stimulates body cells to grow and divide.

Hypothalamus & Pituitary Hormones



Thyroid and Parathyroid

The **thyroid gland** is the largest endocrine organ and is located in front of the trachea.

Three main hormones of the thyroid:

1. **Triiodothyronine (T_3)**: Released in response to **TSH** and increases metabolism in the body. Has a **negative feedback** effect on TSH secretion.
2. **Thyroxine (T_4)**: Performs the same actions as T_3 above. However, T_4 has one more iodine and gets converted into T_3 upon cell uptake. It is much *less* potent than T_3 but is *more* stable in the blood.
3. **Calcitonin**: Secreted by parafollicular **thyroid cells (C cells)** to *decrease* blood calcium levels.

Hypothyroidism describes the under-secretion of T_3 and T_4 hormones, resulting in *reduced* metabolic rates in the body.

Hyperthyroidism describes the over-secretion of T_3 and T_4 hormones, resulting in *increased* metabolic rates in the body.

Both hypothyroidism and hyperthyroidism can lead to a **goiter**, an enlargement of the thyroid gland. In hypothyroidism, low T_3 and T_4 levels cause over-secretion of TRH and TSH. Elevated TSH stimulates thyroid activity, causing enlargement. In hyperthyroidism, a hyperactive thyroid gland can also result in enlargement.

Iodine deficiency is the most common cause of a goiter as iodine is essential for the synthesis of T_3 and T_4 .

The **parathyroid gland** secretes **parathyroid hormone (PTH)**, which *increases* blood calcium levels.

Pancreas

The **pancreas** is a gland that contains **exocrine** and **endocrine** tissue.

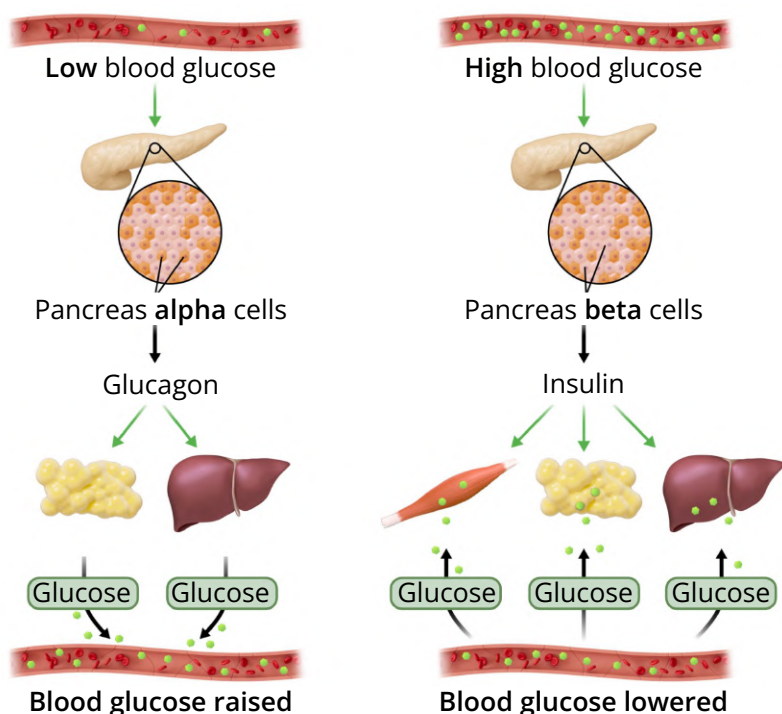
Exocrine tissue secretes digestive enzymes through the pancreatic duct to the small intestine.

The **endocrine tissue** (the **islets of Langerhans**) secretes **glucagon** and **insulin**. These two hormones are each secreted by a different cell type as listed below:

1. **Alpha (α) cells**: Secrete **glucagon** in response to low blood glucose levels. Glucagon *raises* glucose levels by stimulating the liver and fat tissue to release their glucose storages.
2. **Beta (β) cells**: Secrete **insulin** in response to high blood glucose levels. Insulin *lowers* glucose levels by stimulating the liver, muscle, and fat tissue to store glucose.

Insulin is a peptide hormone that triggers an increase in the number of glucose transporters embedded within the cell membrane. This increases the facilitated diffusion of glucose from the bloodstream into cells and *reduces* blood glucose levels.

Blood Glucose Regulation



Adrenal Gland

Our body has two **adrenal glands**. Each adrenal gland has an outer **cortex** and an inner **medulla**. They mainly help the body deal with **stress**.

Adrenal cortex:

- Deals with longer term stress.
- Stimulated by secretion of **ACTH** from the **anterior pituitary**.
- Releases **steroid hormones**.
- Produces **glucocorticoids (e.g., cortisol)** to *raise* blood glucose levels for immediate fuel during periods of long-term stress. However, this also *lowers* our immune response.
- Produces **mineralocorticoids (e.g., aldosterone)** to *increase* blood volume and blood pressure by *raising* reabsorption of Na^+ . Water passively gets reabsorbed with Na^+ due to osmosis.
- Produces a small amount of **male sex hormones (androgens)**.

Adrenal medulla:

- Deals with **short-term stress**.
- Stimulated by the **sympathetic nervous system**.
- Releases **amino acid-derived hormones**.
- Produces **catecholamines (epinephrine and norepinephrine)** to initiate “fight or flight” response by increasing heart rate and the breakdown of glycogen into glucose. Epinephrine binds to adrenergic receptors to cause **vasoconstriction** or **vasodilation**.

Testes and Ovaries

After stimulation by **LH** and **FSH**, the ovaries produce **progesterone** and **estrogen**, while the testes produce **androgens** such as **testosterone**.

Females:

- **LH:** During the menstrual cycle, the LH surge causes **ovulation**. This results in the formation of a corpus luteum, which produces progesterone and estrogen.
- **FSH:** Stimulates **follicle growth** in ovaries, which results in the increased estrogen production.

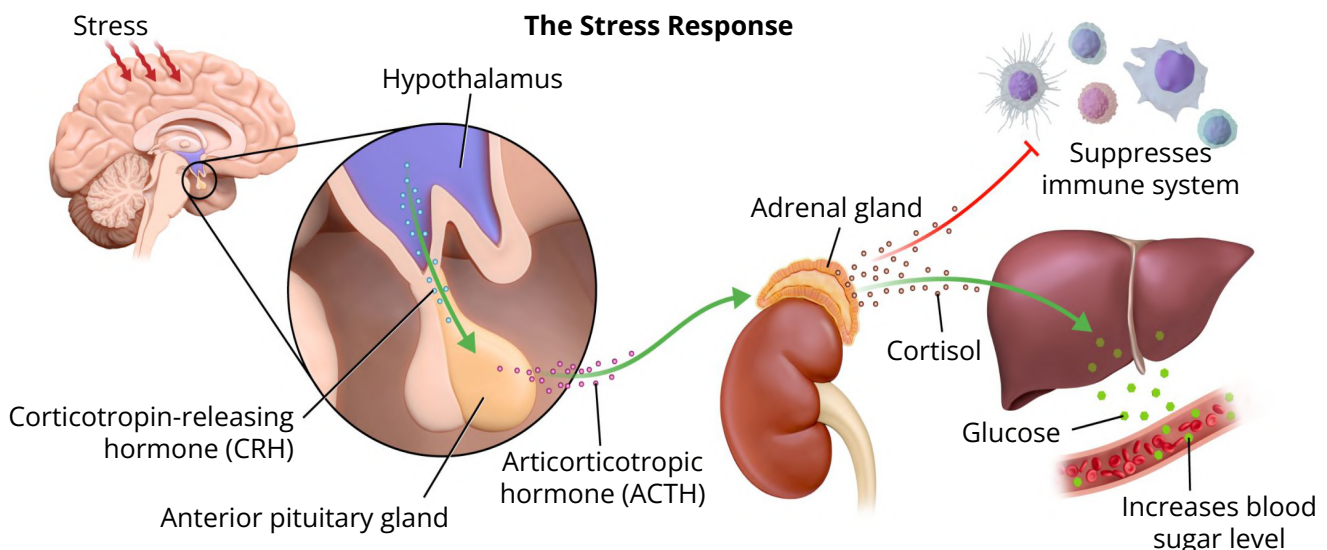
Males:

- **LH:** Triggers **testosterone** production by **Leydig cells**.
- **FSH:** Stimulates **sperm maturation**.

Feedback Loops

Hormonal control relies on positive and negative **feedback systems**.

1. **Positive feedback:** The change causes **amplification** of itself, forming a loop that continues to intensify. Think of it as promoting **exponential growth** (e.g., uterine contractions and oxytocin during labor stimulate each other).
2. **Negative feedback:** The change causes **inhibition** of itself, forming a loop that prevents hormone overproduction. Think of it as promoting **stability** in the body (e.g., high testosterone inhibits GnRH, decreasing LH/FSH and testosterone production).



Endocrine System for the DAT

Source	Hormone	Target	Action
Posterior pituitary (hormones are produced by the hypothalamus and stored/released by the posterior pituitary)	ADH	Kidney (collecting duct)	Increase water reabsorption (adds aquaporins to collecting duct)
	Oxytocin	Mammary gland	Release of milk
Anterior pituitary (FLAT PiG)	FSH	Ovaries, testes	Regulate oogenesis and spermatogenesis
	LH	Ovaries, testes	Regulate oogenesis and spermatogenesis
	ACTH	Adrenal cortex	Secretion of glucocorticoids
	TSH	Thyroid	Secretion of T_3 and T_4
	Prolactin	Mammary glands	Production of milk
	Growth hormone	Bone, muscle	Stimulates growth
Pancreas (alpha cells)	Glucagon	Liver	Increases blood glucose
Pancreas (beta cells)	Insulin	Liver, muscle, fat	Lowers blood glucose
Adrenal gland (medulla)	Epinephrine and norepinephrine	Blood vessels, liver, heart	Increase blood glucose, vasoconstriction, sympathetic response
Adrenal gland (cortex)	Glucocorticoids (cortisol)	General	Increase blood glucose
	Mineralocorticoids (aldosterone)	Kidney	Increase reabsorption of Na^+ and secretion of K^+ in distal convoluted tubule and collecting duct (water follows passively)
Thyroid	T_4 (thyroxine) and T_3 (triiodothyronine)	General	Increase cell metabolism
	Calcitonin	Bone	Lower blood Ca^{2+}
Parathyroid	PTH	Bone	Increase blood Ca^{2+}
Testis	Testosterone	General	Spermatogenesis, secondary sex characteristics
Ovary	Estrogen	Uterus, general	Menstrual cycle, secondary sex characteristics
	Progesterone	Uterus	Menstrual cycle, pregnant
Pineal	Melatonin	General	Circadian rhythms

Ch. 10.8: Digestive System

Ingestion and Inhalation Pathways

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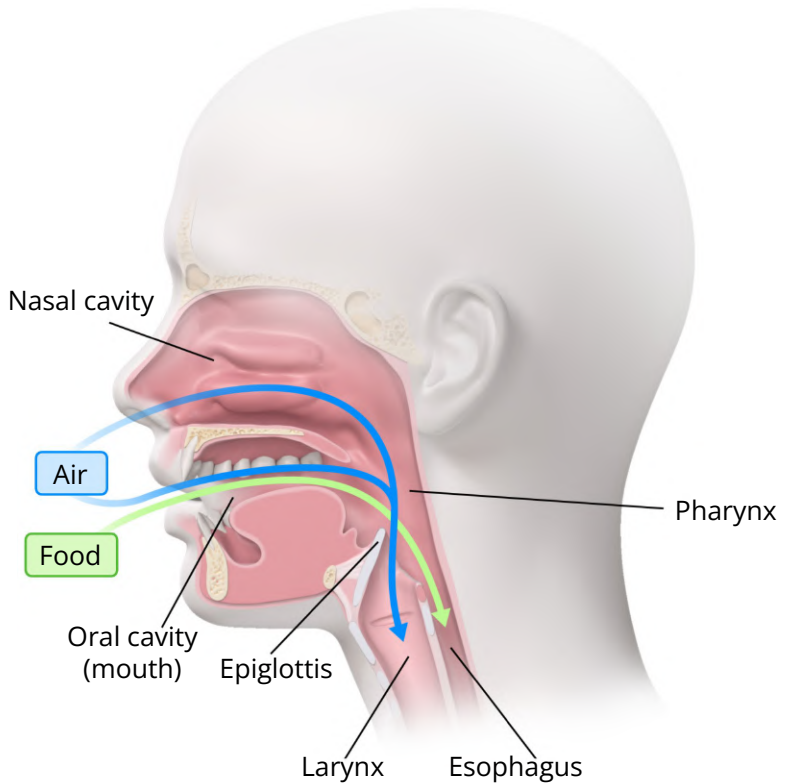
- Digestion in Humans
- Mouth, Pharynx, Esophagus
- Stomach
- Small Intestine
- Liver
- Large Intestine
- Summary: Digestive Hormones & Enzymes

Digestion in Humans

Digestion is the process of breaking down large food into smaller substances for absorption by the body.

The digestive tract has two openings: mouth and anus.

1. **Mechanical digestion:** Physical breakdown of food.
2. **Chemical digestion:** Chemical breakdown of food, using **enzymes**.



Mouth, Pharynx, Esophagus

Mechanical (chewing) and chemical (salivary amylase) digestion begin in the **mouth**. **Salivary amylase** in **saliva** breaks down starch into simpler carbohydrates. Saliva also lubricates the food, creating a **bolus**.

Upon swallowing, food enters the **pharynx** (common to digestive and respiratory systems) which separates to form the trachea and the esophagus. The **epiglottis** blocks the opening to the trachea, preventing **choking**.

Food continues to the **esophagus** (tubular structure guiding food to stomach). The bolus is pushed down via **peristalsis** (rhythmic waves of contraction).

Digestive System

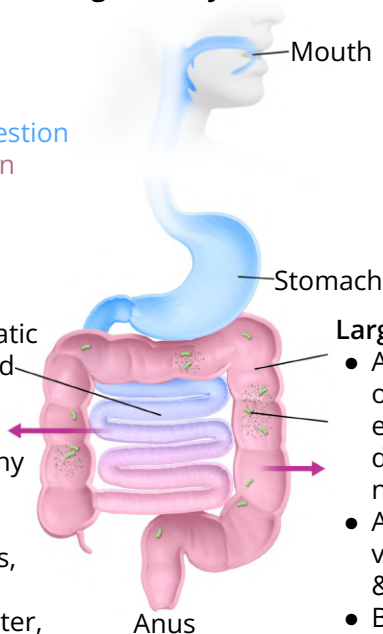
Enzymatic digestion
Absorption

Small intestine

- Both enzymatic digestion and absorption occur
- Absorbs many nutrients (carbs, proteins, fats, minerals, vitamins, water, & salts)

Large intestine

- Absorption occurs, enzymatic digestion does not
- Absorption of vitamins, water, & salts
- Bacteria produce vitamins



Stomach

Food enters the **stomach** via the **cardiac sphincter** (ring of muscles) where mechanical (churning of food) and chemical (enzymatic breakdown of protein and fat) digestion occur.

The stomach lining has **gastric glands** with multiple cell types. **Mucous cells** produce mucus, which lubricates and protects the stomach lining from the acid.

Food entry causes the stomach to **distend**, signaling **G cells** to release **gastrin**, a hormone with two functions:

1. Stimulates **parietal cells** to release very acidic **gastric juice** (high $[HCl]$).
2. Stimulates **chief cells** to secrete gastric lipase and pepsinogen. **Gastric lipase** breaks down fats into fatty acids and glycerol. **Pepsinogen** is a **zymogen** (inactive enzyme precursor) which activates into pepsin in acid. **Pepsin** cleaves peptide bonds (proteins $\rightarrow$ amino acids).

Chyme (acidic, semi-digested food) exits to the small intestine via the **pyloric sphincter**.

Small Intestine

Responsible for 90% of total digestive absorption. Consists of three parts: **duodenum** (digestion), **jejunum** (absorption), and **ileum** (absorption).

Mnemonic: DJ Eye (Duodenum > Jejunum > Ileum)

Goblet cells secrete **mucus** to protect the epithelial lining from acidic chyme. Chyme triggers the release of **secretin** (a hormone), which stimulates the **pancreas** to release basic **bicarbonate ions** (HCO_3^-) into the duodenum via the **pancreatic duct**.

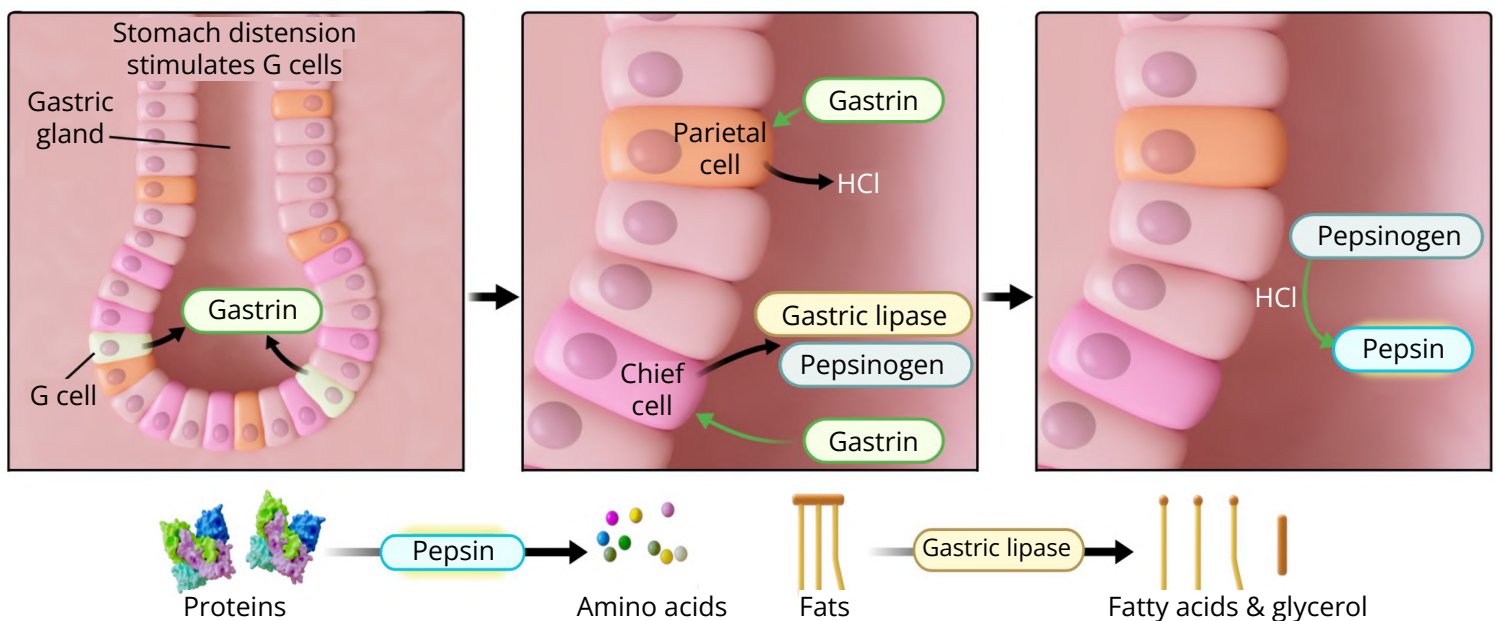
Proteins and fats entering the small intestine stimulate the small intestine to release **cholecystokinin (CCK)**, which slows gastric emptying, stimulates the **pancreas** to release digestive enzymes, and stimulates the **gallbladder** to release **bile** into the duodenum.

The **pancreas** secretes the following ions and digestive enzymes into the duodenum:

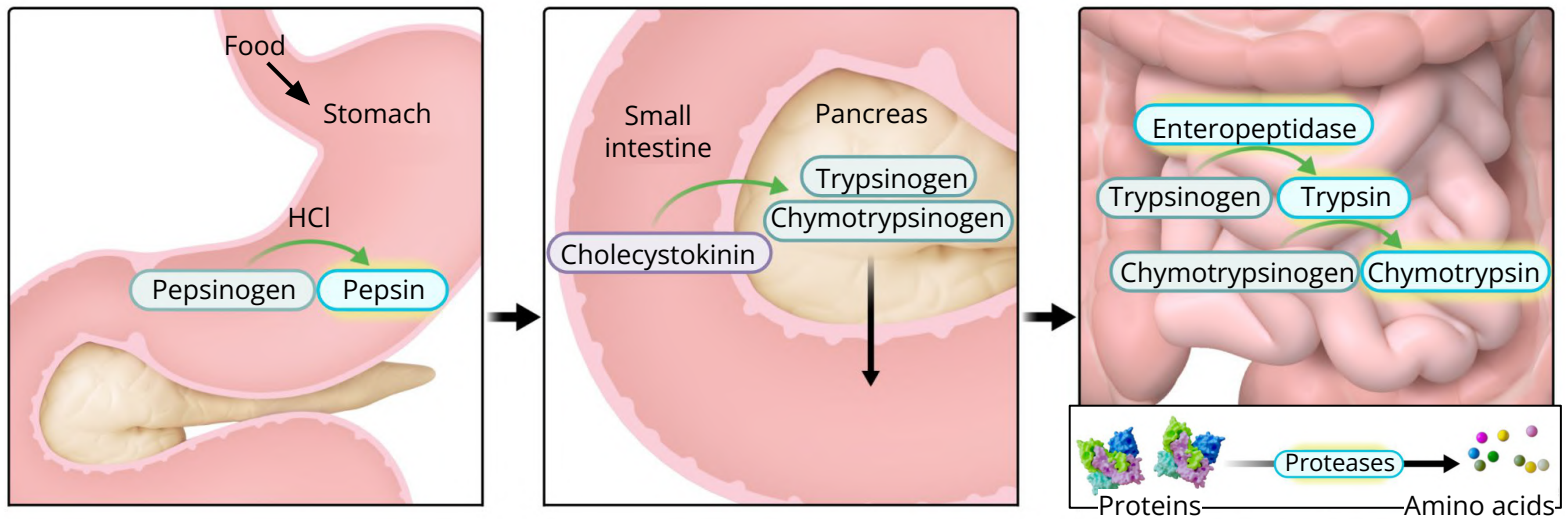
- **Bicarbonate (HCO_3^-)**: Neutralizes acidity.
- **Pancreatic amylase**: Breaks down starch.
- **Proteases**: Proteins $\rightarrow$ amino acids. **Trypsin** and **chymotrypsin** are initially released as zymogens (trypsinogen and chymotrypsinogen).

Enteropeptidase, secreted by the duodenum into its lumen, converts trypsinogen to trypsin, which then converts chymotrypsinogen to chymotrypsin.

Gastric Glands



Digestion of Protein



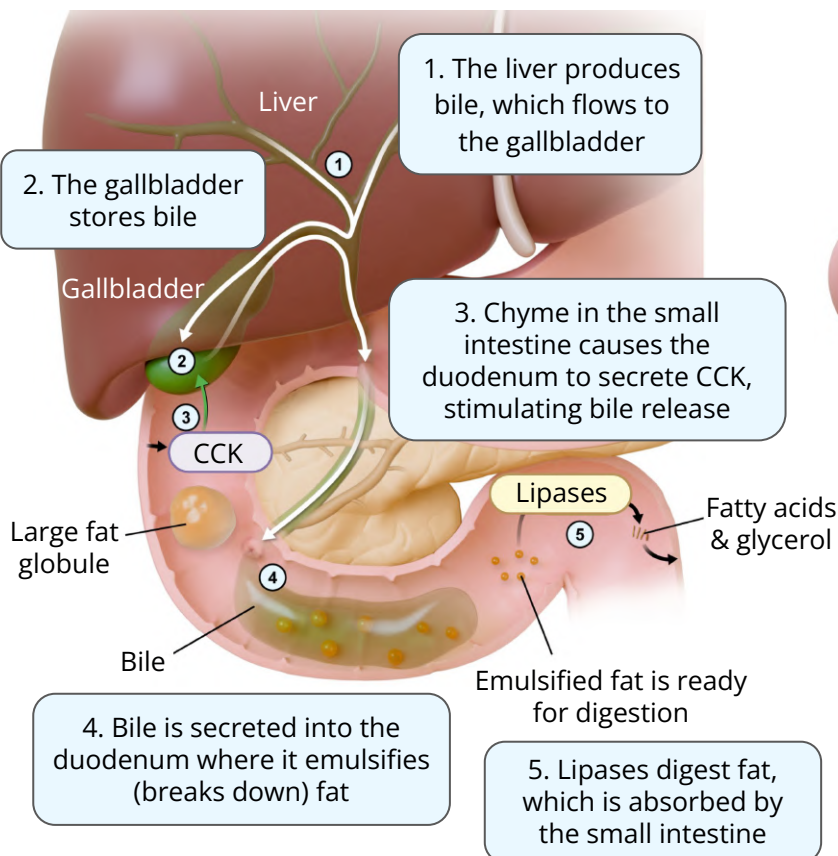
Bile (emulsifies fats) is produced by the **liver** and stored and concentrated by the **gallbladder**. Bile is important for the absorption of **lipids** (fats) from our diets.

Food is moved by **peristalsis** to the **jejunum** and **ileum** for absorption. The **ileocecal sphincter** connects the small intestine (ileum) to the large intestine (colon).

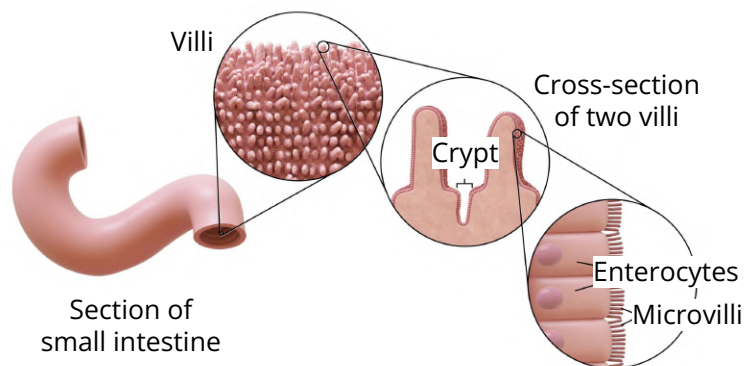
Villi (finger-like projections which increase surface area) are made of **enterocytes** that are lined with **microvilli**. Villi and microvilli increase surface area and absorption efficiency. **Crypts** (invaginations in the intestinal wall) contain cells that secrete enzymes and produce new epithelial cells for the lining.

Inside the villus, nutrients (glucose and amino acids) are absorbed into **blood capillaries** and fats (fatty acids and glycerol) into **lacteals** (specialized lymphatic capillaries).

Fat Absorption



Villi and Microvilli



Liver

In addition to **bile production**, the **liver** is involved in many processes.

Mnemonic: PUSH DoG

Protein synthesis, **U**rea synthesis, **S**torage, **H**ormone synthesis, **D**etoxification, **G**lucose and fat metabolism

1. Blood maintenance
 - **Filters** and **detoxifies** blood coming from the digestive system.
 - Detoxifies the body by metabolizing chemicals and drugs.
 - Destroys erythrocytes and bacteria. Breaks down **hemoglobin** from old RBCs into **bilirubin**, which is secreted in **bile**.
2. Glucose metabolism
 - **Glycogenesis**: Converts excess glucose into glycogen for storage in the liver (after meals). Stimulated by insulin. Inhibited by glucagon and epinephrine.
 - **Glycogenolysis**: Breaks down glycogen to glucose for bodily use (between meals). Stimulated by glucagon and epinephrine. Inhibited by insulin.
 - **Gluconeogenesis**: Converts glycerol and amino acids into glucose when glycogen stores are depleted. Stimulated by glucagon and epinephrine. Inhibited by insulin.
3. Protein metabolism
 - Synthesizes **plasma proteins** from amino acids (**blood clotting factors**).
 - Converts ammonia (dangerous byproduct of protein metabolism) into **urea** (safer) for excretion.

Large Intestine

Water and mineral absorption occur at the **cecum** (small pouch). The **appendix** (projection in the cecum) is a **vestigial structure** with negligible immune function. In the **colon**, water absorption is completed, hardening feces. The feces is stored in the **rectum** and expelled through the **anus**. Many species of bacteria coexist in the large intestine; these bacteria serve a critical function in aiding digestion. Over 99% of bacteria in the large intestine are obligate anaerobes. These bacteria, as well as all the other bacteria that are present in or on the body, are collectively known as a **microbiome**.

The large intestine has three functions:

1. Water absorption.
2. Mineral absorption (salts).
3. Vitamin production and absorption: In a **mutualistic** relationship, bacteria produce and absorb vitamins, metabolize bile acid, and ferment fiber.

Bile acid metabolism is a key indicator of the health of the microbiome. A healthy microbiome will have sufficient levels of all the bacteria needed for bile acid metabolism, while an unhealthy microbiome will not, impairing function.

Digestive Enzymes

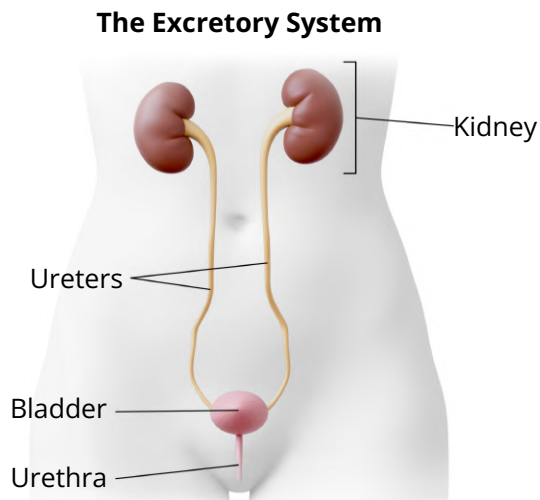
Location	Digestive enzyme(s)	Function
Pancreas	Trypsin, chymotrypsin	Protein breakdown
	Pancreatic amylase	Starch breakdown
Mouth	Salivary amylase	Breakdown of starches into disaccharides
Stomach	Pepsin	Breakdown of proteins into peptides
	Gastric lipase	Breakdown of fats into fatty acids and glycerol
Small intestine	Peptidase	Breakdown of peptides into amino acids
	Enteropeptidase	Activates trypsinogen to trypsin

Ch. 10.9: Excretory System

The Kidney

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- The Nephron
- The Kidney
- Glomerulus
- Proximal Convolved Tubule
- Loop of Henle
- Distal Convolved Tubule & Collecting Duct
- Hormones in Excretory System
- Osmoregulation



Excretion is the filtering out of metabolic wastes from the body's fluids and eliminating them as urine.

Humans have two kidneys. Each kidney consists of a **cortex** (outer portion where blood enters the kidney) and a **medulla** (middle portion). Functions of the kidney include:

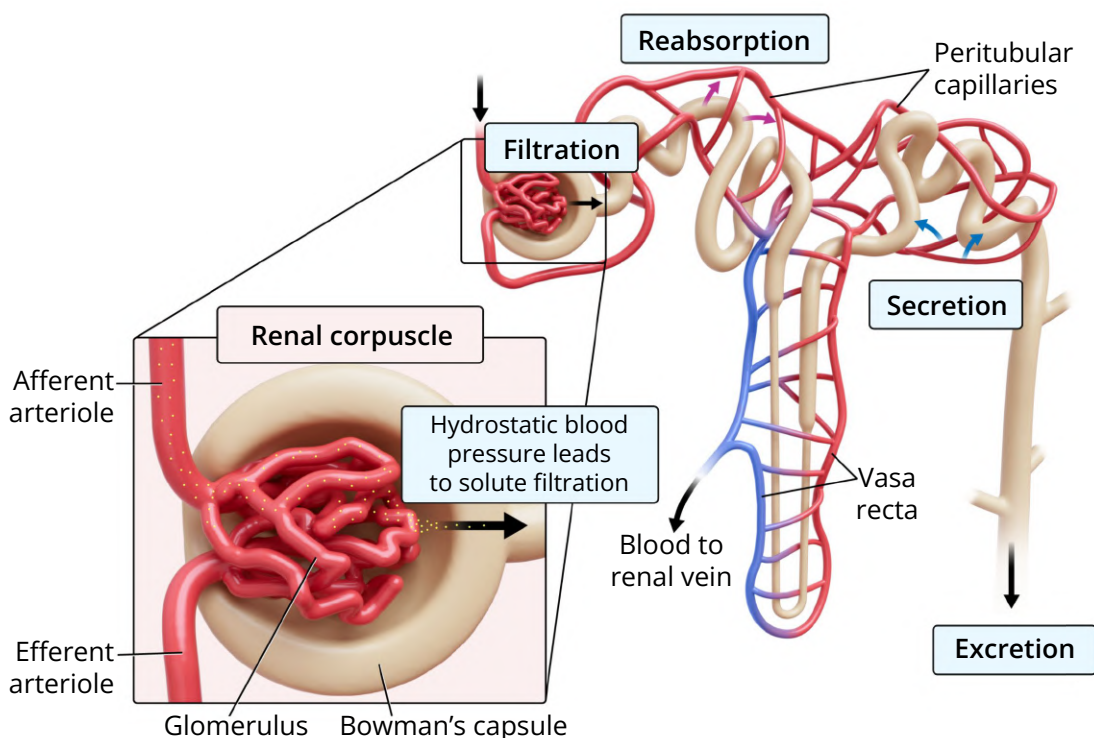
- Regulation of blood pressure
- Regulation of blood pH
- Stimulates the generation of new red blood cells

The Nephron

A **nephron** is a single, functional unit of a kidney. The nephron has four main processes:

- **Filtration:** The process of small substances and water being filtered from the blood (glomerulus) into the Bowman's capsule, forming filtrate.
- **Reabsorption:** The process of specific substances (glucose, ions, water, etc.) being selectively reabsorbed into the bloodstream from the filtrate to maintain homeostasis.
- **Secretion:** The selective transfer of waste, ions, and drugs directly from the bloodstream into the filtrate.
- **Excretion:** The final step where the remaining fluid containing waste products is expelled as urine. The amount excreted equals the amount filtered minus the amount reabsorbed, plus the amount secreted.

Nephron Processes



Glomerulus

The **renal corpuscle** comprises the **glomerulus** and **Bowman's capsule** and is located in the renal cortex.

In the glomerulus, **filtration** occurs. Blood enters from the afferent arteriole into the glomerulus, which acts as a sieve, allowing small substances into the Bowman's capsule. **Hydrostatic pressure** from the blood is the main force driving filtration in Bowman's capsule. The glomerulus exits the Bowman's capsule via the efferent arteriole, which forms the **peritubular capillaries** and the **vasa recta**.

Proximal Convolved Tubule

Most of the reabsorption occurs in the **proximal convoluted tubule** through active transport. The distal convoluted tubule reabsorbs Na^+ and Cl^- . Glucose and amino acids are two molecules the nephron reabsorbs almost completely due to their importance in the body. **Urea**, waste products, and drugs are secreted into the nephron by active and passive transport processes.

Loop of Henle

The **loop of Henle** descends into the medulla and exhibits selective permeability at each limb. Surrounding the loop of Henle is the **vasa recta**, a network of capillaries that runs parallel to it.

The **descending limb** is permeable to water. Water is reabsorbed into the blood as the filtrate travels down the descending limb, becoming *more* concentrated.

The **ascending limb** is impermeable to water. As the filtrate travels up the ascending limb, solutes are reabsorbed, and the filtrate becomes *less* concentrated.

Pro-Tip: This process causes the interstitial fluids surrounding the nephron to become concentrated with salts. The *longer* the loop of Henle, the more concentrated the urine will be.

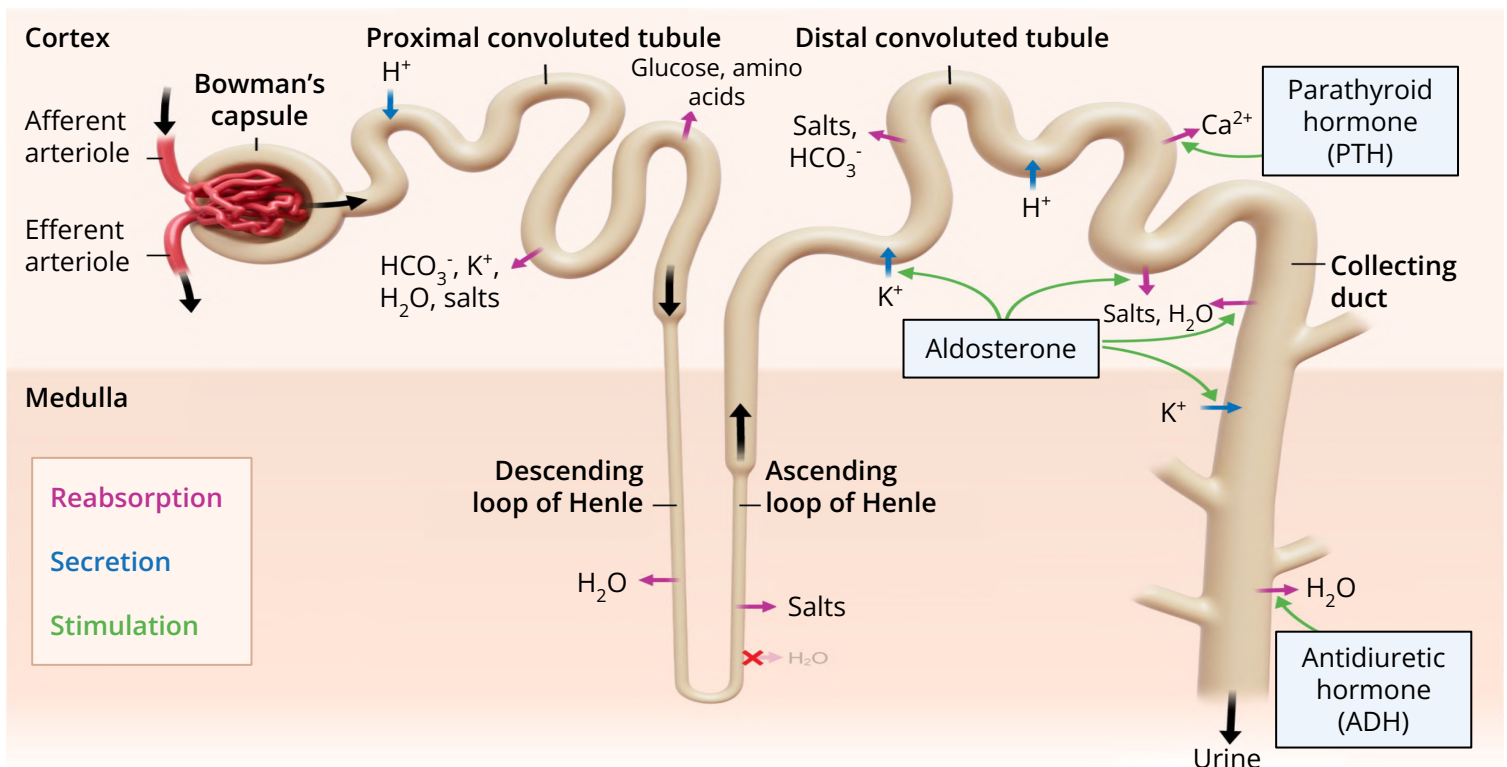
Distal Convolved Tubule & Collecting Duct

From the loop of Henle, the filtrate goes to the **distal convoluted tubule**. Na^+ and Cl^- are reabsorbed here, with water following passively. Waste products are also secreted here.

The filtrate then travels to the **collecting duct**, where water passively moves out and concentrates the urine. The urine travels to the **renal pelvis** and then to the **ureter**.

The ureter connects the kidney to the **bladder**, where urine is stored. When the signal is received, urine is excreted from the bladder and the body via the **urethra**.

The Nephron



Hormones in Excretory System

Renin Angiotensin Aldosterone System (RAAS):

Juxtaglomerular cells release renin when blood pressure is low.

Renin is an enzyme which acts on **angiotensinogen** to activate it to the form **angiotensin I**.

Another enzyme called **angiotensin converting enzyme (ACE)** acts on angiotensin I to convert it to angiotensin II. Angiotensin II is the active hormone.

Angiotensin II has many effects in the body to increase blood pressure and volume, including:

- It stimulates **aldosterone** release from the adrenal gland cortex.
- It *increases* Na^+ reabsorption from the proximal tubule (water follows the salt).
- It is a **systemic vasoconstrictor**, causing vessels to constrict and thereby *increasing* **total peripheral resistance (TPR)**.
- It increases thirst: drinking more *increases* liquid blood volume (and thus TPR).

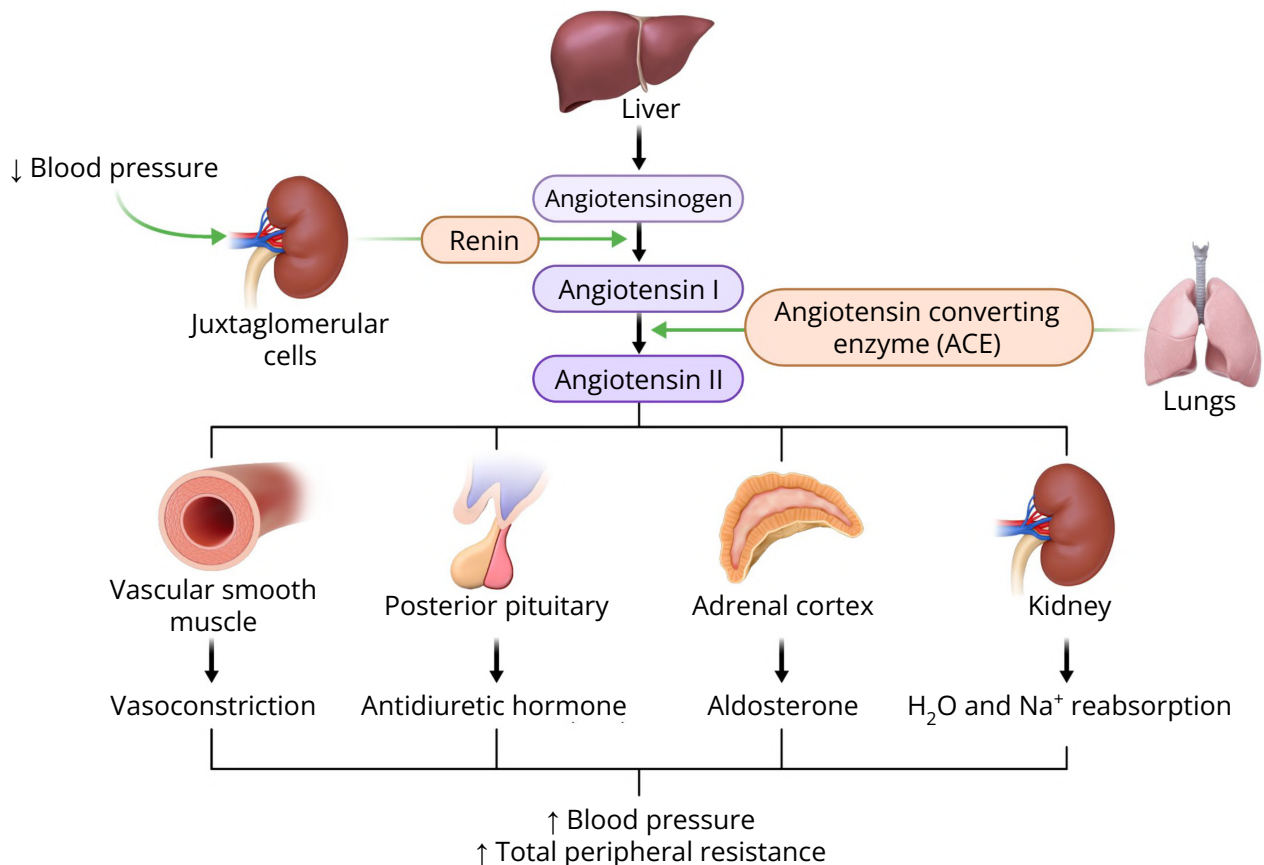
Aldosterone is a mineralocorticoid produced by the adrenal cortex. It *increases* salt and water reabsorption and potassium secretion in the distal tubules and collecting ducts.

Antidiuretic hormone (aka **ADH** or **vasopressin**). Released from the **posterior pituitary** upon stimulation from the hypothalamus. Causes **aquaporins** to insert into the collecting duct of the nephron and *increases* water reabsorption.

Atrial natriuretic peptide (ANP) is produced by atrial cells in response to atrial distension by increased blood volume and pressure. ANP *reduces* blood volume and blood pressure. It accomplishes this by:

- *Increasing* the **glomerular filtration rate (GFR)**, which is the rate at which the kidneys filter blood.
- *Decreasing* sodium reabsorption.
- *Increasing* sodium excretion.
- Inhibiting **renin** and the **renin angiotensin aldosterone system (RAAS)**.

Renin Angiotensin Aldosterone System (RAAS)

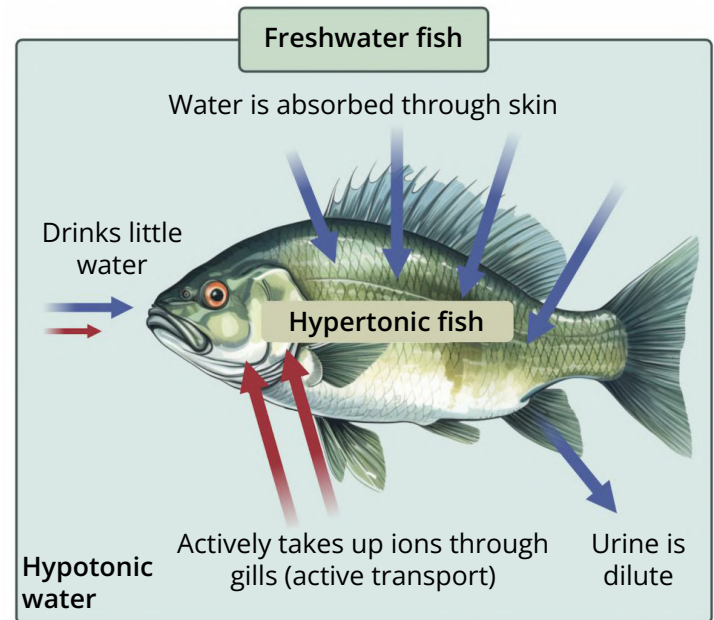
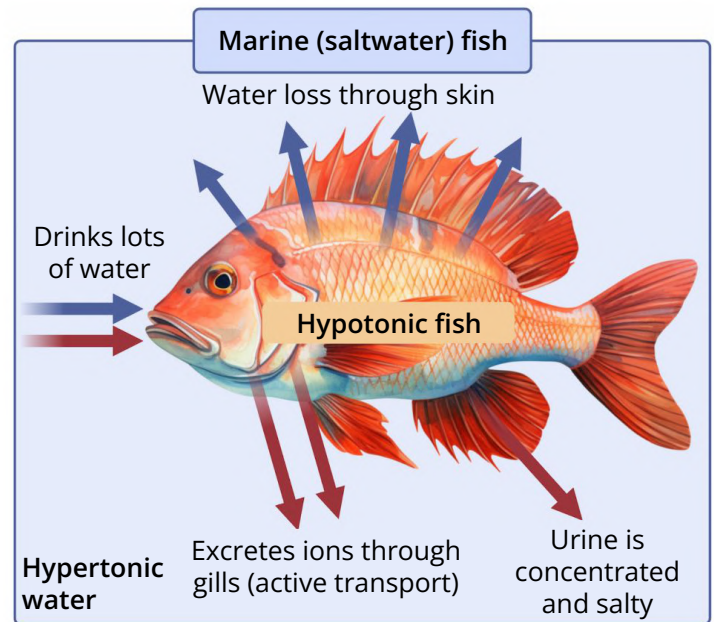


Osmoregulation

Osmoregulation is the absorption and excretion of water and solutes so an organism can maintain a proper water balance. The difference between marine fish and freshwater fish is the most commonly used example:

1. **Marine (saltwater) fish** are hypoosmotic to their environment. They are *less* salty than the surrounding saltwater. Therefore, they're constantly losing water to their environment. As a result, marine fish:
 - Constantly drink water
 - Rarely urinate
 - Secrete accumulated salts through their gills
2. **Freshwater fish** are hyperosmotic to their environment. They are more salty than the surrounding freshwater. Therefore, they're constantly absorbing too much water. As a result, freshwater fish:
 - Rarely drink water
 - Constantly urinate
 - Absorb salts through their gills

Fish Homeostasis



Ch. 10.10: Integumentary System

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- Skin
- Epidermis
- Dermis
- Hypodermis

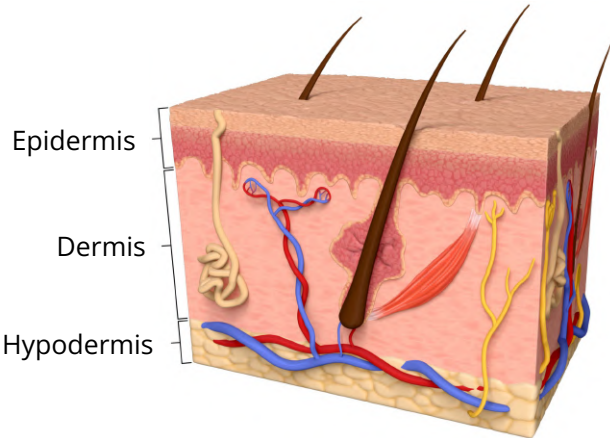
Skin

The **skin** has important functions in the body:

1. **Physical barrier protection:** First line of defense of the innate immune system.
2. **Thermoregulation: Evaporative cooling (sweating)** regulates body temperature. Body heat is transferred to the sweat, which evaporates and cools the body.
3. **Sensory input:** Nerve receptors in skin detect physical stimuli.
4. **Vitamin D synthesis** upon UV exposure.
5. **Excretion of water and salts:** Sweating results in water loss.

It consists of three layers: epidermis, dermis, and hypodermis.

Layers of the Skin



Epidermis

The **epidermis** is the most superficial layer of the skin. It is composed of epithelial tissue and contains **keratinocytes**. Skin epithelial cells are **labile**, meaning they must constantly divide to regenerate and maintain the skin barrier in response to frequent stress.

The **stratum** (layers) of the epithelium from superficial to deep are:

Layer	Description
Corneum	Corneocytes form the outermost, protective layer.
Lucidum	Made of dead keratinocytes that are not yet differentiated into corneocytes. Only present in palms and soles.
Granulosum	Contains keratinocytes that form a water barrier.
Spinosum	Contains keratinocytes with long extensions ("spines"). Important for strength (desmosomes) and flexibility.
Basale (germinativum)	Precursor keratinocyte stem cells proliferate here. Connected to basement membrane (basal lamina), which separates the epidermis and dermis.

Mnemonic: Come Let's Get Some Beers

- Corneum
- Lucidum
- Granulosum
- Spinosum
- Basale

Dermis

The **dermis** is the middle layer of the skin, located between the epidermis and hypodermis. It consists mostly of connective tissue and contains blood vessels and nerve endings. It provides structural support (protection against injury), nourishment, and sensory functions for the skin. In mammals, hair follicles are found in the dermis.

Hypodermis

The **hypodermis** (subcutaneous layer) is the deepest layer and contains larger nerves and blood vessels. It is primarily composed of fat and connective tissue, which function to insulate the body and provide cushioning for underlying structures.

Ch. 11: Reproduction and Developmental Biology

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- Female Anatomy and Oogenesis
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- Extraembryonic Development
- Internal vs. External Reproduction
- Important Animal Reproductive Methods
- Factors Influencing Development

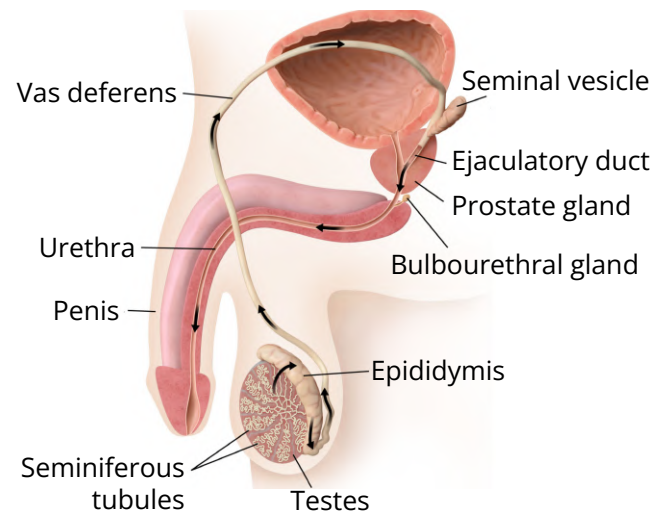
Asexual Reproduction

1. **Binary fission:** Exhibited by prokaryotes, as well as eukaryotic organelles (mitochondria and chloroplasts). Replicated circular DNA migrate to opposite ends of a cell or organelle, and a septum forms in the middle to divide it into two genetically identical products .
2. **Budding:** A parent organism develops a mitotic bud (outgrowth) that becomes a new organism which can detach or remain attached. Exhibited by some fungi (yeast) and animals (hydra).
3. **Fragmentation:** An organism splits into fragments, and each fragment regenerates into a complete organism (e.g., flatworm, starfish).
4. **Parthenogenesis:** An unfertilized egg develops into a viable embryo. Exhibited by honeybees (males develop from haploid eggs, while females are diploid).

Human Reproduction

Sexual reproduction: Fusion of two gametes (male sperm and female egg) to create offspring. **Germ cells** (male spermatogonia, female oogonia) differentiate into gametocytes (spermatocytes, oocytes), which then produce gametes (spermatozoa, ova) via meiosis.

Male Anatomy and Spermatogenesis



Spermatogenesis: The formation of mature haploid gametes (**spermatozoa**) from diploid germ cells (**spermatogonia**) through a series of steps.

Spermatogonia differentiate into spermatocytes, which then undergo two meiotic divisions to become four **spermatids**.

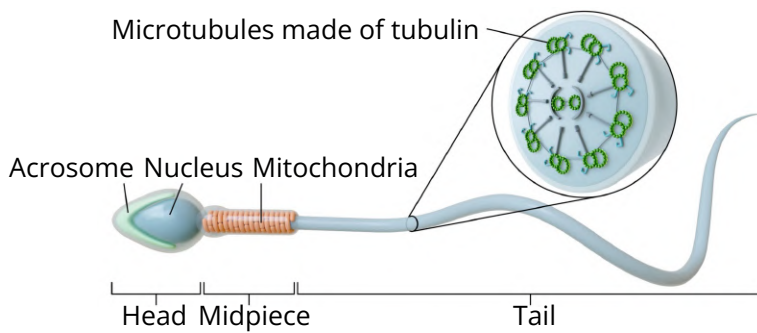
Spermiogenesis is the final stage of spermatogenesis, during which haploid spermatids mature into motile spermatozoa (plural term for mature sperm cells). There is *no* division or change in the amount of genetic material at this stage.

1. **Seminiferous tubules** of the testes are the site of spermatogenesis (sperm production) and contain:
 - **Sertoli cells:** Activated by **follicle stimulating hormone (FSH)**. Surround and nourish sperm.
 - Precursor cells of spermatozoa
2. Immature sperm is transported via peristalsis to the **epididymis** (duct around the testes) for storage and maturation.
3. Sperm move through the **vas deferens** (paired tubules) into the **ejaculatory duct**, which propels sperm into the **urethra** and leads to ejaculation out of the penis as **semen** (sperm & accessory gland secretions).
 - The secretions of accessory glands (seminal vesicles, prostate gland, & bulbourethral glands) nourish sperm, enhance motility, and clean the urethra.

Mnemonic: SEVEN UP sperm pathway

Seminiferous tubules → Epididymis → Vas Deferens
→ Ejaculatory Duct → Urethra → exits the Penis

Sperm Structure



- **Head:** Contains nucleus and acrosome.
- **Midpiece:** Mitochondria (ATP production).
- **Tail:** Long flagellum (microtubules) to propel sperm.

Hormones in Males

1. **Follicle stimulating hormone (FSH):** Stimulates Sertoli cells for spermatogenesis in the seminiferous tubules.
2. **Luteinizing hormone (LH):** Stimulates **Leydig cells** to produce testosterone.
3. **Testosterone:** Responsible for male secondary sex characteristics and supports spermatogenesis.

Female Anatomy and Oogenesis

- **Vagina:** A muscular canal that extends from the cervix to the external environment. Serves as the passageway for menstruation, sperm, and childbirth.
- **Uterus:** Also known as the womb. A pear-shaped, vascular organ that provides an optimal environment for a **blastocyst** to implant (in the innermost endometrial lining) and develop within.
- **Ovary:** A paired organ that produces eggs (singular: ovum; plural: ova) and reproductive hormones estrogen and progesterone. Fertilized or unfertilized eggs travel from the ovary to the uterus through the **oviduct (fallopian tube)**.

Oogenesis:

1. Many **oogonia** are produced during fetal development, but most undergo apoptosis. A small fraction survive and differentiate into **primary oocytes**, which begin meiosis but arrest in prophase I until puberty.
2. Beginning at puberty, one primary oocyte per month completes meiosis I. This produces a large **secondary oocyte** and a polar body. The secondary oocyte is arrested in metaphase II and is released by ovulation.
3. If **fertilization** occurs, the secondary oocyte completes meiosis II. At the end of meiosis II, there are 2-3 polar bodies (non-viable products) and one viable oocyte (containing cytoplasm, mitochondria, and nutrients).

Hormones in Females

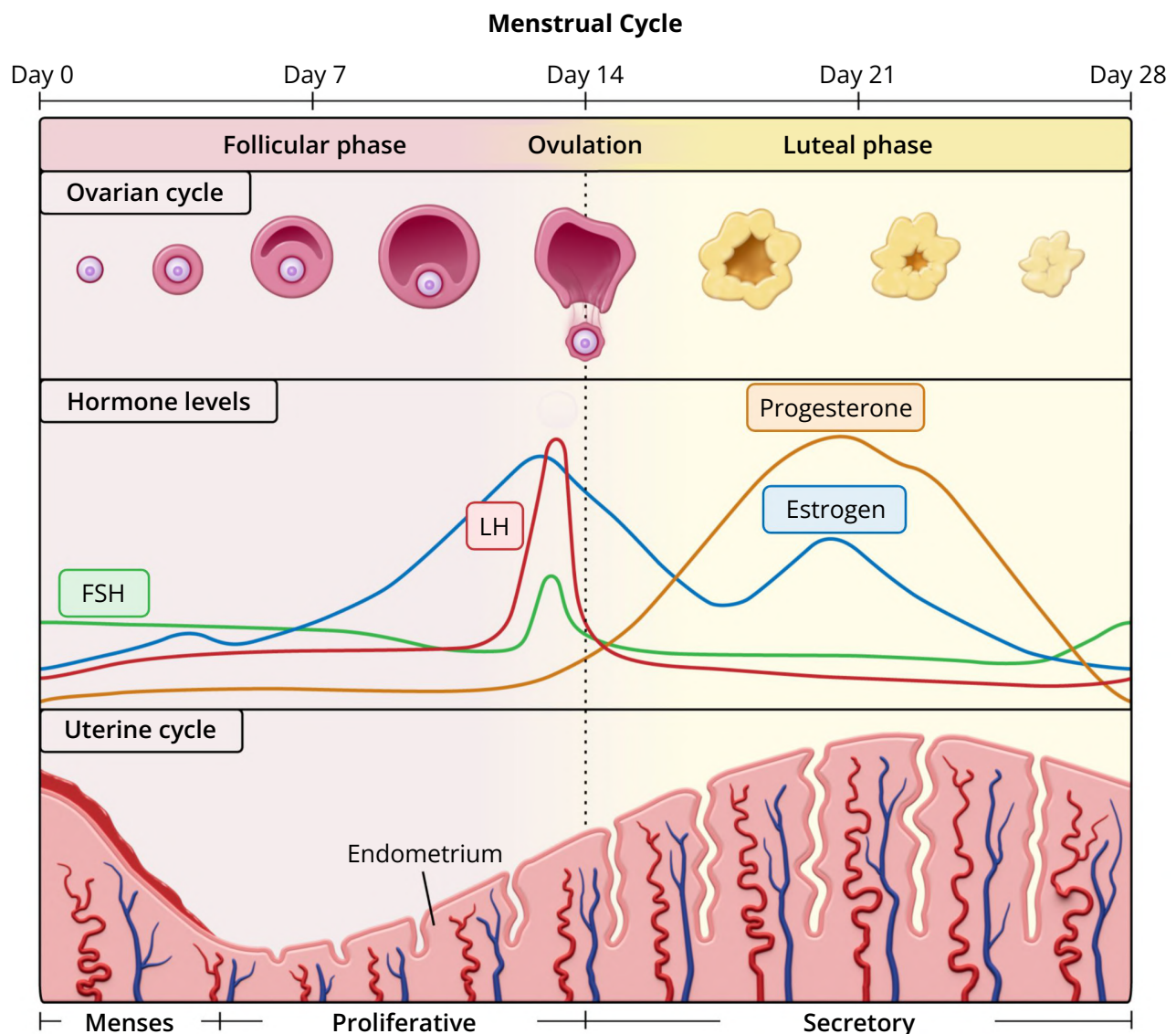
1. **Follicle stimulating hormone (FSH):** Stimulates the development of follicles in the ovary and promotes **estrogen** production.
 - A **follicle** is a fluid filled sac on an ovary that contains an immature oocyte (arrested in prophase I of meiosis). The **corpus luteum** is a temporary endocrine structure formed by an empty follicle after ovulation.
2. **Luteinizing hormone (LH):** Triggers ovulation and promotes corpus luteum formation, which secretes progesterone and estrogen. The ovulated egg is a secondary oocyte arrested in metaphase II of meiosis.
3. **Estrogen and progesterone:** Regulate the menstrual cycle, support reproduction, and are responsible for female secondary sex characteristics.

Menstrual Cycle

- **Follicular phase:** The hypothalamus releases gonadotropin-releasing hormone (GnRH), causing the anterior pituitary to secrete LH and FSH. FSH binds to the ovaries and promotes follicle development. The developing follicle releases estrogen, causing thickening of the **endometrium** (uterine lining) and LH spikes.
- **Ovulation:** Ovulation is triggered by LH. The dominant follicle releases an egg (secondary oocyte). Cilia sweep the egg into the oviduct (fallopian tube) and help move it along, where it may be fertilized by sperm.
- **Luteal phase:** The empty follicle develops into the **corpus luteum**, which is maintained temporarily by LH. The corpus luteum produces progesterone and estrogen, further thickening the uterine lining in preparation for possible implantation.

- **If no implantation occurs:** Progesterone and estrogen inhibit the hypothalamus and pituitary gland, causing LH and FSH levels to drop. The corpus luteum disintegrates, leading to progesterone and estrogen levels falling. The endometrium sheds (menstruation), and the cycle repeats.
- **If implantation occurs:** The placenta produces **human chorionic gonadotropin (hCG)**, which supports the corpus luteum through pregnancy. This allows progesterone and estrogen levels to remain high, ensuring the endometrium remains intact (no menstruation).

hCG can be given to women during fertility treatment to trigger ovulation and enhance progesterone release.

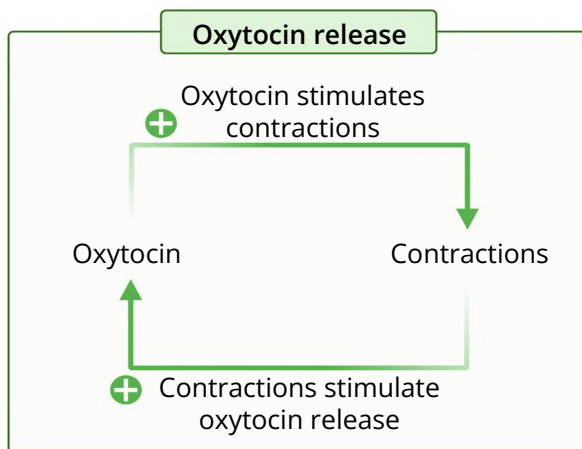


Hormone Feedback Loops

In a **positive feedback loop**, a product reinforces the pathway to *increase* production.

- **Lactation:** Infant suckling *increases* prolactin production. Prolactin causes lactation (milk production) and further *increases* infant suckling. Oxytocin releases milk (**milk let down reflex**).
- **Childbirth:** Oxytocin induces contractions, which push the baby out of the womb. The baby pushes against a nerve in the cervix that signals the hypothalamus and pituitary to release *more* oxytocin.

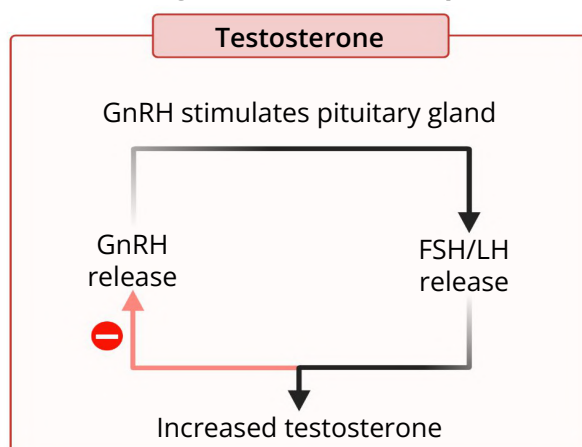
Positive Feedback Loop



In a **negative feedback loop**, a product inhibits the pathway to *decrease* production.

- The hypothalamus releases **GnRH**, causing the pituitary to release **FSH** and **LH**, which *increase* **testosterone** levels. High testosterone levels inhibit the hypothalamus from releasing GnRH, *lowering* FSH/LH and testosterone.
- The same occurs with **estrogen** and **progesterone** in the menstrual cycle.

Negative Feedback Loop



Fertilization

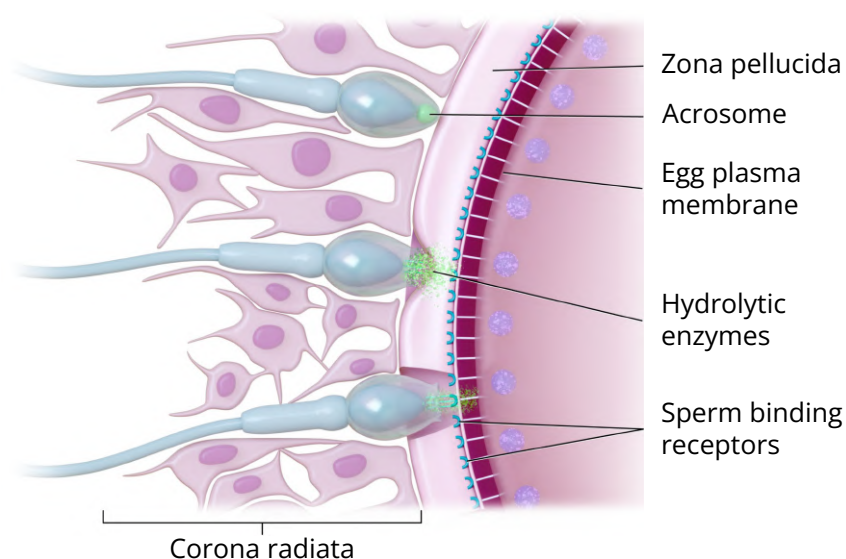
Fertilization occurs in the **oviduct**. It is the joining of a haploid sperm and a haploid egg to form a diploid zygote.

Sperm: Made up of a head (with **acrosome** at its tip), midpiece (contains mitochondria), and tail.

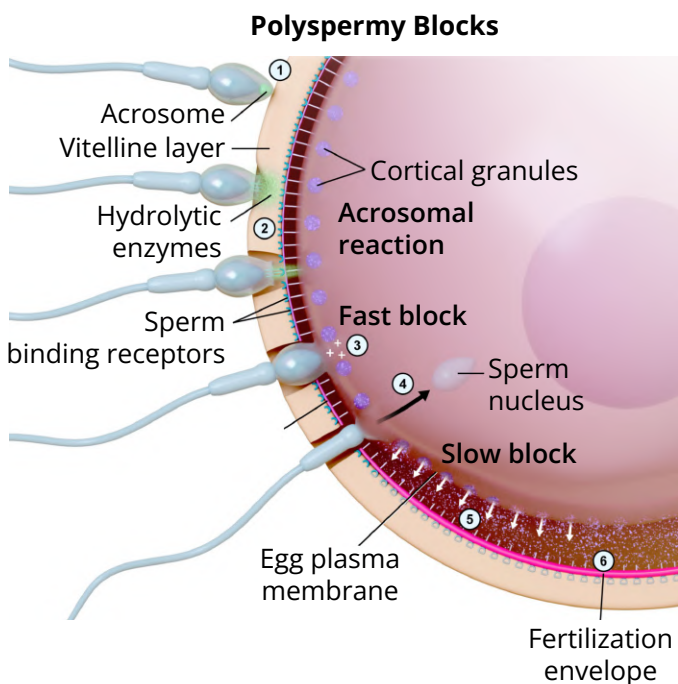
Egg: The outermost layer, **corona radiata**, nourishes the developing egg. Beneath it is the glycoprotein-rich **zona pellucida**, which is analogous to the **vitelline layer** in non-mammalian species. The plasma membrane is deep to the zona pellucida.

1. **Capacitation:** The final maturation step for sperm before fertilization. Triggered by secretions of the uterine wall as sperm travels into the oviduct. It destabilizes the sperm plasma membrane, resulting in:
 - Preparation of the sperm tip for the acrosomal reaction.
 - *Increased* calcium permeability, causing a hyperactive state (flagella beats harder, sperm swims faster).
2. **Acrosomal reaction:** A recognition process between sperm and egg that ensures species-specific fusion. The sperm passes through the corona radiata to reach and bind to the zona pellucida. The sperm's acrosomal and plasma membranes then fuse, releasing **hydrolytic enzymes** that digest the zona pellucida. This allows the sperm to reach and fuse with the egg's plasma membrane, completing **fertilization**.

Acrosomal Reaction



3. **Polyspermy block:** Prevents **polyploidy** by inhibiting polyspermy (multiple sperms penetrating egg).
 - **Fast block:** Immediate and temporary. Upon fertilization, sodium ions diffuse into the egg and depolarize its membrane, preventing additional sperm binding.
 - **Slow block:** Gradual and long-lasting. Calcium ions released within the egg stimulate the **cortical reaction** (exocytosis of **cortical granules**). These granules harden the zona pellucida or vitelline layer and trigger proteases that separate it from the plasma membrane, forming an impenetrable fertilization envelope.



4. **Completion of meiosis II for the secondary oocyte:** During meiosis II, the egg is arrested in metaphase. After penetration, meiosis in the secondary oocyte resumes, creating a haploid ovum and a second polar body.

Pro-Tip: Recall mitochondria have their own DNA, separate from the nucleus. Unlike nuclear DNA, which is passed down from both the mother and the father, mitochondrial DNA is inherited exclusively from the mother.

5. Zygote formation:

- **Monozygotic twins:** Identical twins. Formed when one zygote splits. Two embryos with identical genetic material.
- **Dizygotic twins:** Fraternal twins. Formed when two separate eggs are fertilized by two different sperm. Two genetically unique embryos.

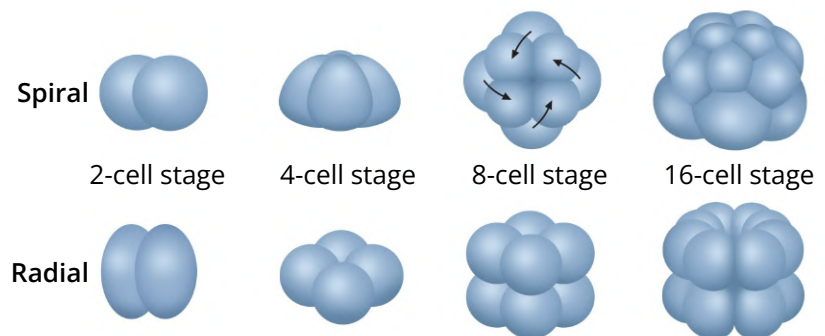
Cleavage, Morula, Blastula

Cleavage: Rapid mitotic cell division that occurs without an increase in total embryo mass. The increasingly smaller cells produced are called **blastomeres**.

1. Axis of cleavage:

- **Radial cleavage:** Cells align vertically (e.g., deuterostomes).
- **Spiral cleavage:** Cells spiral around a vertical axis (e.g., protostomes).

Radial vs. Spiral Cleavage



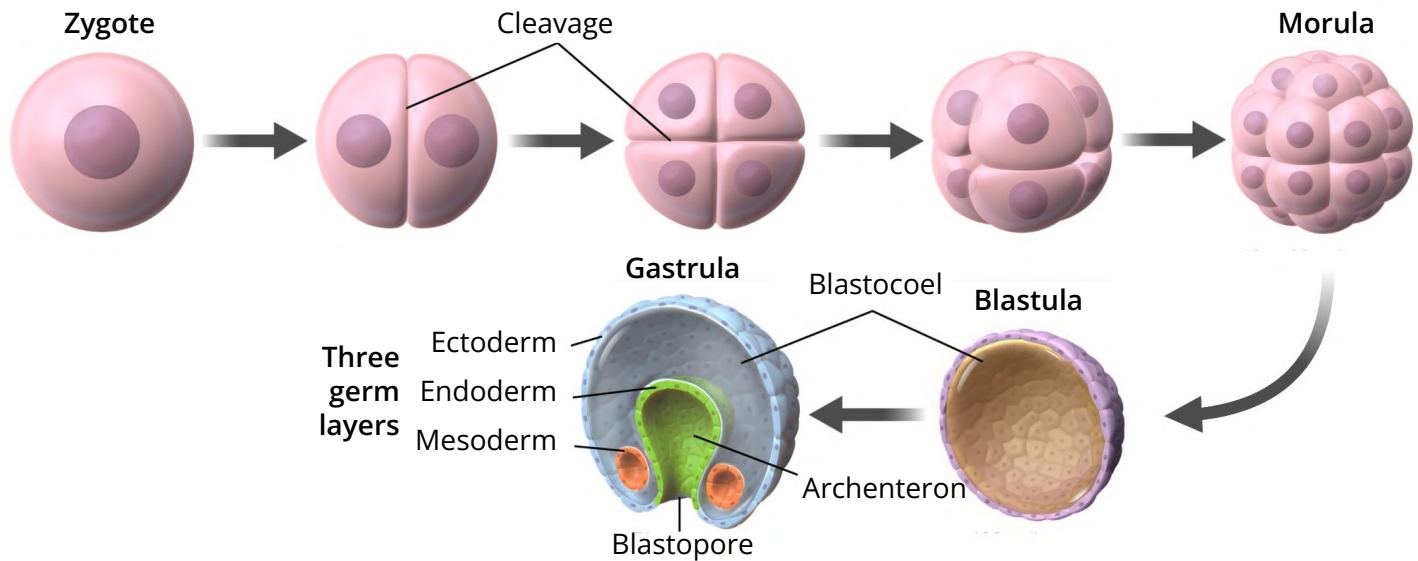
2. Fate of cells:

- **Determinate (mosaic) cleavage:** Cell fates are pre-determined.
- **Indeterminate (regulative) cleavage:** Cell fates are not fixed; cells are **totipotent**.

3. Evenness of embryo division:

- **Holoblastic cleavage:** Even cell division throughout the entire embryo. Occurs in low-yolk animals (e.g., humans, sea urchins).
 - **Exception:** Frogs (moderate-yolk) exhibit uneven holoblastic cleavage with polarity.
- **Meroblastic cleavage:** Partial cleavage of embryo in yolk-rich animals (e.g., birds, fish, reptiles). Exhibits **polarity** with **animal pole** (active division) and **vegetal pole** (yolk-rich, negligible division).

Stages of Embryo Development

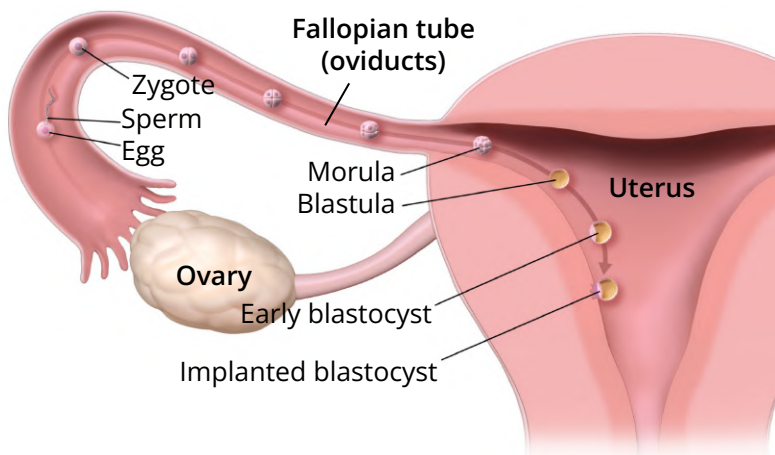


The major stages of embryo development include:

- **Morula:** Ball of blastomeres at 16-32 cell stage.
- **Blastula stage:** 128-cell stage. Contains a **blastocoel**, a hollow, fluid-filled center.
- **Blastocyst stage (mammal-specific):** Cells of blastula begin to differentiate.

Cleavage occurs as the fertilized egg travels to the uterus, and it implants into the uterus at the **blastocyst** stage.

Fertilization, Differentiation, and Implantation Stages



Gastrulation is the formation of a trilaminar embryo known as a **gastrula**. Cells invaginate to form three germ layers: **endoderm**, **mesoderm**, **ectoderm**.

Invagination also creates the **blastopore**, an opening into the **archenteron**, a primitive gut tube (precursor to the digestive tract).

1. **Ectoderm (outer germ layer) forms:**

- CNS (brain and spinal cord) and PNS, as well as neural crest cells and neural ganglia.
- Sensory parts of ear, eye, and nose.
- Epidermis layer of skin, hair, and nails.
- Mammary and sweat glands.
- Pigmentation cells.
- Enamel of teeth.
- Adrenal medulla.

2. **Mesoderm (middle germ layer) forms:**

- Bone and skeleton.
- Skeletal, smooth, and cardiac muscle.
- Cardiovascular system.
- Gonads.
- Adrenal cortex.
- Spleen.
- Notochord (induces spinal cord formation from ectoderm).

3. **Endoderm (inner germ layer) forms:**

- Epithelial lining of digestive, respiratory, and excretory systems.
- **PLTT (Pancreas, Liver, Thyroid and Parathyroid, Thymus).**

Cellular Differentiation

Stem cells are undifferentiated cells with potential (**potency**) to develop into many types of cells.

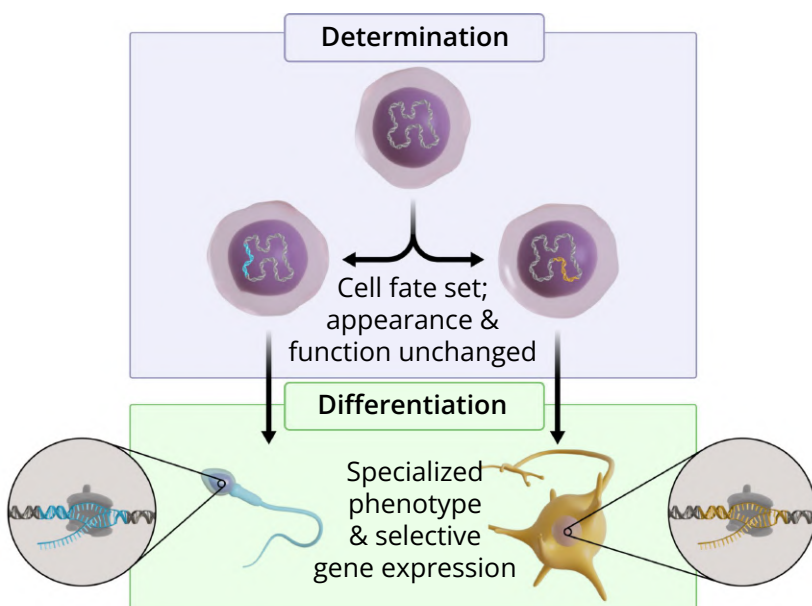
- **Totipotent** stem cells can differentiate into any cell type (e.g., zygote, blastomeres of morula).
- **Pluripotent** stem cells can differentiate into any of the three germ layers (e.g., blastocyst cells → any embryonic stem cell).
- **Multipotent** stem cells can only differentiate into a few cell types of a specific tissue type (e.g., hematopoietic stem cell → many blood cells).

Determination is the process by which a cell becomes committed to a specific fate. At this stage, the cell has not yet changed in appearance or function, but it has irreversibly established the regulatory instructions that will guide future gene expression. This sets the foundation for **selective gene expression**, which leads to differences in cell function.

Differentiation is the subsequent stage in which the cell begins to express those predetermined genes, leading to the production of proteins and functional characteristics that give the cell its **specialized phenotype**.

Differentiated cells retain the entire genome. Via **nuclear reprogramming**, a differentiated cell nucleus can therefore be restored to an earlier state, capable of supporting complete organismal development.

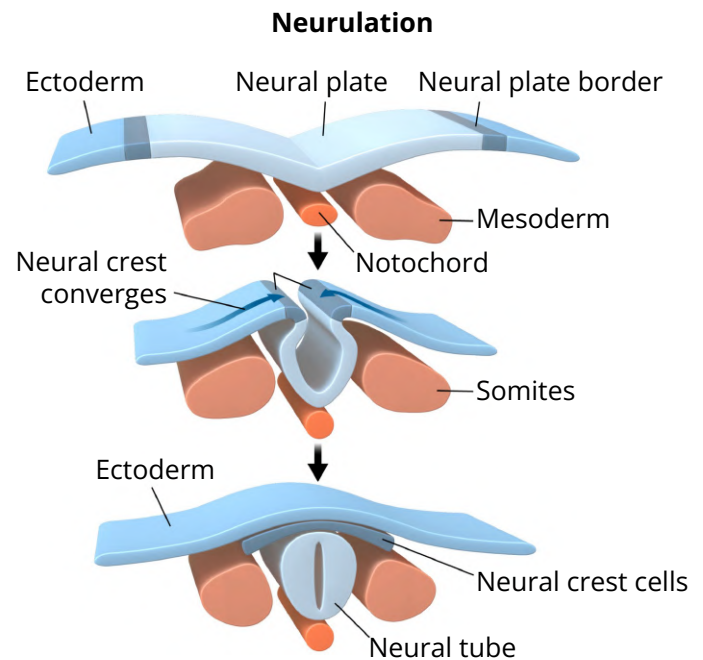
Determination vs. Differentiation



Organogenesis

Organogenesis: Organ development.

Neurulation: Nervous system development. An embryo at this stage is known as a **neurula**.

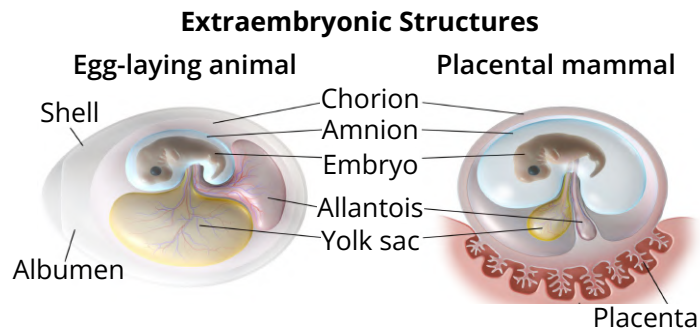


1. The **notochord** (mesoderm-derived) stimulates the ectoderm to thicken, forming the **neural plate**.
2. The neural plate folds onto itself, forming the **neural fold**, or **neural groove**.
3. The neural fold continues to fold until it converges to form a hollow **neural tube**.
 - Some cells roll off to form **neural crest cells** (migrate to form teeth, craniofacial bones, skin pigmentation, etc.).
4. The neural tube differentiates into the central nervous system.
 - **Mesoderm cells (somites)** form two masses alongside the notochord that later develop into vertebrae and skeletal muscles associated with the axial skeleton.

Extraembryonic Development

Placental mammals have internal pregnancies while **egg-laying animals** such as reptiles, birds, and monotremes (egg-laying mammals) lay eggs.

During embryo development, placental and egg-laying animals are supported by four **extraembryonic** structures:



1. **Amnion:** Innermost membrane layer. Secretes amniotic fluid, a protective fluid cushion around the embryo.
 - **Amniotes** (reptiles, mammals, birds, marsupials) have an amnion, while **anamniotes** (amphibians, fish) do *not*.
2. **Chorion:** Outermost layer.
 - **Placental mammals:** Forms the fetal half of the **placenta**, allowing exchange of gases, nutrients, and waste.
 - **Egg-laying animals:** Membrane for gas exchange just underneath egg shell.
3. **Allantois:** Outpouching of the archenteron.
 - **Placental mammals:** Transports waste to placenta, contributes to the **umbilical cord**, and later forms part of the **urinary bladder**.
 - **Egg-laying animals:** Stores uric acid (waste); later fuses with the chorion to aid in gas exchange.
4. **Yolk sac:**
 - **Placental mammals:** No yolk. Functions temporarily until the placenta forms, providing early nutrients and serving as the first site of blood cell formation.
 - **Egg-laying animals:** Contains yolk; supplies all nutrients to the developing embryo.

Internal vs. External Reproduction

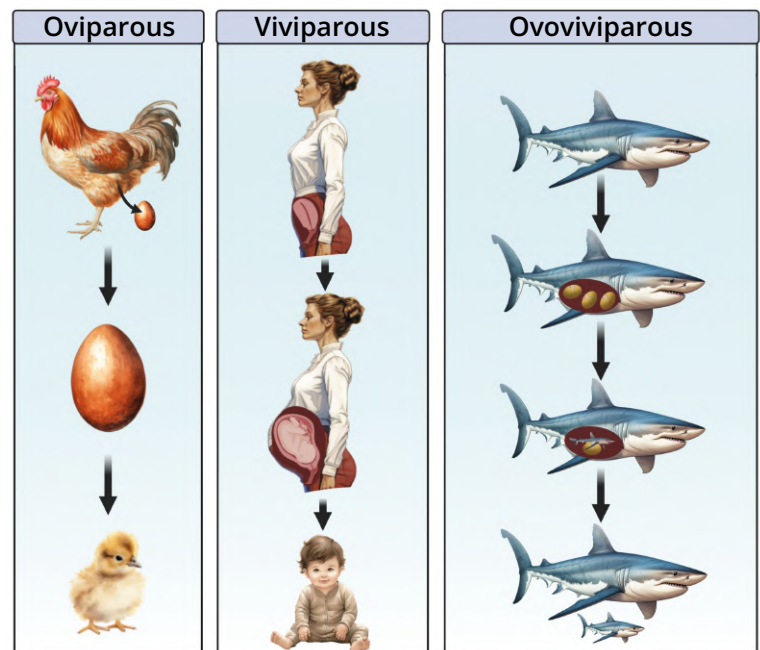
Some organisms exhibit **internal fertilization** (sperm fertilizes egg inside the female's body), while others exhibit **external fertilization** (sperm fertilizes egg outside the female's body). Animals that reproduce via internal fertilization have *more* complex anatomies, produce *fewer* gametes, and have *higher* rates of reproductive success.

Important Animal Reproductive Methods

There are three different types of reproductive methods in animals:

- **Oviparity:** Offspring develop in eggs that hatch outside of the mother's body. There is *no* placental connection; embryos rely on yolk. Common in birds, fish, and reptiles.
- **Viviparity:** Offspring develop inside the mother's body and are nourished via a placental connection. Most common in mammals.
- **Ovoviviparity:** Offspring develop in eggs that hatch within the mother's body. There is *no* placental connection; embryos rely on yolk. Seen in some snakes, fish, and invertebrates.

Reproductive Methods in Animals



Factors Influencing Development

1. **Embryonic induction:** Organizer cells secrete chemicals that influence neighboring cell differentiation.
2. **Homeotic genes:** Act as master controller; turn different gene expressions on/off to regulate correct development of embryonic segments.
 - **HOX genes** are a subset of homeotic genes that are responsible for anterior-posterior (head-to-tail) position of body parts during development.
3. **Apoptosis:** Programmed cell death is important for normal development of the fetus (e.g., removing webbing between fingers) and adults (preventing cancer).
4. **Temperature-dependent sex determination:** Some reptiles determine their sex in development not by chromosomes, but by temperature.
 - **Pattern I** development results in one sex at low temperatures and the opposite sex at high temperatures.
 - **Pattern II** development results in one sex at both low and high temperatures. The other sex develops at intermediate temperatures.

Ch. 12: Diversity of Life

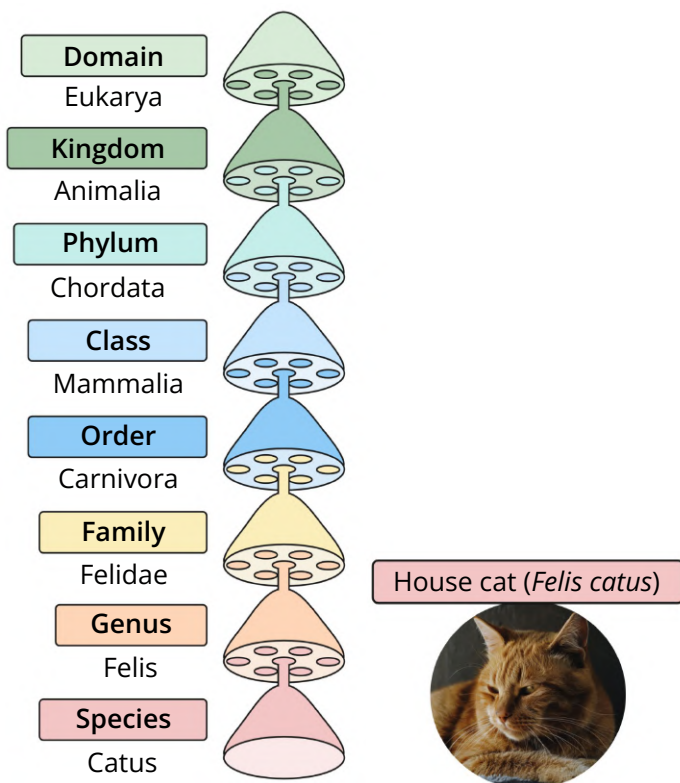
Kingdoms

Table of Contents

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- Domains
- Kingdoms
- Prokaryotes
- Prokaryotes vs. Eukaryotes
- Eukaryotes

Taxonomy

Taxonomy is the science of classifying organisms.



Mnemonic:

Dear **K**ing **P**hillip **C**ame **O**ver **F**or **G**reat **S**oup

Domains

A **domain** is the largest classification of life; the three domains are Archaea, Bacteria (Eubacteria) and Eukarya. **Archaea** are single-celled and prokaryotic. **Bacteria** are also single-celled and prokaryotic. **Eukarya** are classified as having organelles and membrane-bound nuclei.

The six kingdoms are **Archaea, Eubacteria, Protista, Fungi, Plantae, Animalia**.

In the past, bacteria and archaea were grouped in one kingdom, known as **Monera**, due to both groups' lack of membrane-bound organelles.

Prokaryotes

Prokaryotes: Organisms that do *not* have membrane bound nuclei and tend to not have membrane bound organelles (e.g., Eubacteria and Archaea). Even though bacteria and archaea are both prokaryotes, archaea are more closely related to eukaryotes from an evolutionary standpoint.

Prokaryotes do *not* contain steroids. Steroids are only found in the membranes of eukaryotes, not prokaryotes.

Some bacteria can produce an **endospore**, a non-reproductive structure that preserves the cell's genetic material, allowing the cell to survive during times of extreme stress.

Extremophiles are organisms, often archaea, that thrive in extreme environmental conditions (e.g., extreme cold, heat, salinity, pH level) where other life forms cannot survive.

	Archaea	Eubacteria
Type of organism	Prokaryote	Prokaryote
Cell wall & membrane	Lack peptidoglycan in cell wall; contain ether-linkages in membrane	Contain peptidoglycan in cell wall; contain ester-linkages in membrane
Ribosome	70s	70s
DNA organization	Circular with introns and histones	Circular, no introns nor histones
Structural features	Flagella and pili	Flagella and pili
Reproduction	Binary fission	Binary fission

Eubacteria: gram-positive vs. gram-negative

Gram-positive bacteria have a thick peptidoglycan layer in their cell wall, whereas **gram-negative** bacteria have a thin peptidoglycan layer and a second outer membrane. Both are covered by a **capsule** (a virulence factor protecting the bacteria from drying out).

Gram-positive bacteria:

- Stain dark purple.
- Thick peptidoglycan layer in the cell wall.
- No outer membrane.
- Very minor **periplasm** (outside plasma membrane).
- No lipopolysaccharide (LPS - an endotoxin released when bacteria is destroyed).
- Secrete exotoxins.
- Contain **teichoic acids** (polysaccharide connecting peptidoglycan layer and plasma membrane for rigidity and structure).

Mnemonic: PPT

Positive

Peptidoglycan (thick)

Teichoic acid

Gram-negative bacteria:

- Stain pink (due to counterstain).
- Thin peptidoglycan layer in the cell wall.
- Contains periplasm between inner and outer membranes.
- Outer membrane present.
- LPS present.
- Secrete exotoxins.
- No teichoic acids.

Mnemonic: LONG

Lipopolysaccharide

Outer membrane

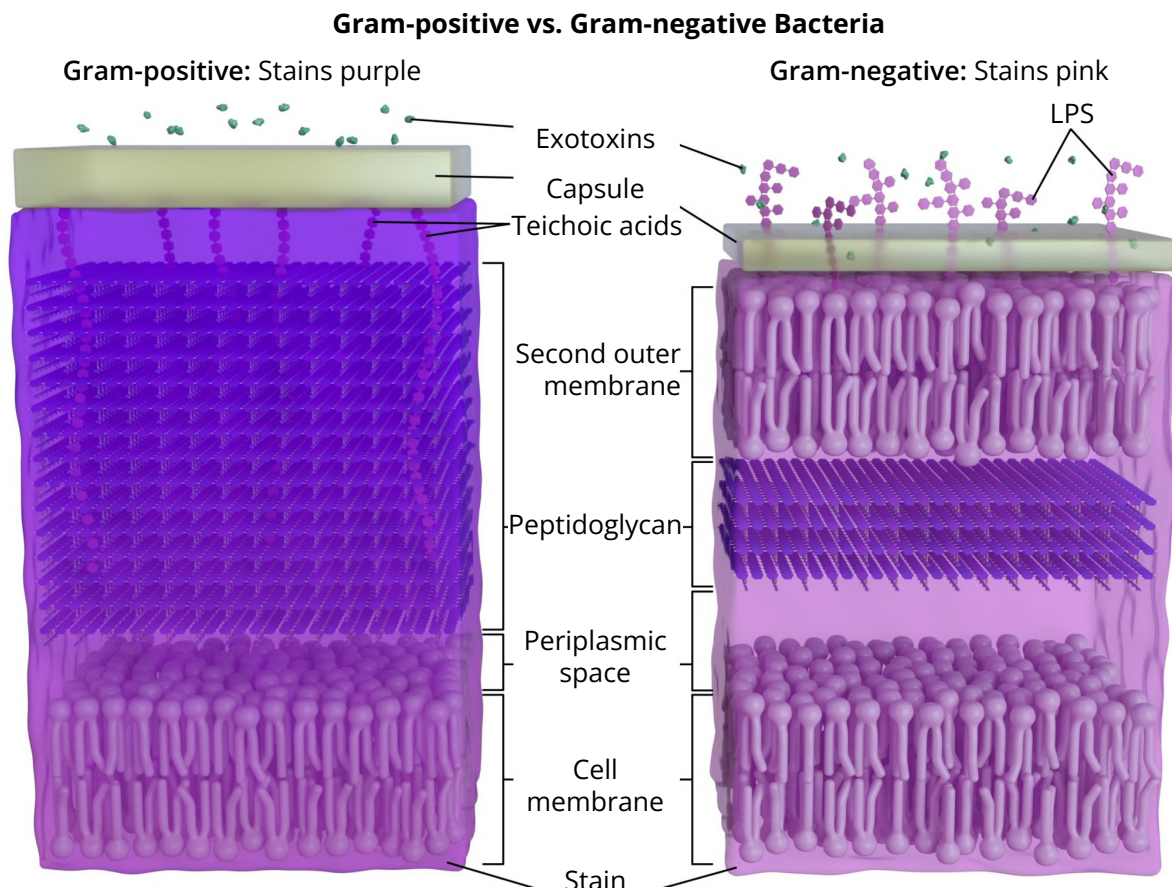
Negative

Gram

Prokaryotes vs. Eukaryotes

Prokaryotes: Organisms that do *not* have membrane bound organelles.

Eukaryotes: Organisms that do have membrane bound organelles.

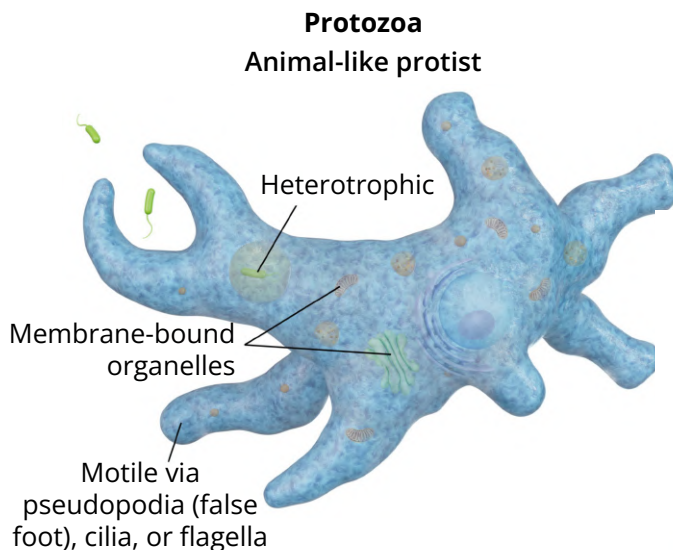


Characteristics	Prokaryotes	Eukaryotes
Membrane-bound organelles	Absent	Present
Cell size (diameter)	Smaller (1 μm)	Larger (10 - 100 μm)
Cell wall	Present	Present only in plants, fungi, and some protists
Cell wall composition	Bacteria: Peptidoglycan Archaea: Polysaccharides	Plants: Cellulose Fungi: Chitin
Steroids in membrane	Absent	Present
Chromosomes	Usually one circular chromosome	Multiple linear chromosomes
Telomere and centromere on chromosome	Absent	Present
Chromosome location	Nucleoid region	Nucleus
Introns and histones	Eubacteria: Absent Archaea: Present	Present
Plasmids	Frequent in eubacteria Sometimes in archaea	Rare
Origin of DNA replication	Eubacteria: One Archaea: Many	Many
DNA replication speed	Fast	Slow
Ribosome	70S (50S and 30S subunits)	80S (60S and 40S subunits)
Transcription and translation	Occur simultaneously in cytoplasm	Transcription first in nucleus, then translation in cytoplasm
Post-transcriptional modification to RNA	No	Yes
Cell cycle length	Short, divide by binary fission	Long, divide by mitosis
Cilia	Absent	Present
Flagella	Present	Present
Cellular respiration location	Glycolysis: Cytosol Pyruvate oxidation: Cytosol Krebs cycle: Cytosol ETC: Cell membrane	Glycolysis: Cytosol Pyruvate oxidation: Mitochondrial matrix Krebs cycle: Mitochondrial matrix ETC: Mitochondrial inner membrane, and thylakoid membrane for plants

Protista

Protists: Kingdom of mostly unicellular eukaryotic organisms.

1. **Fungus-like protists:** Unlike fungi, they lack a cell wall made of chitin and may be motile (e.g., slime molds). Are **saprophytic** and feed via phagocytosis. Reproduce via asexual reproduction and sporulation (resist environmental conditions).
2. **Plant-like protists:** Among the most important primary producers. Algae encompass a large variety of plant-like protists due to having chloroplasts and photosynthetic ability. Include diatoms and dinoflagellates.
3. **Animal-like protists:** Known as **protozoa**, have food vacuoles. Include amoeba and paramecium. Heterotrophic (move via flagella, cilia, or pseudopodia) and are often parasitic pathogens.



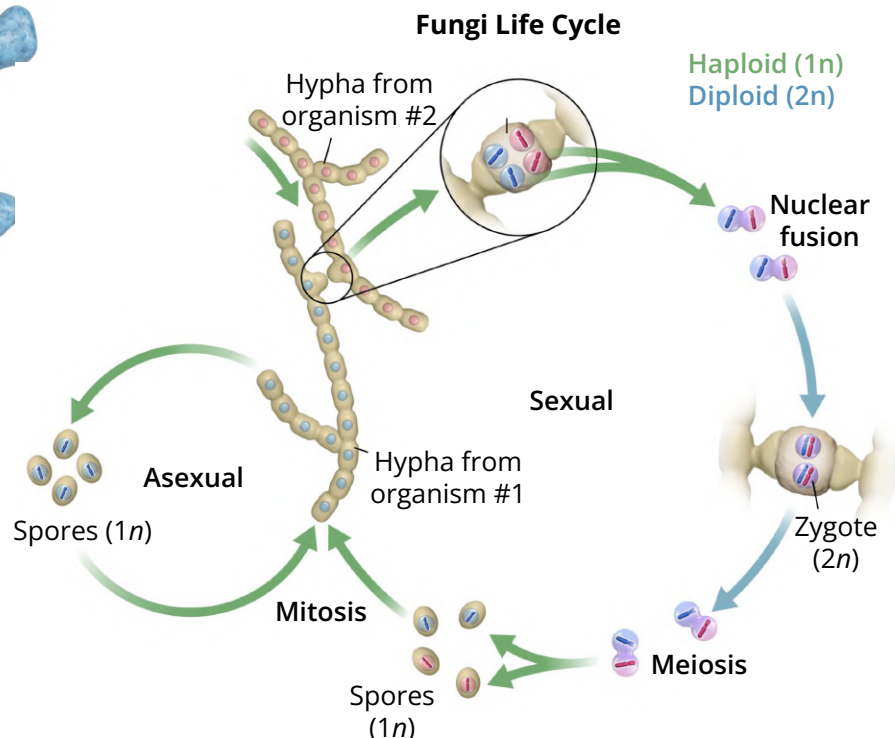
Fungi

Fungi (e.g., mushrooms, molds, yeast) are eukaryotic organisms. They are heterotrophic saprophytes (consume decaying organic matter) that have a predominantly haploid life cycle.

Hyphae are long, branching filaments of fungal cells that extend out to form a network of fungi (mycelium). Fungal cells contain a cell wall made of **chitin**.

Fungi reproduce asexually (by budding or by producing spores via mitosis) in favorable conditions and sexually in unfavorable conditions (to produce genetically different offspring with greater chance of survival). During sexual reproduction, fungi alternate between haploid and diploid stages.

Lichens are **symbiotic autotrophs** where a fungus is paired with either algae or cyanobacteria. The fungus protects the cyanobacteria/algae and provides it with water and nutrients while algae/cyanobacteria photosynthesize, to produce food for the fungi.



Animalia

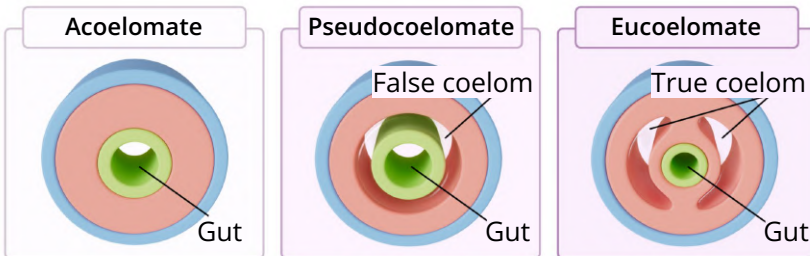
Animals are eukaryotic, diploid, multicellular **heterotrophic aerobes**. They are divided into phyla:

- Porifera
 - Cnidaria
 - Platyhelminthes
 - Nematoda
 - Rotifera
 - Annelida
 - Mollusca
 - Arthropoda
 - Echinodermata
 - Chordata
- Acoelomate
 Pseudocoelomates
 Coelomates

Mnemonic: Privileged Children Play Nicely, Respectfully, And Maturely. Arthur Ensures Cooperation.

Triploblastic animals can be distinguished based on the presence of a **coelom** (fluid-filled cavity). In **coelomates**, mesoderm surrounds the coelom on all sides, whereas in **acoelomates** it does *not*. In **pseudocoelomates**, the coelom is partially surrounded.

Coelom Classifications



Coelomates can be further divided into **schizocoelomates** or **enterocoelomates**. Protostomes are usually schizocoelomates; however, some deuterostomes can also exhibit schizocoely.

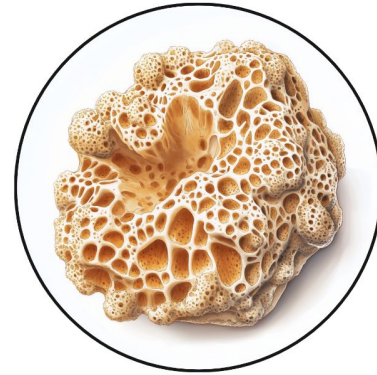
In schizocoelomates, the coelom begins as splits within the mesoderm. In enterocoelomates, the coelom begins as lateral out-pockets of the archenteron.

	Schizocoelomates	Enterocoelomates
Cleavage	Holoblastic, spiral, determinate	Radial, indeterminate
Occurrence	Protostomes	Deuterostomes

Porifera:

- E.g., sponge.
- **Body symmetry:** Asymmetrical.
- **Tissue organization:** No true tissues (parazoan).
- **Other important facts:** Sessile (non-motile), aquatic habitats, earliest animals.

Porifera



Cnidaria:

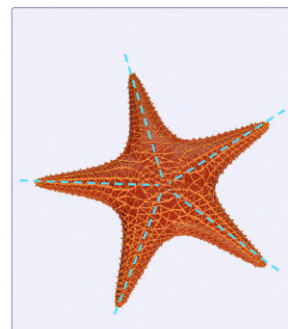
- E.g., hydra, jellyfish.
- **Body symmetry:** Radial (around central axis).
- **Tissue organization:** Diploblast (two cellular layers).
- **Other important facts:** Aquatic habitats.

Cnidaria

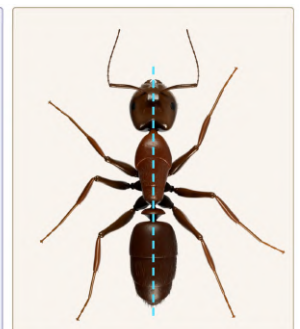


Radial vs. Bilateral Symmetry

Radial



Bilateral



Platyhelminthes:

- E.g., flatworm, tapeworm.
- **Body symmetry:** **Bilateral** (right and left halves).
- **Cephalization** (concentration of nervous and sensory organs in the head) is associated with bilateral symmetry.
- **Tissue organization:** Triploblast (three germ layers); acoelomate.
- **Embryonic development:** **Protostome** (blastopore forms mouth).
- **Other important facts:** Parasitic lifestyles, most primitive triploblast, has organs, non-segmented worms.

Platyhelminthes



Nematoda:

- E.g., roundworm.
- **Body symmetry:** Bilateral.
- **Tissue organization:** Triploblast; pseudocoelomate.
- **Embryonic development:** Protostome.
- **Other important facts:** Parasitic, non-segmented.

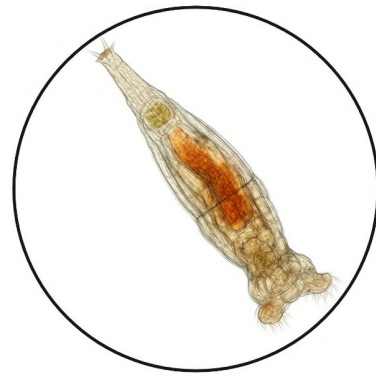
Nematoda



Rotifera:

- E.g., rotifers.
- **Body symmetry:** Bilateral.
- **Tissue organization:** Triploblast; pseudocoelomate.
- **Embryonic development:** Protostome.
- **Other important facts:** Non-segmented, aquatic zooplankton.

Rotifera



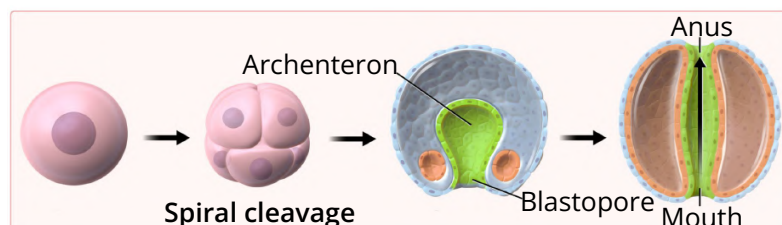
Annelida:

- E.g., earthworm, leech.
- **Body symmetry:** Bilateral.
- **Tissue organization:** Triploblast; coelomate.
- **Embryonic development:** Protostome.
- **Other important facts:** The first phyla to develop segmentation. Closed circulatory system (blood pumped through vessels by heart) with multiple pairs of aortic arches, distinct arteries and veins.

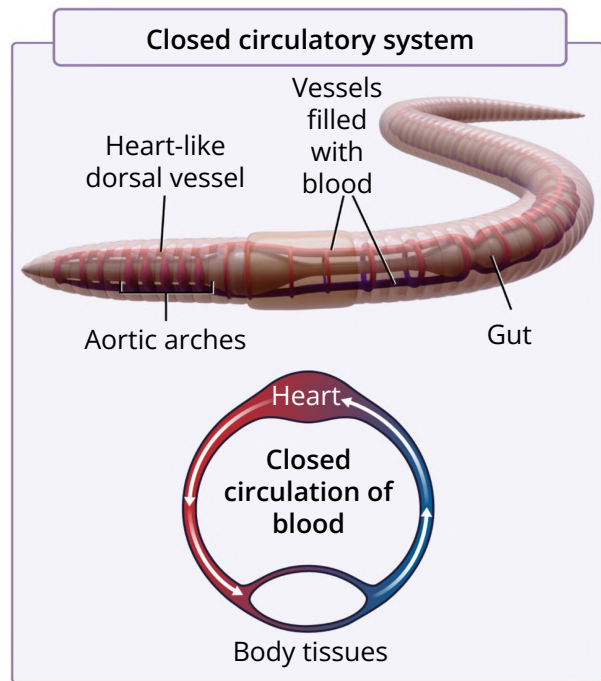
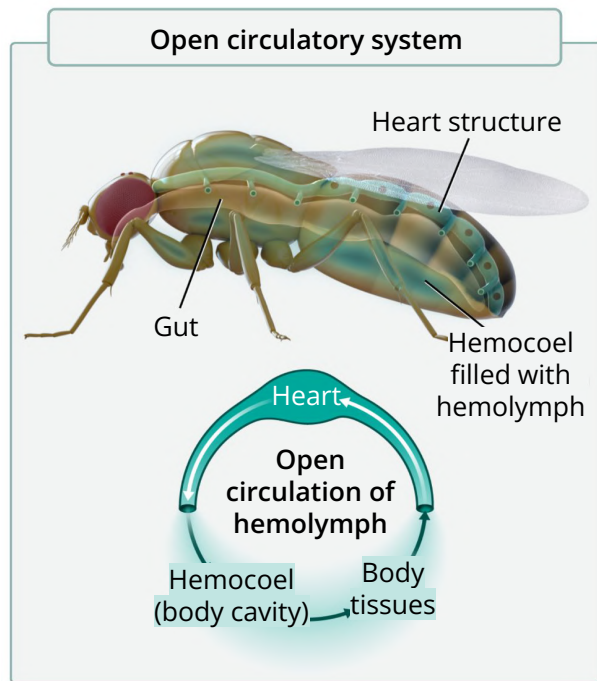
Annelida



Protostome



Open vs. Closed Circulatory Systems



Mollusca:

- E.g., snail, squid, octopus.
- **Body symmetry:** Bilateral.
- **Tissue organization:** Triploblast; coelomate.
- **Embryonic development:** Protostome.
- **Other important facts:** Mostly aquatic, soft-bodied, can have shell.

Mollusca



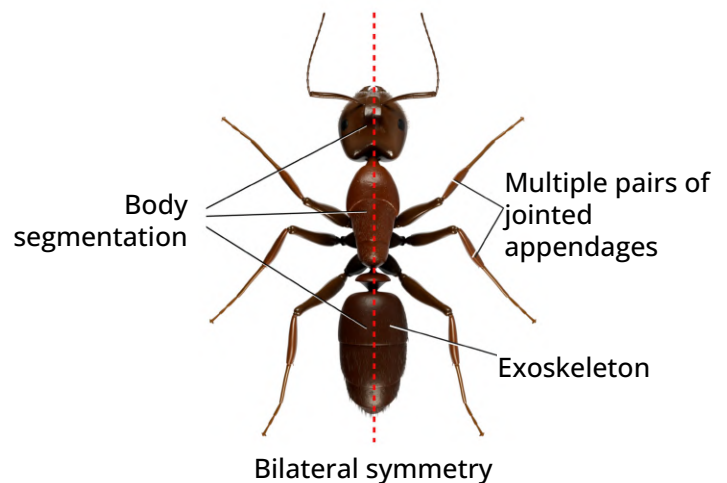
Arthropoda:

- **Body symmetry:** Bilateral.
- **Tissue organization:** Triploblast; coelomate.
- **Embryonic development:** Protostome.
- **Other important facts:** Most diverse phylum, highest number of species, chitin exoskeleton, jointed appendages, segmented bodies. Open circulatory system with hemocoel and hemolymph ("blood").

Types of arthropods:

1. **Insecta** (e.g., ant, grasshopper)
2. **Arachnida** (e.g., spider, scorpion)
3. **Crustacea** (e.g., lobster, crab)

Arthropoda



Echinodermata:

- E.g., starfish, sea urchin.
- **Body symmetry:** Radial (adults).
- **Tissue organization:** Triploblast; coelomate.
- **Embryonic development:** Deuterostome (blastopore forms anus).
- **Other important facts:** Closest related major phyla to chordates.

Echinodermata



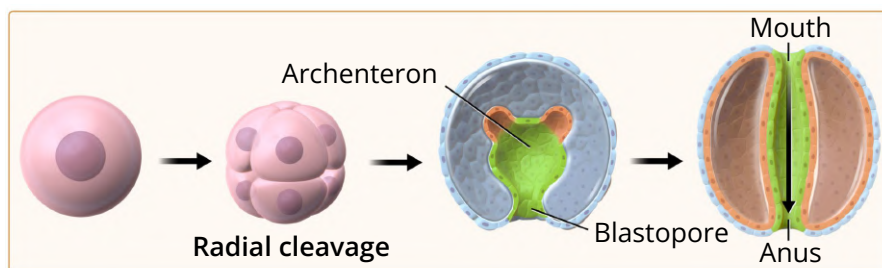
Chordates:

- E.g., vertebrates.
- **Body symmetry:** Bilateral.
- **Tissue organization:** Triploblast; coelomate.
- **Embryonic development:** Deuterostome.
- **Other important facts:** All chordates have segmented bodies.

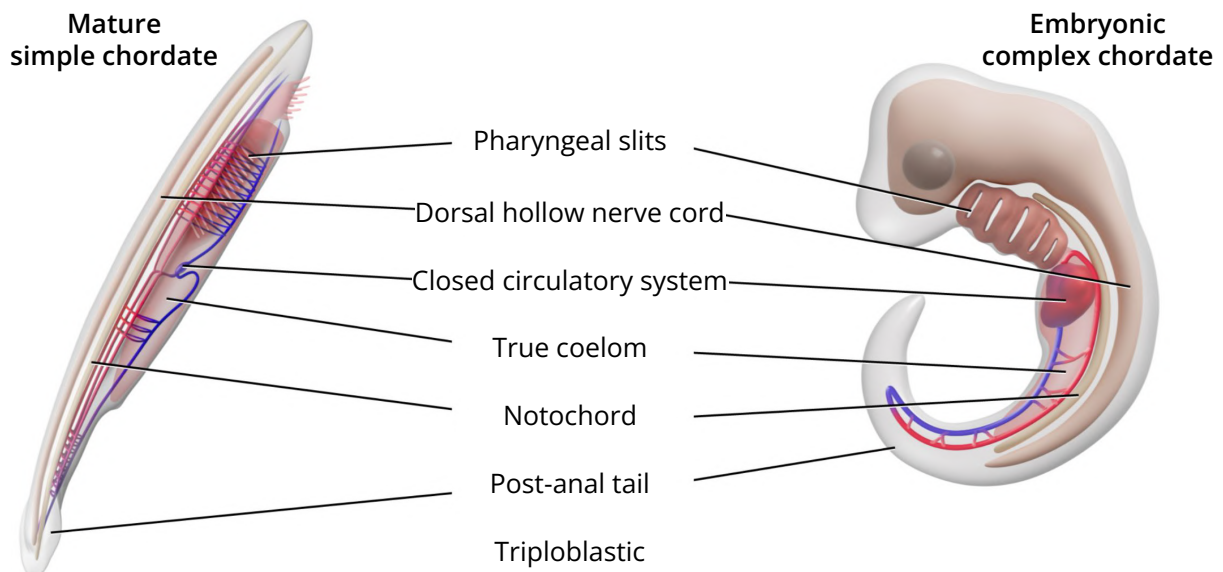
Shared traits of all chordates:

1. **Dorsal hollow nerve cord:** Forms the central nervous system (brain and spinal cord).
2. **Notochord:** Cartilaginous rod derived from mesoderm. Forms the primitive axis and supports the body during embryonic development. Develops into the spinal bones (*not* spinal cord).
3. **Pharyngeal pouch (slits):** Forms pharynx, gills, feeding structures, or head and neck structures.
4. **Post-anal tail:** Lost during embryonic development in humans and many other chordates.

Deuterostome



Chordata Characteristics



Types of chordates:

Invertebrates: Have the same developmental characteristics as other chordates but lack vertebrae. Lancelets and tunicates are early chordates that do not possess a vertebral column.

Vertebrates: Vertebrae observed in muscular and skeletal systems.

Today's living vertebrates evolved from earlier chordates, acquiring novel characteristics over time: vertebrae, jaws, bony skeletons, lungs, limbs, etc.

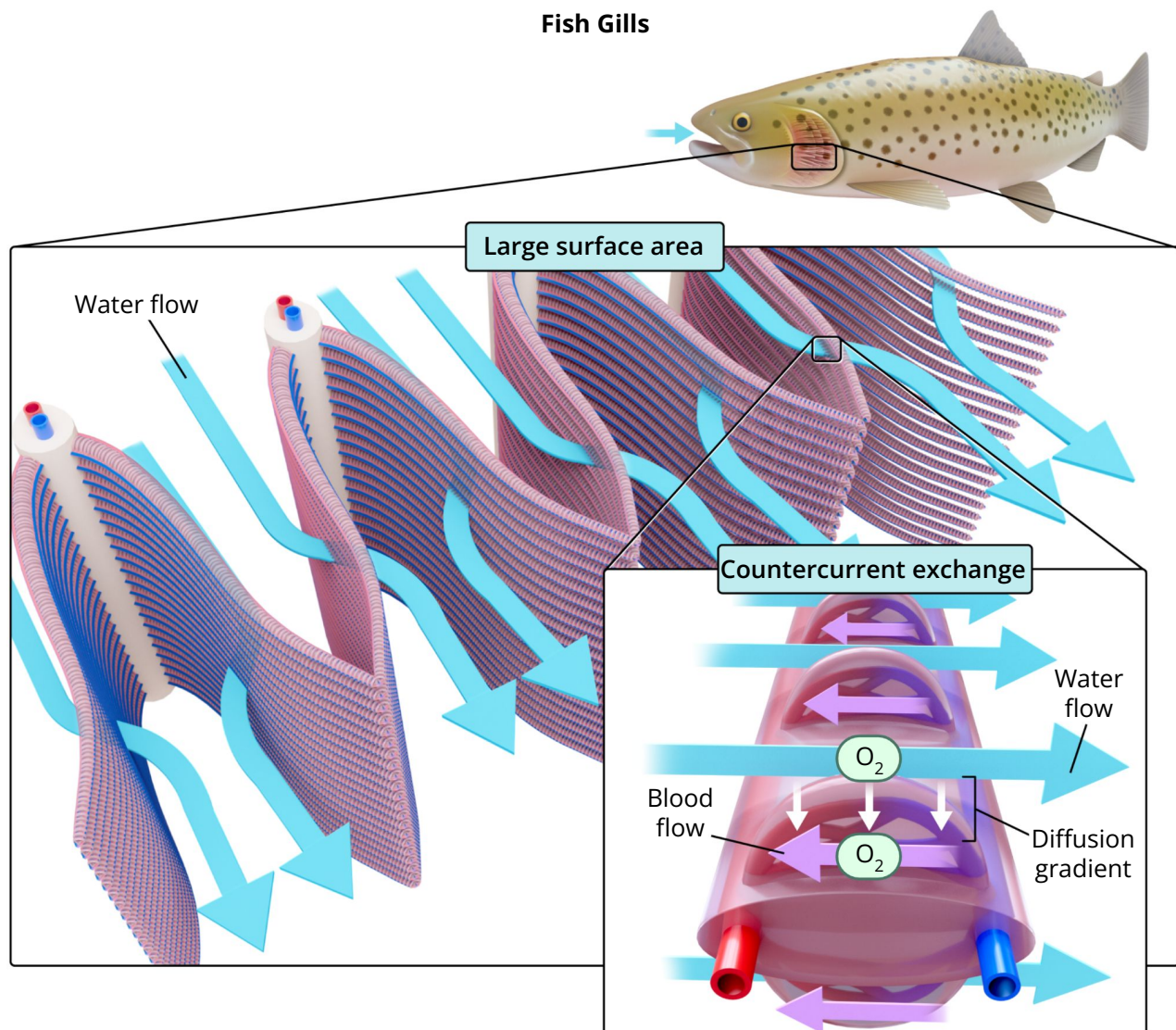
The earliest vertebrates include fish, beginning with jawless fish, followed by cartilaginous fish, or **chondrichthyes** (e.g., shark). Bony fish (**osteichthyes**) soon developed into **lobe-finned** fish, whose fins bore the first resemblance to limbs. Amphibians were the first of the tetrapods to develop proper limbs with digits.

Mammalia are a class of animals in the subphylum Vertebrata. Mammals may be the most evolved vertebrates.

Like all vertebrates, mammals have a closed circulatory system. Along with birds and some reptiles, their circulatory system includes a four-chambered heart. Mammals have several unique characteristics of their own: hair, mammary glands (milk production), and diaphragms.

While mammals and most tetrapods have lungs as part of their respiratory system, fish utilize **countercurrent exchange** to breathe.

Countercurrent exchange describes how oxygen is efficiently absorbed by gills from water. Water flows over gills in an opposite direction to blood flow in vessels. This creates a constant diffusion gradient and optimizes absorption.



Ch. 13: Evolution

Theory of Evolution

Table of Contents

- Evidence of Evolution
- Theory of Evolution
- Natural Selection
- Gene Equilibrium
- Microevolution
- Macroevolution
- Phylogenetic Trees

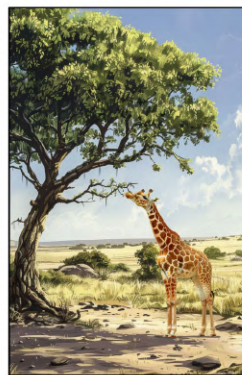
Evolution is the gradual development and change of heritable traits (**allele frequencies**) in populations over successive generations. Evolution *increases* biodiversity.

Evidence of Evolution

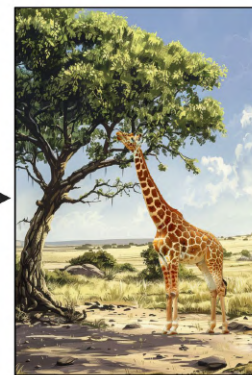
1. **Paleontology** is the study of fossils through actual remains of the animal or their traces. The development of species through time can be observed by comparing the deepest (oldest) fossils to the shallowest (youngest).
2. **Biogeographic evidence**, the distribution of different species around the world, can be observed and analyzed for similarities and differences.
3. **Embryology** allows the observation of embryological similarities and differences between early stages of related organisms (e.g., all chordates have a gill slit during development).
4. **Comparative anatomy** compares different body parts of different animals.
 - **Homologous structures:** May or may not perform the same function but share a common ancestor (e.g., forearm of bird and forearm of human).
 - **Analogous structures:** Same function, do *not* have a common ancestor (e.g., bird wings and bat wings).
 - **Vestigial structures:** Serve *no* purpose but are homologous to functional structures in other organisms (e.g., human appendix and cow cecum).
5. **Biochemical methods** enable DNA sequence comparisons, revealing conserved DNA sequences (more similarity $\Rightarrow$ higher relatedness) and common conserved pathways (e.g., Krebs cycle).

1. **Cuvier** proposed **catastrophism**. Catastrophes lead to mass extinctions in specific areas. Local populations are shaped by the repopulation of organisms that randomly survived the catastrophes.
2. **Lamarck** proposed:
 - **Use and disuse:** Used body parts develop while unused parts weaken, leading to evolution.
 - **Inheritance of acquired traits:** Traits acquired through use and disuse are passed down to offspring (e.g., giraffe stretches neck and produces long-necked offspring). This is incorrect. Acquired characteristics are generally *not* heritable.
3. **Darwin** proposed the theory of **natural selection**. Random genetic variations amongst individuals can lead to differences in fitness.

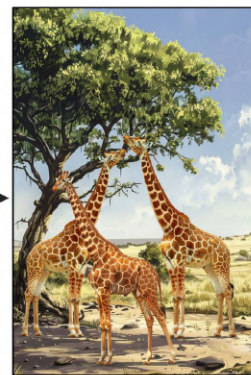
Lamarck's Incorrect Theory



Giraffe must stretch neck to reach high leaves



Develop used traits



Inheritance of acquired traits

Natural Selection



Random mutation causes white pepper pattern



White moths have higher survival



Over time, white moths increase

Natural Selection

Natural selection is the gradual, non-random process where allele frequencies change as a result of environmental interaction. Also known as **survival of the fittest**, individuals with higher **fitness** (ability to survive and produce fertile offspring) pass on more DNA to future generations compared to less fit individuals. Leads to the evolution of the population (*not* individuals).

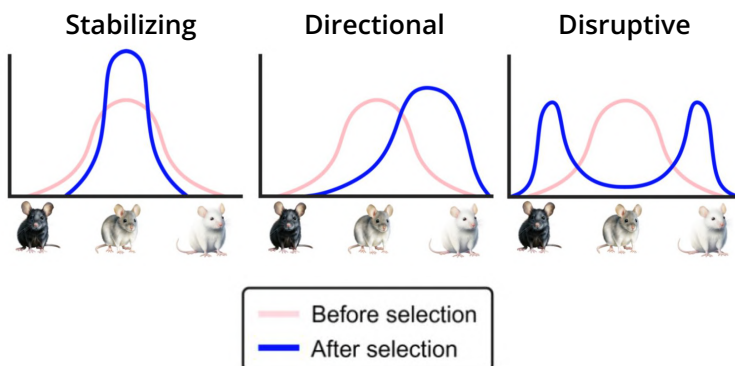
Requirements for natural selection:

1. Resource demand exceeds supply, resulting in competition for survival.
2. Variation of traits exists within a population.
3. Variation in traits must be genetically-influenced (**heritable**).
4. Trait variation must be influential for reproduction and/or survival (fitness). Genes improving fitness are favored and *increase* over generations and vice versa (e.g., during the Industrial Revolution black peppered moths camouflaged better than white moths).

Types of natural selection:

1. **Stabilizing selection:** Mainstream (average) is favored (e.g., birth weight). Diagram follows a standard bell curve.
2. **Directional selection:** One extreme favored (e.g., longest giraffe neck allows access to the most leaves).
3. **Disruptive selection:** Rare traits favored, mainstream is not favored (e.g., snails living in low and high vegetation areas).

Types of Natural Selection



Other types of selection:

Sexual selection: Non-random mating between males and females. Females favor high quality partners, while males prefer high quantity of partners to increase their number of offspring. Traits selected for may be favorable for reproduction but not for survival.

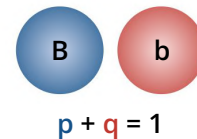
Artificial selection: Carried out by humans to selectively breed for specific traits (e.g., dog breeding).

Gene Equilibrium (No Evolution)

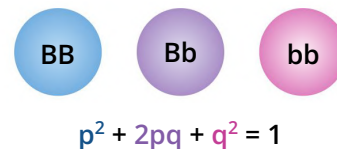
The Hardy-Weinberg formula calculates genetic frequency during **genetic equilibrium** (*no* change in gene frequencies). If both equations hold true, the population is under **Hardy-Weinberg equilibrium**.

Hardy-Weinberg Equilibrium

Allele frequency



Genotype frequency



The requirements for Hardy-Weinberg equilibrium are:

Mnemonic: Large, Random, M&M

- **Large population:** Minimizes genetic drift.
- **Random mating**
- **No mutation**
- **No natural selection**
- **No migration (gene flow):** Population must be isolated.

When conditions are not met, evolution occurs.

Microevolution

Microevolution is the process of gene frequency change within a population over a few generations.

Factors causing microevolution:

1. **Genetic drift:** Random changes in allele frequencies. Stronger effects in small populations. Two key types:
 - **Bottleneck effect:** A sudden reduction in population size (e.g., natural disaster), resulting in the loss of certain alleles.
 - **Founder effect:** A few individuals separate from a larger population, forming a new population with different allele frequencies.
2. **Non-random mating:** Mating is not random; includes inbreeding, outbreeding, and sexual selection.
3. **Mutations:** Random changes to DNA introduce new alleles that may be neutral, beneficial, or harmful.
4. **Natural selection:** Non-random survival. Alleles with a fitness advantage increase in frequency.
5. **Gene flow:** Movement and mixing of alleles between populations via migration and interbreeding. Increases genetic diversity.

Sources of **genetic variation**:

1. **Mutation:** Must *not* be fatal.
2. **Sexual reproduction:** Crossing over, independent assortment, and random joining of gametes.
3. **Balanced polymorphism:** Maintains a variety of phenotypes within a population.
 - **Heterozygote advantage:** Two parents produce an offspring that is more fit than either parent (e.g., heterozygous sickle cell anemia advantage against malaria)
 - **Minority advantage:** Rare phenotypes offer higher fitness. Cycle between high and low frequency (e.g., advantageous against hunters' search images).
 - **Neutral variations:** May become beneficial if the environment changes.
4. **Polyploidy:** Plants have sets of multiple chromosomes introducing more variety and preserving different alleles. Can also mask effects of a harmful recessive allele.

Macroevolution

Macroevolution is long-term and occurs at or above the species level. Species are **reproductively isolated** (via prezygotic and postzygotic isolating mechanisms) resulting in a lack of gene flow between species.

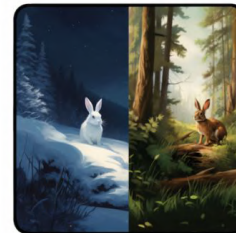
Prezygotic isolation mechanisms are barriers that prevent fertilization and zygote formation from occurring between species:

- **Habitat isolation:** Occupying different habitats.
- **Temporal isolation:** Reproducing at different times/seasons.
- **Behavioral isolation:** Different courtship rituals.
- **Mechanical isolation:** Male and female genitalia are *not* compatible.
- **Gamete isolation (incompatibility):** Gametes do *not* recognize or are unable to fertilize each other.

Prezygotic Isolation



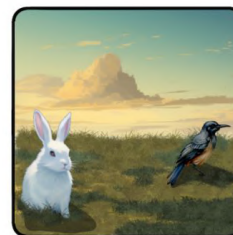
Habitat



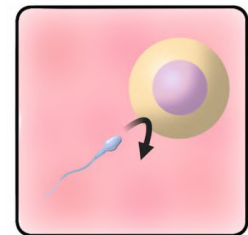
Temporal



Behavioral



Mechanical



Gamete
incompatibility

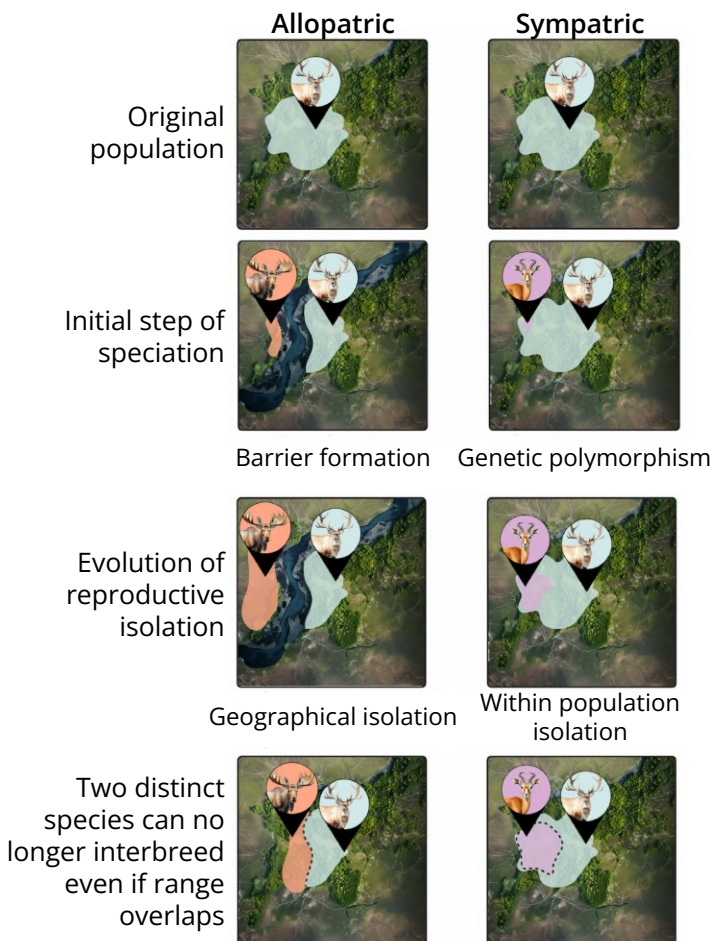
Postzygotic isolation refers to barriers to organism success after a zygote has formed:

- **Hybrid mortality (inviability):** A zygote forms but is not viable and cannot survive past the embryo stage.
- **Hybrid sterility:** Hybrid zygote is sterile (infertile), e.g., mule.
- **Hybrid F2 breakdown:** First hybrid generation is fertile but the following F2 generation has reduced fitness.

Speciation is how species form, starting with **reproductive isolation**, which leads to interruption of gene flow between populations that gradually develop into two species.

1. **Allopatric speciation:** Speciation due to a geographical barrier.
 - **Adaptive radiation:** Many species arise from one ancestor as they adapt differently to their environments. Species can also specialize to fill different niches within the same environment.
2. **Sympatric speciation:** Speciation without a geographical barrier.
 - **Balanced polymorphism:** Occurs when different phenotypes are isolated within the same area.
 - **Polyploidy:** Speciation occurs when two, usually sterile, polyploid organisms (e.g., plant meiosis nondisjunction) meet and are reproductively compatible.
 - **Hybridization:** Occurs when hybrids are *more* fit than purebreds.

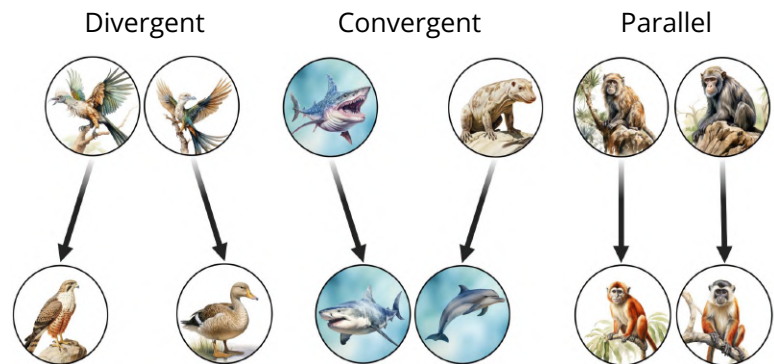
Allopatric vs. Sympatric Speciation



Theories of macroevolution:

1. **Phyletic gradualism:** Evolution happened gradually via accumulation of small intermediary changes. *Not* likely to be true (*not* supported by fossil evidence).
2. **Punctuated equilibrium:** Short spurts of evolutionary changes in between long periods of **stasis** (supported by fossil evidence).

Patterns of Evolution



Patterns of evolution:

1. **Divergent evolution:** Species diverge from a common ancestor.
2. **Convergent evolution:** Unrelated species adapt to similar environments becoming *more* alike (analogous structures).
3. **Parallel evolution:** Species diverge from a common ancestor but undergo similar changes.
4. **Coevolution:** Two species impart selective pressure on each other. Classic examples are hummingbirds and flowers.
 - **Camouflage:** Match appearance to environment to avoid detection. Strictly visual method of concealment.
 - **Aposematic coloration (warning coloration):** Vibrant coloration in poisonous animals to warn predators.
 - **Mimicry:** Evolving to resemble another species. In **Batesian mimicry**, a non-harmful animal resembles a harmful one. In **Mullerian mimicry**, two poisonous animals resemble each other to warn their predator.

Phylogenetic Trees

A **phylogenetic tree** or a **cladogram** is a branched diagram that shows inferred evolutionary relationships between different taxonomic groups. These diagrams may be created based on physical characteristics and/or genetics. The branch lengths may or may not indicate time.

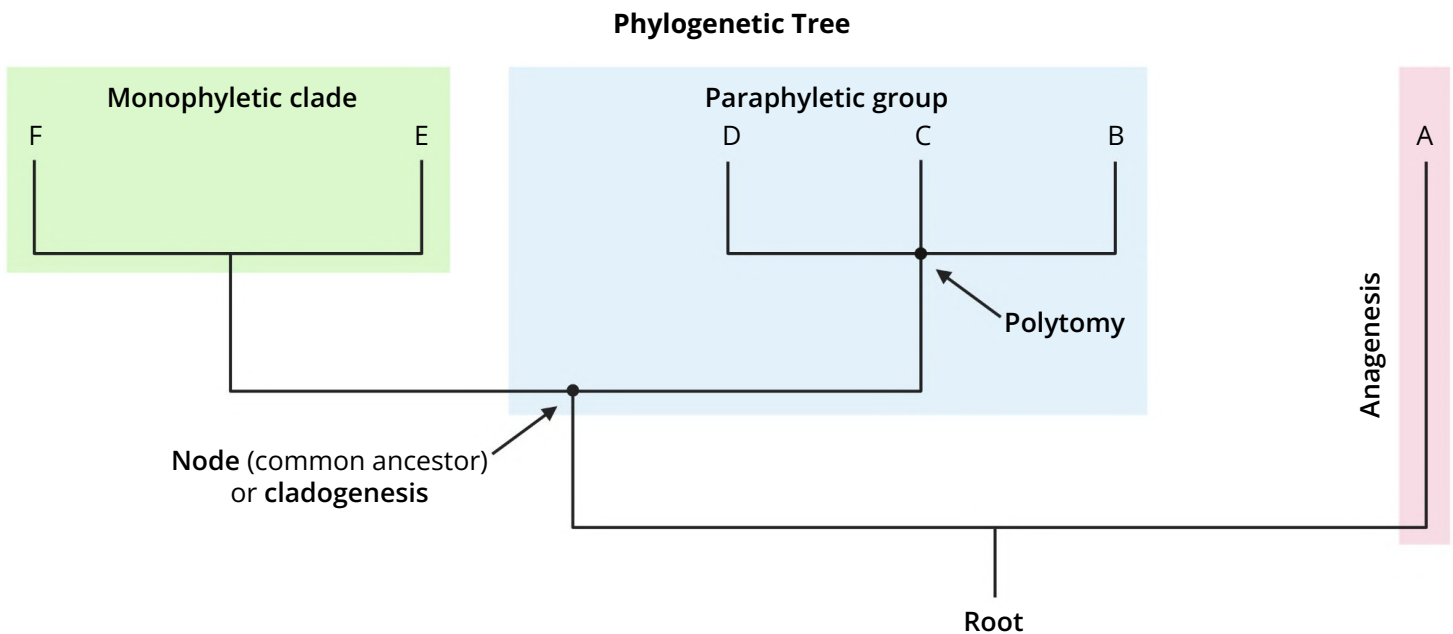
The **root** represents the most common ancestor of all the lineages shown. A **node** is a branch point that represents the splitting (**divergence**) of a single group into (usually) two descendant groups. A **polytomy** is an internal node of a phylogenetic tree that leads to more than two tips.

A **clade** or a **monophyletic group** is a cluster with an ancestor and all of its descendants. A **paraphyletic group** highlights an ancestor and some but not all of its descendants.

Cladogenesis refers to the divergence of evolutionary lineages (formation of new clades).

Anagenesis describes the gradual evolution of an interbreeding population without splitting.

When inferring evolutionary relationships, **parsimony** means the simpler the evolutionary explanation, the better. Phylogenetic trees minimizing evolutionary reversals, convergent evolution and parallel evolution are preferred.



Ch. 14: Ecology

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- Key Terms
- Ecological Niche
- Biological Interactions
- Ecosystem Ecology
- Population Ecology
- Ecological Succession
- Eutrophication
- Biomes

Key Terms

- **Abiotic factors:** Nonliving elements of an ecosystem (e.g., temperature, water, light).
- **Biotic factors:** Living elements of an ecosystem (e.g., plants, animals, etc.).
- **Species:** A group that can interbreed and have viable, fertile offspring.
- **Population:** A specific species living in a specific location.
- **Habitat:** The type of environment where a specific organism lives. Includes other organisms (biotic) and physical aspects (abiotic).
- **Ecological community:** All the populations in a given area.
- **Ecosystem:** System of all the organisms in an ecological community (biotic) and their interactions with the surrounding abiotic factors.
- **Density-dependent factors:** Depend on population density (e.g., disease, resource competition).
- **Density-independent factors:** Do *not* depend on population density (e.g., climate, weather).
- **Biome:** A large area of land or water defined by its biotic factors which are adapted to the climate and geography of that region.

Ecological Niche

A **niche** is a species' unique position and role within its ecosystem, defining how it interacts with its environment and other species to survive and reproduce successfully. A **fundamental niche** encompasses the full range of ecological conditions in which a species could theoretically survive. In contrast, a **realized niche** is the environment in which a species actually lives.

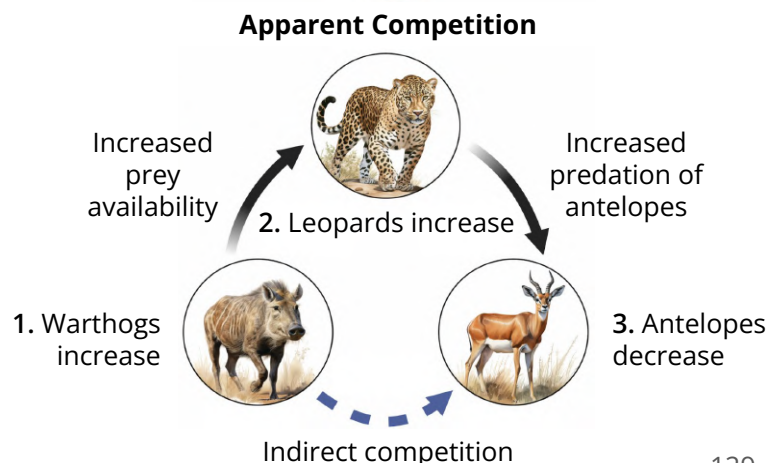
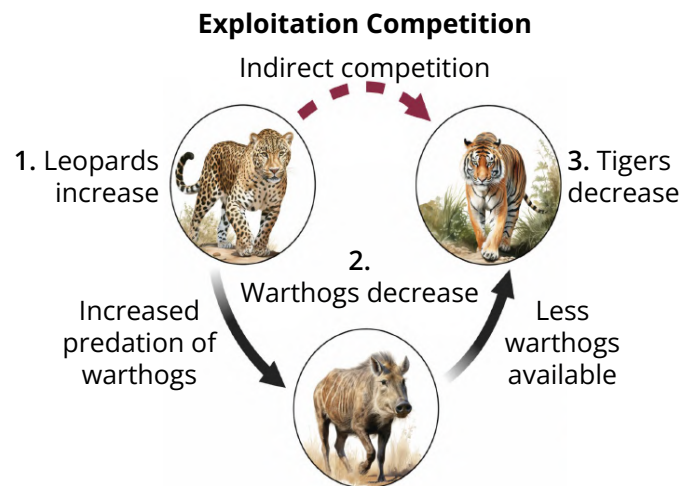
Gause's law (competitive exclusion principle):

Two species *cannot* occupy the same niche and maintain population levels (one will outcompete the other). **Resource partitioning** allows species to coexist by utilizing resources in different ways, times, or places.

Biological Interactions

In **competition**, two species compete for the same resources.

- **Intraspecific competition** occurs between members of the same species (e.g., two rabbits competing for carrots).
- **Exploitation competition** is indirect and occurs when resources are depleted (e.g., cheetahs deplete gazelle population, affecting lions).
- **Apparent competition** is indirect and occurs between two species of prey of one predator.
- **Interference competition** occurs when one species directly prevents the establishment of another species that would compete for a mutual resource.
 - **Allelopathy** is a type of interference competition used by plants where a toxic chemical is secreted to prevent such establishment.



Symbiosis (living together) is a close, long-term interaction between two organisms (symbionts).

- **Mutualism (+/+)**: Both organisms benefit (e.g., oxpecker bird eating ticks off a rhino).
- **Commensalism (+/0)**: One organism benefits and the other is unaffected (e.g., jackal eating a tiger's leftovers).
- **Parasitism (+/-)**: One organism benefits at the other's expense (e.g., tapeworm in human gastrointestinal tract).

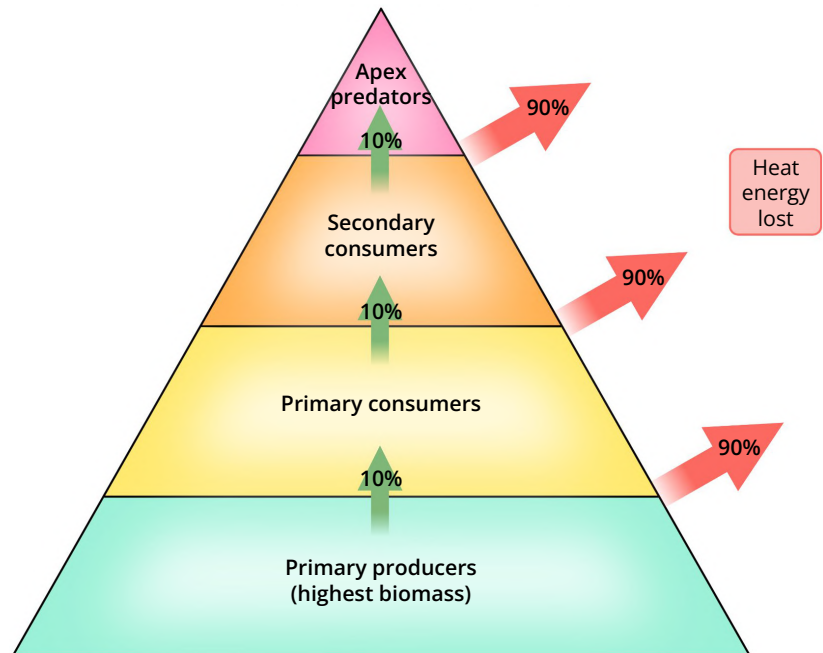
Ecosystem Ecology

- **Food chain**: Linear depiction of what eats what (e.g., carrot → rabbit → fox → lion).
- **Food web**: More complex diagram depicting interconnections between food chains.
- **Trophic level**: An organism's position within a food chain or food web.
- **Ecological pyramid**: Graphical representation of the accumulation of biomass at each trophic level.
- **Autotroph**: Produces organic compounds from abiotic factors (sunlight, water, CO₂, etc.).
- **Heterotroph**: Must ingest organic compounds to generate energy and survive.
- **Predation**: Relationship between **predator** (hunter) and **prey** (hunted - plant or animal).
- **Herbivore**: Organism that eats plants. Has a long intestinal tract and cecum to digest plants.
- **Carnivore**: Organism that eats meat.
- **Omnivore**: Organism that eats plants and meat.
- **Invasive species**: Non-native species that outcompetes native species and overtakes ecosystem.
- **Noninvasive species**: Non-native species that survives in but does *not* overrun an ecosystem.
- **Naturalized species**: A type of non-native species that spreads beyond origin that showcases sufficient reproduction to maintain its population.
- **Exotic species**: A type of non-native species that lives outside of its native-distributional range due to human activity.

Primary producers, at the lowest trophic level, are autotrophs undergoing energy production (e.g., photosynthesis) to generate the **biomass** of an ecosystem. **Consumers** (higher trophic levels) eat producers or other consumers.

Primary consumers (often herbivores) are just above producers. **Secondary consumers** (carnivores) prey on primary consumers. **Tertiary consumers** prey on secondary consumers.

Trophic Levels and Energy Transfer



An **apex predator** is at the top of the chain (tertiary consumer or higher).

Only ~10% of energy stored in a trophic level is converted into organic tissue in the next trophic level, while ~90% of it is lost as heat due to inefficient energy transfer.

Scavengers consume dead plants or animals (e.g., vultures, some beetles). **Decomposers** break down and recycle dead material from dead organisms (e.g., fungi (**saprophytes**), worms (**detritivores**)). **Detritus** refers to decomposing organic matter.

Organisms can also be classified by their energy and carbon source:

Mode of nutrition	Energy source	Carbon source
Photoautotroph	Light	CO ₂
Chemoautotroph	Inorganic compounds	CO ₂
Photoheterotroph	Light	Organic compounds
Chemoheterotroph	Organic compounds	Organic compounds

Autotrophs are able to fix carbon, therefore they use inorganic CO₂ as their primary carbon source.

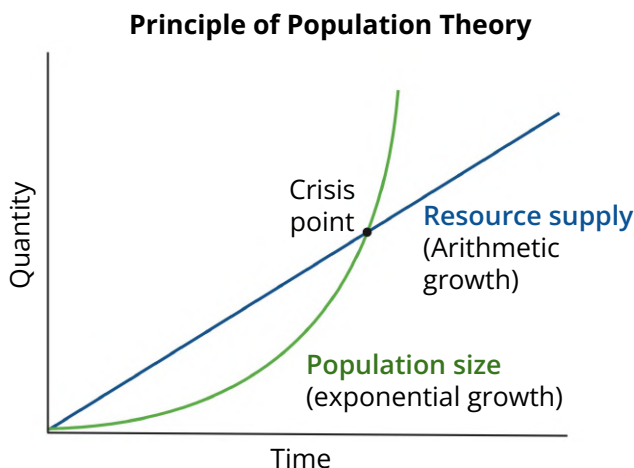
Heterotrophs, however, are unable to fix carbon and derive their carbon from organic compounds.

Population Ecology

Population dynamics explores how populations change in space and time and how they interact with their environment.

- **Biotic potential:** A species' ability to undergo its highest population growth (highest births, lowest deaths) when conditions are ideal.
- **Carrying capacity:** The maximum population size an ecosystem can sustain.

Malthus was a scientist best known for his **principle of population theory**. This theory proposes that any living species increases exponentially (2, 4, 16, 132...) while food production increases linearly (2, 4, 6, 8...) and will *not* be able to keep up with population growth.



r/K selection theory:

K-selected species

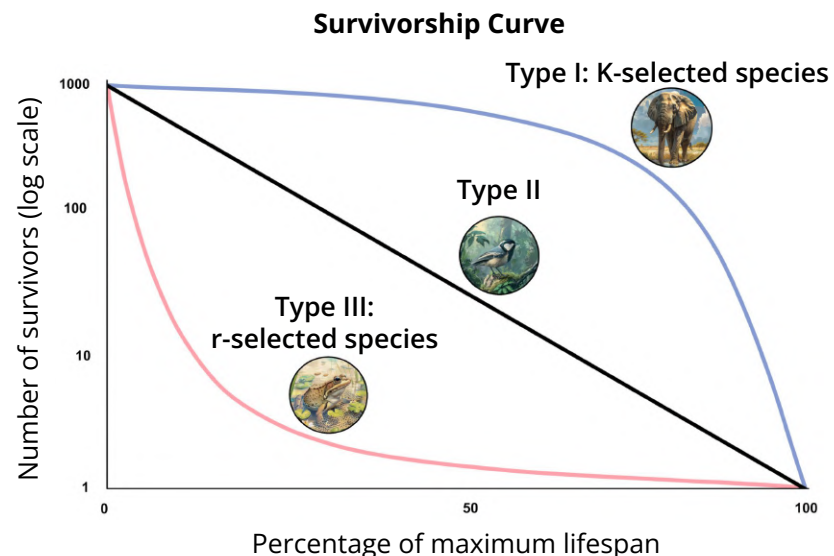
- Type I survivorship curve.
- Produce fewer offspring with higher survival rates.
- Long gestation periods with slow development and late maturation.
- High parental care and investment in offspring to ensure survival and competitive success.
- Population size is relatively stable and near the carrying capacity of the environment.
- Examples: Elephants, humans.

r-selected species

- Type III survivorship curve.
- Produce a high number of offspring with lower survival rates.
- Short gestation periods with quick development and early maturation.
- Greater number of offspring ensures some will survive, despite low parental care and unstable environmental conditions.
- Population size fluctuates wildly. Population densities well below carrying capacity of the environment.
- Examples: Insects, frogs.

Type II survivorship species

- Mortality rate is roughly constant throughout the organism's life span (survival chances do *not* significantly increase or decrease at any stage of life).
- Examples: Many birds, small mammals, and some reptiles.



Ecological Succession

Ecological succession is the predictable process where an ecological community develops and changes over time. Occurs in a new habitat or after a disturbance.

Primary succession occurs after a large disturbance (e.g., volcanic eruption) in an area that has never supported life. Begins with a **pioneer species** (e.g., lichen, fungi, algae).

The order of organisms colonizing is:
pioneer species → thin soil → vascular plants (grasses, shrubs) → larger plants (trees) → animals
Eventually a **climax community** results. A steady state is reached and a balance of species is achieved.

Secondary succession occurs on terrain that has supported life previously, and has had destruction following a disturbance (e.g., flood, fire). Follows a similar pattern as primary succession but begins with grasses and shrubs.

A **keystone species** maintains ecological balance despite low abundance (e.g., **keystone predator** hunts other animals and prevents overabundance).

Primary vs. Secondary Succession

Primary succession



Secondary succession



Eutrophication

Eutrophication is a process whereby a body of water becomes enriched with minerals and nutrients. This can lead to an **algal bloom**, a rapid *increase* in the population of algae (photosynthetic eukaryotes) and/or cyanobacteria (photosynthetic prokaryotes).

Algal blooms lead to the depletion of oxygen in the water, which can kill off aerobic organisms in the area (e.g., fish), damaging natural aquatic ecosystems.

Eutrophication is often a result of industrial nitrogen-rich fertilizer use. Nitrogen-rich fertilizer can dissolve in rain water, eventually accumulating in an aquatic biome, supporting the growth of algal blooms.

Aquatic Biomes

Aquatic biomes are the largest of Earth's biomes (~75% of Earth's surface). Photosynthetic algae contribute most of Earth's atmospheric O₂.

Divided into **freshwater biomes** (~3%) and **saltwater biomes** (~97%).

Layers of the ocean are divided based on the amount of sunlight received:

- **Euphotic zone:** Strong irradiance allows for plant survival and photosynthesis. Closest to surface. The **littoral zone** is the area of the euphotic zone where sunlight penetrates all the way to the ocean floor.
- **Disphotic zone:** Semi-irradiated with sun (not sufficient for plants). **Bioluminescent species** produce light here.
- **Aphotic zone:** No light or photosynthetic species. Some bioluminescent species. Select fish can survive off of dead matter descending to the ocean floor.

Terrestrial Biomes

Land based biomes.

Terrestrial biome	Features	Example
Tropical rainforests	- Consistent, hot temperature - Consistent, heavy rainfall - Biome with the greatest amount of diversity	 Amazon rainforest
Savannas	- High temperatures - Small amount of rainfall (seasonal) - Savannas are tropical grasslands with diffuse trees - Seasonal droughts and fires	 African safari
Temperate grasslands	- Cool winters - Hot summers - Seasonal rain (less rain than savannas) - Seasonal droughts and fires	 North American prairie
Temperate deciduous forests	- Warm summers - Cold winters - Moderate precipitation - Snows during winter - Trees shed leaves in the winter due to poor growing conditions	 Eastern North America
Chaparral	- Mild winters - Hot, dry summers - Scattered vegetation, often with small, tough leaves to conserve water - Common droughts and fires	 Mediterranean coast
Deserts	- Hot days - Cold nights - Extreme temperature fluctuations - Very little rain - Plant growth occurs after rainfall - Animals are often nocturnal and water conservative (concentrated urine) - Plants have leather leaves or spines to conserve water	 Arizona
Taiga (coniferous forest)	- Cold winters with snowfall - Warm, rainy summers - Largest terrestrial biome	 Canada & Alaska
Tundra	- Cold - Very little precipitation (mostly snow) - Ground freezes during winters - Topsoil thaws during summer, but deeper soil is permafrost (stays frozen)	 Iceland

Ch. 15: Animal Behavior

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Ethology: The study of animal behaviors, which are inherited (innate) or learned.

Types of Animal Behaviors

Inherited behaviors (innate behaviors):

1. **Instincts:** Innate behaviors that occur without thought (e.g., birds undergoing migration in response to seasonal changes).
2. **Reflexes:** Involuntary rapid responses to a stimulus. Reflex arcs are controlled by a neural circuit. There are two types:
 - **Simple reflexes** are *most* rapid. Sensory neurons travel from the stimulus to the spinal cord (CNS) and synapse directly onto motor neurons, which travel to muscle.
 - **Complex reflexes** are *slower* because sensory and motor neurons are separated by an **interneuron** and may have a connection to the brain.
3. **Fixed action patterns:** A type of instinct. Hardwired actions initiated by a specific stimulus (**releaser**). Once initiated, they continue to completion even if the stimulus is removed during the behavior. Lead to predictable and appropriate behaviors that do *not* need to be learned (e.g., goose rolling egg back into nest, male insects attacking red bellied males). Fixed action patterns are *not* flexible or adaptable.
4. **Imprinting:** An innate way that animals learn behaviors that will never be forgotten. Occurs during the **critical period** or **critical imprinting stage** (e.g., ducklings treating a moving object as their mother and following it).

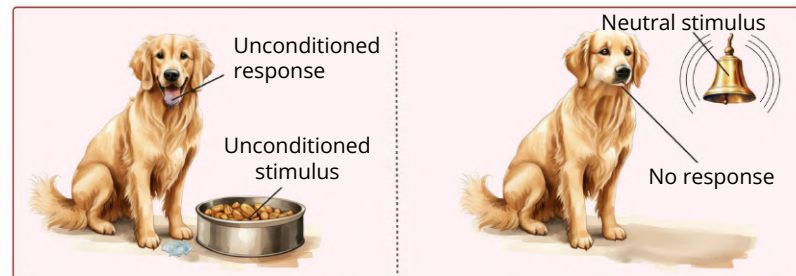
Learned behaviors:

Learned behaviors *increase* an animal's fitness, allowing it to adapt to unexpected events.

1. **Classical conditioning:** Pairing a **neutral stimulus** (elicits *no* physiological response) to an **unconditioned stimulus** (naturally elicits an **unconditioned response**). The unconditioned response is then mentally associated with the neutral stimulus, becoming a **conditioned stimulus** with a **conditioned response**.

Classical Conditioning

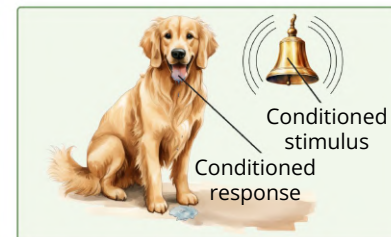
Before conditioning



During conditioning



After conditioning

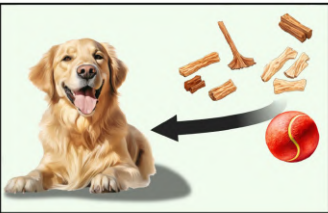
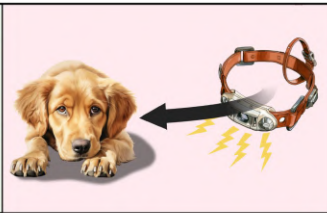


In classical conditioning:

- **Stimulus generalization:** A conditioned animal responds to stimuli that are similar to, but *not* the same as, the conditioned stimulus (e.g., a dog salivates in response to a doorbell that sounds similar to a bell).
- **Stimulus discrimination:** Differentiation between a conditioned stimulus and other similar, but different, non-conditioned stimuli.

2. **Operant conditioning:** Learning to associate a behavior with a **reward** (*increases* behavior) or a **punishment** (*decreases* behavior).

Operant Conditioning

	Reinforcement	Punishment
Positive (add)		
Negative (remove)		

3. **Associative learning:** Learning that two things are connected to each other. *Increases* stimulus response efficiency. Can be forgotten (**extinction**) or remembered via re-association (**recovery**)
4. **Non-associative learning** is a change in response to a stimulus that is *not* associated with a reinforcement or punishment. There are two types:
- **Sensitization:** As stimulus occurs more often, behavioral response *increases*.
 - **Habituation:** *Decreasing* behavioral response in response to repetitive, meaningless stimulus.

Sensitization vs. Habituation

Sensitization

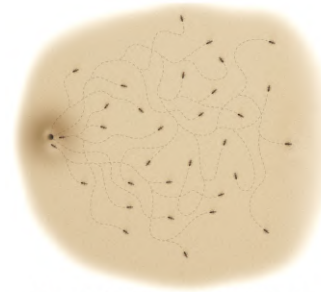


Habituation



Animal Movements

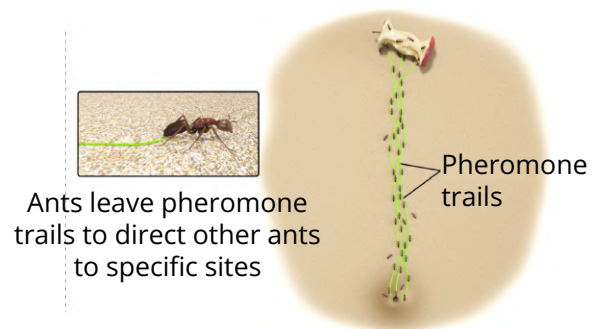
Kinesis



Random organism movement in response to a stimulus

- **Kinesis:** Changing speed in random directions - *no* target. In favorable environments → *reduce* speed; in unfavorable environments → *increase* speed (e.g., flatworm escaping when exposed to light).
- **Taxis:** Movement with a specific direction, towards (positive taxis) or away (negative taxis) from a stimulus. Light stimulus is **phototaxis**; chemical stimulus is **chemotaxis**.
 - **Pheromones**, chemical signals released by organisms to communicate with one another, can be a stimulus for chemotaxis.

Pheromone Chemotaxis



- **Migration:** Long-distance movement from one area to another due to instinct, often seasonal.

Social Behavior

Allows interaction for companionship, finding food, protection, and mating.

Cooperation: Grouping together to better achieve a goal (e.g., coordinated hunting).

Agonistic behaviors: Competing for food, territory, or mates. Include threats, aggression (often detrimental to both parties), and submission. When threatened, an animal may demonstrate **appeasement behavior** and submit, to avoid aggression and prevent injuries.

Dominance hierarchy is pecking order. **Alpha male** is the top ranked male.

Territoriality: Behaviors used to protect an animal's territory or safe space (e.g., employing watchers and defenders and using pheromones to scare off others).

Altruistic behaviors: Sacrifices made for relatives.

- **Inclusive fitness** is the sum of an animal's direct (genes animal passes on) and indirect (genes passed on by relatives) fitness. **Kin selection** describes how animals may use altruism to help their relatives survive (indirect fitness) and therefore *increase* inclusive fitness.

Mating

Sexual selection: How males and females differ in mating behavior to maximize fitness.

- Females generally invest *more* energy in the mating process. To maximize fitness, they focus on selecting high quality mates to increase likelihood of producing quality offspring.
- Males contribute *less* energy to the mating process. To maximize fitness, they focus on producing a high quantity of offspring.
- **Female choice** increases attractive traits in males.
- **Male competition** rewards strongest males with more mating opportunities.
- **Sexual dimorphism:** Males and females of same species look different (e.g., males larger than females).
- **Monogamy:** One mating partner at once.
- **Polygamy:** Multiple partners at once.
- **Semelparity:** Mate once in lifetime (multiple offspring, low survival, harsh conditions, *no* parental care).
- **Iteroparity:** Mate many times in lifetime (one offspring, high survival, dependable environment, parental care).